



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Philippine Copyright 2026

by the Authors

Mariadel B. Falculan

Edil Con L. Gorospe

Joshua D. Javier

Pearl Lorane M. Rafol

Kristine C. Rodeo

and the

Polytechnic University of the Philippines – Bansud Oriental Mindoro Campus

All rights reserved. Portions of this manuscript may be reproduced with proper referencing and due acknowledgement of the authors.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

**E-LIB: A WEB-BASED LIBRARY MANAGEMENT
INFORMATION SYSTEM (LMIS)**

A Capstone Project
Presented to the Faculty of the
Polytechnic University of the Philippines
Bansud Oriental Mindoro Campus
Poblacion, Bansud, Oriental Mindoro

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology

by

**Mariadel B. Falculan
Edil Con L. Gorospe
Joshua D. Javier
Pearl Lorane M. Rafol
Kristine C. Rodeo**

2026



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

CERTIFICATION

This capstone project, ***E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)***, prepared and submitted by MARIADEL B. FALCULAN, EDIL CON L. GOROSPE, JOSHUA D. JAVIER, PEARL LORANE M. RAFOL, and KRISTINE C. RODEO, in partial fulfillment of the requirements for the degree, BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY has been examined and recommended for Oral Examination.

Evaluation Committee

MARVIN M. SANCHO
Adviser

JULIE ANNE D. ABEL, MIT
Member

MONICA C. TOLENTINO
Member

APPROVAL

Approved by the Panel on Oral Examination on _____ with the grade of _____.

MONICA C. TOLENTINO
Chair

RICARDO F. RAMISCAL, MPA
Member

MARIA JOSIEFEL ANN R. MAAPOY, MAEd
Member

JULIE ANNE D. ABEL, MIT
Member

MARVIN M. SANCHO
Member

Accepted in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology

RICARDO F. RAMISCAL, MPA
Campus Director



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

ACKNOWLEDGMENTS

We, the proponents, would like to express our heartfelt gratitude and sincere appreciation to everyone who contributed to the successful completion of our capstone project, "E-Lib: A Web-Based Library Management Information System (LMIS)" for Polytechnic University of the Philippines - Bansud Oriental Mindoro Campus. Their unwavering support, guidance, and encouragement made this achievement possible.

First and foremost, we offer our deepest gratitude to our Almighty God for His endless love, wisdom, strength, and guidance throughout this journey. His blessings gave us the courage and perseverance to overcome every challenge we encountered during the development of this capstone project.

We extend our sincere appreciation to the Polytechnic University of the Philippines – Bansud Campus for providing us with quality education, knowledge, learning opportunities, and the necessary resources that enabled us to complete this capstone project successfully.

To our Campus Director, Assoc. Prof. Director Ricardo F. Ramiscal, we are thankful for his continuous support, encouragement, and commitment to promoting academic excellence. His leadership has inspired us to pursue innovation and achieve our academic goals.

We are also deeply grateful to our capstone adviser, Inst. Monica M. Tolentino, for her invaluable guidance, patience, encouragement, and unwavering support throughout the planning, development, implementation, and completion of this project. Her expertise, constructive suggestions, and dedication greatly contributed to the success of our study.

To our panelists, Inst. Julie Anne B. Abel, Asst. Prof. Maria Josiefel Anne R. Maapoy, and Inst. Marvin M. Sancho, we also extend our sincere appreciation for their valuable



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

comments, insightful recommendations, and constructive criticisms that helped us to improve the overall outcome of our capstone project.

Our heartfelt gratitude also goes to the current Library In-Charge, Ma'am Noeka Shalom B. Supetran, and the former Library In-Charge, Ms. Marlen Mae M. Ugno, for their cooperation, assistance, and willingness to share valuable information regarding the library's existing processes and operations. Their insights and expertise significantly contributed to the analysis, design, and development of the system.

We would also like to thank the faculty members, administrative staff and students of the Polytechnic University of the Philippines – Bansud Campus, especially those who participated in our system evaluation. Their honest responses and active participation greatly contributed to the improvement and validation of our system.

Finally, we extend our deepest appreciation to our beloved parents, families, relatives, friends, classmates, and loved ones for their unconditional love, patience, understanding, prayers, and endless encouragement. Their emotional, moral, and financial support became our source of inspiration and strength throughout this challenging yet rewarding journey.

Mariadel B. Falculan

Edil Con L. Gorospe

Joshua D. Javier

Pearl Lorane M. Rafol

Kristine C. Rodeo



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

CERTIFICATION OF ORIGINALITY

This is to certify that this capstone project, *E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)* for the degree, Bachelor of Science in Information Technology at the Polytechnic University of the Philippines - Bansud Campus embodies the result of original and scholarly work carried out by the undersigned. This capstone project does not contain words or ideas taken from published sources or written works that have been accepted as basis for the award of a degree from any other higher education institution, except where proper referencing and acknowledgment were made.

FALCULAN, MARIADEL B.

GOROSPE, EDIL CON L.

JAVIER, JOSHUA D.

RAFOL, PEARL LORANE M.

RODEO, KRISTINE C.
Proponents

Date Signed



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

ABSTRACT

Title : E-Lib: A Web-Based Library Management Information System (LMIS)
Researcher : Mariadel B. Falculan, Edil Con L. Gorospe, Joshua D. Javier, Pearl Lorane M. Rafol, and Kristine C. Rodeo
Degree : Bachelor of Science in Information Technology
Institution : Polytechnic University of the Philippines – Bansud Oriental Mindoro Campus
Year : 2026
Adviser : Marvin M. Sancho

The E-Lib: A Web-Based Library Management Information System (LMIS) was developed to improve the efficiency, security, and accessibility of library services at the Polytechnic University of the Philippines (PUP) – Bansud Campus by addressing the limitations of its manual library operations. The system provides a centralized web-based platform with role-based access for students, faculty, visitors, and administrators supporting book catalog searching, borrowing and returning transactions, user registration, visitor appointment scheduling, and report generation. It also incorporates SMS and OTP-based authentication to enhance data security and system access. The project supports Sustainable Development Goals (SDGs) 4, 9, and 16.

The system was developed using PHP, MySQL, HTML, CSS, and JavaScript with XAMPP serving as the local development environment. The study adopted the Agile Software Development Methodology, enabling iterative planning, development, testing, and refinement based on user feedback.

The developed system was evaluated by end users such as students, stakeholders, IT experts, and faculty using the ISO/IEC 25010 software quality standards, focusing on functionality, usability, reliability, compatibility, maintainability, and security. The evaluation results indicates that the respondents strongly agreed that the system complied with ISO/IEC 25010 software quality standards. The findings indicate that the system effectively streamlines



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

library operations, improves transaction accuracy, enhances accessibility, strengthens data security, and provides a reliable and user-friendly platform for managing library services. Future enhancements focus on integrating additional academic resources, strengthening security features, and expanding system accessibility to further improve library services.

Keywords: *Library Management Information System, Web-Based System, Library Automation, Appointment Scheduling, SMS and OTP Authentication, ISO/IEC 25010, Agile Software Development, Polytechnic University of the Philippines*



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

TABLE OF CONTENTS

	Page
Title Page	i
Certification of Approval Sheet.....	ii
Acknowledgement	iii
Certificate of Originality	v
Abstract.....	vi
Table of Contents.....	viii
List of Tables	x
List of Figures	xi
1 THE PROBLEM AND ITS BACKGROUND	
Introduction	1
Project Context	3
Purpose and Description	5
Objectives	6
Scope and Limitations	7
Significance of the Study	9
Definition of Terms.....	10
2 REVIEW OF RELATED LITERATURE AND STUDIES	
Foreign Literature	13
Local Literature	15
Foreign Studies.....	16
Local Studies	17
Synthesis of Related Literature	20
3 TECHNICAL BACKGROUND	
Research Approach.....	24
Methodology	25
Requirements Analysis.....	29
System Requirements	31
4 RESULTS AND DISCUSSION	
Requirements Documentation.....	36
Design of the Software/System	39
Development and Testing	42
Implementation Plan	43
Implementation Results	44
5 CONCLUSION AND RECOMMENDATIONS	
Conclusion	85
Recommendation.....	86



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

References	88
Appendices	
Appendix 1: English critic's certification	93
Appendix 2: Ethics clearance	94
Appendix 3: Certificate of originality	95
Appendix 4: Communication/letters.....	96
Appendix 5: Interview guide questions.....	97
Appendix 6: Survey questionnaire	98
Appendix 7: Systems evaluation	101
Appendix 8: Data flow diagram.....	104
Appendix 9: System flowchart	105
Appendix 10: Data dictionary.....	107
Appendix 11: Entity relationship diagram	130
Appendix 12: User's guide.....	131
Appendix 13: Program code listing	192
Appendix 14: Photo documentation.....	294
Appendix 15: Curriculum vitae.....	299



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

LIST OF TABLES

Number	Title	Page
1	Software Specification	32
2	Hardware Specification.....	32
3	Gantt Chart	41
4	Numerical Value, Statistical Limit, Verbal Interpretation.....	43
5	Functionality of the System by the End Users	45
6	Usability of the System by the End Users	47
7	Reliability of the System by the End Users	49
8	Compatibility of the System by the End Users	51
9	Maintainability of the System by the End Users	53
10	Security of the System by the End Users	55
11	Functionality of the System by the IT Experts	57
12	Usability of the System by the IT Experts	59
13	Reliability of the System by the IT Experts	61
14	Compatibility of the System by the IT Experts	63
15	Maintainability of the System by the IT Experts	65
16	Security of the System by the IT Experts	67
17	Functionality of the System by the Faculty	68
18	Usability of the System by the Faculty	70
19	Reliability of the System by the Faculty	72
20	Compatibility of the System by the Faculty	74
21	Maintainability of the System by the Faculty	76
22	Security of the System by the Faculty	78
23	Evaluation of the System by the End Users	79
25	Evaluation of the System by the IT Experts	81
26	Evaluation of the System by the Faculty	82
27	The Result of the System Evaluation	83



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

LIST OF FIGURES

Number	Title	Page
1	Scrum Model	26
2	Organizational Chart	33
3	Conceptual Framework.....	35



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Chapter 1

THE PROBLEM AND ITS BACKGROUND

This chapter introduces the project by presenting its background, purpose, and objectives. It also defines the scope and limitations, highlights the significance, and provides the definition of key terms to ensure clarity and understanding.

Introduction

In the rapidly advancing digital era, libraries remain the heart of academic institutions, serving as vital hubs for information, learning, and research. They support academic and professional growth by providing students with access to essential references, assisting faculty in teaching and scholarly work, and accommodating visitors such as alumni and learners from other institutions in their pursuit of knowledge and lifelong learning. Beyond academics, libraries also contribute to scientific, educational, social, and cultural development. However, traditional manual processes such as book borrowing, catalog searching, and visit scheduling often result in inefficiencies, congestion, and limited accessibility.

At the Polytechnic University of the Philippines – Bansud Campus, the library continues to rely on manual procedures for its daily operations. Tasks such as searching for available books, recording borrowed and returned materials, computing overdue fines, and preparing various library reports are all carried out manually by the librarian. Although this traditional system has been functional for years, it often results in delays, inconsistencies, and inefficiencies in the overall library workflow. Manual record-keeping makes it difficult to ensure accuracy, track transactions, and maintain organized documentation, especially during peak hours when many students request services at the same time. Furthermore, visitors such as alumni and students from other schools who wish to set an appointment or check whether the book they need is available must physically go to the library to inquire. This process is time-



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

2

consuming, as they have to exert time and effort to visit the library, only to discover that the material they need is not available. Such limitations highlight the inconvenience of the current manual system and further emphasize the need for a more efficient and accessible approach to library services.

The continued reliance on manual library management has created significant challenges in performing essential library operations efficiently. Manne (2023) emphasized that manual processes make tasks such as book entry, searching, issuing, returning, and inventory tracking increasingly difficult, preventing librarians and users from accessing real-time information on books availability and limiting the library's ability to analyze borrowing patterns for informed collection development. Similarly, Acasamoso et al. (2021) pointed out that managing libraries with large collections becomes more challenging due to the increasing demand for borrowed materials, particularly during examination and research periods. The study developed a Library Management System to address these issues, and its evaluation by students, faculty, and library staff results indicates that its effectiveness in improving library management. As per Aguirre et al. (2024), the traditional library systems require librarians to manually keep track of inventories, borrowing transactions, due dates, and fines, making it challenging for staff and students to find books due to the lack of an organized catalog. These manual processes are time-consuming, prone to human error, and reduce the overall efficiency of library operations.

A previous study entitled "LibTrack: PUP Bansud Oriental Mindoro Campus Library Information System (LIS)" was developed at Polytechnic University of the Philippines – Bansud Campus to help librarians, students and faculty in managing library process library transactions. However, even though the system improved some manual processes, it did not include an appointment scheduling for visitors like alumni and students from other schools and also did not incorporate enhanced security features and expanded administrative functions



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

3

needed to support the current operational needs of the library. These limitations show the necessity of a more comprehensive and accessible web-based library management system.

To address these limitations, the proponents developed “E-Lib: A Web-Based Library Management Information System (LMIS)” to enhance the efficiency, security, and accessibility of library services at the Polytechnic University of the Philippines Bansud Campus. The system was designed to streamline library operations by providing an integrated platform for managing books, users, and visitors appointments. It allowed students, faculty, and visitor appointments. It allowed students, faculty, visitors to search and view available books, access library policies, view book details, and conveniently use library services through a user-friendly web interface. The system also introduced a secure account registration process for users, with OTP-based email and SMS verification for password recovery to account security. In addition, it enabled visitors, including alumni and students from other institutions to schedule library appointments online. For administrators, the system provided a secure OTP-based login process and a centralized dashboard for efficiently managing users, books, and appointment schedules. Furthermore, it generated reports on book inventory, user records, and appointment schedules to support efficient library management and informed decision-making. The system was developed to promote more efficient, accessible, and secure library services while complying with the ISO/IEC 25010 software quality standards.

Project Context

The E-Lib: A Web-Based Library Management Information System (LMIS) was developed to address the limitations of the traditional manual processes used at the Polytechnic University of the Philippines – Bansud Campus, where tasks such as searching for books, recording borrowed and returned materials, computing fines, and preparing reports are done by hand, resulting in delays, inconsistencies, and a tedious workflow for librarians



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

4

and users. Visitors like alumni and students from other schools must also physically visit the library to inquire about book availability or request appointments, which is time-consuming and inconvenient. To resolve these challenges, the system streamlines library operations by providing an integrated web-based platform that supports real-time transactions, improved data management, and enhanced user accessibility. The system includes role-based modules for students, faculty, visitors, and administrators, each designed with specific functionalities aligned with their needs.

Upon opening the system, the landing page appears with login button at the top left corner. Clicking the Login button opens a form where users can select their account type student, faculty, visitor, or administrator and log in using their respective credentials. For administrators, a verification code which is OTP is sent to their email and SMS for authentication. Users without an account must register by providing their credentials in a form divided into Account Information and Personal Details, with fields such as user type, email address, ID, password, and course, while the system automatically formats IDs based on user type and computes the user's age once the date of birth is entered. Fields marked with an asterisk (*) are required. After registration, users are redirected to the login page where they need to login.

For students and faculty users, the system provides access to a centralized dashboard featuring notifications, book catalog, bookmarks, borrowed books, returned books, and unsettled records. Through the catalog, users can search and filter available books by category, alphabetical order, and Library of Congress Classification (LCC). Each book record includes detailed information such as title, author, description, edition, volume, condition, and availability status. Users may also bookmark books or submit borrowing requests, which are subject to administrator approval and availability rules. The system automatically tracks due dates and applies fines for overdue returns to ensure proper accountability.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

5

For visitors, the system provides controlled access through an appointment-based module. Visitors can request library appointments by selecting available dates, time slots, and purposes of visit through a calendar interface. The system supports a first-come, first-served scheduling policy with a maximum capacity limit, where fully booked or restricted time slots are automatically disabled. Visitors can also view their notifications, pending appointments, and appointment history, allowing them to track the status of their requests and previous transactions.

Lastly, for administrator, the system provides a comprehensive management dashboard that includes admin profile, pending users, user management, notifications, book management, borrowing and return monitoring, replacements, non-returned and paid records, archived data, appointment scheduling, and report analytic. The administrator is responsible for approving or rejecting user registrations, managing book inventories, handling borrowing and return requests, processing replacements and payments, and overseeing visitor appointments. The system also allows the administrator to modify or disable appointment schedules when necessary, such as during emergencies or system maintenance.

Purpose and Description

Purpose. The purpose of this project is to design and develop a system for the Polytechnic University of the Philippines – Bansud Campus to address the inefficiencies of its existing manual library processes. These include challenges in managing book records, user accounts, borrowing and returning transactions, fine computation, and visitor handling, which often lead to delays, inconsistencies, and difficulty in tracking library data. By integrating a computerized system with centralized functions such as book catalog management, user registration, transaction monitoring, and appointment scheduling, the project aims to improve the efficiency, accuracy, and accessibility of library services for students, faculty, and visitors.



Description. The E-Lib: A Web-Based Library Management Information System (LMIS) provides a centralized web-based platform that enables users to conveniently access and manage library services online. It supports role-based access for students, faculty, visitors, and administrators, allowing each user type to perform specific functions such as browsing the catalog, managing bookmarks, tracking borrowed and returned books, and scheduling library appointments. The system also ensures secure administration through SMS and OTP-based email authentication for administrators, strengthening data protection and access control. In addition, the visitor appointment module allows users to check availability, select schedules, and request library visits online, improving convenience and reducing the need for physical inquiries.

Objectives

The main objective of E-Lib: A Web-Based Library Management Information System (LMIS) was to develop a system that enhances the efficiency, security, and accessibility of library services at Polytechnic University of the Philippines – Bansud Campus.

The following are the specific objectives that the system accomplish:

1. To understand how the existing manual process of managing the library operates and to identify its limitations in maintaining efficiency and accessibility;
2. To determine the problems encountered in the current manual process that led to errors and inconsistencies in recording and monitoring library transactions;
3. To develop a web-based system that would:
 - 3.1 allow users to search and view available books in the library;
 - 3.2 provide a secure account registration for students, faculty, and visitors, utilizing OTP based email verification for the forgot password process to enhance account security;



- 3.3 allow students and faculty to conveniently access library services, such as viewing the library policies, searching for books, viewing book details, and borrowing them;
- 3.4 enable visitors to schedule appointments for library access;
- 3.5 provide administrators with a secure OTP based login process and an efficient dashboard for managing users, books, and appointments; and
- 3.6 generate reports on book inventory, user records, and appointment schedules.
- 4. To contribute to the United Nations Sustainable Development Goals (SDGs) by:
 - 4.1 supporting SDG 4 – Quality Education by improving equitable access to library resources and promoting lifelong learning;
 - 4.2 supporting SDG 9 – Industry, Innovation, and Infrastructure by modernizing library operations through technological innovation and real-time data analytics; and
 - 4.3 supporting SDG 16 – Peace, Justice and Strong Institutions by fostering transparency, accountability, and equitable access to information through a secure, well-managed, and data-driven library system.
- 5. To ensure the system complies with ISO/IEC 25010 software quality standards including functionality, usability, reliability, compatibility, maintainability, and security.

Scope and Limitations

Scope. The E-Lib: A Web-Based Library Management Information System (LMIS) is developed to improve the efficiency, organization, and accessibility of library services at the Polytechnic University of the Philippines – Bansud Campus. The system provides a centralized platform for managing essential library operations such as book cataloging, searching, borrowing, returning, replacement, payment tracking, and appointment scheduling. It also supports different user roles, including students, faculty, visitors, and administrators, each with dedicated modules tailored to their specific needs and access levels. The system enhances



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

8

security and usability through a structured authentication process, including secure login through SMS and OTP verification. It features a visitor appointment system that allows users to schedule library visits based on available time slots and approved requests, while administrators manage user accounts, book records, transactions, overdue monitoring, and archived data. In addition, the system provides a comprehensive dashboard and analytics module that presents borrowing trends, user statistics, book distribution, and library utilization, supporting better decision-making and efficient library management.

Limitations. However, the system has certain limitations. One of the main limitations is its dependence on a stable internet connection and email service. Since user registration and login require email verification through a One-Time Password (OTP), interruptions in internet connectivity or email service may prevent students, faculty, visitors, and librarians from accessing the system and performing transactions. In addition, the system is designed solely for the Polytechnic University of the Philippines – Bansud Campus Library. It manages library resources, borrowing and returning transactions, visitor appointments, user accounts, and report generation only within the campus library. The system is not integrated with other library systems, external databases, or inter-library resource-sharing platforms. Furthermore, the system relies on regular maintenance of the web application, server, and database to ensure continuous operation. Failure to perform routine updates, backups, and technical maintenance may affect system availability and performance. As a result, the system requires ongoing technical support from the designated administrator to maintain its reliability, security, and overall functionality.

**Significance of the Study**

The outcome of this project is beneficial to the following:

Librarian. This project will help the librarian to manage and organize the library more efficiently. It reduces manual workload, ensures accurate record-keeping, and improves decision-making in library operations.

Students. This system will benefit students by giving them secure access to their accounts, where they can search and view detailed book information, request borrowing for a specific number of days, and receive feedback or notifications regarding their transactions. This helps improve accessibility, saves time, and enhances their academic and research experience.

Faculty. This system will benefit the faculty members by having access to a reliable system that allows them to search for resources, request book borrowing, and receive timely updates on their transactions. This ensures that their academic and research needs are met efficiently without manual processes.

Visitors. This system will benefit visitors such as alumni and researchers from other universities by providing a reliable platform to schedule appointments. This ensures a smooth and organized library experience without relying on manual logbooks.

PUP Bansud Oriental Mindoro Campus. This project will help to modernize the library through the integration of book transaction management and visitor appointment scheduling. It promotes efficient library services, reduces administrative workload, and supports the university's commitment to digital transformation and academic excellence.

Future Researchers. This study will serve as a reference for future researchers in developing or enhancing similar systems. Its objectives, design, and features may inspire improvements in digital solutions for library and information management.

**Definition of Terms**

To enhance the understanding, the following terms will be used by the proponents were and operationally conceptually defined by the readers of this project.

Appointment Management. The proponents will use this term to refer as a feature of the project that allows librarians to oversee and manage books, users, and appointments for library visits in an organized manner.

Archived Books. The proponents will use this term to refer as a feature of the project that stores records of outdated, lost, or removed books from the active catalog for tracking and reference purposes.

Bookmarks. The proponents will use this term to refer as a feature of the project that allows users to mark or save books of interest for quick access in future sessions.

Borrowed Books. The proponents will use this term to refer as a feature of the project that records and monitors all books borrowed by users, including due dates and borrowing history.

Catalog. The proponents will use this term to refer as a feature of the project that provides users with a searchable collection of all available books in the library.

Capacity Monitor. The proponents will use this term to refer as a feature that tracks and displays the current number of users or appointments, helping ensure the library does not exceed its maximum allowed capacity.

Dashboard. The proponents will use this term to refer as a feature of the project that serves as the main interface displaying an overview of system activities, notifications, and library updates.

E-Lib. The proponents will use this term to refer as the official name of the proposed system, which stands for a web-based Library Management Information System designed for PUP Bansud Campus.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

11

ISO/IEC 25010. The proponents will use this term to refer to an international standard used to evaluate software based on factors like usability, functionality, reliability, and maintainability.

LCC (Library Classification Code). The proponents will use this term to refer as a unique classification or identifier assigned to each book based on its subject category, following library standards.

Library ID. The proponents will use this term to refer as a unique identification number assigned to every registered user for authentication and borrowing purposes.

Library Management Information System (LMIS). The proponents will use this term to refer to a web-based application designed to manage library operations such as book searching, borrowing, returning, and record monitoring in a more systematic and efficient manner.

One-Time Password (OTP). The proponents will use this term to refer to the temporary security code sent via Gmail that verifies a user's identity during registration or login.

Override Control Panel. The proponents use this term to refer as a system feature that allows administrators to manage or modify user access, appointments, or other system settings as needed.

Pending Users. The proponents will use this term to refer as a feature of the project that stores and displays registration requests awaiting librarian approval before being granted full system access.

Replaced Books. The proponents will use this term to refer as a feature of the project that records books that were substituted due to loss, damage, or updates in library resources.



Report and Analytics. The proponents will use this term to refer as a feature of the project that generates summaries and visual insights regarding library operations, book circulation, and user activities.

Returned Books. The proponents will use this term to refer as a feature of the project that tracks books that have been successfully returned by users, updating their borrowing status.

User Management. The proponents will use this term to refer as a feature of the project that allows administrators to update or manage user accounts and their access rights.

Verification. The proponents will use this term to refer to the process of confirming a user's identity and credentials during registration or login, which may include the use of OTP via Email for added security.

Visitor. The proponents will use this term to refer to alumni and other external individuals, such as students or researchers from different universities, who are allowed to schedule appointments.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

One of the significance step is research project is conducting a literature review. Through this chapter, the readers can familiarize to the study of researcher. The related and previous studies presented in this chapter will give the readers and future researcher, an idea of the proposed system of the proponent. The proponent selected and reviewed these studies and previous works to provide information to the readers.

This chapter presents literatures and studies done by experts that have significant bearing to this project.

Foreign Literature

According to Begawadi et al. (2025), Library Management Systems (LMS) are essential to increasing the efficacy and efficiency of library operations. The study uses a literature review from scholarly databases like Scopus, Web of Science, Taylor & Francis, and Google Scholar to investigate the role of LMS in contemporary library functionality. Key functions like cataloging, circulation management, acquisitions, digital resource integration, and data-driven decision-making are found to be supported by LMS. Major implementation obstacles are also identified by the study, including organizational resistance, technical limitations, and human factors like inadequate staff training. The study highlights the evolving nature of LMS, shaped by technologies like artificial intelligence (AI), radio-frequency identification (RFID), and big data analytics, which are increasingly important for the future of knowledge management. To address these issues, the authors suggest thorough needs assessments, continuous professional development, flexible technology solutions, user-centered design, strong data security, and ongoing evaluation.



Similarly, Rathour et al. (2024) explained that libraries must adapt to the growing demand for digital information by implementing modern Library Management Systems that integrate emerging technologies such as artificial intelligence (AI) and machine learning (ML). The study proposed a comprehensive LMS framework that automates library operations, improves resource accessibility, personalizes user experiences, and enhances organizational efficiency through intelligent technologies.

According to Katariya (2023), the Library Database Management System (DBMS) serves as the foundation of modern library operations by enabling the efficient organization, storage, retrieval, and management of large collections of information resources. Through a systematic review of existing literature, the author discussed current developments, challenges, and emerging trends in library database systems while identifying areas that require further research. The review emphasized the importance of well-designed database systems in ensuring reliable and organized library services.

Supporting this concept, Adebayo (2019) discussed the evolution of Integrated Library Systems (ILS) and highlighted the application of topic modeling to improve information retrieval within library management systems. The author explained that topic modeling enhances text classification and search capabilities, allowing users to locate relevant resources more efficiently and make informed decisions when selecting library materials. These developments contribute to more effective knowledge organization and resource discovery in modern libraries.

Moruf and Abu (2024) reviewed the development of next-generation web-based and cloud-based Library Management Systems and emphasized the importance of compatibility, accessibility, scalability, and interoperability in modern library environments. The authors explained that web-based systems enable users to access library services anytime and from various devices, while cloud-based platforms support seamless integration with emerging



technologies and future system enhancements. These characteristics make web-based and cloud-based LMS suitable for educational institutions seeking flexible, efficient, and sustainable library management solutions.

Local Literature

According to official documentation on national information systems, the Philippine eLib Project was established to provide Filipinos with access to digitized information resources through collaboration among major government libraries and educational bodies, including the National Library of the Philippines (NLP), the University of the Philippines (UP), the Department of Science and Technology (DOST), the Department of Agriculture (DA), and the Commission on Higher Education (CHED). The project promotes resource sharing, digital preservation, and online access to library collections, allowing users to retrieve information more efficiently regardless of location (Philippine eLib Project, 2005). This national initiative serves as a foundational benchmark for local web-based library management systems. It demonstrates how integrating digitized catalogs, centralized database management, and web accessibility eliminates geographical barriers for academic research. By transitioning from traditional, localized cataloging to interconnected web platforms, academic institutions can optimize resource allocation, enhance data retrieval speeds, and support continuous learning through online public access systems.

The study by Lagas and Isip (2023), characterizes the maturity and sustainability of digitization in Philippine academic libraries and reports the barriers and recommendations you described. Using a descriptive-comparative design, the authors verified 12 institutions with digitization programs and conducted interviews with five of them. They discovered that the majority of digitization started from external triggers rather than integrated strategies, undermining sustainability, and that staff, space, equipment, and funding were the main



obstacles. In addition to urging libraries to align digitization with short-term and long-term goals, the paper suggests more consulting and training in digitization, teaching digitization ICT in LIS programs, digitization service providers and communities of practice, and consortia for sharing facilities and expertise.

Foreign Studies

Bhati et al. (2024) developed an advanced web-based Library Management System (LMS) to streamline daily library operations, including member registration, inventory management, book transactions, and administrative functions. The system replaced traditional manual procedures with an automated framework that improved data accuracy, operational efficiency, and user experience. It featured modules for member records, book maintenance, issue and return tracking, lost book management, and comprehensive reporting for library administrators. The researchers also incorporated future-ready features such as One-Time Password (OTP) authentication, SMS or email reminders for overdue books, and smart card integration to enhance security, scalability, and resource management.

Similarly, Rahman et al. (2023) developed a web-based Library Management System that automated essential library services, including cataloging, borrowing, returning, renewals, and user account management. Their findings showed that centralizing these processes reduced manual errors while improving the consistency and efficiency of library operations. Likewise, Tsega Weldu Araya (2020) designed a web-based Library Management System that enabled users and librarians to remotely access books, magazines, newspapers, and other learning resources. The study demonstrated that web-based platforms increase accessibility while reducing reliance on traditional paper-based records.

Martirosyan et al. (2022) developed a Library Management System that provides users with a convenient and user-friendly platform through both web and mobile applications. The



system incorporated QR code technology to simplify access to books and library services, making transactions faster and more convenient. The researchers emphasized that an intuitive interface encourages greater user engagement and improves overall satisfaction with library services.

Supporting this idea, Patidar et al. (2024) developed a Library Management System using modern web technologies, including MongoDB, React, Node.js, JavaScript, HTML, and CSS. Their study highlighted that a user-friendly interface enables librarians and users to manage library resources more efficiently while improving accessibility and operational performance. In the same context, Ayo et al. (2023) developed an Integrated Library Management System that incorporated user-centered features such as chatbot assistance, notifications, dashboards, feedback mechanisms, and online library instructions. These features enhanced user interaction, improved accessibility, and increased overall user satisfaction while supporting efficient library services.

Singh (2020) developed a Smart Library Management System that utilized QR code technology to automate library transactions and reduce manual intervention. The study found that automated identification and secure transaction processing improved the accuracy, efficiency, and security of borrowing and returning activities. The integration of smart technologies demonstrated the potential of modern library systems to provide faster, more reliable, and technology-driven services for both librarians and users.

Local Studies

Aguila and Balba (2025) claimed that in order to increase the effectiveness of library operations, they created an RFID-based library management system for the Lyceum of the Philippines University–Laguna (LPUL). Problems like manual attendance recording, ineffective inventory control, and delays in book borrowing and return were all resolved by the



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

18

system. While integrated RFID and barcode technologies allowed for real-time inventory updates and expedited circulation procedures, the system automatically recorded student attendance upon entry and exit by using RFID-enabled identification cards. The system also included a dynamic dashboard that gave administrators operational statistics and visual reports, as well as an Online Public Access Catalog (OPAC) that let users search library resources by title, author, or subject. The system, which was created with Laravel, Bootstrap, MySQL, and XAMPP, had a 100% success rate in terms of functionality and user satisfaction. It was assessed using the ISO/IEC 25010 software quality standards. The results showed that the SmartLibrary system successfully updated library operations by improving accuracy, efficiency, and the general user experience for both students and library employees.

Similarly, Acasamoso (2021) developed the Polytechnic University of the Philippines Ragay Branch Library Management System (PUPRBLMS) to improve the efficiency of library operations and address the difficulties experienced by librarians in managing a growing collection of books and library transactions. The study acknowledged that as the number of books and borrowers increases, manual library management becomes more difficult, especially during exam and research seasons when demand for library resources is at its peak. Common issues found included inefficiencies in tracking library resources, delays in recording transactions, and challenges in keeping an eye on borrowed books. The suggested system automated key library operations, such as the administration of book records, borrower data, and borrowing and returning transactions, in order to solve these problems. The system was created to lessen manual labor, increase the accuracy of library records, and give librarians a more effective means of keeping an eye on the availability and distribution of library materials. The system reduced human error and improved the organization and dependability of library services by computerizing these procedures.



The study by Obsanga and Enierga (2021) developed a web-based automated library management system to improve the efficiency, supervision, and accessibility of library operations in public libraries in the Philippines. The researchers discovered that manual library procedures frequently lead to sluggish transactions, erroneous record-keeping, and challenges in keeping an eye on user activity and library resources. They used PHP, JavaScript, Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), SQL, and the Agile software development methodology to create the system in order to overcome these obstacles. The system automated book cataloging, borrower registration, borrowing and returning transactions, inventory management, and report generation. Additionally, it used QR code technology to enable quicker book and user identification, increasing the speed and precision of library transactions. Additionally, the system's centralized data management and real-time access to library records allowed librarians to oversee library operations more effectively.

Pinto (2024) developed a computerized Library Management System using the ADDIE instructional model and the Waterfall software development model. The study evaluated the system based on ISO/IEC 25010 software quality standards and reported high maintainability, indicating that the system could easily be modified, updated, and improved without affecting its core functions.



Synthesis of Related Literature

Study	Key Findings	Identified Gaps / Challenges	ELib Development Response
Begawadi et al. (2025)	Library Management Systems improve cataloging, circulation, acquisitions, digital resource management, and decision-making.	Implementation is hindered by organizational resistance, technical limitations, and inadequate staff training despite recommending user-centered design.	E-Lib adopts a user-friendly, role-based interface that simplifies library operations while improving accessibility and ease of use for students, faculty, visitors and librarians.
Rathour et al. (2024)	Modern LMS integrates Artificial Intelligence (AI) and Machine Learning (ML) to automate operations and improve library efficiency.	The proposed framework focuses on advanced intelligent technologies that may not be practical for institutions with limited resources.	E-Lib provides practical web-based automation without requiring complex AI technologies, making it suitable for small academic institutions.
Katariya (2023)	Database Management Systems are fundamental for organizing, storing, and retrieving library information efficiently.	The review focuses primarily on database technologies with limited discussion on user-friendly web implementation.	E-Lib integrates a centralized MySQL database with an intuitive web-based interface for efficient and real-time library management.
Adebayo (2019)	Topic modeling enhances information retrieval, text classification, and resource discovery in library systems.	The approach emphasizes information retrieval but provides limited discussion on complete library management functions.	E-Lib incorporates advanced search, filtering, and organized catalog features while supporting complete library management operations.
Moruf and Abu (2024)	Web-based and cloud-based LMS improve accessibility, scalability, compatibility, and interoperability.	The study focuses on general web and cloud architectures without discussing practical security mechanisms for educational institutions.	E-Lib provides secure web-based access through OTP authentication while supporting remote accessibility and centralized management.



Bhati et al. (2024)	Developed a web-based LMS that automates registration, inventory, circulation, reporting, and administrative functions.	Smart card integration and future enhancements were proposed but not fully implemented.	E-Lib implements OTP authentication, automated notifications, and modular library management features while allowing future technology integration.
Rahman et al. (2023)	Automated cataloging, borrowing, returning, renewals, and user account management improved operational efficiency.	Limited emphasis on user role management and access control.	E-Lib implements role-based access for administrators, librarians, students, faculty, and visitors to improve security and system organization.
Araya (2020)	Web-based LMS enables remote access to books and other learning resources.	The system primarily focuses on resource accessibility with limited interactive library services.	E-Lib extends remote accessibility by providing online reservations, notifications, real-time book availability, and enhanced search capabilities.
Martirosyan et al. (2022)	Web and mobile LMS with QR code technology improves accessibility and user convenience.	The study emphasizes accessibility rather than comprehensive library management.	E-Lib provides complete library management functions while supporting future QR code integration.
Patidar et al. (2024)	Modern web technologies create an efficient and user-friendly Library Management System.	Limited implementation of automated reminders and analytical reporting.	E-Lib incorporates automated notifications, overdue reminders, and report analytics to enhance library management.
Ayo et al. (2023)	Integrated LMS includes dashboards, chatbot assistance, notifications, feedback, and online instructions.	The system contains numerous advanced features that may be excessive for smaller institutions.	E-Lib adopts only essential notification and reporting features to provide a simpler, cost-effective, and practical solution.



Singh (2020)	QR code technology automates circulation and improves transaction accuracy and security.	Automation is mainly limited to circulation activities.	E-Lib expands automation beyond circulation by including cataloging, reservations, notifications, inventory management, and user administration.
Philippine eLib Project (2005)	Promotes digitized collections, centralized databases, resource sharing, and online access among participating institutions.	Primarily focuses on digital resource sharing rather than day-to-day library transaction management.	E-Lib complements digital accessibility by automating circulation, reservations, reporting, and library administration within a campus environment.
Lagas and Isip (2023)	Identified staffing, funding, equipment, and sustainability as major challenges in Philippine library digitization.	The study provides recommendations but does not develop an actual library management system.	E-Lib minimizes manual workload through an affordable and practical web-based automation system suitable for educational institutions with limited resources.
Aguila and Balba (2025)	RFID-based LMS improved attendance monitoring, inventory management, circulation, and reporting.	The system depends on RFID infrastructure and specialized hardware.	E-Lib focuses on affordable web-based automation while allowing future integration of RFID technologies.
Acasamoso (2021)	Automated library operations improved record accuracy, transaction efficiency, and monitoring of borrowed materials.	The system mainly automates core library functions with limited interactive services.	E-Lib expands functionality by providing online reservations, automated notifications, OTP authentication, and an improved user experience.
Obsanga and Enierga (2021)	Automated cataloging, circulation, QR code integration, inventory management, and centralized records.	Limited support for fully remote access and online services.	E-Lib enables secure online access for students, faculty, librarians, and visitors while maintaining centralized library management.



Pinto (2024)	Computerized LMS achieved high maintainability based on ISO/IEC 25010 software quality standards.	The study places less emphasis on interactive web interfaces and enhanced user engagement.	E-Lib improves usability through interactive web modules, OTP authentication, real-time notifications, and user-centered system features.
---------------------	---	--	---



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Chapter 3

TECHNICAL BACKGROUND

System design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. This chapter discusses the research method used, software development methodology, different requirement analysis tools, and systems requirements of the software product.

Research Approach

The study used a descriptive quantitative research approach. Lim (2025) defines quantitative research as a methodical approach to studying phenomena by gathering and statistically analyzing numerical data. To guarantee accurate and legitimate results, it places a strong emphasis on objectivity, standardized procedures, and the use of structured research tools. Descriptive quantitative research, one of its many designs, is suitable for studies that use statistical measures like frequencies, percentages, means, and standard deviations to summarize data in order to describe the traits, attitudes, or circumstances of a population. With this method, researchers can accurately and impartially describe the variables they are studying.

This is related to E-Lib: A Web-Based Library Management Information System, since the study is evaluated based on the perceptions of the students, end users such as librarian and stakeholders, IT experts, and faculty members using survey questionnaires and statistical measures such as weighted mean. The descriptive quantitative research approach is appropriate because it enables the researchers to quantify respondents' evaluations and present the findings accurately and systematically.

**Methodology**

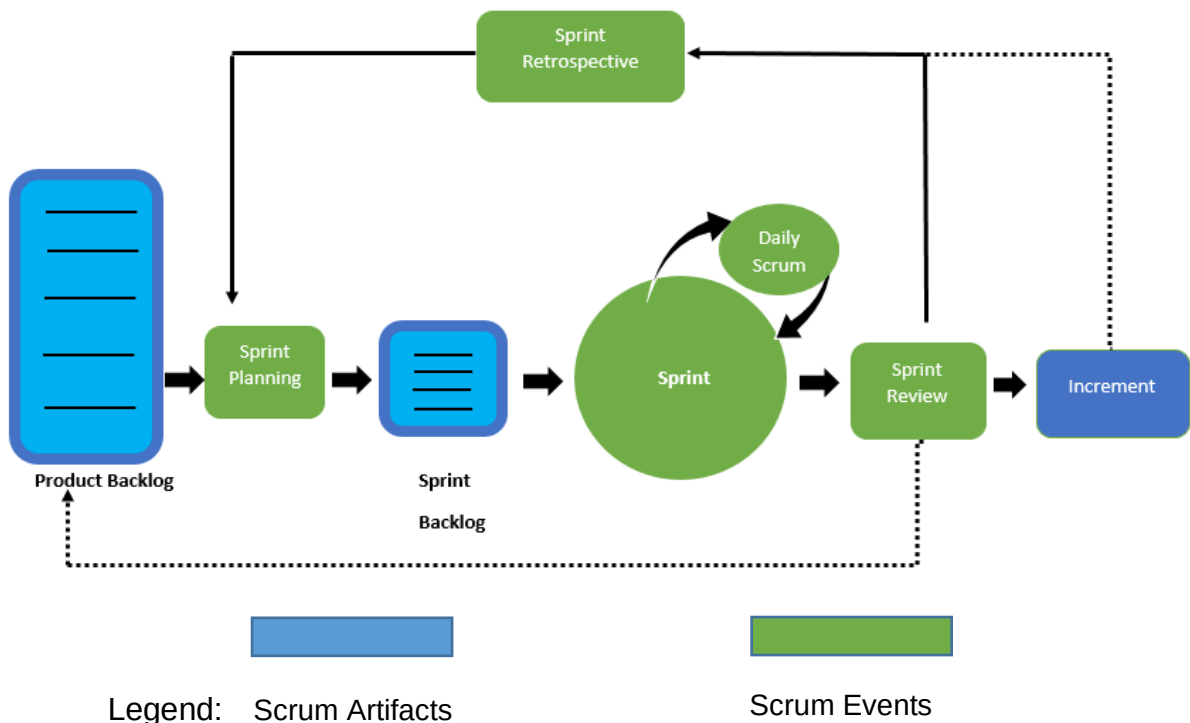
The proponents used Scrum methodology, which is a framework under the Agile methodology. According to Dugbartey and Kehinde (2025), Agile methodologies transform project delivery by emphasizing flexibility, collaboration, and rapid adaptation to change. Unlike traditional project management approaches that rely heavily on rigid planning from the beginning, Agile focuses on iterative development, continuous feedback, and active stakeholder engagement, allowing development teams to respond effectively to evolving requirements. Scrum supports this Agile process by breaking big and complex tasks into smaller and more manageable ones called sprints. The scope and initial requirements are identified at the start, but the number, length, and content of each sprint may be adjusted and improved throughout the development cycle based on team performance and stakeholder input. This approach helps teams deliver solutions incrementally while maintaining clarity, efficiency, and focus on value. These sprints are time-boxed and assigned to team members with the appropriate skills to complete them successfully. During each sprint, the team collaborates, shares updates, checks progress, evaluates what needs improvement, and continuously refines their workflow, ensuring the project remains adaptable and aligned with its goals.

The Agile approach followed by the proponents includes structured phases that guide the development process. The first phase is Sprint Planning, where the team reviews requirements, prioritizes tasks, and sets clear goals for the upcoming iteration. The second phase is the Sprint Execution, where the developers perform coding, testing, and integration activities while maintaining open communication to resolve challenges quickly. The third phase is the Sprint Review, during which the completed features are presented to stakeholders for evaluation and feedback, ensuring alignment with project objectives. The final phase is the Sprint Retrospective, where the team reflects on what worked well and what needs



improvement, proposing adjustments for future sprints. After this reflection, the cycle repeats with the next sprint, allowing continuous improvement, adaptability, and efficient development throughout the project.

Figure 1. Scrum Model



Scrum Events

Sprint Planning. In this phase, the Scrum Team conducted an interview with the Head Librarian of PUP Bansud Campus to gather detailed information regarding project requirements and desired functionalities of the proposed E-Lib: A Web-Based Library Management Information System (LMIS). The scrum team asked the librarian on the existing procedures and the challenges encountered in managing library operations when it comes in managing books. In addition, the proponents also consulted the previous fourth-year BSIT student who developed a similar study which enabled the team to determine the features that



required improvement. Based on the information collected, the team develop a product backlog containing the system's prioritized features and functionalities a roadmap outlining the system features to be developed. Roles and responsibilities were also distributed among scrum members according to their expertise.

Sprint. During the sprint phase, the team focused on developing the prioritized features listed in the product backlog. The development was divided into manageable modules, including user authentication, book catalog, user management, book borrowing and returning, appointment scheduling, overdue monitoring, notification, and report generation. Simultaneously, the technical writer updated the project documentation, and the system analyst also revised the diagrams when necessary. The incremental development approach ensured that every completed functionality contributed to the overall progress of the system while allowing contributions improvement throughout the capstone project.

Sprint Review. At the end of every Sprint, the Scrum Team conducted a Sprint Review to evaluate the completed system functionalities. The developed modules were demonstrated to the Librarian-In-Charge and other intended users to verify whether they satisfied the specified requirements and addressed the identified problems within the library. The feedback was collected regarding the functionality, usability, reliability, compatibility, maintainability, and security of the system. The scrum team, specifically the programmers, also performed a comprehensive testing to identify software defects, inconsistencies, and missing functionalities. The necessary revisions, improvements, and feature enhancements were documented and added to the product backlog to be addressed in the succeeding Sprint.

Sprint Retrospective. The Sprint Retrospective served as an opportunity for the Scrum Team to reflect on the overall Sprint experience after completing the Sprint Review. During this meeting, the scrum team discussed the accomplishments, challenges encountered, and lessons learned throughout the development process. The team also



reviewed unresolved issues, updated the project backlog, and established new priorities and action plans for the succeeding Sprint. By continuously assessing their performance and implementing process improvements, the scrum team enhanced both the quality of the system and the efficiency of the development process, ensuring the successful completion of the E-Lib: A Web-Based Library Management Information System (LMIS).

Scrum Artifacts

These artifacts provide the key information required during the development process.

Product Vision. This represents the long-term direction and ultimate goal of the project. Each Scrum member is assigned specific responsibilities, and all team members collaborate to ensure that the overall project objectives are achieved.

Product Backlog. The product backlog is a comprehensive list of all project requirements provided by the product owner. Items are organized based on priority, as determined by the Scrum Master or Project Manager, ensuring that the most critical features are developed first.

Sprint Backlog. This backlog includes selected items from the product backlog that are planned for implementation during the Sprint. It outlines a clear time frame and provides a real-time visual plan for developers. It also describes how the team will deliver system increments during each iteration.

Increment. An increment refers to the sum of all completed or updated work during a Sprint, including improvements upon previous versions. Each increment is marked as "done" before moving forward with new development, ensuring that every update adds value to the product.

**Requirement Analysis**

The proponents used different data gathering techniques to collect the necessary information for the development of E-Lib: A Web-Based Library Management Information System (LMIS). By identifying the library's current procedures, difficulties, and needs, these methods made it easier to comprehend the needs of users and the features that should be included in the system.

Observation. Observation is a data gathering technique that involves carefully examining the actual processes and activities performed within the library. The proponents conducted an observation of the library's daily operations to understand how books, users, and library transactions are currently managed.

The proponents discovered during the observation that a number of library procedures are still carried out by hand, such as keeping track of overdue materials, recording borrowed and returned books, verifying book availability, and creating inventory reports. Additionally, the proponents noted that librarians spend a lot of time manually updating logbooks and looking for book records, particularly during busy times when a lot of students visit the library. Additionally, as the collection expands, it becomes more challenging to find available books and keep track of borrowed materials. These findings demonstrated the necessity of an automated system that can effectively manage library resources, reduce human error, and raise the standard of library services in general.

Interview. Interview is one of the primary methods of collecting detailed information directly from individuals involved in the library's operations. The proponents prepared an interview guide and conducted interviews with the librarian and designated library personnel to gather information regarding the current library management process and the problems they encounter.



Based on the interviews, the librarian identified several challenges in the existing system, including time-consuming book borrowing and returning procedures, manual inventory management, difficulty in tracking overdue books, and delays in generating reports. The librarian also emphasized the need for a centralized database where book records, borrower information, and transaction histories can be securely stored and easily retrieved. The interview further revealed the importance of having automated notifications for due dates and a searchable online catalog to improve library efficiency and user satisfaction.

Questionnaire. A questionnaire is a structured data gathering instrument used to obtain information regarding the existing library processes, challenges, and system requirements. For this study, the proponents prepared a questionnaire specifically for the librarian, who serves as the primary user and administrator of the library.

The questionnaire concentrated on the librarian's experiences overseeing day-to-day library operations, such as inventory management, report creation, book cataloging, borrower registration, book borrowing and return, and overdue book monitoring. According to the responses, the majority of library transactions are still done by hand, which leads to laborious procedures, challenges keeping track of checked books, delays in report preparation, and a higher risk of human error. The need for a centralized web-based system that can automate library transactions, offer real-time book availability monitoring, streamline inventory management, produce reports effectively, and enhance the overall provision of library services was also underlined by the librarian.

Form and Document Analysis. Form and Document Analysis is a data gathering technique that involves examining existing documents, records, and policies used in the library to better understand its operational procedures and information requirements. The proponents reviewed the library's forms, records, and library policies to ensure that the proposed system complies with the institution's existing rules and regulations.



The borrower registration form, book inventory records, borrowing and return logs, overdue records, and report templates were among the documents that the proponents reviewed. The library's policies regarding borrower eligibility, book borrowing limits, loan periods, due dates, overdue fines, book reservation procedures, and replacement policies for misplaced or damaged books were also examined by the proponents. The key information and business guidelines that should be included in the proposed E-LIB: A Web-Based Library Management Information System (LMIS) were determined by the proponents through this analysis. The results were used to design system modules that improve the efficacy and efficiency of library operations by automating transactions, consistently enforcing library policies, keeping accurate records, and producing trustworthy reports.

System Requirements

Software. The system software provides the platform and development environment necessary for designing, developing, testing, and deploying the system. This project was utilized a various software applications, including operating system, an integrated development environment (IDE), and a local server environment. These software tools supported coding, database management, web server configuration, and system testing throughout the development process. The table below shows the software and its specifications used for the development of the system.



Table 1

Software Specifications

Software	Specification
Operating System	Windows 11 Home Single Language 64-bit
Visual Studio Code	Version 1.54.1
Xampp	MySQL, Apache, PHP

Hardware. The hardware components provide the physical resources required for the development, testing, and implementation of the system. These specifications ensure that the development environment can efficiently support programming, database management, system testing, and other development-related tasks. The table below shows the minimum requirements and recommended specification.

Table 2

Hardware Specifications

Hardware	Specification
Processor	Intel® Core™ i7-1355U (13th Gen)
RAM	16 GB LPDDR5
Monitor	Generic PnP Monitor
Storage	1Tb SSD



Peopleware. During the development process, peopleware is needed. Every member of the team has its own tasks to do. The project manager's role is to manage and monitor the process and development of the system. The programmers along with the system analyst are responsible for the development of the project and the technical writer focused on the documentation.

Figure 2. Organizational Chart

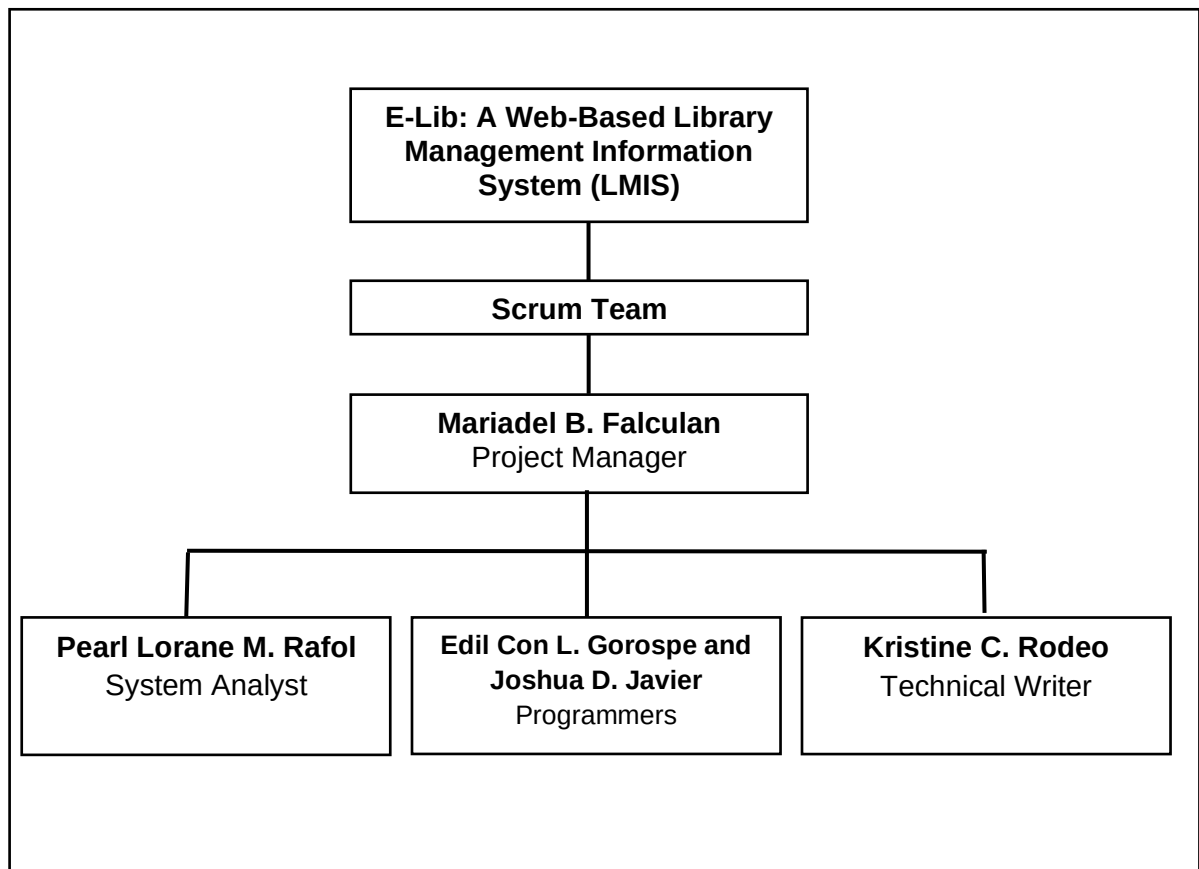


Figure 2 presents the organizational chart of the scrum team responsible for the planning, development, implementation, and documentation of the E-Lib: A Web-Based Library Management Information System (LMIS). At the beginning of the capstone project, the proponents conducted meetings to identify the project requirements and assign roles based



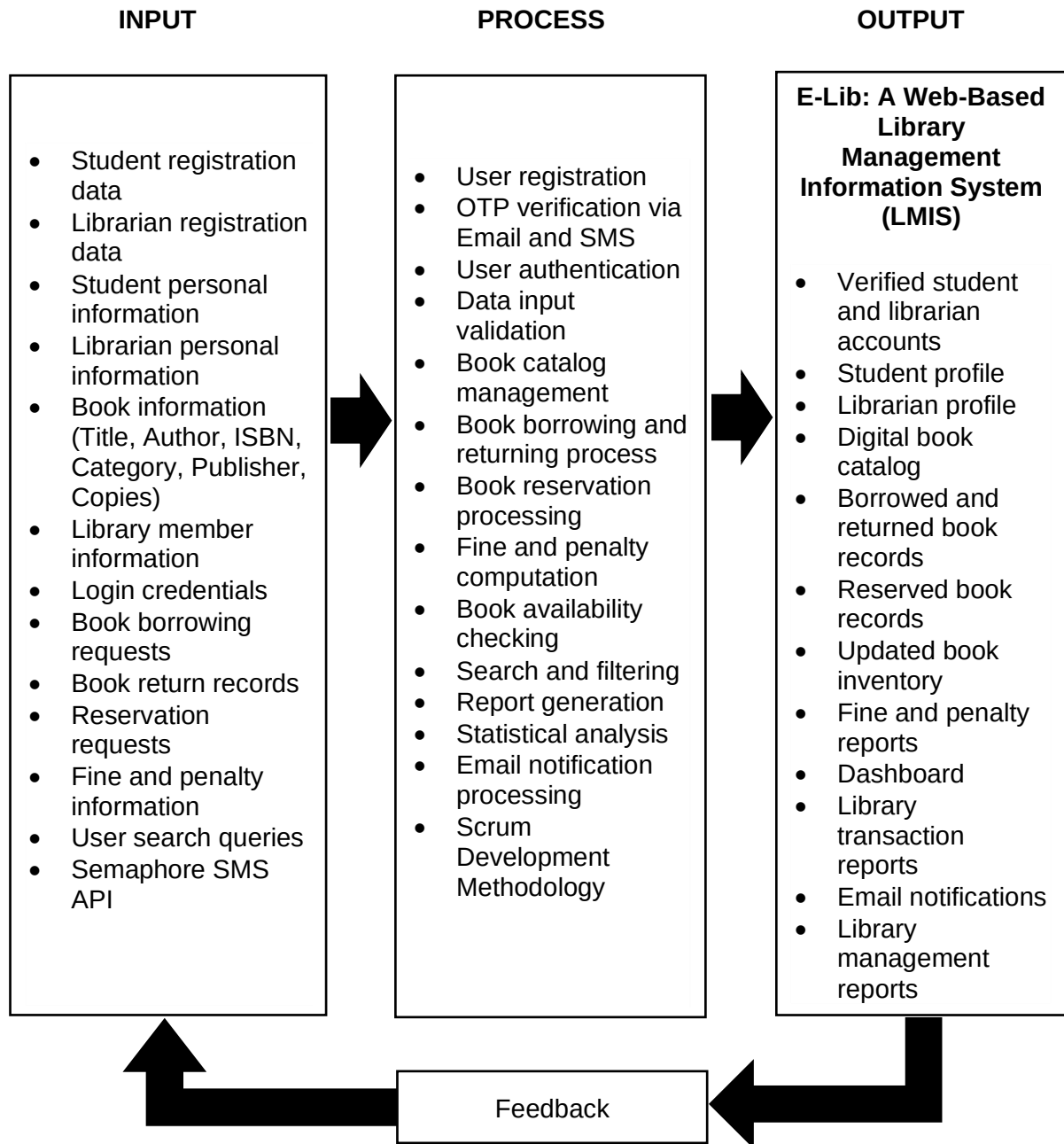
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

34

on each member's knowledge, skills, and area of expertise. This distribution of responsibilities ensured that all tasks were appropriately delegated and completed efficiently through the software development process. Mariadel B. Falculan, who served as the Project Manager was responsible for leading the team and monitoring the progress of the system. Pearl Lorane M. Rafol, the System Analyst was assigned with data gathering and analyzing the project requirements, evaluating the existing library processes, and preparing the necessary system analysis and design documents including Flowchart, Data Flow Diagram, Entity Relationship Diagram, and other charts required for the development of the system. The programming responsibilities were assigned to Edil Con L. Gorospe and Joshua D. Javier. As the lead programmer, Edil Con L. Gorospe was primarily responsible for designing, developing, integrating, and implementing the system, ensuring that all modules functioned properly and met the specified project requirements. Meanwhile, Joshua D. Javier, who served as the assistant programmer, supported the lead programmer by developing assigned modules, implementing necessary modifications and enhancements, fixing identified system errors, and assisting in testing and debugging activities to ensure the reliability and functionality of the system. Lastly, Kristine C. Rodeo, the Technical Writer, was responsible for preparing, organizing, and maintaining the capstone project's documentation. Her responsibilities included writing and revising the manuscript, documenting every phase of the system development process, preparing technical reports, and ensuring that all required project documents were accurate, complete, and properly formatted in accordance with the prescribed manual. Together, the coordinated efforts of each scrum team contributed significantly to the successful development and completion of the system.



Figure 3. Conceptual Framework





POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Chapter 4

RESULTS AND DISCUSSION

The E-Lib: A Web-Based Library Management Information System (LMIS) was developed to improve the efficiency, security, and accessibility of library services at the Polytechnic University of the Philippines – Bansud Campus. The system enables students and faculty to search for available books, request book borrowing, monitor borrowed books, and access library services online. It also allows visitors such as alumni and students from other school to check book availability and schedule library appointments through an online appointment module. The ratings from the respondents were treated statistically using a numerical scale and recorded based on the weighted mean or average. The study presents, analyzes, and interprets the data gathered based on the ISO/IEC 25010 Software Quality Standard, which includes functionality, usability, reliability, compatibility, maintainability, and security of the system, as well as requirements documentation.

Requirements Documentation

The proponents explained and show the system how it works to the librarian of the Polytechnic University of the Philippines – Bansud Oriental Mindoro Campus, faculty members, visitors, and student beneficiaries of the E-Lib: A Web-Based Library Management Information System (LMIS). They also answered the requirements documentation questionnaire to validate whether the developed system met the identified requirements and provided feedback based on its features and functionalities.

Feasibility Studies. The proponents conducted an analysis and evaluation of the proposed E-Lib: A Web-Based Library Management Information System (LMIS) to assess its



economic and technical feasibility for enhancing the existing library processes at the Polytechnic University of the Philippines – Bansud Oriental Mindoro Campus.

1. Economic Feasibility. The E-Lib: A Web-Based Library Management Information System (LMIS) offers significant advantages to librarians, students, faculty members, and visitors at the Polytechnic University of the Philippines – Bansud Oriental Mindoro Campus. The system enhances the library's management operations by streamlining processes such as user registration, book cataloging, searching, borrowing, returning, appointment, and report generation, thereby ensuring a more organized and efficient library service.

2. Technical Feasibility. The proponents used different software applications and programming languages to develop a user-friendly system, including HTML, CSS, JavaScript, PHP, and MySQL. The front-end of the system utilizes HTML, CSS, and JavaScript to create the user interface and enhance the overall user experience. Meanwhile, PHP serves as the server-side scripting language that handles the system's logic and functionality, while MySQL is used to store, manage, and retrieve the system's data. These technologies work together to provide an efficient, reliable, and easy-to-use E-Lib: A Web-Based Library Management Information System (LMIS) for both developers and administrators.

3. Schedule Feasibility. The E-Lib: A Web-Based Library Management Information System (LMIS) offers a comprehensive framework and timeline for managing the library operations. This ensures a realistic and efficient approach for book cataloging, searching, borrowing, returning, appointment, and report generation.

Systems Flowchart. To illustrate the flow of the E-Lib: A Web-Based Library Management Information System (LMIS), the library administrator has complete control over



the system, enabling them to manage user registrations, book records, borrowing and returning transactions, replacement requests, overdue books, appointments, archived books and users, payments, notifications, and reports. Meanwhile, users such as students and faculty can browse the book catalog, bookmark titles, search and filter available books, manage borrowed items, view unsettled transactions, and receive notifications. Visitors can schedule library appointments according to available time slots, monitor their appointment history, and receive related notifications. The system flow depicts the overall process, structure, data movement, and communication between the administrator, students, and visitors throughout the library management process. It demonstrates how information is processed from user registration to book management, borrowing and returning transactions, appointment scheduling, payment recording, and report generation. The system flowchart identifies the major components of the system, illustrating how data flows within the system and how decisions are made during each transaction.

Data Flow Diagrams. The Data Flow Diagrams illustrates how users interact with the various functions of the E-Lib: A Web-Based Library Management Information System (LMIS) and how the data is collected, processed, stored, and retrieved within the system. The diagram begins with the external entities, such as the administrator, students, faculty, and visitors, who provide inputs including user information, book records, book borrowing and returning transactions, appointment scheduling, and account requests. User authentication, user account management, book catalog management, borrowing and returning management, reservation management, inventory management, report generation, and notification management are just a few of the functional modules that the system uses to process this data. All data is verified during processing and kept in the system database to guarantee data security, accuracy, and consistency. Following processing, the system produces outputs like



transaction summaries, inventory reports, updated user profiles, book availability status, borrowing and return records, reservation details, and overdue notifications. While faculty and students can easily access library resources and keep track of their transactions, these outputs allow librarians and administrators to effectively monitor library operations. The Data Flow Diagram guarantees effective data processing, dependable record management, and prompt access to accurate information for all authorized users by clearly depicting the flow of information throughout the system.

Design of Software/Systems

Hierarchical Input-Process-Output. For the input, the knowledge requirements include programming skills, tutorials, PDFs, and literatures. The software requirements include HTML, CSS, and JavaScript for front-end development, while PHP is used for back-end development and MySQL serves as the database management system. The system is developed and tested using the XAMPP local server, with Visual Studio Code as the source code editor and Google Chrome as the web browser. For deployment, the system is hosted on an online web hosting platform, making it accessible through the internet. The hardware requirements include a laptop with a stable internet connection for system development, testing, and deployment.

For the process, the system design includes the system flowchart, data flow diagram (DFD), and entity-relationship diagram (ERD). The system development follows the Scrum methodology, which is a framework under the Agile. These inputs and processes resulted in the development of E-Lib: A Web-Based Library Management Information System (LMIS) for the Polytechnic University of the Philippines – Bansud Campus, which improves the efficiency, security, and accessibility of library resources.



Program Flowchart. It shows the logical sequence of events, starting with user authentication and continuing through data entry, processing, storage, report and visualization development.

Entity-Relationship Diagram. The Entity-Relationship Diagram (ERD), depicts the logical structure of the E-Lib: A Web-Based Library Management Information System (LMIS) database, illustrating the entities, attributes, and relationships among the different data components. It specifies how data is arranged and linked within the database with regard to users, books, authors, categories, borrowings, returns, appointments, fines, notifications, and reports. By guaranteeing data integrity, reducing data redundancy, and facilitating effective library information storage, retrieval, updating, and management across the system, the ERD acts as a guide for database design.

Data Dictionary. This showed the definitions of the database contents, format, and structure of the E-Lib: A Web-Based Library Management Information System (LMIS) at the Polytechnic University of the Philippines – Bansud Oriental Mindoro Campus. It outlines the relationships among its elements, which are utilized to control access and manipulate the database effectively. The data dictionary serves as a key reference for understanding the various data elements involved in the system, ensuring clarity and consistency in database management and system development.



Gantt Chart

[illegible]

**Development and Testing**

The proponents used Scrum framework under Agile methodology to guide the development of the system. The development process was carried out through the Scrum phases which include Sprint Planning, Sprint Execution, Sprint, Sprint Review, and Sprint Retrospective. By breaking down the development process into this phases made it easier to evaluate each part of the software development process. The data gathered were analyzed using descriptive statistical method. The responses were carefully recorded, computed, and interpreted using weighted mean. To evaluate the functionality, usability, reliability, compatibility, maintainability, and security of the system, testing was conducted with students, IT experts, faculty, and visitors such as alumni and student from other schools. The results were used to determine whether the system met its intended objectives and satisfied the user's needs.

1. The weighted mean was used to analyze the data gathered from the respondents in evaluating the system's functionality, usability, reliability, compatibility, maintainability, and security. It was also used to determine the overall evaluation of the system. The formula used in computing the weighted mean is:

Formula:

$$\bar{x} = \frac{\sum x}{N}$$

Where:

Σ - the sum of all values

N - the total number of values

In analyzing and computing the results, the weighted mean was used to determine the overall evaluation of the E-Lib: A Web-Based Library Management Information System (LMIS).



The weighted mean formula was applied to all responses gathered from end users, IT experts, and faculty and staff to ensure an accurate representation of their evaluations. This method enabled the proponents to summarize the respondents' ratings into a single value that reflected the overall quality and acceptability of the system based on the evaluation criteria. The scale used to interpret the computed weighted mean is presented in the table below.

Table 4

Numerical Value, Statistical Limit, Verbal Description, Verbal Interpretation

Numerical Value	Statistical Limit	Verbal Description	Verbal Interpretation
5	4.51-5.00	Strongly Agree	The respondents always agree to the following statement.
4	3.51-4.50	Agree	The respondents often agree to the following statement.
3	2.51-3.50	Moderately Agree	The respondents moderately agree to the following statement.
2	1.51-2.50	Disagree	The respondents disagree to the following statement.
1	1.00-1.50	Strongly Disagree	The respondents strongly disagree to the following statement.

Table 4 presents the numerical values, statistical limits, verbal description, and verbal interpretation used in evaluating the E-Lib: A Web-Based Library Management Information System (LMIS). For the survey questionnaire, the proponents used a Likert Scale consisting of Strongly Agree, Agree, Moderately Agree, Disagree, and Strongly Disagree.

Implementation Plan. The E-Lib: A Web-Based Library Management Information System (LMIS) was tested during the Sprint Review phase to verify its functionality and ensure that it met the project's objectives. During this stage, each module of the system was carefully evaluated to confirm that it operated according to the specified requirements. Key features such as user registration and login, book borrowing and returning, appointment scheduling,



notifications, and report generation were tested for accuracy, reliability, and ease of use. The administrator dashboard and its functions for managing users, books, and appointments were also evaluated to ensure that the system effectively supports the library operations of the Polytechnic University of the Philippines – Bansud Campus.

Along with evaluating the system's functionality, its interface was also assessed to ensure that it is user-friendly, responsive, and easy to navigate for end users, IT experts, and faculty and staffs. The system was tested across different web browsers and devices to verify its compatibility and accessibility. Performance testing was also conducted to ensure that the system could process multiple transactions efficiently and provide reliable service during regular library operations.

Implementation Results. The E-Lib: A Web-Based Library Management Information System (LMIS) was evaluated by eighty-five (86) respondents composed of seventy-one (71) end users such as students, stakeholders, and librarian, five (5) IT experts, ten (10) faculty and staffs from the Polytechnic University of the Philippines – Bansud Campus. The tabular representation of the evaluation is presented as follows:



Table 5

**Functionality of the System by the End Users
(Student/Stakeholder/Librarian)**

Items	Mean	Verbal Description
The system can manage book records without errors.	4.72	Strongly Agree
The system accurately processes book borrowing and returning transactions.	4.89	Strongly Agree
The system correctly manages visitor appointments and scheduling.	4.77	Strongly Agree
The system allows librarians to manage user accounts such as students, faculty, and visitors.	4.92	Strongly Agree
The system generates accurate library reports for book borrowing and visitors' appointment.	4.86	Strongly Agree
Overall Mean	4.83	Strongly Agree

The table 5 shows the evaluation results of the system's functionality as assessed by the end users such as students, stakeholders, and librarian. The overall mean of 4.83 in the table, which is verbally described as Strongly Agree, shows that the respondents strongly believed that the system was functional and capable of carrying out its intended functions effectively. Among the indicators, the Item 4 stating that the system allows the librarians to manage user accounts such as students, faculty, and visitors, received the highest mean, 4.92, indicating that the user account management feature was considered as the most effective functionality of the system. On the other hand, the item stating that system can



manage book records without errors, received the lowest mean, 4.72, which still described as Strongly Agree. Although it ranked the lowest among the indicators, the result still reflects a very high level of agreement, indicating that the end users were highly satisfied with the system's ability to maintain accurate book records.

The study by Guo et al. (2023), which highlighted the need for an automated library management information system to replace ineffective manual procedures due to the increasing number of library resources and users, supports the findings. The authors discovered that a web-based library management system satisfies the functional needs of both librarians and library users while efficiently managing book records, user accounts, and borrowing and returning transactions. Similarly, the current study found that the respondents firmly agreed that the system effectively performs this core library tasks, proving its capacity to enhance operational effectiveness, reduce human error, and support precise and reliable library services.

The results show that the functionality of the E-Lib: A Web-Based Library Management Information System (LMIS) meets the operational requirements of the library and the expectations of its end users. The consistently high ratings across all indicators indicate that the system can support librarian in performing administrative tasks more efficiently while providing students, faculty, and visitors with a reliable and convenient library services. Furthermore, the findings suggest that implementing the system can reduce manual workload, improve the accuracy of library records and transactions, and contribute to a more organized and efficient library management process.



Table 6

**Usability of the System by the End Users
(Student/Stakeholder/Librarian)**

Items	Mean	Verbal Description
1. The system interface is easy to understand and navigate.	4.87	Strongly Agree
2. The system navigation menu is easy to use.	4.85	Strongly Agree
3. The system notifies users about important transaction updates, such as book borrowing, book returning, and visitor appointments.	4.87	Strongly Agree
4. The system displays the current condition of the book.	4.9	Strongly Agree
5. The system displays the available time slots during which visitors may schedule a library appointment.	4.89	Strongly Agree
Overall Mean	4.88	Strongly Agree

As presented in Table 6, the end users rated the usability of the system an overall mean of 4.88, which is verbally described as Strongly Agree, indicating that the respondents perceived the system as highly usable, user-friendly, and easy to operate. Among the indicators, the item stating that the system displays the available time slots during which visitors may schedule a library appointment, obtained the highest mean of 4.9, indicating that the respondents highly appreciated the system's ability to provide updated and accessible information about book availability and condition. Meanwhile, item stating that the system navigation menu is easy to use, received the lowest mean of 4.85. Although it obtained the



lowest rating among the indicators, the result still reflects a very high level of user satisfaction, suggesting that the end users found the navigation intuitive and convenient.

The findings align with the work of Martirosyan et al. (2022), who developed a web and mobile library management information system. Their study emphasized that highly accessible, well-designed interfaces drive user engagement and overall library satisfaction. By successfully integrating feature like automated scheduling and clear transaction notifications, the system achieves exactly a seamless and a highly used platform that fundamentally improves the overall library experience.

Results imply that the E-Lib: A Web-Based Library Management Information System (LMIS) provides an effective and user-centered interface that promoted ease of use and enhances the overall user experience. The high ratings across all usability indicators suggest that the system enables users to navigate efficiently, receive timely updates on library transactions, and easily access relevant information such as book conditions and appointment schedules. These findings demonstrate that the system can improve user productivity, encourage greater utilization of library services, and support a more efficient and convenient library management environment.



Table 7

Reliability of the System by the End Users
(Student/Stakeholder/Librarian)

Items	Mean	Verbal Description
1. The system reliably validates user input and prevents incorrect data entries.	4.75	Strongly Agree
2. The system ensures that stored data remains consistent and error-free.	4.72	Strongly Agree
3. The system maintains stable performance when multiple users access it simultaneously.	4.73	Strongly Agree
4. The system ensures that library policies are consistently implemented and followed.	4.86	Strongly Agree
5. The system consistently performs book borrowing, returning, and visitor appointment tasks without errors or failures.	4.73	Strongly Agree
Overall Mean	4.76	Strongly Agree

As shown in the table above, the assessment of the respondents regarding the reliability of the system obtained an overall mean of 4.76, which is verbally described as Strongly Agree, indicating that the respondents perceived the system as highly reliable in performing its intended functions consistently and accurately. Among the indicators, the item stating that the system ensures that library policies are consistently implemented and followed, obtained the highest mean of 4.86, suggesting that respondents recognized the system's effectiveness in enforcing library rules and maintaining standardized procedures. Meanwhile,



the item stating that the system ensures that stored data remains consistent and error-free, received the lowest mean of 4.72. Although it obtained the lowest rating among the indicators, the result still reflects a very high level of agreement, indicating that the respondents considered the system dependable in maintaining the accurate and consistent library data.

Rahman et al. (2023) highlighted a web-based library management system that automates traditional library processes through the centralized management of catalog, borrowing, returning, renewals, and user accounts. The study emphasized that automation minimizes processing errors while improving the consistent with the present study where respondents strongly agreed that the system reliably validates user inputs, maintains consistent data, enforces library policies, and performs borrowing, returning, and visitor appointment transactions without errors or failures.

Results imply that the E-Lib: A Web-Based Library Management Information System (LMIS) is a dependable platform for managing daily library operations. The consistently high ratings across all reliability indicators suggest that the system can accurately process transactions, maintain data integrity, validate user inputs, and consistently enforce library policies. These findings indicate that implementing the system can reduce operational errors, strengthen the accuracy of library records, and improve users' confidence in the reliability of library services, ultimately contributing to a more efficient and trustworthy library management environment.



Table 8

**Compatibility of the System by the End Users
(Student/Stakeholder/Librarian)**

Items	Mean	Verbal Description
1. The system works properly on different web browsers/devices.	4.89	Strongly Agree
2. The system operates smoothly in an online environment without causing conflicts with other software applications.	4.73	Strongly Agree
3. The system seamlessly integrates with external services, including email and SMS gateways, to facilitate automated notifications and communication.	4.9	Strongly Agree
4. The system displays information correctly regardless of the platform being used.	4.79	Strongly Agree
5. The system maintains stable performance during simultaneous operation with other applications.	4.77	Strongly Agree
Overall Mean	4.82	Strongly Agree

Table 8 presents the assessment of the respondents regarding the compatibility of the system. The table obtained the overall mean of 4.82, which is verbally described as Strongly Agree, indicating that the respondents perceived that the system is highly compatible across different platforms, devices, and software environments. Among the indicators, the item stating that the system seamlessly integrates with external services, including email and SMS gateways, to facilitate automated notifications and communication, received the highest mean



of 4.90, suggesting that the respondents highly appreciate the system's capability to integrate with external communication services for efficient and timely notifications. On the other hand, the item stating that the system operates smoothly in an online environment without causing conflicts with other software applications, received the lowest mean of 4.73. Although it obtained the lowest rating among the indicators, the result still reflects a very high level of agreement, indicating that the respondents considered the system capable of operating reliably alongside other software applications.

Findings supported by the study of Moruf and Abu (2024), who reviewed next-generation web-based and cloud-based library management systems and emphasized that compatibility, accessibility, scalability, and interoperability are essential characteristics of modern library systems. Their study explained that web-based systems should support access across different devices and platforms while ensuring seamless integration with other technologies to enhance library services. These findings are consistent with the present study, where respondents strongly agreed that the system functions properly across various web browsers and devices, integrates with email and SMS gateways, accurately displays information regardless of the platform used, and maintains stable performance while operating with other applications.

The results imply that the E-Lib: A Web-Based Library Management Information System (LMIS) provides a high level of compatibility, enabling users to access library services efficiently across different devices, web browsers, and computing environments. The consistently high ratings across all compatibility indicators suggest that the system can seamlessly integrate with external communication services, display information consistently across platforms and maintain stable performance alongside other software applications. These findings demonstrate that the system offers flexible, accessible, and dependable library



services, thereby improving the overall user experience and supporting efficient library operations.

Table 9

Maintainability of the System by the End Users**(Student/Stakeholder/Librarian)**

Items	Mean	Verbal Description
1. The system structure complies with software standards making it easy to adapt to changes.	4.86	Strongly Agree
2. The system can be easily updated to add new features or improvements.	4.83	Strongly Agree
3. The system can be enhanced and maintained with minimal downtime.	4.76	Strongly Agree
4. The system includes clear and well-documented code, making it easier for developers to understand, debug, and modify.	4.83	Strongly Agree
5. The system allows administrators to easily monitor and maintain system operations.	4.92	Strongly Agree
Overall Mean	4.84	Strongly Agree

As shown in the table above, the results of the evaluation by the end users regarding the compatibility of the system received the overall mean of 4.84, which is verbally interpreted as Strongly Agree. This result indicates that the respondents perceived the system as highly maintainable. The proponents found that the highest mean was 4.92, which was obtained by the item stating that the system allows administrators to easily monitor and maintain system



operations. This shows that the respondents firmly concur that the system provides administrators effective tools for keeping an eye on and maintaining library operations, which improves the efficiency and organization of system management. However, the items that stated that the system can be enhanced and maintained with minimal downtime had the lowest mean of 4.76, but despite having the lowest mean, these items are verbally described as Strongly Agree, indicating that users believe that the system enhancements can be implemented with minimal disruption to library operations.

The result is consistent with the findings of Pinto (2024), who evaluated a computerized library management system and evaluated it using the ISO/IEC 25010 software quality standards. The study reported high maintainability, emphasizing that the system could be modified, updated, and improved efficiently without compromising its core functionalities. Such findings reinforce the present study, as the respondents likewise recognized the system's ability to support future enhancements, facilitate system maintenance, and provide administrators with effective tools for managing system operations.

All in all, the evaluation demonstrates that the developed system possesses strong maintainability characteristics that can support its long-term implementation. The positive ratings indicate that future updates, feature enhancements, and corrective maintenance can be carried out efficiently while ensuring the continuity of library services. This adaptability enables the system to remain responsive to changing user requirements and technological advancements, contributing to the sustainability and effectiveness of library management operations.



Table 10

Security of the System by the End Users
(Student/Stakeholder/Librarian)

Items	Mean	Verbal Description
1. The system sends One-Time Password (OTP) notifications via Gmail and SMS during registration and login to verify user identity.	4.92	Strongly Agree
2. The system requires users to create strong passwords to enhance security.	4.86	Strongly Agree
3. The system limits login attempts to three (3) consecutive tries to prevent unauthorized access.	4.94	Strongly Agree
4. The system automatically generates and assigns unique credentials to each user to maintain secure and personalized account access.	4.86	Strongly Agree
5. The system ensures that users' personal information is accessible only to authorized personnel, maintaining confidentiality and privacy.	4.94	Strongly Agree
Overall Mean	4.90	Strongly Agree

Table 10 revealed a strong positive assessment of the system's security features. The respondents gave the system an overall mean of 4.90 which is verbally described as Strongly Agree indicating a very high level of confidence in the protection of user accounts and personal information. The highest mean of 4.94 was shared by the items stating that the system limits login attempts to three (3) consecutive tries to prevent unauthorized access and ensures that users' personal information is accessible only to authorized personnel, maintaining confidentiality and privacy. This shows that the respondents firmly concur that the system successfully protects user accounts and confidential information by ensuring protections in



place that prevent illegal access and protect sensitive information. However, the item that stated that the system requires users to create strong passwords to enhance security and the system automatically generates and assigns unique credentials to each user to maintain secure and personalized account access had the lowest mean of 4.86. It is verbally described as Strongly Agree despite having the lowest mean, suggesting that respondents also recognized these security measures as effective components of the system.

The results are aligned with the study of Bhati et al. (2024) who emphasized that a modern web-based library management system should provide secure, scalable, and user-centered account management. Their study highlighted the importance of authentication mechanisms such as OTP based verification, secure user identification, and automated validation in strengthening the safety and reliability of library operations. These findings support the present evaluation as the respondents strongly agreed that the developed system effectively verifies user identity, restricts unauthorized login attempts, assigns secure credentials, and protects personal information.

Taken together, the findings indicate that the E-Lib: A Web-Based Library Management Information System provides a high level of security for its users and administrators. The consistently high ratings across all indicators demonstrate that the implemented security features contribute to protecting user accounts, maintaining data confidentiality, and preventing unauthorized access. As a result, the system can foster greater trust among students, stakeholders, and librarians while supporting a secure and reliable environment for library transactions and information management.



Table 11

Functionality of the System by the IT Experts

Items	Mean	Verbal Description
1. The system can manage book records without errors.	4.60	Strongly Agree
2. The system accurately processes book borrowing and returning transactions.	4.80	Strongly Agree
3. The system correctly manages visitor appointments and scheduling.	4.40	Agree
4. The system allows librarians to manage user accounts such as students, faculty, and visitors.	5.00	Strongly Agree
5. The system generates accurate library reports for book borrowing and visitors' appointment.	4.20	Agree
Overall Mean	4.60	Strongly Agree

Table 11 shows the evaluation of the system's functionality as assessed by the IT experts. The system obtained an overall mean of 4.60, which is interpreted as Strongly Agree, indicating that the experts found that the system is capable of performing its intended functions effectively. The highest mean score of 5.00 was obtained by the indicator stating that the system allows librarians to manage user accounts such as students, faculty members, and visitors, demonstrating that the system provides efficient and reliable user account management. On the other hand, the indicator stating that the system generates accurate library reports for book borrowing and visitors' appointments received the lowest mean score of 4.20, which is interpreted as Agree. Although the indicator received the lowest rating, the



result suggests that the report generation feature is functional but still has room for improvement to further enhance its accuracy and usefulness.

The results imply that the developed system is functionally reliable and capable of supporting the daily operations of the library. Its ability to manage book records, process circulation transactions, organize visitor appointments, and maintain user accounts contributes to a more efficient and organized library management process. While the report generation module received a comparatively lower rating, it remains acceptable and highlights an opportunity for further enhancement. All in all, the findings demonstrate that the system can reduce manual processing, improve operational efficiency, and provide librarians with a dependable tool for managing library services.



Table 12

Usability of the System by the IT Experts

Items	Mean	Verbal Description
1. The system interface is easy to understand and navigate.	4.60	Strongly Agree
2. The system navigation menu is easy to use.	4.60	Strongly Agree
3. The system notifies users about important transaction updates, such as book borrowing, book returning, and visitor appointments.	4.80	Strongly Agree
4. The system displays the current condition of the book.	4.60	Strongly Agree
5. The system displays the available time slots during which visitors may schedule a library appointment.	4.80	Strongly Agree
Overall Mean	4.68	Strongly Agree

Table 12 presents the evaluation of the system's usability as assessed by the IT experts. The system obtained an overall mean of 4.68, which is interpreted as Strongly Agree, indicating that the experts found that the system is highly usable and easy to operate. The highest mean score of 4.80 was obtained by two indicators stating that the system notifies users about important transaction updates, such as book borrowing, book returning, and visitor appointments, and the system displays the available time slots during which visitors may schedule a library appointment. These results demonstrate that the IT experts strongly believe that the notification and appointment scheduling features improve user interaction by providing timely information and making the scheduling process more convenient and accessible. Meanwhile, the indicators stating that the



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

60

system interface is easy to understand and navigate, the system navigation menu is easy to use, and the system displays the current condition of the book obtained the lowest mean score of 4.60. Despite being the lowest scores, all three indicators are still interpreted as Strongly Agree, showing that the IT experts positively evaluated the system's interface, navigation, and presentation of book information. These findings suggest that the users can easily understand the system and complete their tasks efficiently without significant difficulty.

The results imply that the developed system is highly usable and capable of supporting efficient library operations. Its intuitive interface, easy navigation, timely transaction notifications, accurate display of book information, and convenient appointment scheduling features can help users accomplish tasks quickly and with minimal confusion. Since all indicators received high ratings and were interpreted as Strongly Agree, the system has the potential to enhance user satisfaction, reduce operational difficulties, and provide a more accessible and user-friendly approach to managing library services.



Table 13

Reliability of the System by the IT Experts

Items	Mean	Verbal Description
1. The system reliably validates user input and prevents incorrect data entries.	4.60	Strongly Agree
2. The system ensures that stored data remains consistent and error-free.	4.40	Agree
3. The system maintains stable performance when multiple users access it simultaneously.	4.60	Strongly Agree
4. The system ensures that library policies are consistently implemented and followed.	4.60	Strongly Agree
5. The system consistently performs book borrowing, returning, and visitor appointment tasks without errors or failures.	4.60	Strongly Agree
Overall Mean	4.56	Strongly Agree

Table 13 presents the evaluation of the system's reliability as assessed by the IT experts. The system obtained an overall mean of 4.56 which is interpreted as Strongly Agree, indicating that the experts found that the system is dependable in performing its intended operations consistently and accurately. The highest mean score of 4.60 was obtained by four indicators which states that the system reliably validates user input and prevents incorrect data entries, maintains stable performance when multiple users access it simultaneously, ensures that library policies are consistently implemented and followed, and consistently performs book borrowing, returning, and visitor appointment tasks without errors or failures. These results



demonstrate that the IT experts strongly believe that the system reliably performs its core functions, supports concurrent users efficiently, and consistently enforces library policies while minimizing processing errors. Meanwhile, the indicator stating that the system ensures that stored data remains consistent and error-free obtained the lowest mean score of 4.40, which is interpreted as Agree. Although this indicator received the lowest rating, it still reflects a positive evaluation, suggesting that the system effectively maintains data consistency while indicating that further improvements may enhance data integrity and overall reliability.

The results imply that the developed system is reliable and capable of supporting the daily operations of the library with consistency and accuracy. Its ability to validate user inputs, process borrowing and returning transactions, manage visitor appointments, and enforce library policies contributes to dependable library services. Although the data consistency feature received a comparatively lower rating, it remains acceptable and identifies an opportunity for further enhancement. All in all, the findings demonstrate that the system can reduce processing errors, maintain operational stability, and provide librarians and users with a trustworthy platform for managing library services.



Table 14

Compatibility of the System by the IT Experts

Items	Mean	Verbal Description
1. The system works properly on different web browsers/devices.	4.80	Strongly Agree
2. The system operates smoothly in an online environment without causing conflicts with other software applications.	4.60	Strongly Agree
3. The system seamlessly integrates with external services, including email and SMS gateways, to facilitate automated notifications and communication.	4.80	Strongly Agree
4. The system displays information correctly regardless of the platform being used.	4.80	Strongly Agree
5. The system maintains stable performance during simultaneous operation with other applications.	4.60	Strongly Agree
Overall Mean	4.72	Strongly Agree

Table 14 shows that the evaluation of the system's compatibility as assessed by the IT experts. The system obtained an overall mean of 4.72 which is interpreted as Strongly Agree indicating that the experts found that the system is highly compatible across different platforms, devices, and operating environments. The highest mean score of 4.80 was obtained by three indicators including the system works properly on different web browsers and devices, the system seamlessly integrates with external services, including email and SMS gateways, to facilitate automated notifications and communication, and the system displays information



correctly regardless of the platform being used. These results demonstrate that the IT experts strongly believe that the system provides consistent performance across various platforms while effectively supporting external communication services. Meanwhile, the indicators stating that the system operates smoothly in an online environment without causing conflicts with other software applications and the system maintains stable performance during simultaneous operation with other applications obtained the lowest mean score of 4.60. Despite being the lowest scores, bot indicators are still interpreted as Strongly Agree, indicating that the IT experts positively evaluated the system's ability to operate reliably alongside other software applications and maintain stable performance in different computing environments.

The results imply that the developed system is highly compatible and capable of supporting library operations across various technological environments. Its ability to function properly on different browsers and devices, integrate with email and SMS gateways, display information consistently across platforms, and operate alongside other applications contributes to a reliable and accessible user experience. Since all indicators received high ratings and were interpreted as Strongly Agree, the system has the potential to improve accessibility, ensure dependable interoperability, and provide flexible and efficient library services.



Table 15

Maintainability of the System by the IT Experts

Items	Mean	Verbal Description
1. The system structure complies with software standards making it easy to adapt to changes.	4.80	Strongly Agree
2. The system can be easily updated to add new features or improvements.	4.80	Strongly Agree
3. The system can be enhanced and maintained with minimal downtime.	4.80	Strongly Agree
4. The system includes clear and well-documented code, making it easier for developers to understand, debug, and modify.	4.80	Strongly Agree
5. The system allows administrators to easily monitor and maintain system operations.	4.80	Strongly Agree
Overall Mean	4.80	Strongly Agree

Table 15 presents the evaluation of the system's maintainability as assessed by the IT experts. The system obtained an overall mean of 4.80 which is verbally interpreted as Strongly Agree, indicating that the experts perceived the system as highly maintainable. The consistent ratings across all indicators suggest that the system was designed with a well-structured architecture, making it easier to update, enhance, monitor, and manage over time. All five indicators received the same mean score of 4.80 including the system's compliance with software standards, ease of implementing updates and new features, ability to undergo maintenance with minimal downtime, availability of clear and well-documented code, and support for efficient administrative monitoring and maintenance. These results indicate that



the IT experts strongly agreed that the system possesses the essential characteristics of maintainable software, enabling future modifications and improvements to be carried out efficiently while minimizing disruptions to system operations.

Results show that the E-Lib: A Web-Based Library Management Information System (LMIS) complies with software standards, supports future enhancements, reduces downtime during maintenance, offers well-documented code for easier modification, and permits administrators to efficiently monitor and maintain system operations. The IT experts gave the system's maintainability a high rating. These findings show that the system is built to support long-term sustainability and ongoing improvement without compromising its essential features. In overall, the system received an average mean of 4.80, which is verbally defined as Strongly Agree. This indicates that the IT experts are quite pleased with the system's maintainability.



Table 16

Security of the System by the IT Experts

Items	Mean	Verbal Description
1. The system sends One-Time Password (OTP) notifications via Gmail and SMS during registration and login to verify user identity.	5.00	Strongly Agree
2. The system requires users to create strong passwords to enhance security.	5.00	Strongly Agree
3. The system limits login attempts to three (3) consecutive tries to prevent unauthorized access.	5.00	Strongly Agree
4. The system automatically generates and assigns unique credentials to each user to maintain secure and personalized account access.	5.00	Strongly Agree
5. The system ensures that users' personal information is accessible only to authorized personnel, maintaining confidentiality and privacy.	5.00	Strongly Agree
Overall Mean	5.00	Strongly Agree

As shown in the table above, the system's security assessed by the IT experts obtained an overall mean of 5.00, which is verbally described as Strongly Agree, indicating that the experts perceived that the system is highly secure. Meanwhile, every security indicator received a perfect mean of 5.00, including the use of One-Time Password (OTP) verification, strong password enforcement, login attempt limitations, automatic generation of unique user credentials, and protection of users' personal information. The uniform ratings highlight the IT experts' confidence in the system's ability to safeguard user accounts and confidential data through comprehensive security measures.

The results show that the IT experts gave the E-Lib: A Web-Based Library Management Information System (LMIS) a high security rating, proving that the system offers



strong authentication procedures, safe user account management, restricted access to sensitive data, and efficient defense against unauthorized system access. These findings show that the system offers a secure and dependable environment for library transactions while guaranteeing the confidentiality, integrity, and security of user data. All in all, the system received an average mean of 5.00, which is verbally described as Strongly Agree. This indicates that the IT experts are completely satisfied with the security of the developed system and acknowledge its high level of dependability and security.

Table 17

Functionality of the System by the Faculty

Items	Mean	Verbal Description
1. The system can manage book records without errors.	4.70	Strongly Agree
2. The system accurately processes book borrowing and returning transactions.	4.90	Strongly Agree
3. The system correctly manages visitor appointments and scheduling.	4.60	Strongly Agree
4. The system allows librarians to manage user accounts such as students, faculty, and visitors.	5.00	Strongly Agree
5. The system generates accurate library reports for book borrowing and visitors' appointment.	4.60	Strongly Agree
Overall Mean	4.76	Strongly Agree



Table 17 presents the evaluation of the system's functional suitability as assessed by the IT experts. The system obtained an overall mean of 4.76 which is verbally interpreted as Strongly Agree, indicating that the experts found the system to be highly capable of performing its intended functions effectively and accurately. The results suggest that the system successfully supports the core library operations required by its users. Among the indicators, the item stating that the system allows librarians to manage user accounts such as students, faculty, and visitors, obtained the highest mean score of 5.00 indicating that the experts unanimously agreed that the system effectively supports user account management. Meanwhile, the indicators stating that the system correctly manages visitor appointments and scheduling and generates accurate library reports for book borrowing and visitors' appointments both obtained the lowest mean score of 4.60. Although these indicators received the lowest ratings, they are still interpreted as Strongly Agree, reflecting the experts' positive evaluation of these system functions.

The findings imply that the developed E-Lib: A Web-Based Library Management Information System (LMIS) effectively fulfills its intended purpose by providing reliable support for essential library operations. Its ability to accurately manage user accounts, process borrowing and returning transactions, maintain book records, schedule visitor appointments, and generate library reports contributes to efficient library administration. The consistently high ratings across all indicators demonstrate that the system possesses a high level of functional suitability, making it a dependable solution for improving library management processes.



Table 18

Usability of the System by the Faculty

Items	Mean	Verbal Description
1. The system interface is easy to understand and navigate.	4.80	Strongly Agree
2. The system navigation menu is easy to use.	4.80	Strongly Agree
3. The system notifies users about important transaction updates, such as book borrowing, book returning, and visitor appointments.	4.90	Strongly Agree
4. The system displays the current condition of the book.	4.80	Strongly Agree
5. The system displays the available time slots during which visitors may schedule a library appointment.	4.90	Strongly Agree
Overall Mean	4.84	Strongly Agree

The table above presents the evaluation of the system's usability as assessed by the IT experts. As a result, system obtained an overall mean of 4.84, which is verbally interpreted as Strongly Agree, indicating that the experts found the system easy to understand and capable of helping users recognize and utilize its features effectively. The results suggest that the system provides a clear interface and presents relevant information that supports efficient user interaction. Among the indicators, the statements that the system notifies users about important transaction updates, such as book borrowing, book returning, and visitor appointments and the system displays the available time slots during which visitors may



schedule a library appointment obtained the highest mean score of 4.90. These findings indicate that the IT experts highly appreciated the system's ability to keep users informed and provide clear scheduling information. Meanwhile, the indicators stating that the system interface is easy to understand and navigate, the system navigation menu is easy to use, and the system displays the current condition of the book each obtained a mean score of 4.80. Although these indicators received slightly lower ratings, they are still interpreted as Strongly Agree, reflecting the experts' positive assessment of the system's interface and usability.

The findings imply that the developed E-LIB: A Web-Based Library Management Information System (LMIS) effectively enables users to recognize and access its functions with minimal difficulty. Its intuitive interface, organized navigation, timely transaction notifications, book status display, and appointment scheduling features contribute to a more convenient and user-friendly library experience. The consistently high ratings across all indicators demonstrate that the system successfully supports users in understanding and utilizing its functionalities, thereby enhancing the overall effectiveness of library services.



Table 19

Reliability of the System by the Faculty

Items	Mean	Verbal Description
1. The system reliably validates user input and prevents incorrect data entries.	4.80	Strongly Agree
2. The system ensures that stored data remains consistent and error-free.	4.70	Strongly Agree
3. The system maintains stable performance when multiple users access it simultaneously.	4.80	Strongly Agree
4. The system ensures that library policies are consistently implemented and followed.	4.70	Strongly Agree
5. The system consistently performs book borrowing, returning, and visitor appointment tasks without errors or failures.	4.80	Strongly Agree
Overall Mean	4.76	Strongly Agree

Table 19 shows the system's evaluation as perceived by the faculty. As the overall mean, the system obtained 4.76 which is verbally described as Strongly Agree. This indicates that the faculty perceived the system as highly reliable in performing its intended functions consistently and accurately. The results demonstrate that the system is capable of maintaining dependable performance while supporting essential library operations. The highest mean of 4.80 was shared by the items related to user input validation, stable performance during simultaneous access, and error-free execution of borrowing, returning, and visitor appointment transactions. These findings highlight the system's ability to perform its core functions



consistently. The items concerning data consistency and the implementation of library policies received the lowest mean of 4.70, however, both remained within the Strongly Agree range, indicating positive evaluations while suggesting opportunities for continuous enhancement.

The findings imply that the developed E-Lib: A Web-Based Library Management Information System (LMIS) provides dependable and consistent performance in supporting daily library operations. Its ability to validate user inputs, maintain data integrity, support multiple users simultaneously, enforce library policies, and accurately process library transactions contributes to a reliable and efficient library management environment. The consistently high ratings further indicate that the system can be trusted to deliver stable and accurate services that meet the operational needs of the library.



Table 20

Compatibility of the System by the Faculty

Items	Mean	Verbal Description
1. The system works properly on different web browsers/devices.	4.90	Strongly Agree
2. The system operates smoothly in an online environment without causing conflicts with other software applications.	4.80	Strongly Agree
3. The system seamlessly integrates with external services, including email and SMS gateways, to facilitate automated notifications and communication.	4.90	Strongly Agree
4. The system displays information correctly regardless of the platform being used.	4.70	Strongly Agree
5. The system maintains stable performance during simultaneous operation with other applications.	4.80	Strongly Agree
Overall Mean	4.82	Strongly Agree

Table 20 shows the evaluation of the system's compatibility as assessed by the faculty respondents. The system obtained an overall mean of 4.82 which is verbally interpreted as Strongly Agree indicating that the faculty members perceived the system as highly compatible with different technological environments. Among the indicators, the statements that the system works properly across different web browsers and devices and seamlessly integrates with external services such as email and SMS gateways both obtained the highest mean of 4.90 reflecting the system's ability to provide consistent access and support automated



communication. Meanwhile, the indicator stating that the system displays information correctly regardless of the platform received the lowest mean of 4.70. Despite having the lowest rating, it is still interpreted as Strongly Agree, demonstrating that the respondents remained satisfied with the system's cross-platform performance.

The findings imply that the developed E-Lib: A Web-Based Library Management Information System (LMIS) is capable of operating efficiently across various devices and web browsers while maintaining reliable integration with external communication services. Its ability to provide consistent functionality in different computing environments enhances accessibility and supports uninterrupted library operations. The favorable ratings across all indicators further suggest that the system can effectively meet users' needs regardless of the platform they use, making it a dependable solution for library management.



Table 21

Maintainability of the System by the Faculty

Items	Mean	Verbal Description
1. The system structure complies with software standards making it easy to adapt to changes.	4.90	Strongly Agree
2. The system can be easily updated to add new features or improvements.	4.90	Strongly Agree
3. The system can be enhanced and maintained with minimal downtime.	4.90	Strongly Agree
4. The system includes clear and well-documented code, making it easier for developers to understand, debug, and modify.	4.90	Strongly Agree
5. The system allows administrators to easily monitor and maintain system operations.	4.90	Strongly Agree
Overall Mean	4.90	Strongly Agree

As shown in the table above, the system's maintainability obtained an overall mean of 4.90, verbally described as Strongly Agree, indicating that faculty members strongly agree that the system is easy to maintain, modify, and improve. Notably, all five indicators received an identical mean of 4.90, reflecting a uniformly high level of agreement across all aspects of maintainability. These include the system's compliance with software standards for easy adaptation to changes, its capacity to be updated with new features, its ability to be enhanced and maintained with minimal downtime, the availability of clear and well-documented code for easier debugging and modification, and its support for administrators in monitoring and



maintaining system operations. The consistency of these ratings suggests that faculty members perceive no notable difference in strength among these maintainability indicators, viewing the system as equally reliable across all of them.

These results imply that the E-Lib: A Web-Based Library Management Information System (LMIS) is well-equipped to support long-term use, as its uniformly high maintainability ratings suggest it can accommodate future upgrades, minimize service interruptions, and reduce the technical burden on administrators. This further suggests that the system was designed with sustainability in mind, enabling the library to adapt to evolving user needs and technological demands without compromising operational continuity. In overall, the findings affirm that the system's architecture supports efficient long-term management, contributing to the continued delivery of reliable library services.



Table 22

Security of the System by the Faculty

Items	Mean	Verbal Description
1. The system sends One-Time Password (OTP) notifications via Gmail and SMS during registration and login to verify user identity.	5.00	Strongly Agree
2. The system requires users to create strong passwords to enhance security.	5.00	Strongly Agree
3. The system limits login attempts to three (3) consecutive tries to prevent unauthorized access.	5.00	Strongly Agree
4. The system automatically generates and assigns unique credentials to each user to maintain secure and personalized account access.	5.00	Strongly Agree
5. The system ensures that users' personal information is accessible only to authorized personnel, maintaining confidentiality and privacy.	4.90	Strongly Agree
Overall Mean	4.98	Strongly Agree

As shown in the table above, the system's security received an overall mean of 4.98, verbally described as Strongly Agree indicating that faculty members strongly agree that the system provides reliable protection for user accounts and data. Among the indicators, four items obtained the highest mean of 5.00: the system's sending of One-Time Password (OTP) notifications via Gmail and SMS during registration and login, its requirement for users to create strong passwords, its limitation of login attempts to three consecutive tries to prevent unauthorized access, and its automatic generation and assignment of unique credentials to



each user. This suggests that faculty members are fully confident in these specific security measures. Meanwhile, the item stating that the system ensures users' personal information is accessible only to authorized personnel obtained the lowest mean of 4.90, which, while still verbally interpreted as Strongly Agree, indicates a slightly lower, though still strong, level of agreement compared to the other indicators.

These results imply that the E-Lib: A Web-Based Library Management Information System (LMIS) provides a strong layer of protection for its users, particularly through its multi-factor verification process, enforced password strength, restricted login attempts, and automatically assigned credentials, all of which help prevent unauthorized access and account compromise. The slightly lower rating on restricting personal information to authorized personnel may suggest a relatively smaller, though still minor, area for improvement — such as further clarifying or reinforcing access-control protocols for sensitive user data. Overall, the findings affirm that the system maintains a high standard of security, helping to safeguard user identities and protect the confidentiality of library records.

Table 23

**Summary Table Result of Evaluation of System by the End Users
(Students/Stakeholders/Librarian)**

Items	Mean	Verbal Description
Functionality	4.83	Strongly Agree
Usability	4.88	Strongly Agree
Reliability	4.76	Strongly Agree
Compatibility	4.82	Strongly Agree
Maintainability	4.84	Strongly Agree
Security	4.90	Strongly Agree
Overall Mean	4.84	Strongly Agree

The summary table presents the overall evaluation of the E-Lib: A Web-Based Library Management Information System (LMIS) by the end users, including students, stakeholders,



and the librarian. The system obtained an overall mean of 4.84, verbally interpreted as Strongly Agree, indicating that the respondents highly evaluated the system's overall quality based on the ISO/IEC 25010 software quality characteristics. Among the evaluated criteria, Security received the highest mean of 4.90, demonstrating that respondents strongly agreed that the system effectively protects user information, controls access, and ensures data confidentiality. On the other hand, Reliability obtained the lowest mean of 4.76, which is still verbally interpreted as Strongly Agree, indicating that the respondents perceived the system as dependable and capable of performing its intended functions consistently with minimal errors.

These findings imply that the E-Lib: A Web-Based Library Management Information System (LMIS) successfully satisfies the quality expectations of its end users. The highest rating for security emphasizes the respondents' confidence in the system's ability to safeguard sensitive information and provide a secure environment for library transactions. Meanwhile, the consistently high ratings across functionality, usability, reliability, compatibility, and maintainability indicate that the system is efficient, user-friendly, adaptable, easy to maintain, and dependable for daily library operations. Overall, the evaluation confirms that the developed system effectively meets the needs of students, stakeholders, and the librarian, making it a reliable and secure solution for improving library management services.



Table 25

Summary Table Result of Evaluation of System by the IT Experts

Items	Mean	Verbal Description
Functionality	4.60	Strongly Agree
Usability	4.68	Strongly Agree
Reliability	4.56	Strongly Agree
Compatibility	4.72	Strongly Agree
Maintainability	4.80	Strongly Agree
Security	5.00	Strongly Agree
Overall Mean	4.73	Strongly Agree

The summary table presents the overall evaluation of the E-Lib: A Web-Based Library Management Information System (LMIS) by the IT experts. The system obtained an overall mean of 4.73, verbally interpreted as Strongly Agree indicating that the experts highly evaluated the system based on the ISO/IEC 25010 software quality characteristics. Among the evaluated criteria, Security received the highest mean of 5.00, demonstrating that the IT experts strongly agreed that the system provides excellent protection for user data, ensures secure access, and safeguards system operations. Meanwhile, Reliability obtained the lowest mean of 4.56, which is still verbally interpreted as Strongly Agree, indicating that the experts perceived the system as dependable and capable of performing its intended functions consistently with minimal issues.

These findings imply that the E-Lib: A Web-Based Library Management Information System (LMIS) complies with the ISO/IEC 25010 software quality standards in terms of functionality, usability, reliability, compatibility, maintainability, and security. The highest rating for security reflects the experts' confidence in the system's ability to protect sensitive information and maintain secure operations, while the consistently high ratings across the other quality characteristics indicate that the system is efficient, user-friendly, reliable,



adaptable, and maintainable. In overall, the evaluation confirms that the developed system possesses the necessary software quality attributes for deployment and can effectively support library operations while meeting professional and industry standards.

Table 26

Summary Table Result of Evaluation of System by the Faculty

Items	Mean	Verbal Description
Functionality	4.76	Strongly Agree
Usability	4.84	Strongly Agree
Reliability	4.76	Strongly Agree
Compatibility	4.82	Strongly Agree
Maintainability	4.90	Strongly Agree
Security	4.98	Strongly Agree
Overall Mean	4.84	Strongly Agree

The summary table above presents the overall evaluation of the E-Lib: A Web-Based Library Management Information System (LMIS) by the faculty members. The system obtained an overall mean of 4.84, verbally interpreted as Strongly Agree indicating that the respondents highly evaluated the system based on the ISO/IEC 25010 software quality characteristics. Among the evaluated criteria, Security received the highest mean of 4.98, demonstrating that the faculty members strongly agreed that the system effectively safeguards user information and ensures secure access to library resources. Meanwhile, Functionality and Reliability both obtained the lowest mean of 4.76 which is still verbally interpreted as Strongly Agree indicating that the respondents perceived the system as capable of performing its intended functions consistently, with only minimal areas for improvement.

These findings imply that the E-Lib: A Web-Based Library Management Information System (LMIS) complies with the ISO/IEC 25010 software quality standards in terms of functionality, usability, reliability, compatibility, maintainability, and security. The highest rating



for security reflects the faculty members' confidence in the system's ability to protect sensitive information and provide secure access to library services. Furthermore, the consistently high ratings across all quality characteristics indicate that the system is functional, user-friendly, dependable, compatible, and easy to maintain. All in all, the evaluation confirms that the developed system effectively supports library operations while meeting the quality expectations and professional standards of its faculty users.

Table 27

The Result of the System Evaluation

Items	End Users	IT Experts	Faculty	Overall Mean	Verbal Description
Functionality	4.83	4.60	4.76	4.73	Strongly Agree
Usability	4.88	4.68	4.84	4.8	Strongly Agree
Reliability	4.76	4.56	4.76	4.69	Strongly Agree
Compatibility	4.82	4.72	4.82	4.79	Strongly Agree
Maintainability	4.84	4.80	4.90	4.85	Strongly Agree
Security	4.90	5.00	4.98	4.96	Strongly Agree
Overall Mean	4.84	4.73	4.84	4.80	Strongly Agree

As shown in the table above, the end users, IT experts, and faculty members all gave the system evaluation a Strongly Agree rating, as indicated by the overall mean score of 4.80. This suggests that the intended users' expectations are successfully met by the E-Lib: A Web-Based Library Management Information System (LMIS). Based on the ISO/IEC 25010 software quality standards, the respondents' favorable evaluation of the system's functionality, usability, reliability, compatibility, maintainability, and security is reflected in the overall rating.

Among the assessed software quality standards, Security received the highest overall mean score of 4.96. With features like One-Time Password (OTP) verification through email and SMS, strong password enforcement, login attempt restrictions, and controlled access to sensitive data, the system's ability to protect user accounts, safeguard confidential information,



and provide secure access was highly recognized by the respondents, according to this result. These results imply that the system successfully satisfies the security needs of a contemporary library management system and gives users assurance regarding the security of their transactional and personal information. However, despite still being in the Strongly Agree category, Reliability received the lowest overall mean score of 4.69. While acknowledging that there is still space for improvement in maintaining stable performance and data consistency under various operating conditions, this result indicates that respondents thought the system was reliable and capable of consistently carrying out its intended functions, such as book borrowing, returning, and visitor appointment transactions.

All in all, the findings imply that the E-Lib: A Web-Based Library Management Information System (LMIS) successfully complies with the ISO/IEC 25010 software quality standards in terms of functionality, usability, reliability, compatibility, maintainability, and security. The consistently high ratings across all quality characteristics demonstrate that the system is functional, user-friendly, reliable, compatible across different platforms, easy to maintain, and highly secure. Furthermore, the excellent evaluation from the end users, IT experts, and faculty members confirms that the system effectively supports library operations, improves the efficiency of library services, enhances the overall user experience, and satisfies the quality expectations of its intended users. Overall, the results indicate that the developed system is suitable for implementation at the Polytechnic University of the Philippines – Bansud Campus Library.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Chapter 5

CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the conclusions drawn and necessary recommendations based on the findings stated in the preceding chapter.

Conclusion

Based on the findings of the study, the proponents conclude that the objectives of the research were successfully achieved.

1. The system successfully examined the existing manual library management process and identified its limitations, including inefficient record management, time-consuming transactions, difficulty in monitoring borrowed books, and limited accessibility of library services, which justified the development of the system.
2. The study identified the problems encountered in the manual library process such as errors and inconsistencies in recording and monitoring library transactions, delays in retrieving information, and challenges in managing library resources, thereby establishing the need for an automated library management system.
3. The developed E-Lib: A Web-Based Library Management Information System (LMIS) successfully achieved its intended functionalities by providing a comprehensive platform that allows users to search and view available books, securely register accounts with OTP-based email verification for password recovery, and conveniently access library services such as viewing library policies, searching for books, viewing book details, and borrowing books. The system also enables visitors to schedule library appointments, provides administrators with a secure OTP-based login and an efficient dashboard for managing users, books, borrowing and returning transactions, and



appointments, and generates accurate reports on books, users, borrowing transactions, and appointment schedules.

4. The developed system contributes to the attainment of the United Nations Sustainable Development Goals (SDGs) by supporting SDG 4 (Quality Education) through improving access to library resources and promoting lifelong learning; SDG 9 (Industry, Innovation, and Infrastructure) by modernizing library operations through web-based technologies and automated information management; and SDG 16 (Peace, Justice, and Strong Institutions) by promoting transparency, accountability, and secure access to library information through a centralized and well-managed system.

5. Based on the findings of evaluation conducted by the end users, IT experts, and faculty and staff, the developed system successfully complied with the ISO/IEC 25010 software quality standards. The respondents strongly agreed that the functionality, usability, reliability, compatibility, maintainability, and security of the E-Lib: A Web-Based Library Management Information System (LMIS) meets the expected software quality standards. This indicates that the system is effective, user-friendly, reliable, compatible across different platforms, easy to maintain, and capable of protecting sensitive library information making it suitable for implementation at the Polytechnic University of the Philippines – Bansud, Oriental Mindoro Campus.

Recommendations

Based on the result, the following recommendations are made:

1. The librarian may fully utilize the E-Lib: A Web-Based Library Management Information System (LMIS) in managing library resources, monitoring borrowing and returning transactions, generating reports, and organizing user records. Proper training



and continuous system maintenance are also recommended to ensure efficient, accurate, and reliable library resources.

2. Users such as students, faculty and staff, and visitors are encouraged to maximize the features of the system by searching available library resources, requesting book borrowing, monitoring their transactions, and utilizing the appointment scheduling feature when necessary. Regular use of the system will improve accessibility to library services, reduce processing time, and provide a more convenient and organized library experience that supports academic, research, and information needs.

3. The Polytechnic University of the Philippines – Bansud Campus may adopt and continuously support the implementation of the system by providing regular maintenance, software updates, user training, and the necessary technological infrastructure. The university is also enhancements to ensure its long-term sustainability and further improve the quality and efficiency of library services.

4. The system may be further enhanced by integrating external academic databases, such as Google Scholar, JSTOR, and ResearchGate, to provide users with broader access to scholarly resources. In addition, CAPTCHA verification may be implemented to strengthen system security by preventing automated attacks and unauthorized access.

5. Future researchers and developers may use this study as a reference in developing or enhancing similar web-based library management systems by exploring emerging technologies, improving existing system functionalities, and integrating innovative features, such as a mobile application version, to enhance accessibility, improve user experience, and further strengthen digital library and information management services.

**REFERENCES**

- Acasamoso, D.M., Avila, E.C., & Ficundo, N.T. (2021). PUPRBLMS: Development And Acceptability of a Proposed Library Management System. 2021 IEEE 13th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), 1-5. [10.1109/HNICEM54116.2021.9731869](https://doi.org/10.1109/HNICEM54116.2021.9731869)
- Adebayo, B. R. (2019). Library management system with topic modelling and its adaptability to open and distance learning libraries. Library Philosophy and Practice (e-journal), Article 2709. <https://digitalcommons.unl.edu/libphilprac/2709>
- Aguila, J. C. & Balba, N. P. (2025). Smartlibrary: Comprehensive Rfid-Based Library System For Attendance Tracking And Inventory Management. Asia Pacific Journal of Management and Sustainable Development, 13(No. 1, Part 2), 127-132. <https://doi.org/10.70979/MZAC6221>
- Aguirre, R. G. R., Data, A. R. M., Elloran, M. R., Lubis, H. M., & Suarez, A. J. (2024). LIBTRACK: PUP Bansud Campus Library Information System (LIS).
- Araya, T. W., & Mengsteab, A. (2020). Designing web-based library management system. International Journal of Engineering Research & Technology (IJERT), 9(10). <http://www.ijert.org>
- Ayo, E. B., Jotic, R. N., Raqueño, A. R., Loresca, J. V. G., Mendoza, I. F., & Baroña, P. V. M. (2023). Development of an integrated library management system (ILMS). International Journal of Interactive Mobile Technologies (IJIM), 17(10), 242–256. <https://doi.org/10.3991/ijim.v17i10.37509>



- Bagewadi, P., Mathapati, D., Gangur, B., Madiwalar, A., & Patil, S. S. (2025). Automating libraries: The role of library management systems in modern library operations. *International Journal of Research in Library Science (IJRLS)*, 11(2), 105–114. <https://doi.org/10.26761/IJRLS.11.2.2025.1877>
- Bhati, P., Komal, Prince. & Saxena, S. (2024). Modern Library Management System: A Web-Based Framework For Resource Automation And User Oversight. *International Journal of Research -Granthaalayah*, 12(6), 160–172. <https://doi.org/10.29121/granthaalayah.v12.i6.2024.6099>
- Borgoña, C. A., Bose, J., Cadag, N. A., & Trinidad, D. R. (2023). Library management system for Abra Valley Colleges [Capstone project, Abra Valley Colleges]. <https://www.studocu.com/ph/document/abra-valley-colleges/capstone-project/thesis-library-management-system/69020201>
- Doromal, W. G., & Soberano, K. T. (2023). Capstone archiving management system with citation and reference generator. *International Journal of Research in Engineering and Science (IJRES)*, 11(9), 128-133. <https://www.ijres.org>
- Dugbartey, A. N., & Kehinde, O. (2025). Optimizing project delivery through agile methodologies: Balancing speed, collaboration and stakeholder engagement. *World Journal of Advanced Research and Reviews*, 25(01), 1237-1257. <https://doi.org/10.30574/wjarr.2025.25.1.0193>
- Guo, X., Hong, T., Chen, Z., & Kadoch, M. (2023). Design and implementation of web-based library management system. *In 2023 4th International Conference on Image Processing and Network Protocol (ICIPNP)* (pp. 270–273). IEEE. <https://doi.org/10.1109/ICIPNP62754.2023.00050>



- Katariya, B. (2023). Library database management system - A systematic literature review. *International Journal of Humanities Social Science and Management (IJHSSM)*, 3(5), 501–505. <https://www.ijhssm.org>
- Lagas, S., & Isip, J. (2023). Challenges to Digital Services in Philippine Academic Libraries. *Philippine Journal of Librarianship and Information Studies*, 43(1), 27–38. Retrieved from <https://phjlis.org/index.php/phjlis/article/view/133>
- Lim, W. M. (2025). What is quantitative research? An overview and guidelines. *Australasian Marketing Journal*, 33(3), 325–348. <https://doi.org/10.1177/14413582241264622>
- Manne, K. (2023). Library management system (Capstone project No. 676). All Capstone Projects. <https://opus.govst.edu/capstones/676>
- Moruf, H. A., & Abu, I. Z. (2024). Next generation platforms in library management systems: A review of web-based and cloud-based software. *FUW Trends in Science & Technology Journal*, 9(2), 86–93. <https://www.ftstjournal.com/Digital%20Library/92%20Article%2014.php>
- Obsanga, A. P., & Enierga, R. R. (2021). Automated library management system for public libraries in the Philippines. *Library Hi Tech News*, 38(9), 17–22. <https://doi.org/10.1108/LHTN-10-2021-0072>
- Padilla, R. C. (2022). Assessment of library users' problems on transactional procedures: Basis for library management system development. *International Journal of Scientific and Management Research*, 5(6), 10–17. <https://doi.org/10.37502/IJSMR.2022.5602>
- Patidar, D., Maheshwari, H., & Garg, I. (2024). Library management system. *International Journal of Information Technology and Computer Engineering*, 04(03). <https://doi.org/10.55529/ijitc.43.9.13>



- Pinto, J. M. (2024). Efficient library solutions: Developing a library management system for FL Vargas College. *AIDE Interdisciplinary Research Journal*, 8(1), 174–181. <https://doi.org/10.56648/aide-irj.v8i1.119>
- Rahman, A. (2023). Library Management System: A complete solution for an online library. <https://doi.org/DOI>
- Rathour, S. S., & Dewangan, O. (2024). Library management system. *International Journal of Novel Research and Development*, 9(5). <https://ijnrd.org>
- Singh, A. (2020). Smart library management system using QR code (Senior Capstone Project). Earlham College. https://portfolios.cs.earlham.edu/wp-content/uploads/2020/05/SLMS_paper_adarsh.pdf
- Tahil, S. K. (2022). Library automation: An emerging technology for state university and colleges in Sulu Province. *Natural Sciences Engineering & Technology Journal*, 2(1). <https://doi.org/10.37275/nasetjournal.v2i1.16>
- Udo-Okon, T. N., & Esiere, E. U. (2022). Assessment of the steps in designing a web base digital library management system for schools: Opportunities and challenges. *International Journal of Eminent Scholars*, 8(1), 1–12. <https://www.globalacademicstar.com/download/article/assessment-of-the-steps-in-designing-a-web-base-digital-library-management-system-for-schools-opportunities-and-challenges.pdf>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

APPENDICES



Appendix 1
ENGLISH CRITIC'S CERTIFICATION

This is to certify that the undersigned has reviewed all the pages of capstone project, **"E-Lib: A Web-Based Library Management Information System (LMIS)"**, prepared by Mariadel B. Falculan, Edil Con L. Gorospe, Joshua D. Javier, Pearl Lorane M. Rafol, and Kristine C. Rodeo. The manuscript has been reviewed for compliance with the structural rules governing the composition of sentences, phrases, and words in the English language. Furthermore, all corrections and recommendations have been incorporated into the final paper.

Signed:

MARIA JOSIEFEL ANN R. MAAPOY, MAEd
Editor



Appendix 2

ETHICS CLEARANCE



Republic of the Philippines
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Office of the Vice President for Campuses
Bansud Oriental Mindoro Campus
UNIVERSITY RESEARCH ETHICS COMMITTEE



Date: March 12, 2026

TO/FOR: **MARIADEL B. FALCULAN**
EDIL CON L. GOROSPE
JOSHUA D. JAVIER
PEARL LORANE M. RAFOL
KRISTINE C. RODEO
Ethical Clearance

Subject: Ethical Clearance

From: 
RHEA A. MAGANDA, MAEd
Panel Chair

Recommendation From: Aileen Camille B. Dimatatac, MC (sgd)
Chief, Research Ethics Section

This is to inform you that the documentary requirements you submitted for your research project titled "**E-Lib: A Web-Based Library Management Information System**" passed the evaluation of the PUP Research Ethics Committee (REC) through the **College/Campus Research Ethics Committee** under the requirements set by the Philippine Health Research Ethics Board (PHREB).

UREC Code	CM-BAN-BSIT 2026-002
Type of Review	EXPEDITED
Approval Date	March 12, 2026
Expiry Date	March 13, 2027
PUP-UREC Decision	Approved

The standard conditions of this approval are as follows:

1. Conduct the project following the submitted and approved research protocol and other documentary requirements.
2. If changes are to be made in the research project/study that will affect the research participants, an amendment of the research protocol must be submitted to **CLREC/CMREC DESIGNATED EMAIL ADD** before implementing such changes.
3. When ethical clearance is about to expire, the researcher must apply to resubmit the research protocol.
4. A final report/terminal report must be submitted when the research project/study is complete.
5. Researchers must advise the CLREC/CMREC DESIGNATED EMAIL ADD in writing if the research project/study has been discontinued.
6. You may now commence your research project/study. Good luck.

S423, 4th Floor South Wing, PUP A. Mabini Campus, Anonas Street, Sta. Mesa, Manila 1016
Trunk Line: 335-1787 or 335-1777 local 235/357
Website: www.pup.edu.ph | Email: urec@pup.edu.ph

THE COUNTRY'S 1st POLYTECHNIC





Appendix 3

CERTIFICATION OF ORIGINALITY

This certifies that the capstone project titled:

E-Lib: A Web-Based Library Management Information System (LMIS)

by

Mariadel B. Falculan

Edil Con L. Gorospe

Joshua D. Javier

Pearl Lorane M. Rafol

Kristine C. Rodeo

has passed the originality check with a detected text-matching similarity of 9%

using Turnitin

This certificate is issued on _____, upon the request of the above for

whatever legal purpose, it may serve.

JULIE ANN D. ABEL, MIT

Program Coordinator



Appendix 4

COMMUNICATION/LETTERS

 **Republic of the Philippines**
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Office of the Vice President for Campuses
BANSUD ORIENTAL MINDORO CAMPUS 

September 12, 2025

RICARDO F. RAMISCAL, MPA
Campus Director
This University

Thru: **MS. NOEKA SHALOM B. SUPETTRAN, PhD**
Cultural Affairs Officer/ Library In-charge

Dear Sir Ramiscal,

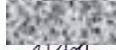
Greetings of peace!

We, the 3rd year Bachelor of Science in Information Technology students at Polytechnic University of the Philippines Bansud Oriental Mindoro Campus, are currently conducting a Capstone Project entitled "E-Lib: A Web-Based Library Management Information System (LMIS)". In accordance with our project, we would like to request an interview regarding the background information of the PUP Bansud Campus Library and other information that can help our study.

Rest assured that all collected information will be kept strictly confidential and will only be used for academic purposes. If you have any questions or need more details, feel free to contact us at 09291706231 or through email at gorospeedilcon@gmail.com

Your favorable approval of this request would be of great help in this endeavor. Thank you very much and more power. God Bless!

Respectfully yours,


Edil Con L. Gorospe
Programmer

Noted by:


MARVIN M. SANCHO
Capstone Adviser


DINA GRACE T. MAGNAYE, PhD
Head, Academic Programs

Recommending Approval:


RICARDO F. RAMISCAL, MPA
Campus Director

PUP A. Mabini Campus, Anonas Street, Sta. Mesa, Manila 1016
Trunk Line 335-1787 or 335-1777
Website www.pup.edu.ph | Inquiries <https://bit.ly/PUPSINTA>
A LEADING COMPREHENSIVE
POLYTECHNIC UNIVERSITY IN ASIA

**Appendix 5****INTERVIEW GUIDE QUESTIONS****Interview Questions for the Librarian**

Purpose: To identify the current library management procedures, determine the challenges encountered in managing library records and transactions, and gather the system requirements for the development of the E-Lib: A Web-Based Library Management Information System (LMIS).

1. What challenges do you experience in managing library records manually?
2. What is the current step-by-step process of managing library transactions, such as borrowing and returning books?
3. What information is required when registering a new library member?
4. How do you currently record and organize book information in the library?
5. How do you monitor the availability and status of books?
6. Approximately how many books are borrowed and returned each day?
7. How many library users do you usually accommodate on a daily, weekly, monthly, and yearly basis?
8. What challenges do you encounter in managing library records and transactions?
9. What types of library materials are available for borrowing?
10. What are the most frequently borrowed books or library resources?
11. How do you handle damaged or lost books?
12. During peak hours, approximately how many users are accommodated in the library?
13. What features or improvements would you like to see to improve library services?



Appendix 6

SURVEY QUESTIONNAIRE

 POLYTECHNIC UNIVERSITY OF THE PHILIPPINES																																											
SURVEY QUESTIONNAIRE E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)																																											
Dear Evaluator,																																											
Good day!																																											
We, Bachelor of Science in Information Technology students from Polytechnic University of the Philippines Bansud, Oriental Mindoro Campus, currently conducting our Capstone Project entitled "E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)". This project aims to develop a web-based system that will enhance the efficiency, security, and accessibility of library services. The system will streamline library operations, improve record keeping, and provide easy access to library resources for students, faculty, and visitors. It will also generate reports and dashboards to help administrators make informed decisions regarding library management. As part of our data gathering, we would like to ask for your participation in this survey by answering this questionnaire. In accordance with RA 10173 of the Data Privacy Act of 2012, all personal and/or sensitive information solicited and disclosed from this survey questionnaire shall only be used for the study alone. Rest assured that your responses to this instrument will be kept confidential. Thank you for taking the time to complete this questionnaire. Respectfully yours, The Proponents Name of Evaluator (Optional): _____ Date: <u>6-19-21</u> I. Direction: Kindly put a check (✓) on the box that signifies your answer. ☒ Expert ☐ Students ☐ Faculty ☐ Stakeholders <table border="1"><thead><tr><th>Numerical Scale</th><th>Verbal Description</th></tr></thead><tbody><tr><td>5</td><td>Strongly Agree</td></tr><tr><td>4</td><td>Agree</td></tr><tr><td>3</td><td>Moderately Agree</td></tr><tr><td>2</td><td>Disagree</td></tr><tr><td>1</td><td>Strongly Disagree</td></tr></tbody></table> II. Direction: Please put a check (✓) on the column from the scale above that will correspond to your answer. 1. FUNCTIONALITY <table border="1"><thead><tr><th>Items</th><th>5</th><th>4</th><th>3</th><th>2</th><th>1</th></tr></thead><tbody><tr><td>1. The system can manage book records without errors.</td><td></td><td>✓</td><td></td><td></td><td></td></tr><tr><td>2. The system accurately processes book borrowing and returning transactions.</td><td></td><td>✓</td><td></td><td></td><td></td></tr><tr><td>3. The system correctly manages visitor appointments and scheduling.</td><td></td><td></td><td>✓</td><td></td><td></td></tr><tr><td>4. The system allows librarians to manage user accounts such as students, faculty, and visitors.</td><td>✓</td><td></td><td></td><td></td><td></td></tr></tbody></table>		Numerical Scale	Verbal Description	5	Strongly Agree	4	Agree	3	Moderately Agree	2	Disagree	1	Strongly Disagree	Items	5	4	3	2	1	1. The system can manage book records without errors.		✓				2. The system accurately processes book borrowing and returning transactions.		✓				3. The system correctly manages visitor appointments and scheduling.			✓			4. The system allows librarians to manage user accounts such as students, faculty, and visitors.	✓				
Numerical Scale	Verbal Description																																										
5	Strongly Agree																																										
4	Agree																																										
3	Moderately Agree																																										
2	Disagree																																										
1	Strongly Disagree																																										
Items	5	4	3	2	1																																						
1. The system can manage book records without errors.		✓																																									
2. The system accurately processes book borrowing and returning transactions.		✓																																									
3. The system correctly manages visitor appointments and scheduling.			✓																																								
4. The system allows librarians to manage user accounts such as students, faculty, and visitors.	✓																																										



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES					
5. The system generates accurate library reports for book borrowing and visitors' appointment.			✓		
2. USABILITY					
Items	5	4	3	2	1
1. The system interface is easy to understand and navigate.		✓			
2. The system navigation menu is easy to use.		✓			
3. The system notifies users about important transaction updates, such as book borrowing, book returning, and visitor appointments.		✓			
4. The system displays the current condition of the book.			✓		
5. The system displays the available time slots during which visitors may schedule a library appointment.		✓			
3. RELIABILITY					
Items	5	4	3	2	1
1. The system reliably validates user input and prevents incorrect data entries.		✓			
2. The system ensures that stored data remains consistent and error-free.		✓			
3. The system maintains stable performance when multiple users access it simultaneously.		✓			
4. The system ensures that library policies are consistently implemented and followed.		✓			
5. The system consistently performs book borrowing, returning, and visitor appointment tasks without errors or failures.		✓			
4. COMPATIBILITY					
Items	5	4	3	2	1
1. The system works properly on different web browsers/devices.	✓				
2. The system operates smoothly in an online environment without causing conflicts with other software applications.		✓			
3. The system seamlessly integrates with external services, including email and SMS gateways, to facilitate automated notifications and communication.		✓			
4. The system displays information correctly regardless of the platform being used.		✓			
5. The system maintains stable performance during simultaneous operation with other applications.		✓			
5. MAINTAINABILITY					
Items	5	4	3	2	1
1. The system structure complies with software standards making it easy to adapt to changes.	✓				
2. The system can be easily updated to add new features or improvements.	✓				



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES					
3. The system can be enhanced and maintained with minimal downtime.	✓				
4. The system includes clear and well-documented code, making it easier for developers to understand, debug, and modify.	✓				
5. The system allows administrators to easily monitor and maintain system operations.	✓				
6. SECURITY					
Items	5	4	3	2	1
1. The system sends One-Time Password (OTP) notifications via Gmail and SMS during registration and login to verify user identity.	✓				
2. The system requires users to create strong passwords to enhance security.	✓				
3. The system limits login attempts to three (3) consecutive tries to prevent unauthorized access.	✓				
4. The system automatically generates and assigns unique credentials to each user to maintain secure and personalized account access.	✓				
5. The system ensures that users' personal information is accessible only to authorized personnel, maintaining confidentiality and privacy.	✓				



Appendix 7
SYSTEM EVALUATION



As an IT expert, Ma'am Julie Anne D. Abel evaluated and analyzed our system. During the evaluation, she identified the issues she found in the system and provided recommendations for improvement.



As an IT expert, Ma'am Jerlie D. Minay evaluated and analyzed our system, providing valuable comments and suggestions for improvement.



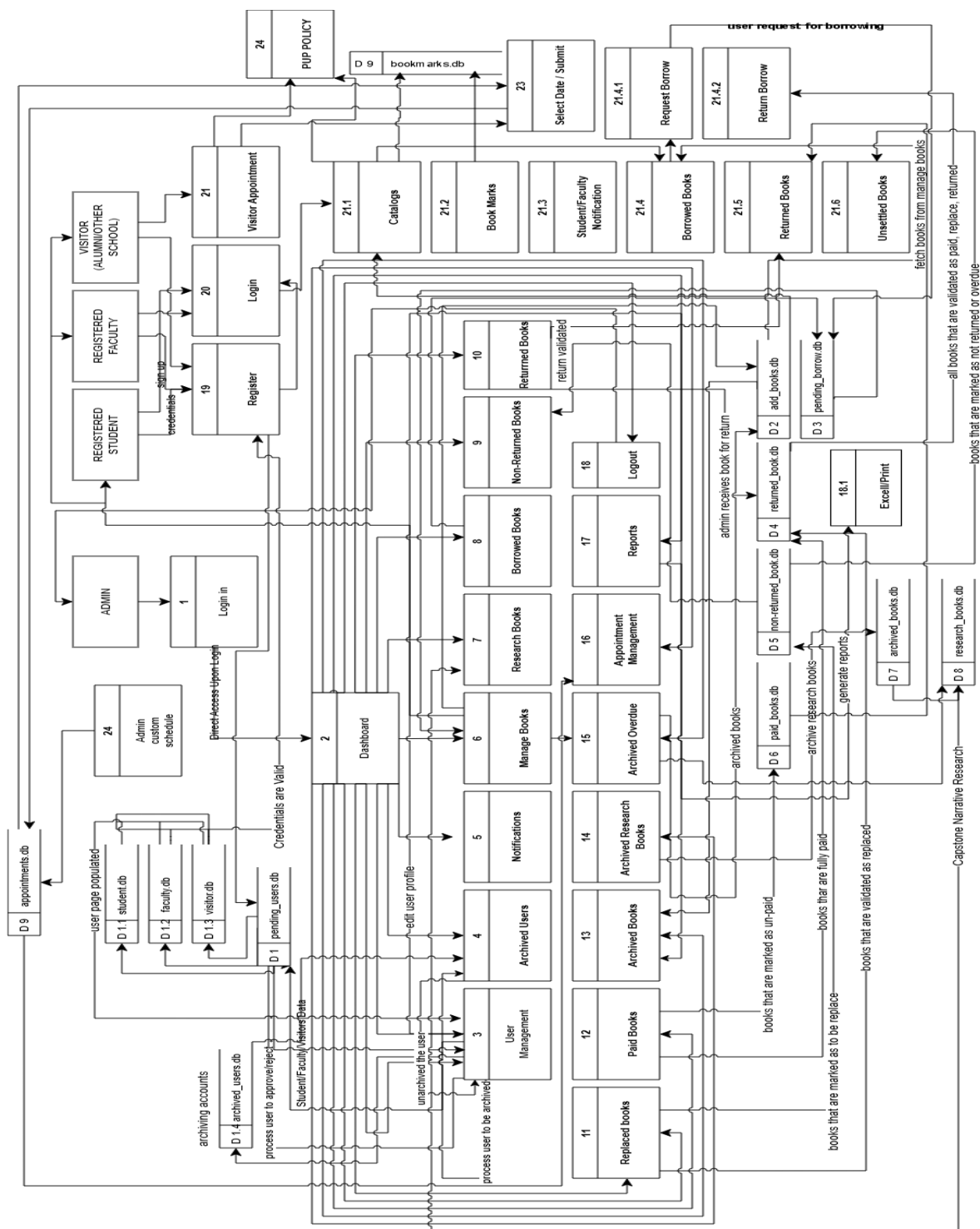
The scrum team presented the system to the students and the programmer was the one who explained and show the system.



The faculty members also participated in the evaluation of the system to assess its functionality, usability, reliability, compatibility, maintainability, and security.



DATA FLOW DIAGRAM

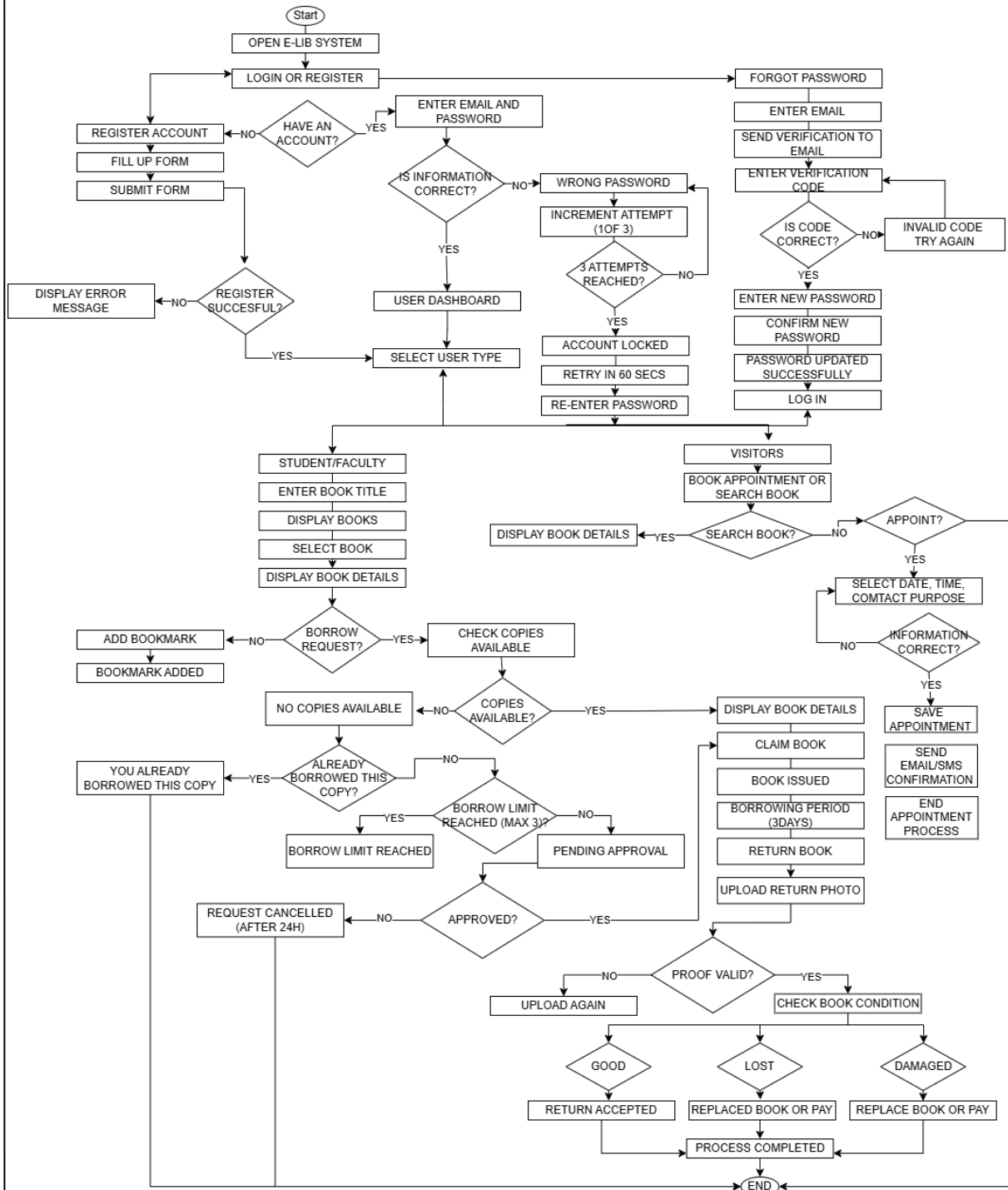




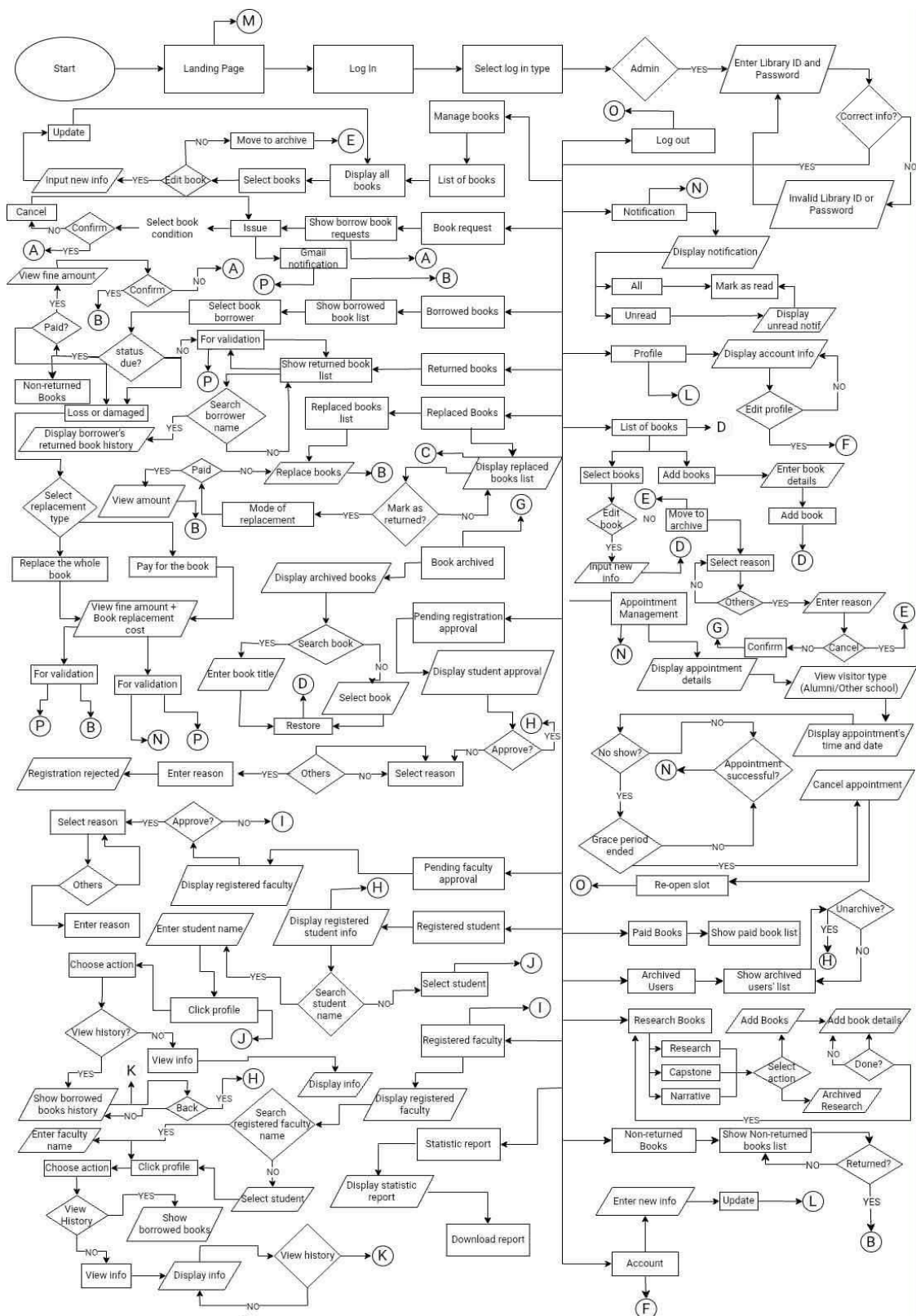
Appendix 9

SYSTEM FLOWCHART

User Side



Admin Side





Appendix 10

DATA DICTIONARY

add_books

Field	Data Type	Description
book_id	int(11)	Unique ID for each book record.
title	varchar(255)	Title of the book.
author	varchar(255)	Name of the book's author.
description	text	Brief summary or description of the book.
lcc	varchar(50)	Library of Congress Classification (LCC) code used to classify the book.
volume	varchar(50)	Volume number of the book, if applicable.
edition	varchar(100)	Edition of the book (e.g., 1st Edition, 2nd Edition).
category	varchar(100)	Category or genre of the book.
image_path	varchar(255)	File path of the book cover image.
book_condition_image	varchar(255)	File path of the image showing the book's physical condition.
stock_quantity	int(11)	Current number of available copies in the library.
original_quantity	int(11)	Original total number of book copies acquired by the library.
status	enum('Available', 'Not Available')	Availability status of the book (Available or Not Available).
book_condition	varchar(100)	Physical condition of the book (e.g., New, Good, Fair, Damaged).
book_price	decimal(10,2)	Purchase price or estimated value of the book.
date_added	datetime	Date and time when the book was added to the library system.
last_updated	datetime	Date and time when the book record was last updated.
is_archived	tinyint(1)	Archive status



admin

Field	Data Type	Description
id	int(11)	Unique ID for each administrator record.
admin_id	varchar(50)	Unique administrator identification number.
password	varchar(255)	Encrypted password used for administrator authentication.
email	varchar(255)	Administrator's email address used for communication and account recovery.
phone_number	varchar(20)	Administrator's contact number.
username	varchar(255)	Username used by the administrator to log in to the system.
profile_image	varchar(255)	File path of the administrator's profile picture.

admin_notifications

Field	Data Type	Description
id	int(11)	Unique ID for each notification record.
notification_id	varchar(100)	Unique identifier assigned to each notification.
notification_type	varchar(50)	Type or category of the notification (e.g., Borrowing, Return, Appointment, Reservation).
message	text	Detailed notification message displayed to the recipient.
source	varchar(50)	Module or feature that generated the notification.
full_name	varchar(255)	Full name of the user associated with the notification.
user_type	varchar(50)	Type of user receiving or related to the notification (e.g., Student, Faculty, Librarian).
identifier	varchar(255)	User identifier such as student number, employee ID, or library ID.
course	varchar(100)	Course or academic program of the user, if applicable.
book_id	varchar(50)	Identifier of the book associated with the notification.
book_title	varchar(255)	Title of the book related to the notification.
user_id	varchar(100)	Unique identifier of the user involved in the notification.
appointment_date	date	Scheduled date of the appointment, if applicable.



start_time	varchar(10)	Scheduled start time of the appointment.
end_time	varchar(10)	Scheduled end time of the appointment.
contact_number	varchar(20)	Contact number of the user associated with the notification.
booking_type	varchar(100)	Type of booking or reservation made by the user.
purpose	varchar(255)	Purpose or reason for the appointment or booking.
status	varchar(50)	Current status of the related transaction or request.
return_condition	varchar(50)	Condition of the returned book (e.g., Good, Damaged).
replacement_status	varchar(50)	Status of the book replacement process, if applicable.
is_read	tinyint(1)	Read status of the notification .
admin_id	varchar(50)	Identifier of the administrator responsible for or associated with the notification.
created_at	datetime	Date and time when the notification was created.

admin_notification_dismissed

Field	Data Type	Description
id	int(11)	Unique ID for each dismissed notification record.
admin_id	varchar(50)	Identifier of the administrator who dismissed the notification.
notification_id	varchar(100)	Unique identifier of the dismissed notification.
dismissed_at	timestamp	Date and time when the notification was dismissed by the administrator.

admin_notification_reads

Field	Data Type	Description
id	int(11)	Unique ID for each read notification record.
admin_id	varchar(50)	Identifier of the administrator who read the notification.
notification_id	varchar(100)	Unique identifier of the notification that has been marked as read.



read_at	timestamp	Date and time when the notification was marked as read by the administrator.
---------	-----------	--

admin_verifications

Field	Data Type	Description
id	int(11)	Unique ID for each administrator verification record.
admin_id	varchar(50)	Unique identifier of the administrator associated with the verification request.
email	varchar(255)	Administrator's email address where the verification code is sent.
otp	varchar(6)	One-Time Password (OTP) generated for administrator verification.
token	varchar(255)	Unique verification token used to validate the administrator's identity.
is_verified	tinyint(1)	Verification status of the administrator (1 = Verified, 0 = Not Verified).
created_at	datetime	Date and time when the verification record was created.
expires_at	datetime	Date and time when the OTP or verification token expires.
is_used	tinyint(1)	Indicates whether the OTP or verification token has already been used.

appointments

Field	Data Type	Description
appointment_id	int(11)	Unique ID for each appointment record.
visitor_id	varchar(255)	Unique identifier of the visitor who scheduled the appointment.
booking_id	varchar(50)	Unique booking reference number assigned to the appointment.
contact_number	varchar(20)	Contact number of the visitor.
appointment_date	date	Scheduled date of the library appointment.
purpose	text	Purpose or reason for the library visit.
additional_notes	text	Additional information or special requests provided by the visitor.



Status	enum('Pending', 'Cancelled', 'Finished')	Current status of the appointment (Pending, Cancelled, or Finished).
grace_period_start	datetime	Date and time when the appointment grace period begins.
cancel_available_at	datetime	Date and time when the appointment becomes eligible for cancellation.
created_at	timestamp	Date and time when the appointment was created.
check_in_time	datetime	Date and time when the visitor checked in for the appointment.
grace_period_end	datetime	Date and time when the appointment grace period ends.
no_show_notified	tinyint(1)	Indicates whether a no-show notification has been sent (1 = Yes, 0 = No).

appointment_books

Field	Data Type	Description
id	int(11)	Unique ID for each appointment-book record.
appointment_id	int(11)	Unique identifier of the appointment associated with the reserved or requested book.
book_id	int(11)	Unique identifier of the book linked to the appointment.
created_at	timestamp	Date and time when the appointment-book association was created.

appointment_notifications

Field	Data Type	Description
id	int(11)	Unique ID for each appointment notification record.
recipient_id	varchar(255)	Unique identifier or email address of the notification recipient.
recipient_type	enum('visitor', 'admin')	Specifies whether the notification is intended for a visitor or an administrator.
message	text	Notification message sent to the recipient.



notification_type	enum(...)	Type of notification generated by the system (e.g., appointment requested, approved, cancelled, completed).
appointment_id	int(11)	Unique identifier of the appointment associated with the notification.
is_read	tinyint(1)	Indicates whether the notification has been read (1 = Read, 0 = Unread).
created_at	datetime	Date and time when the notification was created.
additional_data	text	Stores additional information related to the notification, such as appointment details or system-generated metadata.

appointment_time_slots

Field	Data Type	Description
id	int(11)	Unique ID for each appointment time slot record.
appointment_id	int(11)	Unique identifier of the appointment associated with the time slot.
time_slot	varchar(50)	Scheduled time slot assigned to the appointment (e.g., 9:00 AM – 10:00 AM).
slot_status	enum('pending', 'attended', 'no_show')	Current status of the appointment time slot (Pending, Attended, or No Show).
created_at	timestamp	Date and time when the appointment time slot record was created.

archived_books

Field	Data Type	Description
archive_id	int(11)	Unique ID for each archived book record.
book_id	int(11)	Unique identifier of the original book record before it was archived.
title	varchar(255)	Title of the archived book.
author	varchar(255)	Name of the book's author.
description	text	Brief summary or description of the archived book.
lcc	varchar(50)	Library of Congress Classification (LCC) code assigned to the book.
volume	varchar(50)	Volume number of the book, if applicable.



Edition	varchar(100)	Edition of the archived book.
category	varchar(100)	Category or genre of the archived book.
image_path	varchar(255)	File path of the archived book's cover image.
stock_quantity	int(11)	Number of copies recorded at the time the book was archived.
status	varchar(20)	Availability status of the book before it was archived.
book_condition	varchar(100)	Physical condition of the book at the time of archiving.
book_price	decimal(10,2)	Purchase price or estimated value of the archived book.
archived_date	datetime	Date and time when the book was archived.
archived_by	varchar(100)	Username of the administrator who archived the book.
archive_reason	text	Reason provided for archiving the book.
original_date_added	datetime	Original date and time when the book was first added to the library system.

archived_overdue_books

Field	Data Type	Description
archive_id	int(11)	Unique ID for each archived returned book record.
return_id	int(11)	Reference to the original returned book record.
borrow_id	int(11)	Unique identifier of the corresponding borrowing transaction.
book_id	int(11)	Unique identifier of the returned book.
user_id	varchar(50)	Unique identifier of the user who borrowed and returned the book.
user_type	enum('student', 'faculty')	Type of user associated with the transaction (Student or Faculty).
book_title	varchar(255)	Title of the returned book.
book_author	varchar(255)	Author of the returned book.
borrow_date	datetime	Date and time when the book was borrowed.
due_date	datetime	Scheduled due date and time for returning the book.
returned_date	datetime	Actual date and time when the book was returned.



return_condition	enum('good', 'damaged', 'lost')	Physical condition of the book upon return (Good, Damaged, or Lost).
replacement_type	enum('replace_whole_book', 'pay_for_book')	Method selected to settle a lost or damaged book (Replace Whole Book or Pay for Book).
replacement_deadline	datetime	Original deadline for replacing the lost or damaged book.
grace_period_end	datetime	End date and time of the allowed grace period for replacement.
base_replacement_cost	decimal(10,2)	Maximum replacement cost or penalty assigned to the book.
total_accumulated_penalty	decimal(10,2)	Total penalty accumulated at the time the record was archived.
hours_overdue_at_archive	int(11)	Total number of overdue hours when the transaction was archived.
archived_date	datetime	Date and time when the returned book record was archived.
archived_by	varchar(50)	Administrator ID or system account that archived the record.
archive_reason	text	Reason for archiving the returned book record.
settlement_status	enum('unsettled', 'settled')	Indicates whether the replacement or payment has been settled.
settlement_date	datetime	Date and time when the settlement was completed.
settlement_processed_by	varchar(50)	Identifier of the administrator who processed the settlement.
settlement_notes	text	Additional remarks regarding the settlement process.
original_notes	text	Original notes or comments associated with the returned book transaction before archiving.

archived_users

Field	Data Type	Description
archive_id	int(11)	Unique ID for each archived user record.
original_id	int(11)	Original ID of the user from the source table before archiving.
user_type	enum('student', 'faculty', 'visitor')	Type of user whose account was archived (Student, Faculty, or Visitor).
user_identifier	varchar(50)	Unique user identifier such as student ID, faculty ID, or visitor email address.



library_id	varchar(50)	Unique library identification number assigned to the user.
email	varchar(255)	Email address of the archived user.
phone_number	varchar(20)	Contact number of the archived user.
first_name	varchar(100)	First name of the archived user.
last_name	varchar(100)	Last name of the archived user.
middle_name	varchar(100)	Middle name of the archived user.
suffix	varchar(20)	Name suffix of the archived user (e.g., Jr., Sr., III).
birthdate	date	Birthdate of the archived user.
age	int(11)	Age of the user at the time of archiving.
sex	enum('male', 'female', 'other')	Biological sex of the archived user.
gender_type	varchar(50)	Gender identity of the archived user, if specified.
profile_picture	varchar(255)	File path of the user's profile picture.
course	varchar(100)	Academic course or program of the student, if applicable.
department	varchar(100)	Department of the faculty member, if applicable.
visitor_type	varchar(50)	Classification of the visitor (e.g., Alumni, Researcher, Guest).
institution	varchar(255)	Institution or organization of the visitor.
original_status	varchar(20)	User account status before it was archived.
registration_date	datetime	Date and time when the user registered in the library system.
archived_date	datetime	Date and time when the user account was archived.
archived_by	varchar(50)	Administrator who archived the user account.
archive_reason	text	Reason for archiving the user account.
is_active	tinyint(1)	Indicates whether the archived account has been restored (1 = Restored/Active, 0 = Archived).

bookmarks

Field	Data Type	Description
bookmark_id	int(11)	Unique ID for each bookmark record.
user_id	int(11)	Unique identifier of the user who bookmarked the book.



book_id	int(11)	Unique identifier of the bookmarked book.
user_type	enum('student', 'faculty')	Type of user who created the bookmark (Student or Faculty).
created_at	datetime	Date and time when the bookmark was created.
is_read	tinyint(1)	Indicates whether the bookmarked item has been viewed or acknowledged by the user (1 = Read, 0 = Unread).

book_borrowings

Field	Data Type	Description
borrow_id	int(11)	Unique ID for each borrowing transaction.
book_id	int(11)	Unique identifier of the borrowed book.
user_id	varchar(50)	Unique identifier of the user who borrowed the book.
user_type	enum('student', 'faculty')	Type of user who borrowed the book (Student or Faculty).
borrow_date	datetime	Date and time when the book was borrowed.
due_date	datetime	Scheduled date and time when the borrowed book must be returned.
returned_date	datetime	Actual date and time when the book was returned.
status	enum('pending', 'ready_for_pickup', 'borrowed', 'due_soon', 'overdue', 'returned', 'cancelled')	Current status of the borrowing transaction.
fine_amount	decimal(10,2)	Total fine incurred for late return or policy violations.
notes	text	Additional remarks or notes regarding the borrowing transaction.
return_image	varchar(255)	File path of the uploaded image documenting the returned book's condition.
warning_sent	tinyint(1)	Indicates whether an overdue warning notification has been sent (1 = Yes, 0 = No).



last_fine_update	datetime	Date and time when the fine amount was last updated.
due_date_warning_sent	tinyint(1)	Indicates whether a due date reminder has been sent to the borrower (1 = Yes, 0 = No).
grace_period_warning_sent	tinyint(1)	Indicates whether a grace period warning has been sent (1 = Yes, 0 = No).
non_returned	tinyint(1)	Indicates whether the borrowing record has been classified as a non-returned book (1 = Yes, 0 = No).
moved_to_non_returned	tinyint(1)	Indicates whether the borrowing record has been transferred to the non_returned_books table (1 = Yes, 0 = No).
moved_to_non_returned_date	datetime	Date and time when the borrowing record was transferred to the non_returned_books table.

book_notifications

Field	Data Type	Description
id	int(11)	Unique ID for each borrowing notification record.
user_id	varchar(50)	Unique identifier of the user receiving the notification.
user_type	enum('student', 'faculty')	Type of user receiving the notification (Student or Faculty).
message	text	Notification message related to the borrowing transaction.
notification_type	enum('borrow_request', 'borrow_approved', 'borrow_rejected', 'ready_for_pickup', 'borrowed', 'due_date_reminder', 'overdue', 'return_confirmed', 'fine_updated', 'borrow_cancelled')	Type of borrowing notification generated by the system.
reference_id	int(11)	Reference ID of the related borrowing transaction or associated record.



book_id	int(11)	Unique identifier of the book associated with the notification.
is_read	tinyint(1)	Indicates whether the notification has been read by the user (1 = Read, 0 = Unread).
created_at	datetime	Date and time when the notification was created.

capacity_logs

Field	Data Type	Description
id	int(11)	Unique ID for each library capacity log record.
date	date	Date when the library capacity was recorded.
hour_interval	time	Specific hour or time interval associated with the recorded capacity.
previous_count	int(11)	Number of occupants in the library before the update.
new_count	int(11)	Updated number of occupants in the library after the change.
admin_id	varchar(20)	Identifier of the administrator who recorded or updated the library capacity.
notes	text	Additional remarks or comments regarding the capacity update.
created_at	datetime	Date and time when the library capacity log record was created.

capacity_settings

Field	Data Type	Description
id	int(11)	Unique ID for each library setting record.
setting_key	varchar(50)	Unique name or key that identifies a specific system setting.
setting_value	int(11)	Value assigned to the corresponding system setting.
updated_by	varchar(50)	Identifier of the administrator who last updated the setting.
updated_at	timestamp	Date and time when the system setting was last updated.



categories

Field	Data Type	Description
category_id	int(11)	Unique ID for each book category record.
category_name	varchar(100)	Name of the book category or genre.
description	text	Brief description of the book category.
created_at	datetime	Date and time when the category was created.
is_active	tinyint(1)	Indicates whether the category is active and available for use (1 = Active, 0 = Inactive).

courses

Field	Data Type	Description
id	int(11)	Unique ID for each course record.
course_code	varchar(50)	Short code assigned to the academic course (e.g., BSIT, BSEd).
course_name	varchar(150)	Complete name of the academic course or program.
year_section	varchar(20)	Year level and section associated with the course (e.g., 3A, 2B).
display_name	varchar(200)	Formatted display name combining the course, year, and section for easy identification.
is_active	tinyint(1)	Indicates whether the course record is active and available for use (1 = Active, 0 = Inactive).
created_at	datetime	Date and time when the course record was created.
updated_at	datetime	Date and time when the course record was last updated.

faculty

Field	Data Type	Description
id	int(11)	Unique ID for each faculty record.
library_id	varchar(50)	Unique library identification number assigned to the faculty member.
faculty_id	varchar(50)	Official faculty identification number.
password	varchar(255)	Encrypted password used by the faculty member to access the system.
email	varchar(255)	Faculty member's email address.
phone_number	varchar(20)	Faculty member's contact number.



Status	enum('pending', 'approved', 'rejected')	Registration status of the faculty account (Pending, Approved, or Rejected).
profile_picture	varchar(255)	File path of the faculty member's profile picture.
first_name	varchar(100)	Faculty member's first name.
last_name	varchar(100)	Faculty member's last name.
middle_name	varchar(100)	Faculty member's middle name.
suffix	varchar(20)	Name suffix of the faculty member (e.g., Jr., Sr., III).
birthdate	date	Birthdate of the faculty member.
age	int(11)	Age of the faculty member.
sex	enum('male', 'female', 'other')	Biological sex of the faculty member.
gender_type	varchar(50)	Gender identity of the faculty member, if specified.
department	varchar(100)	Academic department where the faculty member belongs.
registration_date	datetime	Date and time when the faculty member registered in the library system.
reject_reason	text	Reason provided if the faculty registration request was rejected.

non_returned_books

Field	Data Type	Description
id	int(11)	Unique ID for each non-returned book record.
borrow_id	int(11)	Reference to the original borrowing transaction.
book_id	int(11)	Unique identifier of the non-returned book.
user_id	varchar(50)	Unique identifier of the user who borrowed the book.
user_type	enum('student', 'faculty')	Type of user associated with the borrowing transaction (Student or Faculty).
book_title	varchar(255)	Title of the non-returned book.
book_author	varchar(255)	Author of the non-returned book.
book_price	decimal(10,2)	Original purchase price of the book, which also serves as the maximum fine limit.
borrow_date	datetime	Date and time when the book was borrowed.



due_date	datetime	Scheduled date and time when the book should have been returned.
grace_period_end	datetime	Date and time when the grace period for returning the book ends.
fine_start_date	datetime	Date and time when overdue fines began accumulating after the grace period.
cap_reached_date	datetime	Date and time when the accumulated fine reached the maximum amount equal to the book price.
total_fine	decimal(10,2)	Total accumulated fine charged for the non-returned book.
hours_overdue	int(11)	Total number of overdue hours when the maximum fine was reached.
working_hours_overdue	int(11)	Total overdue working hours, excluding Sundays and holidays.
settlement_status	varchar(50)	Current settlement status of the non-returned book (e.g., Unsettled or Settled).
settlement_date	datetime	Date and time when the settlement was completed.
settlement_processed_by	varchar(50)	Identifier of the administrator who processed the settlement.
settlement_notes	text	Additional remarks regarding the settlement process.
created_at	datetime	Date and time when the non-returned book record was created.
updated_at	datetime	Date and time when the record was last updated.
settlement_type	varchar(50)	Method used to settle the non-returned book (e.g., Payment or Replacement).
move_to_replacement	tinyint(1)	Indicates whether the record has been transferred to the replacement process (1 = Yes, 0 = No).
replacement_deadline	datetime	Deadline for replacing the lost or non-returned book.

paid_books

Field	Data Type	Description
paid_id	int(11)	Unique ID for each validated payment record.



return_id	int(11)	Reference to the returned book record associated with the payment.
borrow_id	int(11)	Unique identifier of the original borrowing transaction.
book_id	int(11)	Unique identifier of the book associated with the payment.
user_id	varchar(50)	Unique identifier of the user who made the payment.
user_type	enum('student', 'faculty')	Type of user who made the payment (Student or Faculty).
borrow_date	datetime	Date and time when the book was borrowed.
due_date	datetime	Scheduled date and time for returning the book.
returned_date	datetime	Actual date and time when the book was returned.
original_condition	enum('damaged', 'lost')	Original condition of the book that required payment (Damaged or Lost).
payment_amount	decimal(10,2)	Total amount paid by the user for the damaged or lost book.
base_cost	decimal(10,2)	Original replacement cost of the book before additional fines.
overdue_fine	decimal(10,2)	Additional overdue fine included in the total payment, if applicable.
payment_validated_date	datetime	Date and time when the payment was verified and validated by the administrator.
validated_by	varchar(50)	Identifier of the administrator who validated the payment.
admin_notes	text	Additional remarks or notes recorded by the administrator regarding the payment.
created_at	datetime	Date and time when the payment record was created in the system.

password_reset_tokens

Field	Data Type	Description
id	int(11)	Unique ID for each password reset OTP record.
email	varchar(255)	Email address of the user requesting a password reset.
otp	varchar(6)	One-Time Password (OTP) generated to verify the password reset request.



expires_at	datetime	Date and time when the OTP expires and becomes invalid.
user_type	enum('student', 'faculty', 'visitor')	Type of user requesting the password reset (Student, Faculty, or Visitor).
created_at	timestamp	Date and time when the OTP record was created.
used	tinyint(1)	Indicates whether the OTP has already been used (1 = Used, 0 = Not Used).

pending_borrows

Field	Data Type	Description
request_id	int(11)	Unique ID for each book borrowing request record.
book_id	int(11)	Identifier of the book being requested, linked to the book information record.
user_id	varchar(50)	Identification number or account ID of the user requesting the book.
user_type	enum('student', 'faculty')	Type of user who submitted the request (Student or Faculty).
request_date	datetime	Date and time when the borrowing request was submitted.
requested_due_date	datetime	Requested return date of the borrowed book.
status	enum('pending', 'approved', 'rejected', 'cancelled', ...)	Current status of the borrowing request (Pending, Approved, Rejected, or Cancelled).
notes	text	Additional remarks or information provided by the user regarding the borrowing request.
admin_notes	text	Remarks or comments added by the administrator regarding the request processing.
processed_date	datetime	Date and time when the request was reviewed or processed by the administrator.
processed_by	varchar(50)	Identification of the administrator who processed the borrowing request.
claimed	tinyint(1)	Indicates whether the requested book has already been claimed by the user (1 = Claimed, 0 = Not Claimed).



pending_users

Field	Data Type	Description
id	int(11)	Unique ID for each user account record.
email	varchar(100)	Email address of the user used for account registration and communication.
phone_number	varchar(20)	Contact number of the user.
user_type	enum('student', 'faculty', 'visitor')	Type of user account registered in the system (Student, Faculty, or Visitor).
id_number	varchar(100)	Unique identification number assigned to the user.
full_name	varchar(255)	Complete name of the user.
password_hash	varchar(255)	Encrypted password used for user authentication and security.
profile_picture	varchar(255)	File path or filename of the user's uploaded profile picture.
first_name	varchar(100)	First name of the user.
last_name	varchar(100)	Last name or surname of the user.
middle_name	varchar(100)	Middle name of the user.
suffix	varchar(20)	Name extension of the user, if applicable (e.g., Jr., III).
birthdate	date	Date of birth of the user.
age	int(11)	Age of the user.
sex	enum('male', 'female', 'other')	Biological sex classification of the user.
gender_type	varchar(50)	Gender identity or preferred gender classification of the user.
course	varchar(100)	Academic course or program of the student user.
section	varchar(50)	Class section or group designation of the student user.
current_status	varchar(100)	Current status of the user (e.g., enrolled, graduated, active).
occupation	varchar(255)	Profession or job designation of the visitor or faculty user.
visitor_type	varchar(50)	Classification or category of the visitor user.
registration_date	datetime	Date and time when the user account was registered.
status	enum('pending', 'approved', 'rejected')	Current approval status of the user registration request.
reject_reason	text	Reason provided by the administrator when a registration request is rejected.



department	varchar(100)	Department or office where the faculty member or user belongs.
institution	varchar(255)	School, organization, or institution associated with the user.
referral_letter	varchar(255)	File path or filename of the visitor's uploaded referral letter or supporting document.

returned_books

Field	Data Type	Description
return_id	int(11)	Unique ID for each book return transaction record.
borrow_id	int(11)	Identifier of the borrowing transaction associated with the returned book.
book_id	int(11)	Identifier of the returned book linked to the book information record.
user_id	varchar(50)	Identification number or account ID of the user who returned the book.
user_type	enum('student', 'faculty')	Type of user who borrowed and returned the book (Student or Faculty).
borrow_date	datetime	Date and time when the book was borrowed.
due_date	datetime	Expected date and time when the book should be returned.
returned_date	datetime	Actual date and time when the book was returned.
return_condition	enum('good', 'damaged', 'lost')	Condition of the returned book (Good, Damaged, or Lost).
replacement_image	varchar(255)	File path or filename of the uploaded image related to book replacement or damage evidence.
replacement_type	enum('replace_whole_book', 'pay_for_book')	Type of replacement action required for damaged or lost books (Replace Whole Book or Pay for Book).
replacement_deadline	datetime	Date and time limit given to the user for completing the replacement requirement.
replacement_status	enum('pending', 'completed', 'overdue', 'pending_verification')	Current status of the replacement process.
base_replacement_cost	decimal(10,2)	Original replacement cost of the damaged or lost book.



replacement_completed_date	datetime	Date and time when the replacement process was completed.
processed_by	varchar(50)	Identification of the administrator or staff member who processed the return transaction.
fine_amount	decimal(10,2)	Amount of fine charged to the user for late return, damage, or loss of the book.
notes	text	Additional remarks or comments regarding the book return transaction.

slot_closures

Field	Data Type	Description
id	int(11)	Unique ID for each library closure schedule record.
closure_date	date	Specific date when the library closure is scheduled.
time_slot	varchar(50)	Specifies the affected time period of the closure; NULL indicates a full-day closure.
closure_type	enum('fullday', 'halfday_am', 'halfday_pm', '1week', ...)	Type or duration of the library closure schedule (Full Day, Half Day Morning, Half Day Afternoon, One Week, etc.).
start_date	date	Starting date of the library closure period.
end_date	date	Ending date of the library closure period.
reason	text	Explanation or purpose of the scheduled library closure.
created_by	varchar(50)	Identification of the administrator or staff member who created the closure schedule.
created_at	timestamp	Date and time when the closure record was created.
is_active	tinyint(1)	Indicates whether the closure schedule is currently active (1 = Active, 0 = Inactive).



sms_messages

Field	Data Type	Description
id	int(11)	Unique ID for each SMS notification record.
request_id	int(11)	Identifier of the related request or transaction associated with the SMS notification.
user_id	varchar(64)	Identification number or account ID of the recipient.
user_type	varchar(16)	Type of user who received the SMS notification (e.g., Student or Faculty).
phone	varchar(32)	Mobile phone number of the SMS recipient.
message	text	Content of the SMS notification sent to the user.
provider_message_id	varchar(128)	Unique message ID assigned by the SMS service provider for tracking purposes.
provider_response	longtext	Complete response returned by the SMS service provider after processing the message.
status	varchar(32)	Current delivery status of the SMS notification (e.g., Queued, Sent, Delivered, or Failed).
created_at	datetime	Date and time when the SMS notification record was created.
updated_at	datetime	Date and time when the SMS notification record was last updated.

student

Field	Data Type	Description
id	int(11)	Unique ID for each student account record.
library_id	varchar(50)	Unique library identification number assigned to the student.
student_id	varchar(50)	Official student identification number used for account registration.
password	varchar(255)	Encrypted password used for student account authentication.
email	varchar(255)	Email address of the student.
phone_number	varchar(20)	Contact number of the student.
status	enum('pending', 'approved', 'rejected')	Current approval status of the student's registration request.



profile_picture	varchar(255)	File path or filename of the student's uploaded profile picture.
first_name	varchar(100)	First name of the student.
last_name	varchar(100)	Last name or surname of the student.
middle_name	varchar(100)	Middle name of the student.
suffix	varchar(20)	Name extension of the student, if applicable (e.g., Jr., III).
birthdate	date	Date of birth of the student.
age	int(11)	Age of the student.
sex	enum('male', 'female', 'other')	Biological sex classification of the student.
gender_type	varchar(50)	Gender identity or preferred gender classification of the student.
course	varchar(100)	Academic course or program in which the student is enrolled.
registration_date	datetime	Date and time when the student account was registered.
department	varchar(100)	Academic department to which the student belongs.
reject_reason	text	Reason provided by the administrator if the student's registration request is rejected.

visitor

Field	Data Type	Description
email	varchar(255)	Unique email address used as the primary identifier for the visitor account.
phone_number	varchar(20)	Contact number of the visitor.
booking_id	varchar(50)	Unique booking or reservation identification number assigned to the visitor.
password	varchar(255)	Encrypted password used for visitor account authentication.
profile_picture	varchar(255)	File path or filename of the visitor's uploaded profile picture.
last_name	varchar(100)	Last name or surname of the visitor.
first_name	varchar(100)	First name of the visitor.
middle_name	varchar(100)	Middle name of the visitor.
status	enum('pending', 'approved', 'rejected')	Current approval status of the visitor registration request.
suffix	varchar(20)	Name extension of the visitor, if applicable (e.g., Jr., III).
birthdate	date	Date of birth of the visitor.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

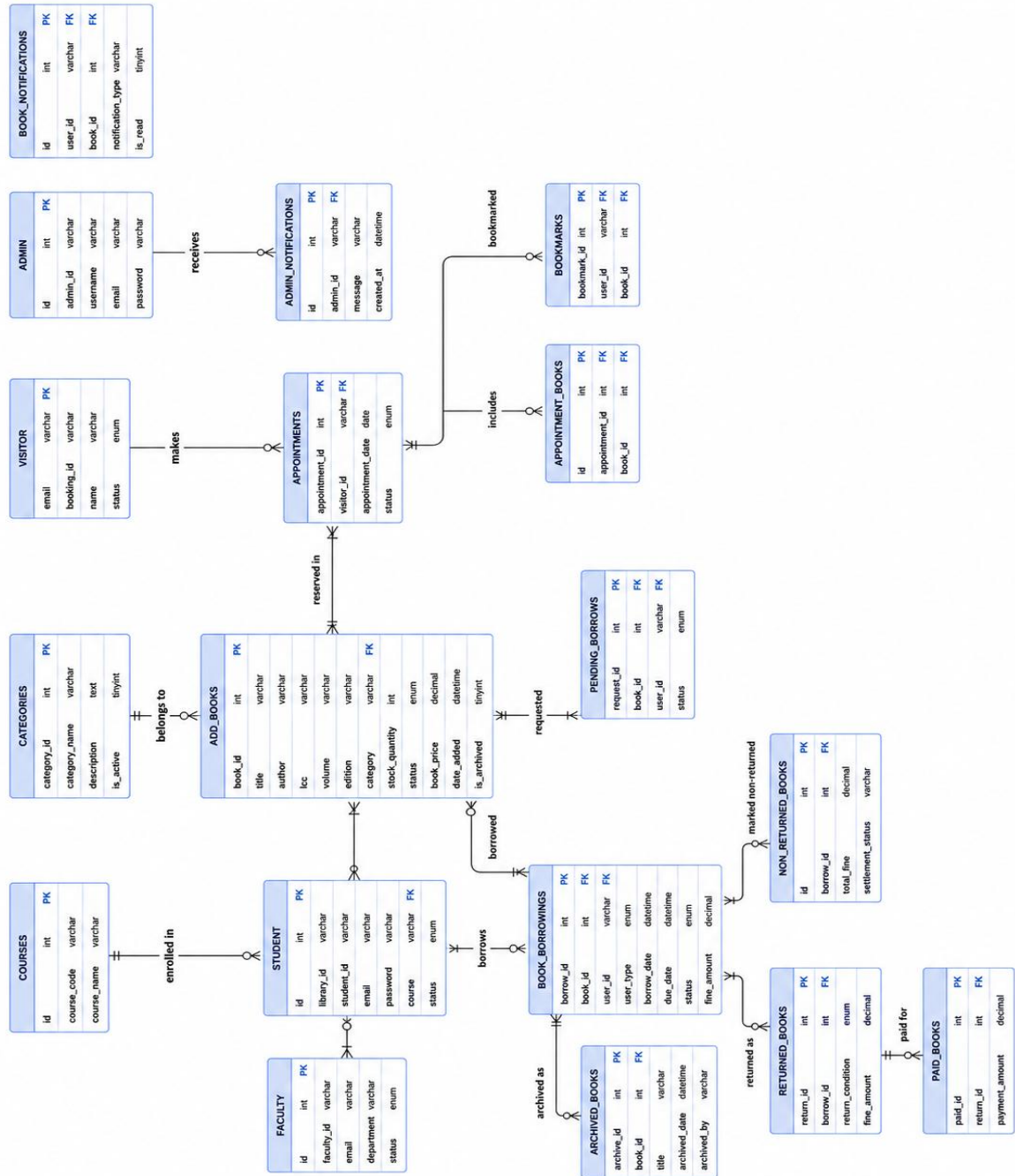
129

age	int(11)	Age of the visitor.
sex	enum('male', 'female', 'other')	Biological sex classification of the visitor.
gender_type	varchar(50)	Gender identity or preferred gender classification of the visitor.
visitor_type	varchar(50)	Category or classification of the visitor.
course	varchar(100)	Academic course or program of the visitor, if applicable.
section	varchar(50)	Class section of the visitor, if applicable.
current_status	varchar(100)	Current academic or employment status of the visitor.
occupation	varchar(255)	Occupation or profession of the visitor.
institution	varchar(255)	School, organization, or institution affiliated with the visitor.
referral_letter	varchar(255)	File path or filename of the uploaded referral letter or supporting document.
registration_date	datetime	Date and time when the visitor account was registered.
reject_reason	text	Reason provided by the administrator if the visitor registration request is rejected.
id_number	varchar(100)	Identification number of the visitor, if applicable.



Appendix 11

ENTITY RELATIONSHIP DIAGRAM

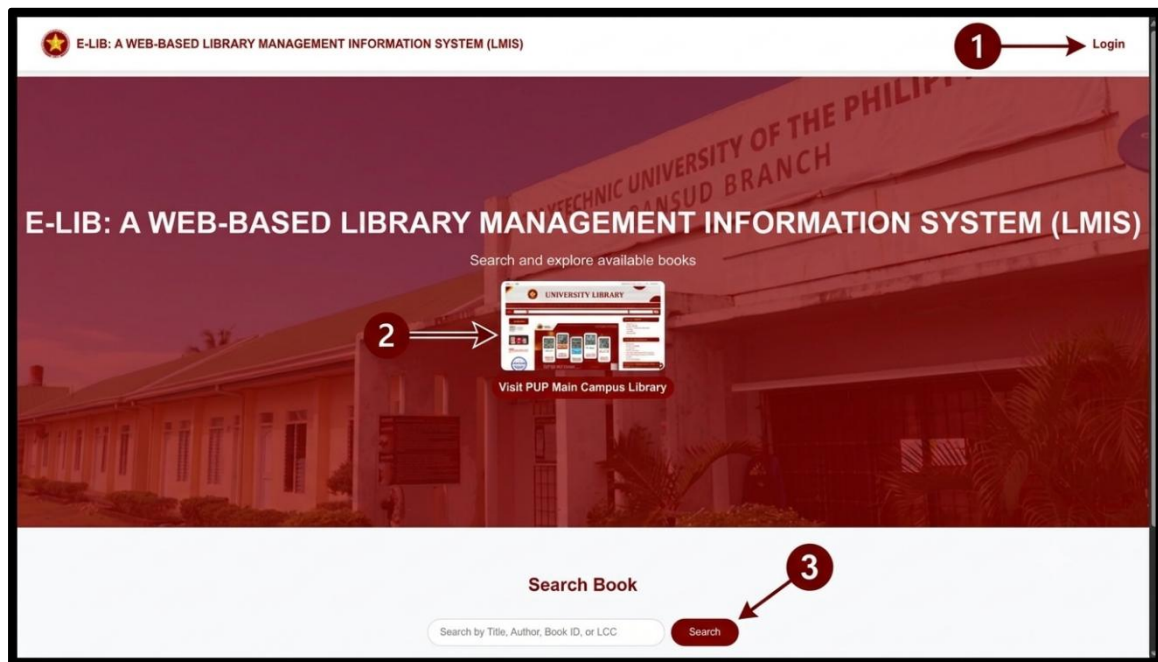




Appendix 12

USER'S GUIDE

Landing Page/ Home Page



The Landing Page serves as the main entry point of the E-Lib: A Web-Based Library Management Information System (LMIS). Upon accessing the system, users are greeted with the name of the system displayed prominently on the navigation bar, along with the PUP Bansud Campus building as the background image, providing a familiar and welcoming interface for all users.

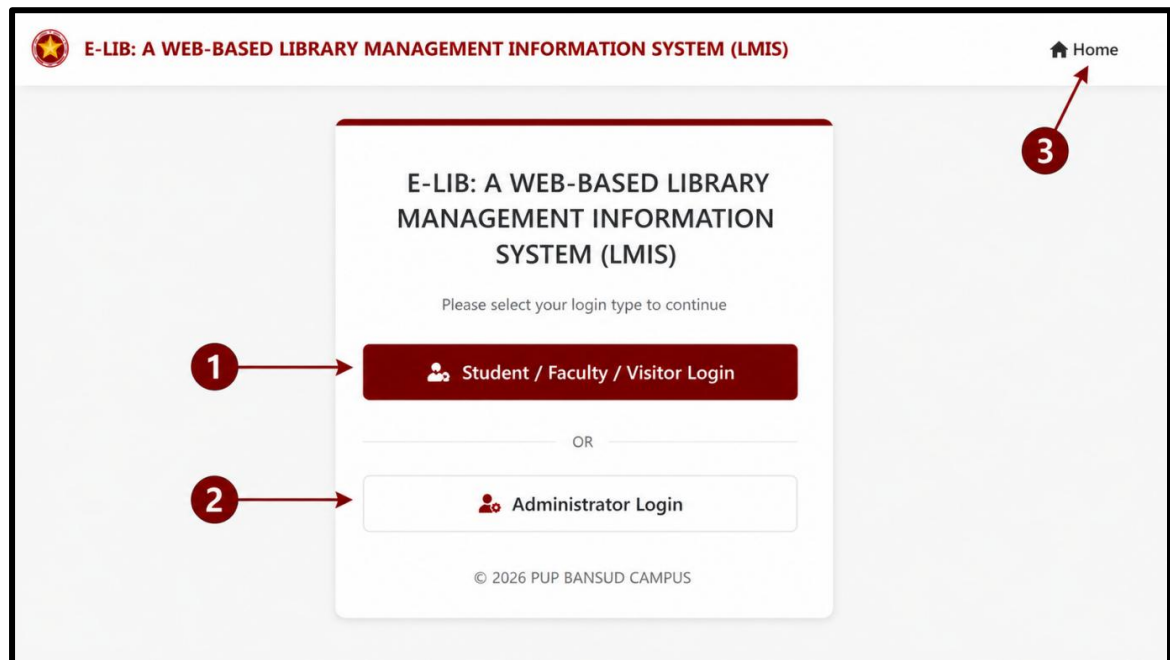
1. Located at the upper-right corner of the navigation bar is the "Login" button. Clicking this button will redirect the user to the "Login Page", where authorized users such as the Administrator and Librarian may enter their credentials to access the system.
2. At the center of the page, a "Visit PUP Main Campus Library" button is displayed alongside a preview image of the PUP Main Campus Library website. Clicking this



button will redirect the user to the official PUP Main Campus Library website for additional library resources and references.

3. Below the hero section is the “Search Book” feature, which is accessible to all users without requiring login. Users may type a book's Title, Author, Book ID, or LCC classification in the search field and click the Search button to look up available books in the library's collection.

Login Type Selection Page



Upon clicking the “Login” button, the Login Type Selection page is displayed, allowing users to choose their preferred login type. The page provides three options:

1. Student / Faculty / Visitor Login - Users who are registered as a Student, Faculty, or Visitor must click the Student / Faculty / Visitor Login button. This will redirect them to the corresponding login page where they can enter their credentials to access their respective accounts.



2. Administrator Login - Users with administrative privileges must click the “Administrator Login” button. This will redirect them to a separate login page designated for system management and administrative functions.
3. Home Button - The “Home’ button is located at the upper-right corner of the navigation bar. Clicking this button will redirect the user back to the Landing Page at any time without proceeding with the login process.

STUDENT/ FACULTY

Login Page

The screenshot shows the login interface for the E-LIB system. At the top, a red header bar contains the university logo, the text "E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)", and a "← Back to Selection" button. The main content area is white and features a central login form. The form has a title "Sign In to your account" and a section "Please Select User Type" with three buttons: "Student" (highlighted with a red circle and arrow labeled 1), "Faculty", and "Visitor". Below these are input fields for "Library ID" (with a hint "e.g., 2026-001") and "Password" (with a hint "eg. password123" and a toggle icon). A link "Forgot your password?" is positioned below the password field. A large red button labeled "Sign In as Student" is below the inputs, with a red circle and arrow labeled 2 pointing to it. Below this button is a link "Don't have an account yet? Register here", with a red circle and arrow labeled 3 pointing to it. A red circle and arrow labeled 4 points to the "Forgot your password?" link. A red circle and arrow labeled 5 points to the "Back to Selection" button in the top right corner. The footer of the page is red and contains the copyright notice "© 2026 Polytechnic University of the Philippines".

Upon clicking the “Student / Faculty / Visitor Login” button from the Login Type Selection page, the User Login page is displayed. This page allows users to select their user type and securely access the system. It includes the following features:



1. User Type Selection – From the Login Type Selection Page, the user selects their account type (Student, Faculty, or Visitor) by clicking the corresponding button. The selected user type is highlighted to indicate the active selection before proceeding with the login process.
2. Sign In Button – After entering the required Library ID and Password, the user clicks the Sign In button to access the system. Students and Faculty are redirected to the Book Borrowing Page, while Visitors are redirected to the Appointment Scheduling Page.
3. Register Here – Users who do not yet have an account may click the Register Here link to create a new account. This redirects them to the Registration Page, where they can provide the required information to register in the system.
4. Forgot Password – If users forget their password, they may click the Forgot Password link to initiate the password recovery process and regain access to their account.
5. Back to Selection – The Back to Selection button allows users to return to the Login Type Selection Page if they wish to choose a different login option before signing in.



Registration Page

The screenshot shows the 'Register Now' page of the E-LIB system. It features two main sections: 'Account Information' and 'Personal Details'. The 'Account Information' section includes fields for User Type (a dropdown menu), Phone Number, Email Address, ID Number, Password, and Confirm Password. The 'Personal Details' section includes fields for First Name, Last Name, Middle Name (optional), Suffix (optional), Sex (Male, Female, Other), Date of Birth, Age, and a Profile Picture upload button. At the bottom, there is a checkbox for terms and conditions, a 'Register Now' button, and a link for existing users. Numbered annotations are as follows: 1 points to the 'Account Information' header; 2 points to the 'Personal Details' header; 3 points to the terms and conditions checkbox; 4 points to the 'Register Now' button; and 5 points to the 'Already have an account? Log in' link.

If users do not yet have an account, they must first complete the registration process by clicking the Register Here link on the User Login page. This action redirects them to the Register Now page, where they are required to provide the necessary information to create a new account.

1. **Account Information** – Upon clicking the Register Here link from the Login Page, the user is redirected to the Registration Page. Under the Account Information section, the user must select their User Type (Student, Faculty, or Visitor) and provide the required information, including their Phone Number, Email Address, ID Number, Password, and Confirm Password, to create an account.
2. **Personal Details** – Under the Personal Details section, the user must provide the required personal information, including their First Name, Last Name, Middle Name (optional), Suffix (optional), Sex, Date of Birth, Age, and Profile Picture. These details



are required to complete the registration process and establish the user's profile in the system.

3. Certification Checkbox – Before submitting the registration form, the user must select the Certification Checkbox to confirm that all information provided is true and correct to the best of their knowledge and belief.
4. Register Now Button – After completing all required fields and selecting the Certification Checkbox, the user clicks the Register Now button to submit the registration form. The system validates the entered information, and if all requirements are satisfied, the user's account is successfully created. The user is then redirected to the Login Page, where they can sign in using their newly registered credentials.
5. Log In Link – Users who already have an existing account may click the Log In link to return to the Login Page and access the system using their registered credentials instead of creating a new account.

Registration Page Email Verification

The screenshot displays the 'Email Verification' interface of the E-LIB system. At the top, a red header bar contains the system logo and the title 'E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)'. The main content area is white and features a central card with a red envelope icon and the title 'Email Verification'. Below the title, it states 'We've sent a 6-digit verification code to:' followed by a redacted email address. A green progress bar indicates the code expires in '481 : 45'. Below this, there are six empty input boxes for the verification code, with a red circle and arrow '1' pointing to the first box. A red 'Verify Email' button with a checkmark icon is positioned below the input boxes, with a red circle and arrow '2' pointing to it. Below the button is a 'Resend verification code' link with a circular arrow icon, with a red circle and arrow '3' pointing to it. At the bottom of the card is a 'Back to Registration' link with a left-pointing arrow, with a red circle and arrow '4' pointing to it. A blue information box at the very bottom provides instructions: 'Didn't receive the email? Check your spam/junk folder. If the email address doesn't exist, the OTP will not be delivered.'



After successfully completing the registration process, the user is redirected to the Email Verification page. The system automatically sends a six-digit One-Time Password (OTP) to the registered email address, which is partially masked for security purposes.

1. The user must enter the verification code into the designated input fields before the countdown timer expires.
2. Once the correct OTP is entered, the user clicks the Verify Email button to validate the code and activate the account.
3. If the verification code expires or is not received, the user may request a new code by clicking the Resend Verification Code link.
4. The page also provides a Back to Registration link, allowing the user to return to the Registration Page if necessary.

Additionally, an information notice reminds users to check their Spam or Junk folder if the verification email is not found in their inbox and informs them that no OTP will be delivered if the provided email address is invalid.



Reset Password

The screenshot shows a web interface for the 'E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)'. The page has a red header with the university logo and title. The main content area is light blue. In the center, there is a white card titled 'Password Reset Verification'. Inside the card, it states: 'We've sent a 6-digit password reset code to:'. Below this, it shows an email address 'g*****n@gmail.com' and a mobile number '0929****231' with a green 'SMS sent' status. There is a section labeled 'Enter 6-Digit Code' with six input boxes. Below the input boxes is a red button labeled 'Verify Code'. At the bottom of the card, there are two links: 'Resend verification code' and 'Back to Forgot Password'. The footer of the page is red and contains the copyright notice '© 2020 Polytechnic University of the Philippines'.

Upon clicking the Forgot Password link on the Login Page and submitting the required account information, the user is redirected to the Password Reset Verification page. The system sends a 6-digit password reset verification code to the user's registered email address and mobile number. The user must verify the code before being allowed to create a new password.

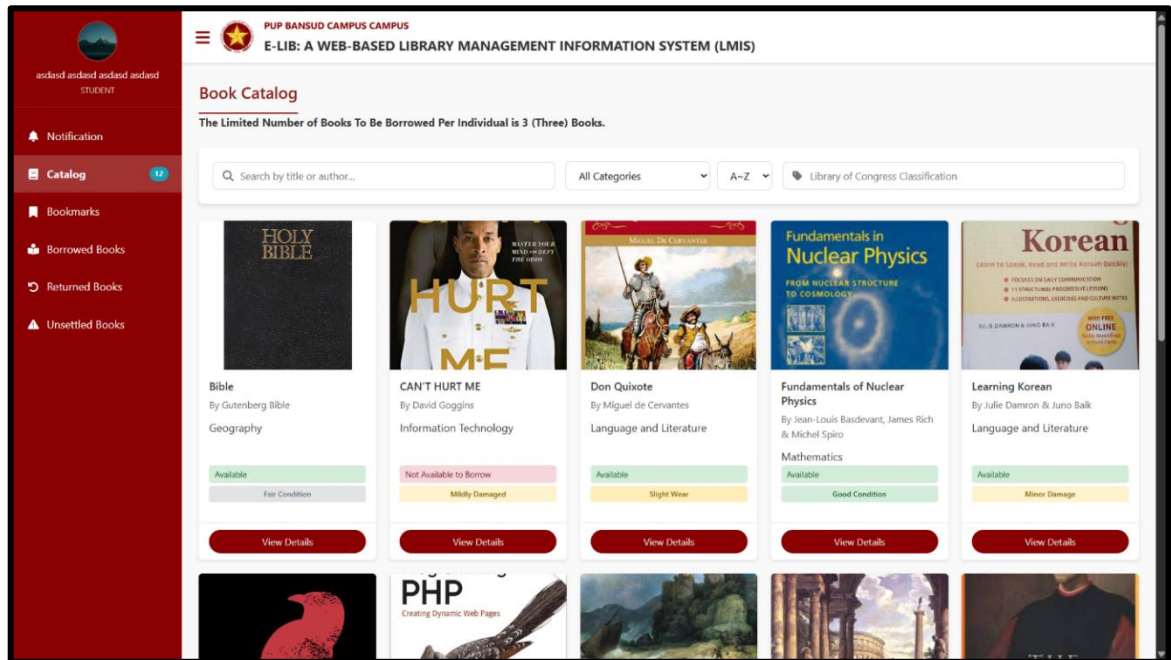


Upon successfully verifying the 6-digit verification code, the user is redirected to the Reset Your Password page. This page allows the user to create a new password for their account.

1. The user must enter a new password that meets the system's password requirements. The password must contain at least six (6) characters, including one uppercase letter, one lowercase letter, one number, and one special character. The user may click the eye icon to show or hide the entered password.
2. The user must re-enter the new password in the “Confirm Password” field to ensure that both passwords match.
3. After entering and confirming a valid password, the user clicks the “Update Password” button to save the new password. The system validates the entered passwords to ensure they match and satisfy the required password criteria. If the validation is successful, the password is updated, and the user is redirected to the “Login Page”, where they can sign in using the newly created password.



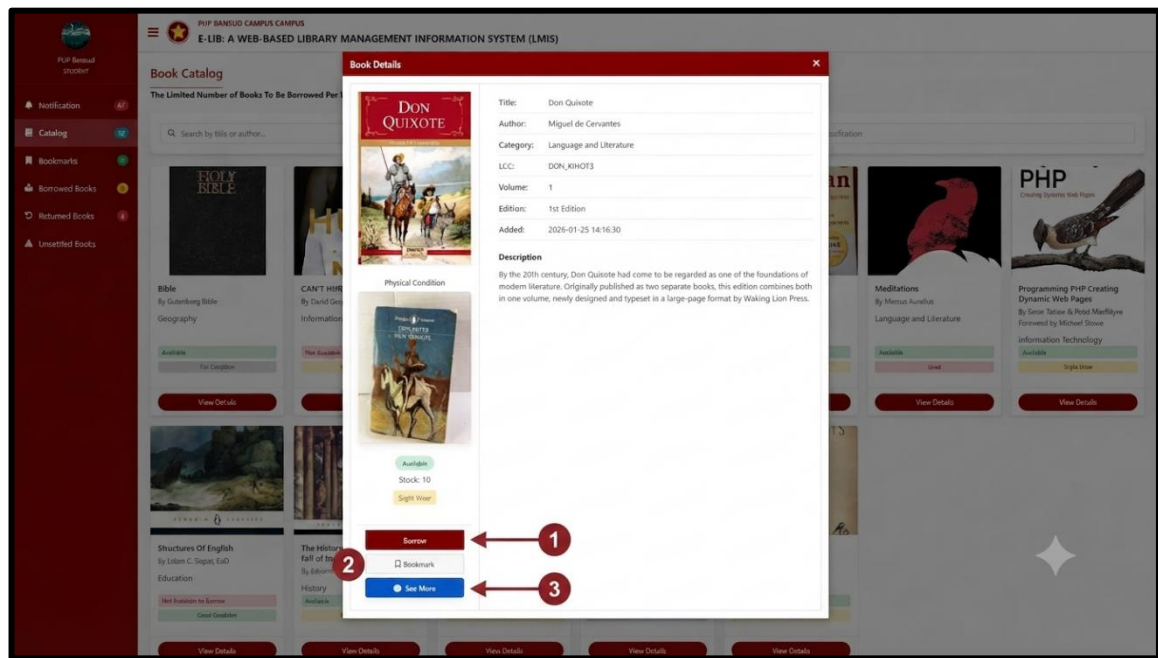
Book Catalog



After successfully logging into the system, the user may access the Book Catalog page through the navigation menu. This page displays all available library books and allows users to search, filter, and browse the collection before requesting to borrow a book. The system also displays the maximum borrowing limit, which is three (3) books per individual.



Book Details



Upon clicking the View Details button from the Book Catalog page, the system displays the Book Details window. This window provides complete information about the selected book, allowing the user to review its bibliographic details, physical condition, availability, and borrowing options before proceeding with a transaction.

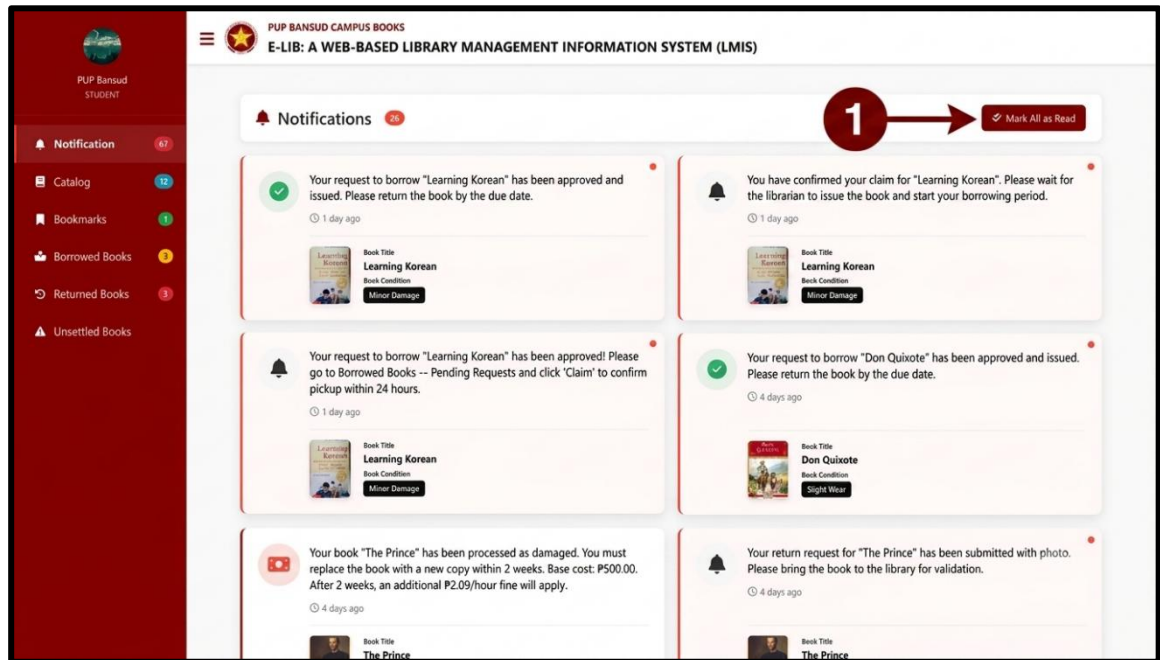
It has three (3) buttons below:

1. Borrow – If the user wishes to borrow the selected book, they may click the Borrow button. The system processes the borrowing request and proceeds with the borrowing transaction, provided that the book is available and the user meets the library's borrowing requirements.
2. Bookmark – If the user does not wish to borrow the book immediately but wants to save it for future reference, they may click the Bookmark button. The selected book is then added to the My Bookmarked Books page for easy access at a later time.
3. See More – If the user clicks the See More button, the system displays additional information about the selected book. This includes the book description, table of



contents, and references, allowing the user to review more comprehensive details before deciding whether to borrow the book.

Notifications

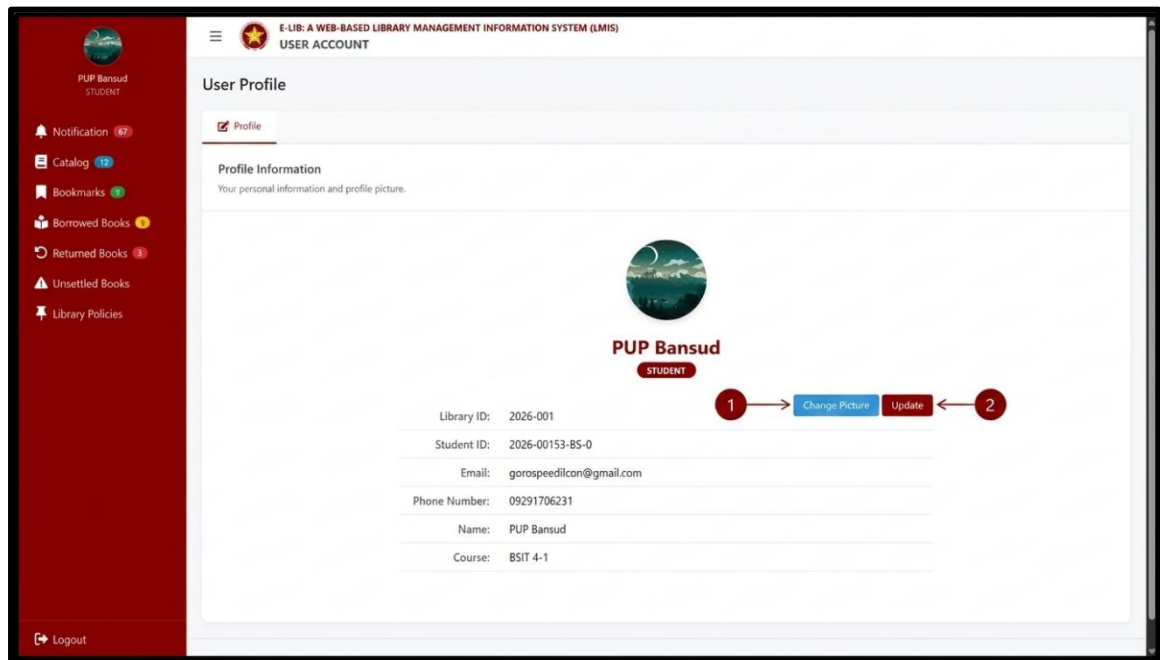


Upon clicking the “Notification” module from the navigation menu, the user is redirected to the “Notifications” page, which displays all system-generated notifications related to the user's library transactions. The notifications are presented in chronological order, with the most recent updates appearing first. Each notification contains the transaction status, a brief description of the activity, the corresponding book information, and the date or time it was generated. The system provides notifications for various library activities, including borrow request approvals, claim confirmations, book issuance, return requests, due date reminders, overdue notices, damaged or lost book reports, and other important library-related updates. Each notification is accompanied by a status icon for easy identification, while unread notifications are marked with a red indicator, enabling users to quickly distinguish notifications that require their attention until they are opened or marked as read.



1. The “Mark All as Read” button allows the user to mark all unread notifications as read with a single click. Once selected, the system updates the notification status by removing the unread indicators, helping users organize their notifications and easily distinguish newly received notifications from those that have already been viewed.

User’s Profile



Upon clicking the “Profile” icon, the user is redirected to the User Profile page. This page displays the user's registered account information, including the profile picture, full name, user type, Library ID, Student ID, email address, phone number, and course. It also provides options for updating the user's profile picture and saving changes to the account information.

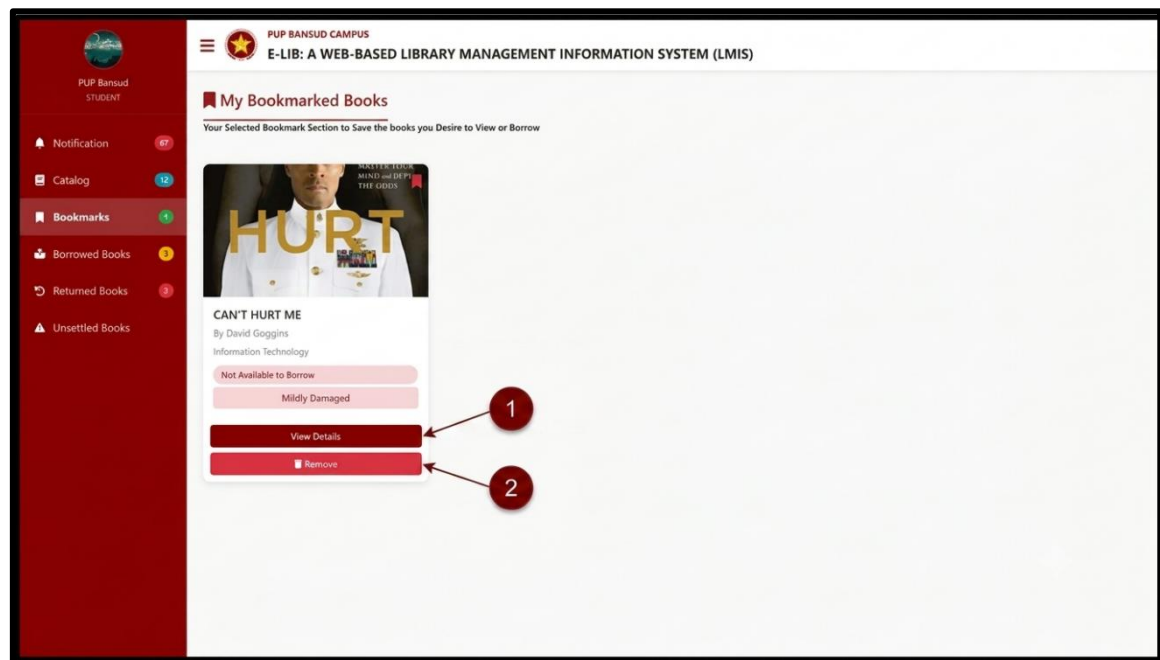
The User Profile page contains two (2) action buttons:

1. The “Change Picture” button allows the user to upload a new profile picture from their device. After selecting an image, the newly chosen profile picture is displayed on the page, allowing the user to preview it before saving the changes.



2. After selecting a new profile picture or modifying any editable account information, the user clicks the “Update” button to save the changes. The system validates the updated information, and if all requirements are met, the user's profile is successfully updated. The latest profile information is then displayed on the User Profile page.

Bookmarks



Upon clicking the “Bookmarks” module from the navigation menu, the user is redirected to the “My Bookmarked Books” page. This page displays all books that the user has bookmarked from the “Book Catalog” for future reference or borrowing. Each bookmarked book is presented as a book card containing the cover image, title, author, category, availability status, and physical condition. The page also provides two (2) action buttons for each bookmarked book.

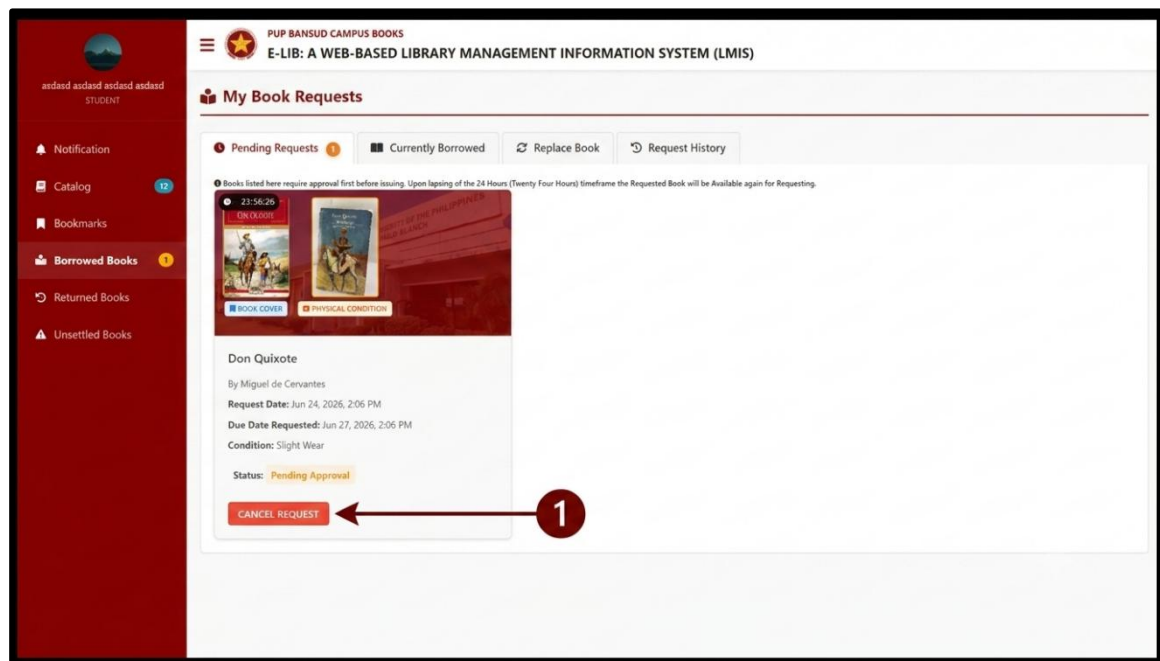
1. View Details Button – The “View Details” button allows the user to view the complete information about the selected book, including its bibliographic details, availability,



physical condition, and borrowing information. If the book is available, the user may also proceed with the borrowing process.

2. Remove Button – The “Remove” button allows the user to delete the selected book from their bookmarked list. Once clicked, the system removes the book from the “My Bookmarked Books” page, and it will no longer appear in the user's bookmarks unless it is bookmarked again from the Book Catalog.

Borrowed Books



Upon clicking the “Borrowed Books” module from the navigation menu, the user is redirected to the “My Book Requests” page. This page allows users to monitor the status of their borrowing transactions. It is divided into four (4) tabs: Pending Requests, Currently Borrowed, Replace Book, and Request History. By default, the Pending Requests tab displays all book borrowing requests that are awaiting librarian approval before the books can be issued.



The “Pending Requests” tab displays all borrowing requests submitted by the user that are still awaiting approval from the librarian or system administrator. A notice at the top of the page informs the user that all requested books require librarian approval before they can be issued. It also states that if the request is not processed within the 24-hour reservation period, the request will automatically expire, and the book will become available for other users to request. Each pending request card displays the book cover and physical condition images, along with the book title, request date, requested due date, book condition, current request status (e.g., Pending Approval), and a countdown timer indicating the remaining time before the request expires.

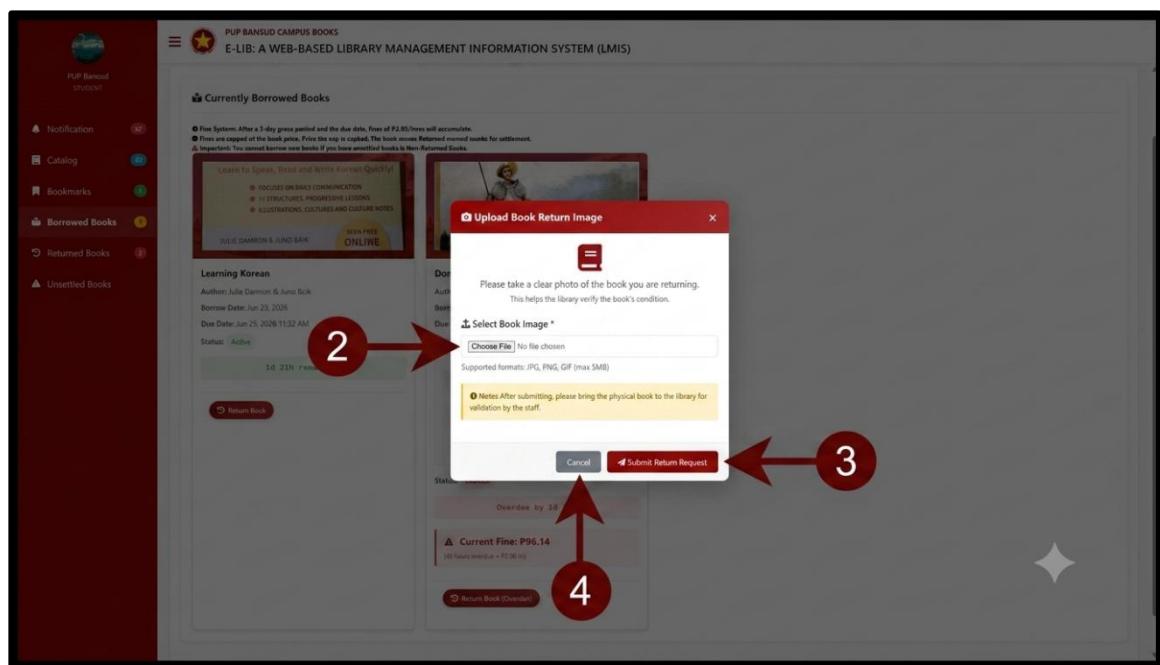
1. If the user no longer wishes to borrow the selected book, they may click the “Cancel Request” button to withdraw the borrowing request. Once the request is successfully cancelled, the reservation is removed, and the book becomes available for other users to request in accordance with the library's borrowing policies.

Tab 2. Currently Borrowed



The Currently Borrowed tab displays all books that have been successfully issued to the user and are currently on loan. Each borrowed book card presents the book cover, title, author, borrowing date, due date, current borrowing status, and the physical condition of the book at the time it was issued. For books that are still within the allowed borrowing period, the system displays the remaining number of days before the due date. If a book becomes overdue, the system automatically changes its status to Overdue, displays the duration of the overdue period, and calculates the corresponding overdue fine based on the library's policy. Users may return a borrowed book at any time by clicking the Return Book button.

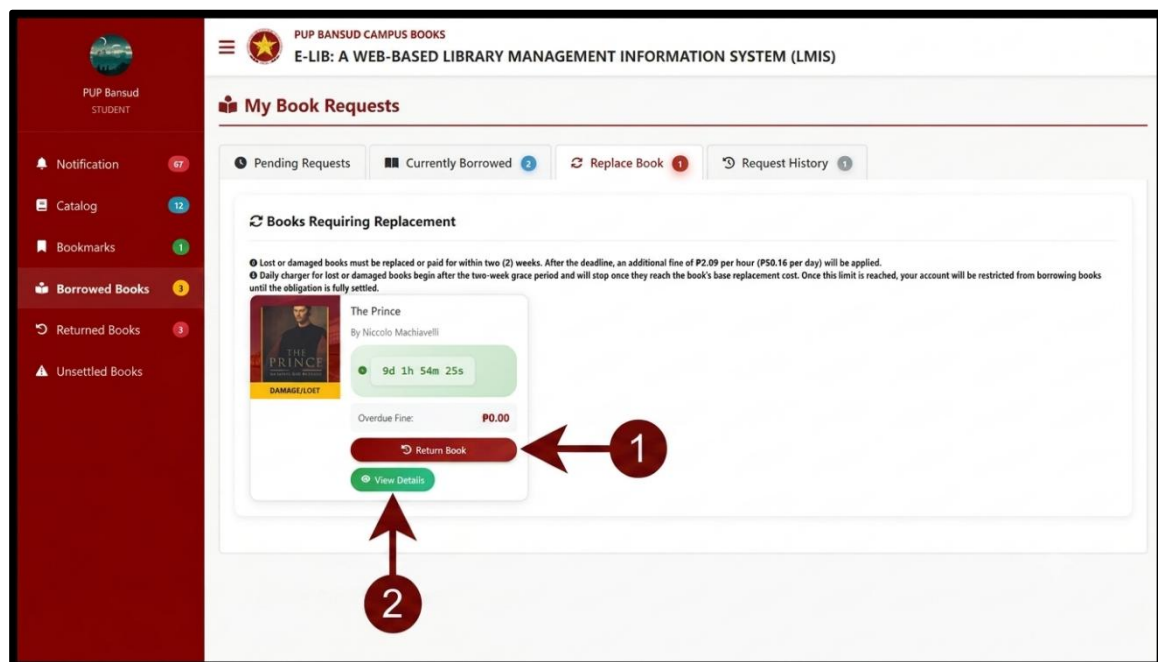
1. For overdue books, the Return Book (Overdue) button is provided to initiate the return process. Upon clicking the appropriate return button, the system records the return request and notifies the librarian for validation and processing.



Upon clicking the Return Book or Return Book (Overdue) button, the system displays the Upload Book Return Image modal window.



2. This window requires the user to upload an image showing the current physical condition of the borrowed book as part of the return request process. The user must select an image file in JPG, PNG, or GIF format with a maximum file size of 5 MB using the file upload control. A notice within the modal reminds the user that, after submitting the return request, they must personally bring the physical book to the library for inspection and validation by the librarian.
3. Submit Return Request – After selecting the required image, the user clicks the Submit Return Request button to upload the image and submit the return request. The system records the request and notifies the librarian to verify the book's condition and complete the return transaction.
4. Cancel – If the user decides not to proceed with the return request, they may click the Cancel button. This action closes the modal without submitting the return request, and no changes are made to the borrowing transaction.

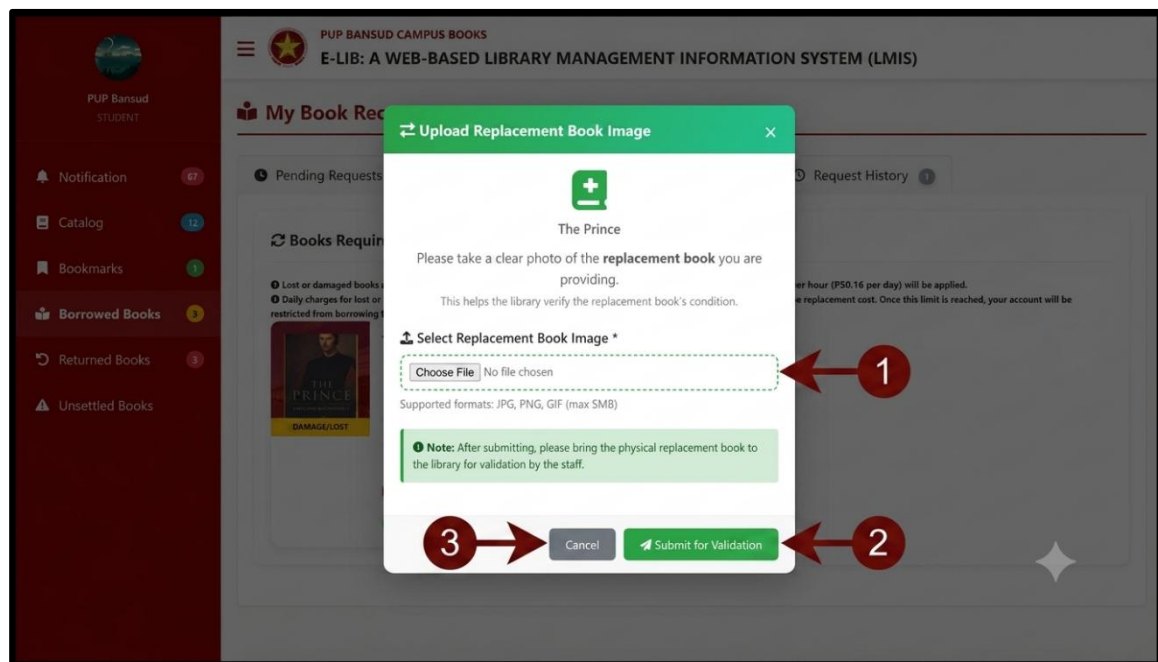
Tab 3: Replace Book



The Replace Book tab displays all books that have been assessed as damaged or lost and are required to be replaced by the user. A notice at the top of the tab informs the user that lost or damaged books must be replaced or paid for within two (2) weeks, after which an additional fine of ₱2.09 per hour (₱50.16 per day) will be applied. Daily charges for lost or damaged books begin after the two-week grace period and will stop once they reach the book's base replacement cost, after which the user's account will be restricted from borrowing books until the obligation is fully settled. Each book requiring replacement displays a DAMAGE/LOST badge on the book cover, a countdown timer showing the remaining time to comply with the replacement obligation, the current Overdue Fine.

In the Replace Book Tab, there are two (2) action button:

1. Return Book



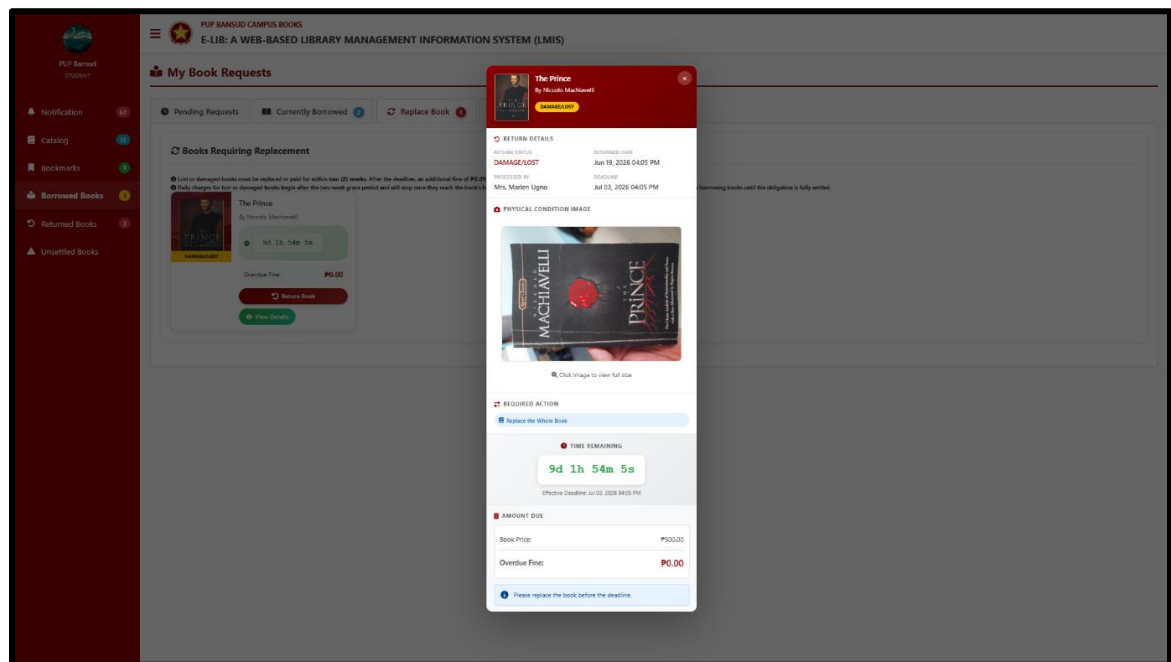
Upon clicking the "Return Book" button, the "Upload Replacement Book Image" modal appears.



1. The student is required to upload a clear photo of the replacement book. The system accepts JPG, PNG, and GIF image formats with a maximum file size of 5 MB.
2. After selecting the image, the student can click "Submit for Validation" to send the replacement book request to the library staff for verification.
3. If the student decides not to proceed, they can click the "Cancel" button to close the modal without submitting the replacement book request.

A note is displayed reminding the student to bring the physical replacement book to the library for final validation by the staff.

2. View Details

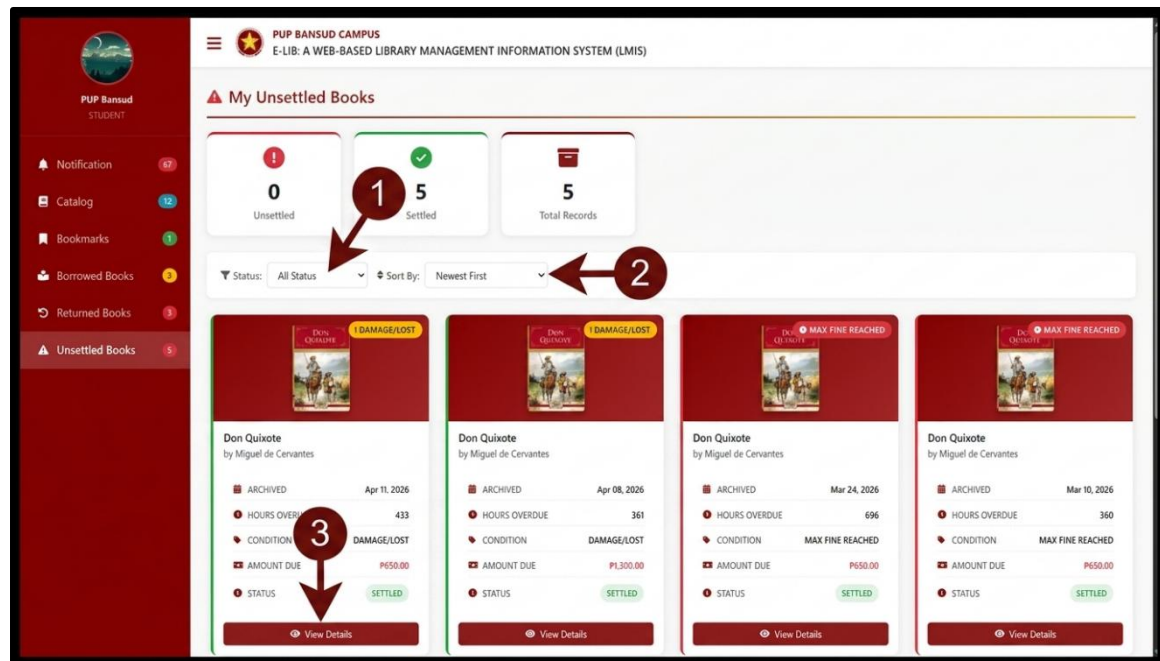


Upon clicking the View Details button, a modal window will appear displaying the complete replacement details, including the Return Status (e.g., DAMAGE/LOST), the Returned Date, the Deadline for replacement, the name of the librarian who Processed the assessment, the Physical Condition Image uploaded by the user upon return showing the damage, the Required Action (e.g., Replace the Whole Book), the Time Remaining countdown



before the deadline, and the Amount Due breakdown showing the Book Price (e.g., ₱500.00) and any applicable Overdue Fine. A reminder at the bottom of the modal instructs the user to replace the book before the deadline.

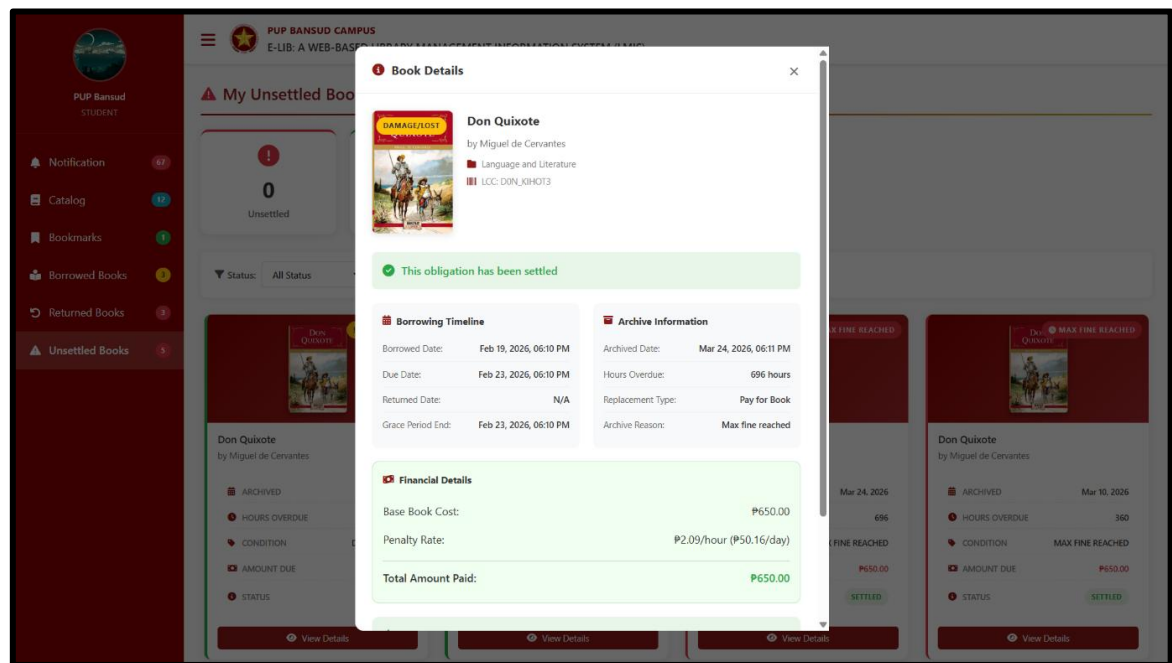
Unsettled Books



The My Unsettled Books page is accessible by selecting the Unsettled Books option from the left sidebar. This page provides a comprehensive overview of all borrowing transactions that have resulted in financial obligations due to overdue, damaged, or lost books, while also indicating whether each obligation has been settled or remains outstanding. The notification badge beside the Unsettled Books menu displays the total number of records available in this section. At the top of the page, three summary counters provide a quick overview of the user's records: Unsettled, which shows the number of transactions with pending obligations; Settled, which indicates the number of transactions that have already been resolved; and Total Records, which displays the combined total of all unsettled book transactions. Above the list of records are two filtering options to help users quickly locate specific transactions.



1. Status – a dropdown that allows the user to filter records by their settlement status (e.g., All Status, Unsettled, Settled).
2. Sort By – a dropdown allows the user to arrange the records by date (e.g., Newest First) for easier navigation.



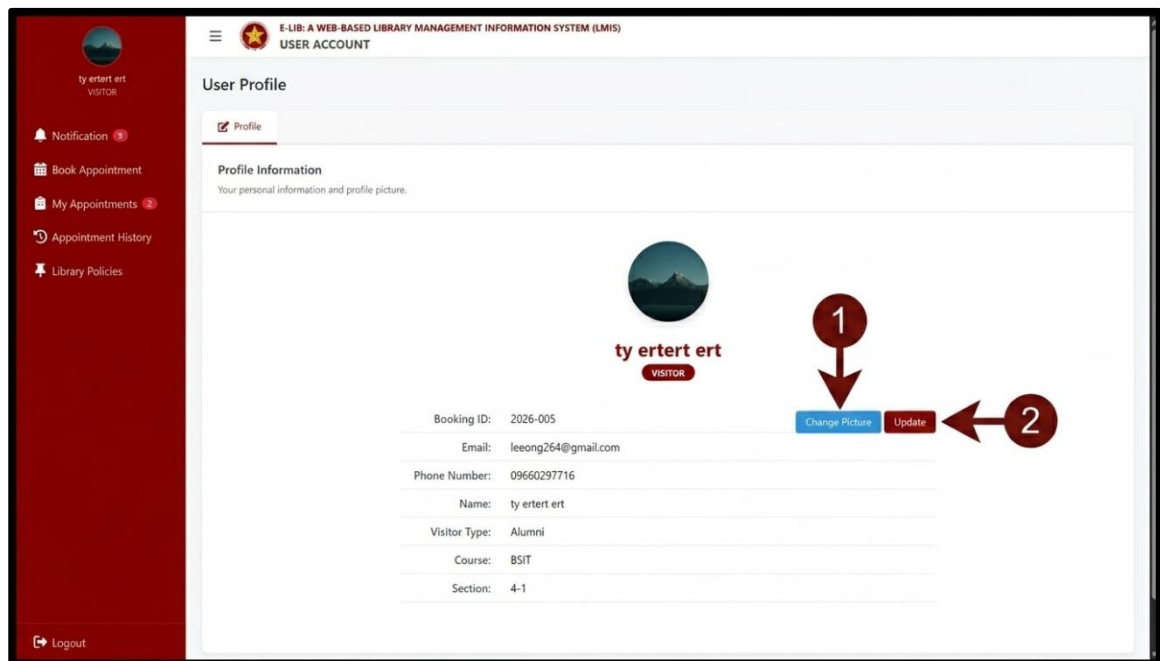
3. Upon clicking the "View Details" button, the Book Details modal appears, providing comprehensive information about the selected unsettled book transaction. The modal displays the book's cover image, title, author, category, and library call number, along with a condition badge indicating the reason for the obligation (e.g., DAMAGE/LOST). A status message is also shown to indicate whether the obligation has been settled. The Borrowing Timeline section presents the Borrowed Date, Due Date, Returned Date (if applicable), and the Grace Period End. The Archive Information section includes the Archived Date, Hours Overdue, Replacement Type, and the Archive Reason, such as Max Fine Reached. At the bottom of the modal, the Financial



Details section summarizes the Base Book Cost, Penalty Rate, and the Total Amount Paid, providing users with a complete breakdown of the financial obligation associated with the transaction. The modal can be closed by clicking the Close (×) button at the top-right corner after reviewing the information.

VISITORS

Visitor Profile



Upon clicking the Profile icon, the visitor is redirected to the User Profile page. The Profile Information section displays the Visitor user's complete account details, which include the following: Booking ID, which is the unique reference number assigned to the visitor's account in the system (e.g., 2026-005); Email, the registered email address linked to the account; Phone Number, the registered mobile number of the user; Name, the full name of the account holder; Visitor Type, which indicates the classification of the visitor (e.g., Alumni); Course, which reflects the academic program the visitor completed (e.g., BSIT); and Section,



which indicates the visitor's former year and section (e.g., 4-1). Note that the specific fields displayed under the profile may vary depending on the Visitor Type selected during registration.

The Visitor Profile page contains two (2) action buttons:

1. The Change Picture button allows the user to upload a new profile picture from their device. After selecting an image, the newly chosen profile picture is displayed on the page, allowing the user to preview it before saving the changes.
2. After selecting a new profile picture or modifying any editable account information, the user clicks the Update button to save the changes. The system validates the updated information, and if all requirements are met, the user's profile is successfully updated.

The latest profile information is then displayed on the User Profile page.

Book Appointment

PUP BANSUD CAMPUS APPOINTMENT
E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)

Appointment

Select Date
Choose an available date for your appointment

June 2026

Sun	Mon	Tue	Wed	Thu	Fri	Sat
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Available Books

Books Available for Viewing

Search by title, author, or description

Filter by Category: All Categories

Bible
Gutenberg Bible
Religion
View Details See More
Select for consultation

Don Quixote
Miguel de Cervantes
Language and Literature
Good Condition
View Details See More
Select for consultation

Appointment Details

Fill in your information to schedule the appointment

Full Name * Juan Dela Cruz Email Address * juandelacruz@gmail.com

Contact Number * 09123456789 Booking ID Number * 2026-001

Select Time Slot * Library Hours: 8:00 AM - 5:00 PM

Time Slot	Slots Left
8:00 AM - 9:00 AM	3
9:01 AM - 10:00 AM	2
10:01 AM - 11:00 AM	1
11:01 AM - 12:00 NN	3
12:01 PM - 1:00 PM	2
1:01 PM - 2:00 PM	1
2:01 PM - 3:00 PM	2
3:01 PM - 4:00 PM	2
4:01 PM - 5:00 PM	1

Purpose of Visit * Select a purpose

Additional Notes (Optional)
Share any specific requests or additional information...

Submit Appointment Request

Fundamentals of Nuclear Physics
John-Louis Bessire, James Rich & Michel Spino
Mathematics
Good Condition
Good Condition
View Details See More
Select for consultation

Learning Korean
Jaeil Dymally & Juno Bae
Language and Literature
Like New
View Details See More
Select for consultation

Meditations
Marcus Aurelius
Personal Growth
Good Condition
View Details See More
Select for consultation

Programming PHP Creating Dynamic Web Pages
Kevin Tatum, Peter Muchinger & Rasmus Lerdorf
Computer Science
Good Condition
Good Condition
View Details See More
Select for consultation

The History of the Greeks and Fall of the Roman Empire (Vol. 1-8)
Edward Gibbon
History
Like New
View Details See More
Select for consultation



Upon successful login, the Visitor is redirected to the Appointment Page, where they can schedule a library visit for on-site reading. This page serves as the main dashboard for Visitor users.

2. Select Date - The user selects an available appointment date using the calendar. Only available dates can be selected, while grayed-out dates are unavailable for booking.

3. Available Books - The Available Books section displays books that may be consulted during the visit. Users can search, filter by category, view book details, and select books for consultation.

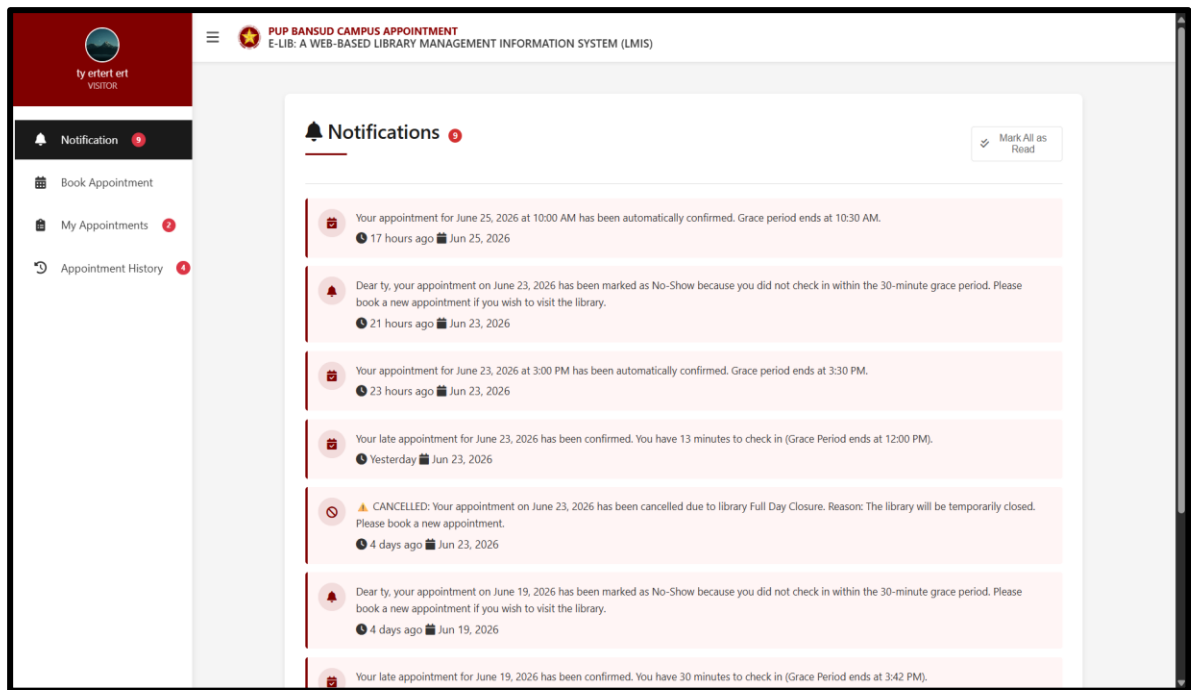
4. Appointment Details - After selecting a date, the Appointment Details section becomes active. The user reviews or completes the required information, including their name, email address, contact number, and system-generated booking ID.

5. Time Slot - The user selects an available time slot within the library's operating hours. Time slots become available only after an appointment date has been selected.

6. Purpose of Visit - The user selects the Purpose of Visit, may enter additional notes if necessary, and clicks the Submit Appointment Request button to send the appointment request for administrator approval.



Notification

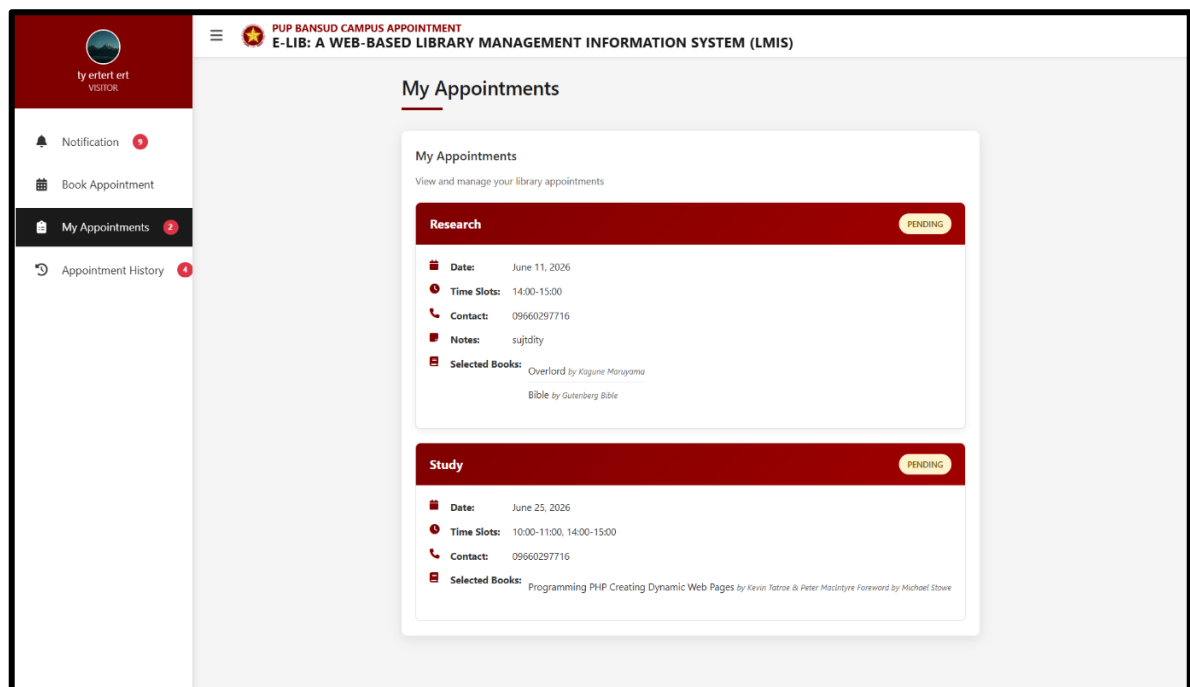


Upon clicking the Notification option from the left sidebar, the Visitor user is redirected to the Notifications page, which displays all system-generated alerts and updates related to the user's appointment activities within the E-LIB system. The total number of unread notifications is indicated by a notification badge displayed beside the Notification menu in the sidebar and on the page header. Users may click the Mark All as Read button located at the upper-right corner of the page to mark all unread notifications as read simultaneously. Each notification includes a status icon, the complete notification message, the time elapsed since the notification was received, and the corresponding appointment date. Notifications are arranged in chronological order, with the most recent entries displayed first. The system generates various notification types throughout the appointment process, including appointment confirmation notifications indicating the scheduled date, time, and grace period end time; late appointment notifications showing the remaining time available for check-in within the grace period; no-show notifications informing the user that the appointment has been marked as



missed due to failure to check in within the 30-minute grace period; and appointment cancellation notifications issued when the library is closed for the day or under other circumstances, along with the reason for the cancellation and instructions to schedule a new appointment. Visitors are encouraged to check this page regularly to remain informed about the status of their appointments, including confirmations, grace period reminders, no-show notices, and cancellations.

My Appointments

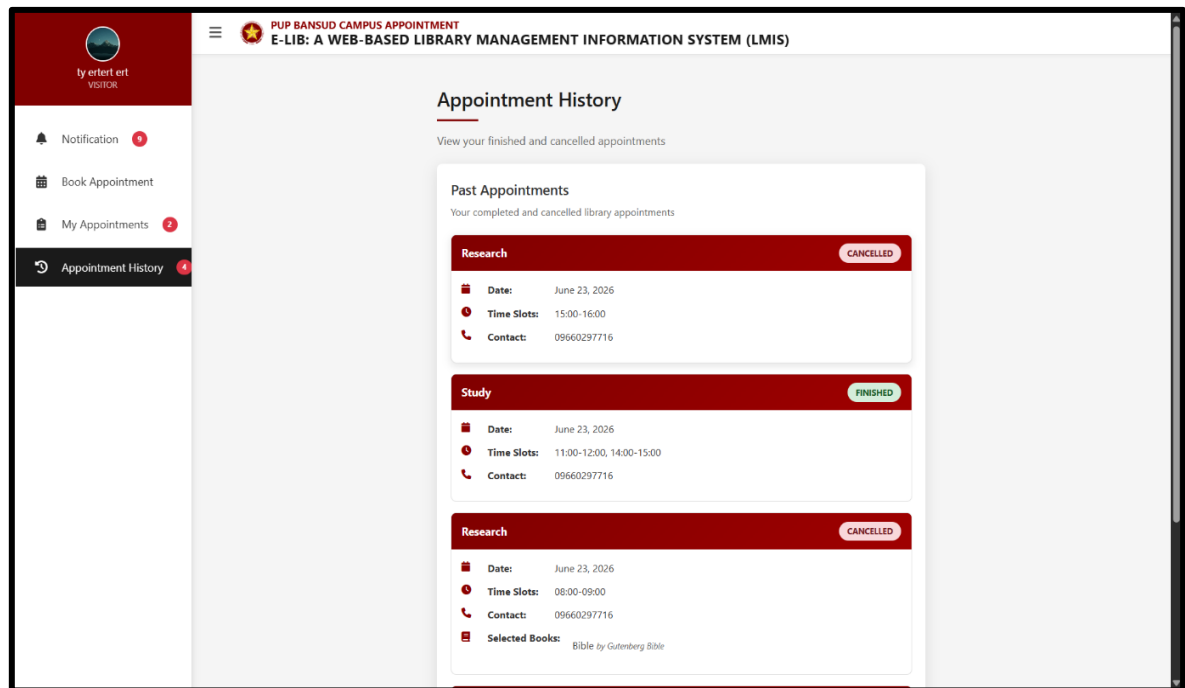


Upon clicking the My Appointment module from the left sidebar, the Visitor user is redirected to the My Appointment page, where all active and pending library appointment requests are displayed and managed. The badge beside the My Appointment menu indicates the total number of active appointment records associated with the user's account. Each appointment is presented as a card showing the Purpose of Visit (e.g., Research or Study) as the card header, together with a status badge indicating the current appointment status (e.g.,



Pending). Each appointment card also displays the scheduled Date of the visit, the selected Time Slot(s), the user's Contact Number, any Notes or special requests provided during the appointment booking (if applicable), and the list of Selected Books for consultation, including their titles and authors. As shown in the image, two appointment requests are currently displayed, both with a Pending status, indicating that they have been successfully submitted and are awaiting review and confirmation by the library administrator before the scheduled visit.

Appointment History

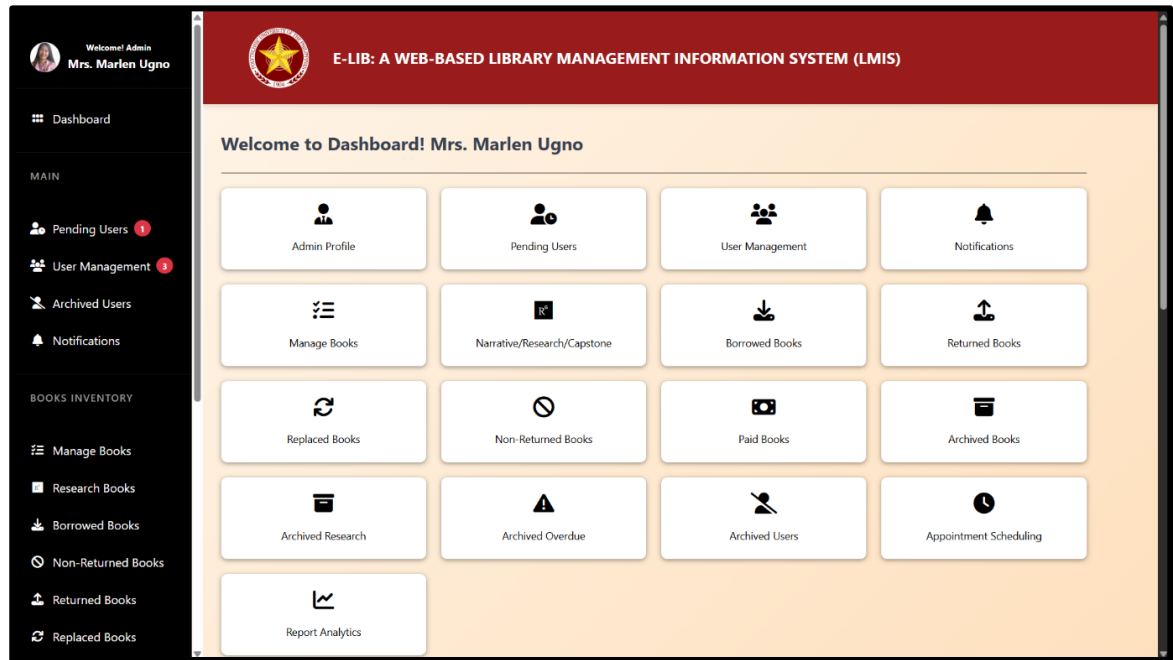


Upon clicking the "Appointment History" option from the navigation menu, the system displays the user's past library appointments. The page presents detailed information for each appointment, including the appointment type, scheduled date, reserved time slot(s), contact information, selected books, and the appointment status. Finished appointments are marked with a green status badge, while cancelled appointments are marked with a red status badge, allowing users to easily review their previous library transactions.



ADMIN SIDE

Dashboard



Upon logging into the administrator account, the user is redirected to the Admin Dashboard page, which serves as the main control panel of the E-Lib: A Web-Based Library Management Information System (LMIS). The dashboard provides administrators with an overview of the overall library operations through quick-access modules, statistical summaries, and graphical reports. It allows administrators to efficiently manage users, books, borrowing transactions, appointments, archives, and system analytics from a centralized interface.



Upon viewing the Overview Statistics section of the Admin Dashboard, the administrator can use the available time-filter buttons located above the statistical overview cards to adjust the displayed data based on a selected time period. These filters allow administrators to analyze library activities and monitor system performance according to specific time ranges.

1. The This Week button displays library activities and statistics recorded within the current week, including recent transactions, user activities, and book-related operations.
2. The This Month button shows the accumulated data and transactions recorded within the current month, providing a summary of monthly library activities.
3. The This Year button displays yearly statistics and serves as the currently selected filter for viewing annual system data, allowing administrators to monitor overall performance throughout the year.



4. The All Time button displays the complete historical records stored in the system since its implementation, providing a comprehensive overview of all recorded library activities.

The Analytics and Visualization section provides administrators with graphical representations and summarized data to effectively monitor and evaluate library performance.

5. The Overview Statistics cards display important numerical information, including:
 - Total Books, which shows the total number of available library materials in the system;
 - Active Borrows, which displays the number of books currently borrowed by users;
 - Registered Users, which indicates the total number of registered accounts, including students, faculty members, and visitors;
 - Today's Appointments, which shows the number of scheduled library visits for the current day;
 - Pending Users, which displays registration requests awaiting administrator approval;
 - Returned Books, which shows the number of successfully returned library materials;
 - Non-Returned Books, which displays overdue or unreturned books requiring monitoring; and Paid Books, which records completed payments related to fines, lost books, or replacement fees.
6. The dashboard also includes several graphical reports to help administrators interpret library data more effectively.



- The Borrowing Trends Line Chart presents the movement of book borrowing activities over a selected period, allowing administrators to identify borrowing patterns and changes in library usage.
- The Book Categories Distribution Pie Chart displays the percentage distribution of library materials according to categories, such as Language and Literature, Information Technology, History, Mathematics, and Science Fiction, helping administrators analyze the composition of the library collection.
- The User Type Distribution Donut Chart illustrates the proportion of registered users based on classification, including students, faculty members, and visitors.
- Lastly, the Library Capacity Utilization Bar Chart compares the number of daily library visitors against the maximum facility capacity from Monday to Friday, enabling administrators to monitor library occupancy and manage available resources efficiently.



Admin Profile

Welcome! Admin
Mrs. Marlen Ugno

Dashboard

MAIN

- Pending Users 1
- User Management 1
- Archived Users
- Notifications 64

BOOKS INVENTORY

- Manage Books 12
- Borrowed Books 6
- Non-Returned Books
- Returned Books 3
- Replaced Books 2
- Paid Books 1

E-LIB: A WEB-BASED LIBRARY MANAGEMENT INFORMATION SYSTEM (LMIS)

Admin Profile Management

Mrs. Marlen Ugno
System Administrator

Admin ID
LIB-2026-0001

Admin Email

https://e-libran.online/AdminDashboard/components/Admin_Notification.php

Welcome! Admin
Mrs. Marlen Ugno

Dashboard

MAIN

- Pending Users 1
- User Management 3
- Archived Users
- Notifications

BOOKS INVENTORY

- Manage Books
- Research Books
- Borrowed Books
- Non-Returned Books
- Returned Books
- Replaced Books

Admin ID
LIB-2026-0001

Admin Email
nsbsupetran@pup.edu.php

Phone Number
09291706231

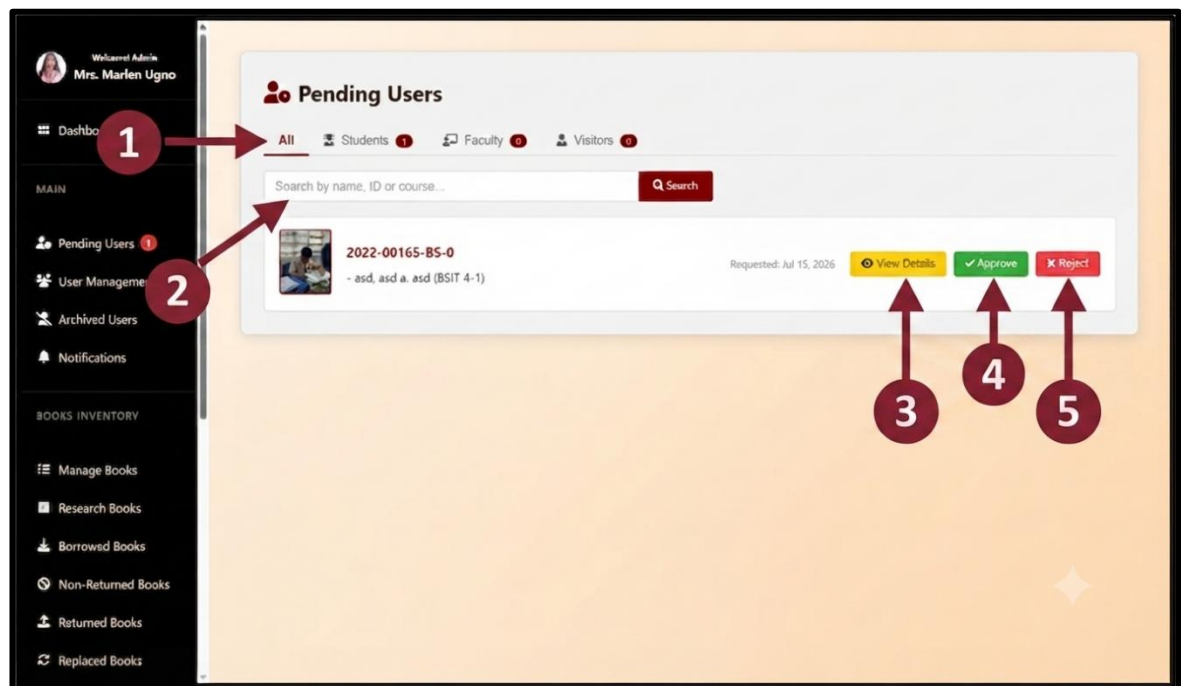
Password

[Save All Profile Changes](#)



Upon clicking the Admin Profile, the administrator is redirected to the Admin Profile page, which allows the administrator to view and manage their account information. The Profile Information section displays the administrator's complete account details stored in the system, including the following: Name, which indicates the full name of the administrator; Admin ID, which serves as the unique identification number assigned to the administrator account; Admin Email, which shows the registered email address associated with the administrator account; Phone Number, which displays the registered contact number of the administrator; and Password, which allows the administrator to manage their account security credentials. The displayed information enables the administrator to verify, update, and maintain their personal account details within the system.

Pending Users



Upon clicking the Pending Users module from the navigation menu, the administrator is redirected to the Pending Users page, which serves as the review queue for all newly

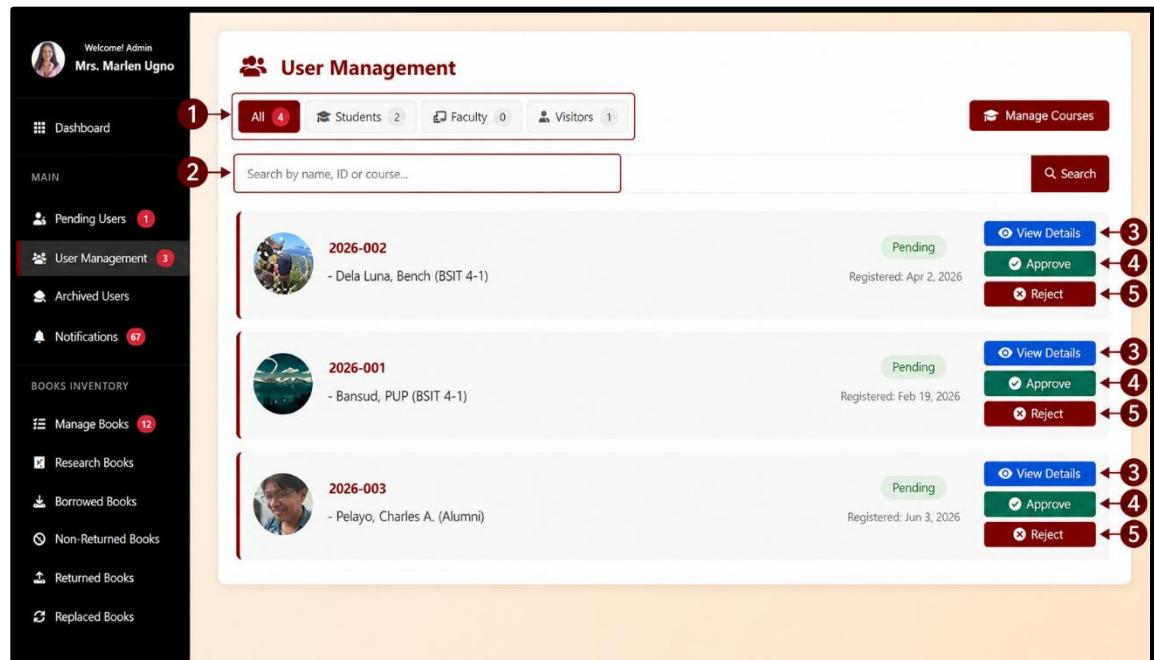


registered users awaiting account approval. The page allows the administrator to review and manage pending registration requests before users are granted access to the system.

1. At the top of the page are filter tabs that categorize pending requests into All, Students, Faculty, and Visitors, with each tab displaying a badge indicating the total number of pending requests for that user type.
2. A search bar is also provided to enable the administrator to quickly locate a specific pending user by entering the user's name, ID number, or course. Each pending registration is displayed as a card containing the user's profile picture, ID number, full name, course or section (if applicable), and the date the registration request was submitted.
3. The administrator may review each request by clicking the View Details button to display the applicant's complete registration information.
4. After reviewing the details, the administrator may click the Approve button to activate the user's account and grant access to the system.
5. However, the admin can also click the Reject button to deny the registration request.



User Management



Upon clicking the Pending Users module from the navigation menu, the administrator is redirected to the Pending Users page, which serves as the review queue for all newly registered users awaiting account approval. This page enables the administrator to review and manage registration requests before granting users access to the system.

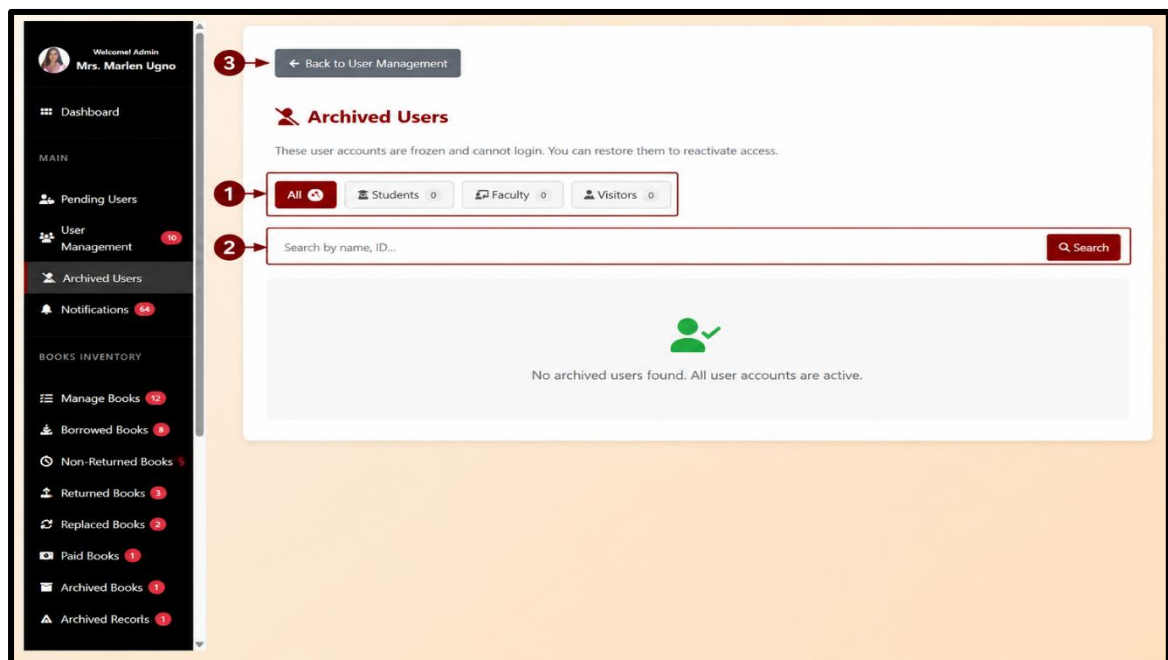
1. At the top of the page are filter tabs that categorize pending registration requests into All, Students, Faculty, and Visitors, with each tab displaying a badge indicating the total number of pending users under each category.
2. A Search bar is also provided, allowing the administrator to quickly locate a specific pending user by entering the user's name, ID number, or course.

Each pending registration request is displayed as a user card containing the user's profile picture, ID number, full name, course or user type, and the date the registration request was submitted.



3. The View Details button allows the administrator to view the applicant's complete registration information before making a decision.
4. The Approve button allows the administrator to approve the registration request. Once approved, the user's account is activated and transferred to the User Management module, enabling the user to access the system.
5. The Reject button allows the administrator to deny the registration request. Once rejected, the pending registration is removed from the Pending Users page, and the user is not granted access to the system.

Archived Users

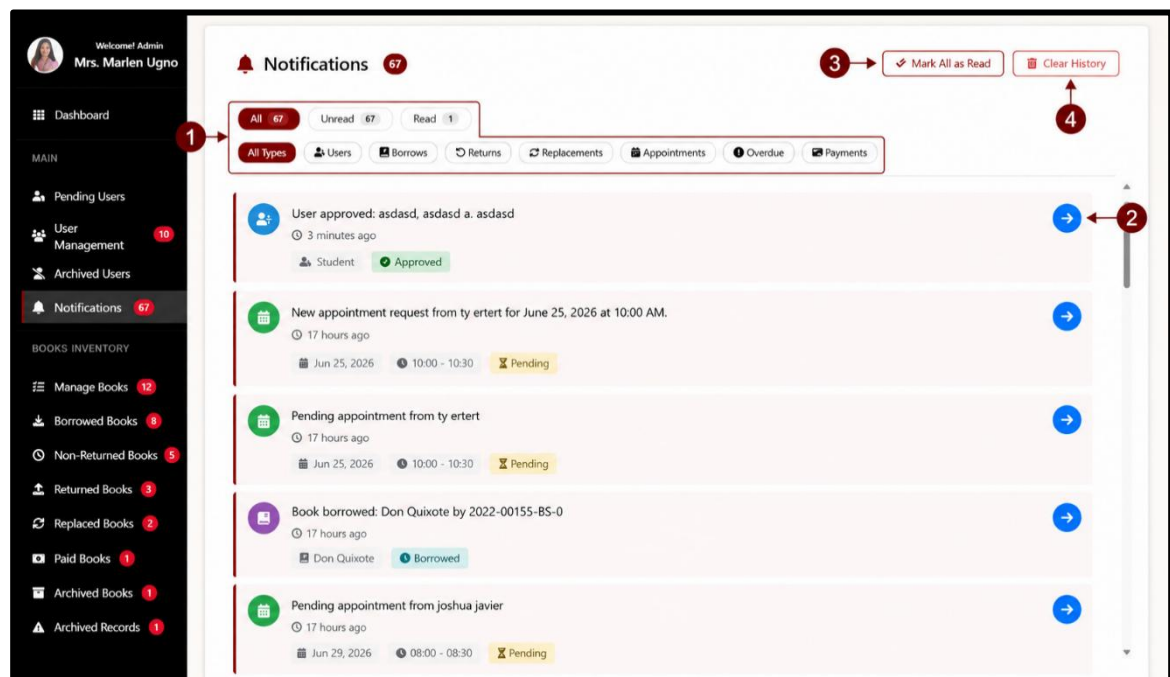


Upon clicking the Archived Users module from the navigation menu, the administrator is redirected to the Archived Users page, which serves as the repository of all archived or deactivated user accounts in the system. This page enables the administrator to view and manage user accounts that have been temporarily suspended. Archived users are unable to log into the system unless their accounts are restored.



1. At the top of the page are filter tabs that categorize archived user accounts into All, Students, Faculty, and Visitors, with each tab displaying a badge indicating the total number of archived users under each category.
2. A Search bar is also provided, allowing the administrator to quickly locate a specific archived user by entering the user's name or ID number.
3. The Back to User Management button, located at the upper-left corner of the page, allows the administrator to return to the User Management page to continue managing active user accounts.

Notifications



Upon clicking the Notifications module from the navigation menu, the administrator is redirected to the Notifications page, which serves as the central activity feed of the system. This page displays all system-generated notifications related to user registrations, borrowing transactions, book returns, replacements, appointments, overdue books, and payments, enabling the administrator to monitor and manage all significant system activities in one place.



1. The page provides filter tabs that allow the administrator to organize and view notifications based on their status or type. Selecting a filter displays only the notifications associated with the chosen classification, enabling the administrator to quickly locate and monitor specific system activities without viewing the entire notification feed.
2. The View (→) button allows the administrator to open the selected notification and view its complete details or proceed to the corresponding module to perform the necessary administrative action, depending on the type of notification.
3. The Mark All as Read button, located at the upper-right corner of the page, allows the administrator to mark all unread notifications as read simultaneously, updating their status and removing the unread indicators.
4. The Clear History button, located beside the Mark All as Read button, allows the administrator to remove all notification records from the notification history, keeping the activity feed organized.



Manage Books

Book Management

Search by title, author or LCC...

+ Add New Category + Add New Book

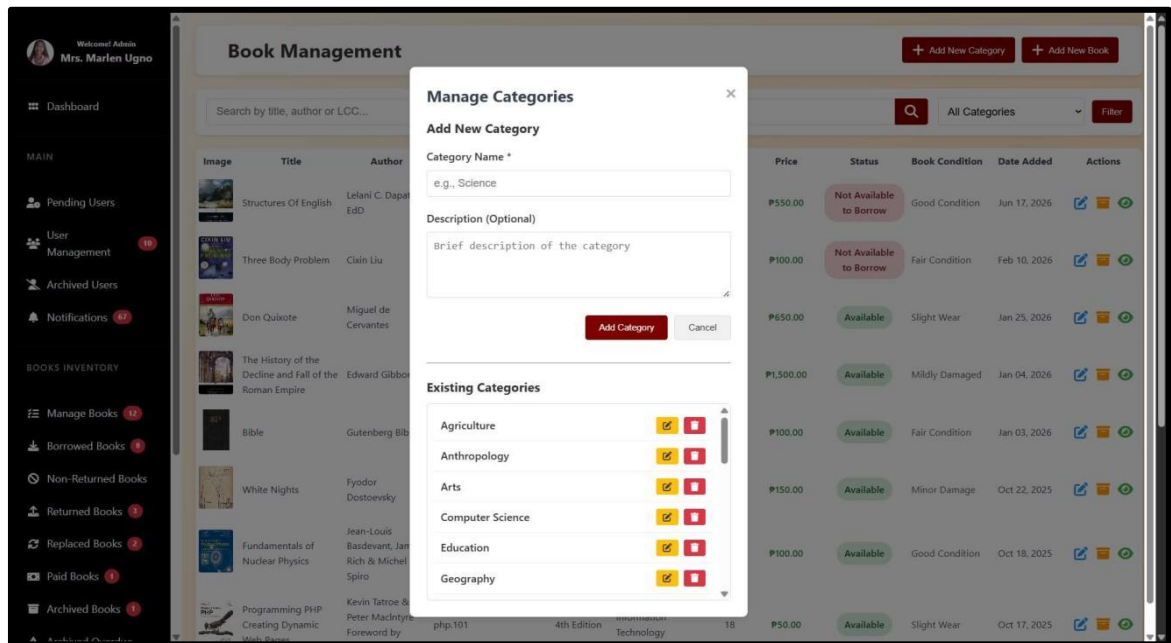
Image	Title	Author	LCC	Volume	Edition	Category	Stock	Price	Status	Book Condition	Date Added	Actions
	Structures Of English	Lelani C. Dagat, EDD	PE1.3728	-	-	Education	1	P150.00	Not Available to Borrow	Good Condition	Jan 17, 2026	
	Three Body Problem	Cixin Liu	ZBDYPRBLM	-	-	Science Fiction	1	P100.00	Not Available to Borrow	Fair Condition	Feb 10, 2026	
	Don Quixote	Miguel de Cervantes	DON_QUIXOT3	1	1st Edition	Language and Literature	10	P450.00	Available	Slight Wear	Jan 25, 2026	
	The History of the Decline and Fall of the Roman Empire	Edward Gibbon	ROMAN_3MPIR3	2	1st Editions	History	10	P1,500.00	Available	Mildly Damaged	Jan 04, 2026	
	Bible	Gutenberg Bible	B1BL3	-	-	Geography	15	P100.00	Available	Fair Condition	Jan 03, 2026	
	White Nights	Fyodor Dostoevsky	F2D1721.W	-	-	Geography	20	P150.00	Available	Minor Damage	Oct 22, 2025	
	Fundamentals of Nuclear Physics	Jean-Louis Basdevant, Jannie Rich & Michel Spigler	Nuc1541	1	1st Edition	Mathematics	16	P100.00	Available	Good Condition	Oct 18, 2025	
	Programming PHP Creating Dynamic Web Pages	Kevin Tatroe & Peter MacIntyre Foreword by	php.101	-	4th Edition	Information Technology	18	P50.00	Available	Slight Wear	Oct 17, 2025	

Upon clicking the "Manage Books" option in the sidebar, the system will display the "Book Management" page. This page contains a complete list of books stored in the library system. It allows the administrator to view and manage book records, including details such as the book image, title, author, LCC (Library of Congress Classification), volume, edition, category, number of available copies, price, availability status, book condition, date added, and available actions.

1. Each book record includes action buttons that allow the administrator to View, Edit, or Archive a book.
2. The Book Management page includes a search function that enables the administrator to quickly locate books by entering the title, author, or Library of Congress Classification (LCC) in the search bar. Additionally, the administrator can use the Category filter to display books belonging to a specific category, making it easier to organize and manage the library's inventory.



3. Add New Category



Upon clicking the Add New Category, it allows the administrator to create, edit, or delete book categories, while the Add New Book feature enables the administrator to add new books by entering details such as the book title, author, LCC, volume, edition, category, description, table of contents, and other required information before saving the record.



4. Add New Book

The screenshot displays the 'Book Management' interface. A sidebar on the left contains navigation links: Dashboard, Pending Users, User Management (16), Archived Users, Notifications (57), and a 'BOOKS INVENTORY' section with links for Manage Books (12), Borrowed Books (4), Non-Returned Books, Returned Books (3), Replaced Books (2), Paid Books (1), and Archived Books (1). The main content area features a search bar and a table of books. A 'Add New Book' modal is open, showing the following fields:

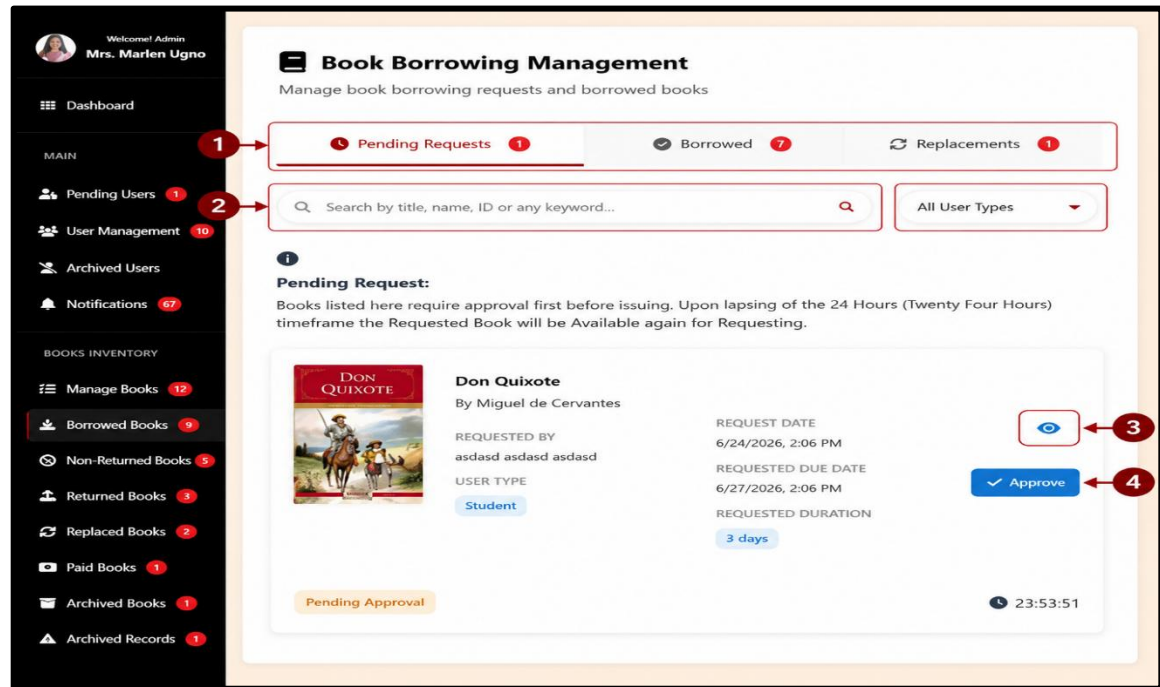
- Book Title*
- Author*
- LCC (Library of Congress Classification)*
- Volume (optional)
e.g. Vol. 1, Volume II
- Edition (optional)
e.g. 2nd Edition, Revised Edition
- Category*
Select Category
- Description
- Table of Contents (HTML)
Enter table of contents in HTML format...

The background table lists books with columns: Image, Title, Author, Price, Status, Book Condition, Date Added, and Actions. Visible entries include 'Structures Of English' (Leland), 'Three Body Problem' (Cixin Liu), 'Don Quixote' (Miguel Cervantes), 'The History of the Decline and Fall of the Roman Empire' (Edward Gibbon), 'Bible' (Gutenberg), 'White Nights' (Fyodor Dostoevsky), 'Fundamentals of Nuclear Physics' (Jean-Louis Bardegey), and 'Programming PHP' (Kevin Foreman).

After clicking the Add New Book button on the Book Management page, the system displays the Add New Book form in a pop-up window. The administrator or librarian is required to enter the necessary book information, including the Book Title, Author, Library of Congress Classification (LCC), and Category, while optional fields such as Volume, Edition, Description, and Table of Contents may also be completed to provide additional details about the book.



Borrowed Books

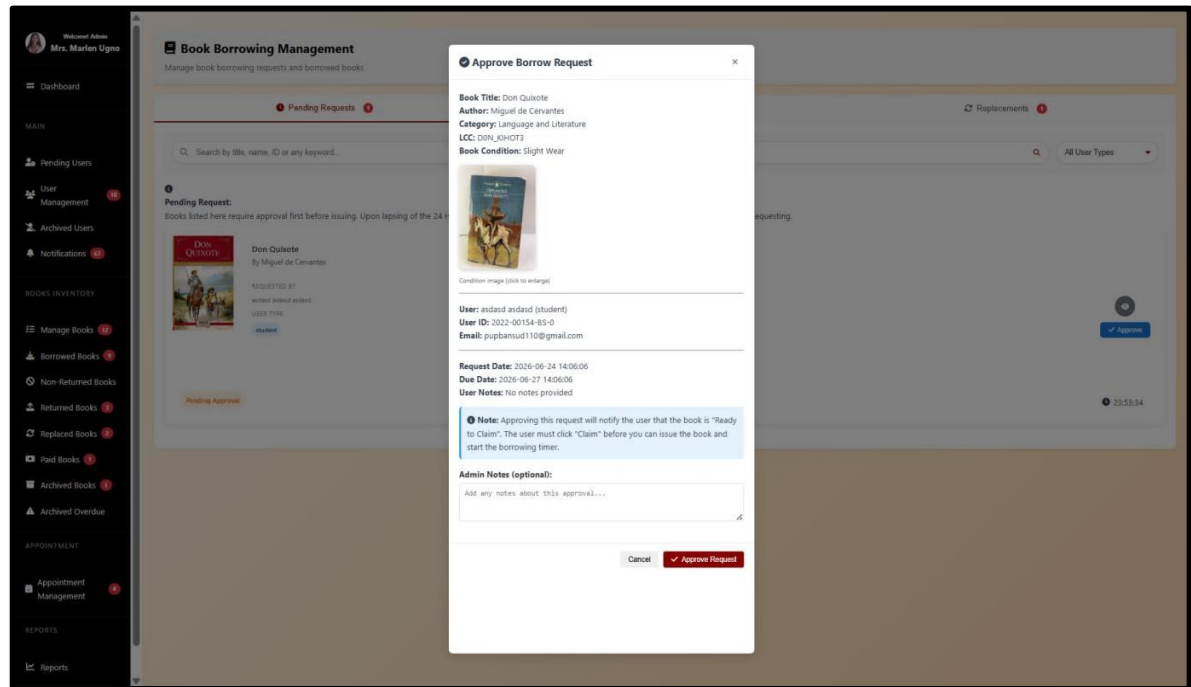


Upon clicking the Borrowed Books menu, the system redirects the administrator to Pending Request where all book borrowing requests awaiting approval are displayed.

1. In the Borrowed Books, there are three (3) tabs: Pending Requests, which displays borrowing requests awaiting approval; Borrowed, which lists books that have been successfully borrowed by users; and Replacements, which contains requests for replacing lost or damaged books.
2. This page includes a search bar that enables the administrator to quickly find borrowing records by entering the book title, borrower's name, ID, or any related keyword. Additionally, the administrator can use the User Type filter to display borrowing records for specific user groups, making it easier to organize and manage borrowing transactions efficiently.



Pending Users



In the Pending Requests tab, the system displays all book borrowing requests submitted by users that are awaiting the administrator's approval. Each request contains the book information, borrower details, request date, and borrowing duration.

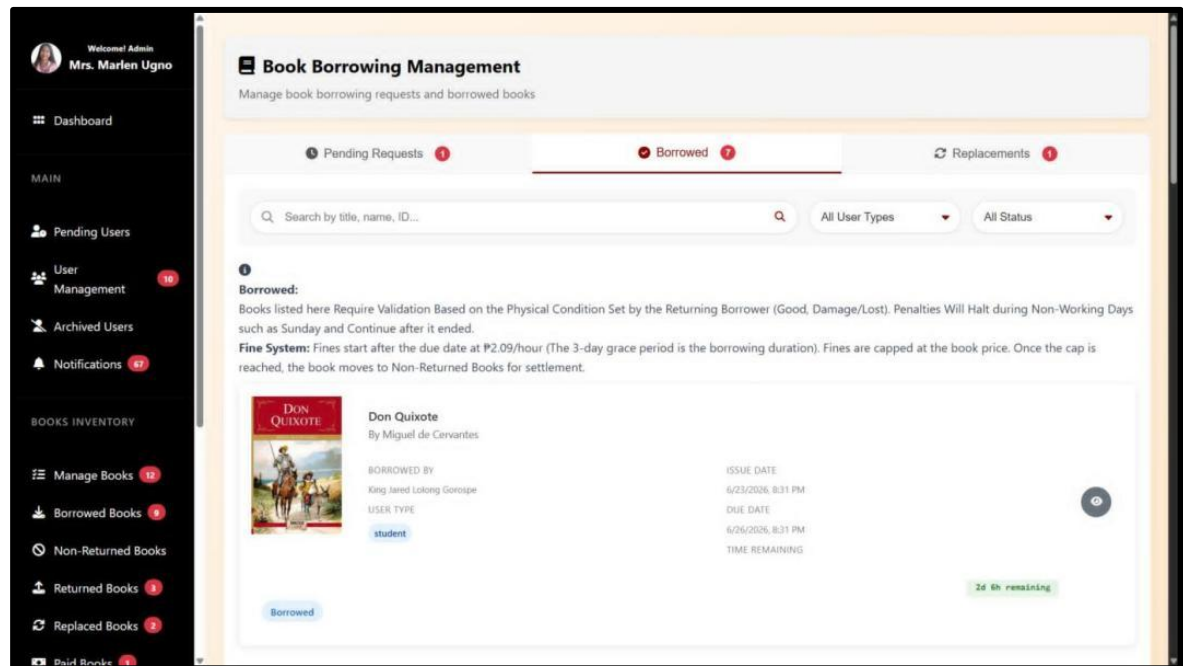
3. Upon clicking the View (Eye) icon, the system opens the Approve Borrow Request window, which provides detailed information about the selected request, including the book title, author, category, LCC, book condition, borrower information, request and due dates, and any additional notes. The administrator may enter optional approval notes in the designated text box before clicking the Approve Request button to authorize the borrowing request. If the administrator decides not to proceed, they may click the Cancel button or close the pop-up window without making any changes.

4. By clicking the Approve Request button in the Pending Requests tab, the system validates and approves the borrowing request. Once approved, the borrower is automatically notified that the request has been accepted and that the book is ready for



claiming. The approved request is then transferred from the Pending Requests tab to the Borrowed tab, where the administrator can monitor the book's borrowing status, issue date, due date, and remaining borrowing period.

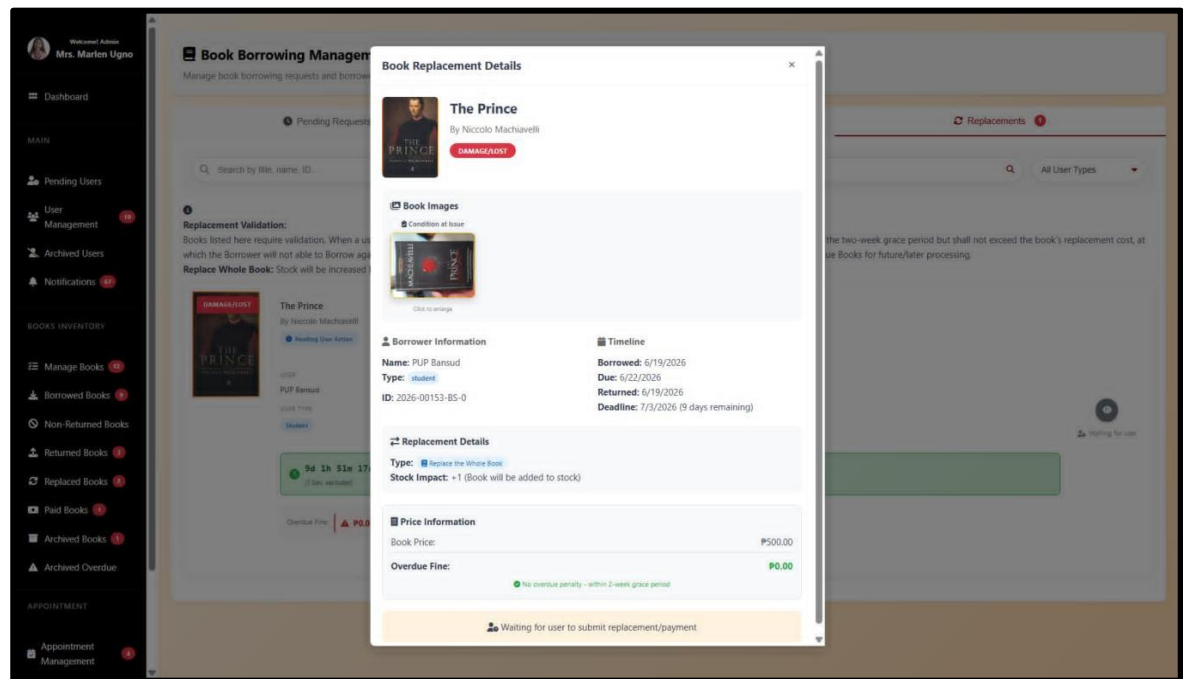
Borrowed



After the administrator clicks the Approve button in the Pending Requests tab, the selected borrowing request is approved and automatically transferred to the Borrowed tab of the Book Borrowing Management page. This tab displays all books that have been successfully issued to users, along with relevant information such as the book title, borrower, user type, issue date, due date, and the remaining borrowing time. Upon approval, the system automatically sets the maximum borrowing period of three (3) days and starts the countdown timer. As the due date approaches, the system continuously updates the remaining time. Once the borrowing period reaches its final stage or expires, the system automatically changes the timer to a red warning indicator, notifying both the administrator and the borrower that the book is due or overdue and requires immediate attention.



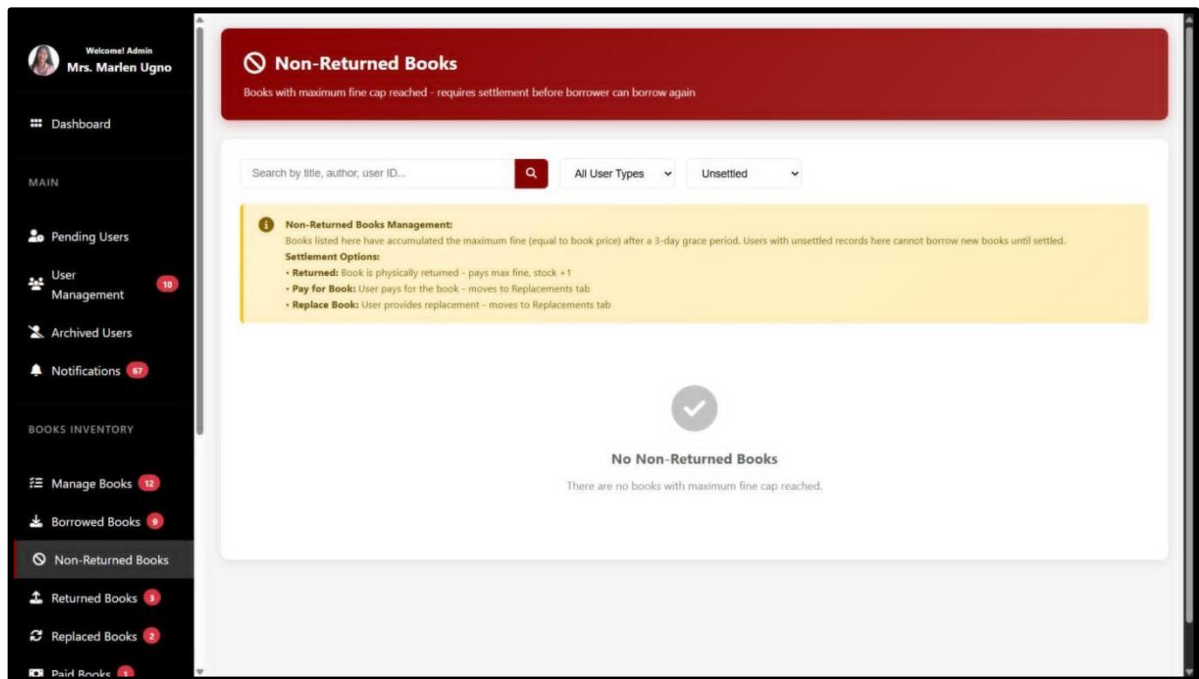
Replacements



The Replacements tab contains the lists of all books that require replacement due to damage or loss. The page displays the affected book, borrower information, replacement type, deadline, remaining replacement period, and any accumulated overdue fines. The administrator can monitor the replacement status and verify whether the borrower has replaced the entire book or settled the replacement cost. Once the replacement or payment has been validated, the administrator may complete the process, allowing the system to update the inventory and borrower records accordingly.



Non-Returned Books



Upon clicking the Non-Returned Books menu, the system redirects the administrator to the Non-Returned Books page, which displays all borrowed books that have exceeded the maximum borrowing period and accumulated fines up to the maximum fine cap, equivalent to the book's retail price. Borrowers listed on this page are automatically restricted from borrowing additional books until their outstanding record has been settled.



Returned Books

Manage Returned Books

Students 3 | Faculty 0 | All Returns 3

Search by title, name, ID...

Book Title	Author	Borrower	Student ID	Requested	Return Date	Settlement Date	Fine Paid
Don Quixote	Miguel de Cervantes	PUP Bansud	2026-00153-BS-0	Jun 2, 2026	Jun 2, 2026		
Don Quixote	Miguel de Cervantes	PUP Bansud	2026-00153-BS-0	Feb 19, 2026	Feb 27, 2026		P650.00
Don Quixote	Miguel de Cervantes	PUP Bansud	2026-00153-BS-0	Feb 19, 2026	Feb 19, 2026		

FROM NON-RETURNED

Upon clicking the Returned Books menu, the system redirects the administrator to the Manage Returned Books page, which displays all books that have been successfully returned by borrowers. The page categorizes returned records based on borrower type, including students and faculty members, and provides details such as the book title, author, borrower information, return date, settlement date, and any applicable fines paid. Records that were previously marked as non-returned but have been settled are also displayed with an indication that they originated from the Non-Returned Books list. This page allows the administrator to verify completed returns, monitor fine settlements, and maintain accurate records of returned library materials.



Replaced Books

Manage Replaced Books

Refresh

Students 2 Faculty 0 All Returns 2

Search by title, name, ID...

Book Title	Author	Borrower	Student ID	Borrow Date	Return Date	Processed By	Status
The Prince	by Niccolo Machiavelli	PUP Bansud	2026-00153-BS-0	Jun 19, 2026	Jun 19, 2026	Mrs. Marlen Ug...	Pending
Don Quixote	by Miguel de Cervantes	PUP Bansud	2026-00153-BS-0	Feb 19, 2026	Feb 19, 2026	Mrs. Marlen i	Completed

The Replaced Books page displays records of lost or damaged books that have been settled through book replacement instead of monetary payment. It enables the administrator to monitor replacement transactions, verify borrower information, and track the status of each replacement request. Administrators can search records by book title, borrower name, or ID, filter records by user type (Students or Faculty), and view detailed information for each case. Each record also indicates whether the replacement process is Pending or Completed, allowing administrators to monitor unresolved and finalized replacements. If there are no replacement records available, the page displays a message indicating that there are currently no replaced books.



Paid Books

Paid Books
View records of books paid for by users (DAMAGE/LOST)

Students 1 | Faculty 0 | All Records 1

Search by title, name, ID...

Overlord
by Kagune Maruyama
BORROWER: PUP Bansud
REPLACEMENT: Pay for Book
PAID DATE: Feb 19, 2026, 06:09 PM

TYPE: Student
AMOUNT PAID: ₱650.00

PAID
[View Details](#)

Upon clicking the Paid Books menu, the system redirects the administrator to the Paid Books page, which displays all records of books with settled payment transactions, including books that were previously marked as non-returned and resolved through payment.

Book Paid Details

Overlord
by Kagune Maruyama
DAMAGE/LOST
Pay for Book

BORROWER ID: 2026-00153-B5-0
USER TYPE: Student
BORROW DATE: Feb 19, 2026, 06:07 PM
RETURNED DATE: Feb 19, 2026, 06:07 PM

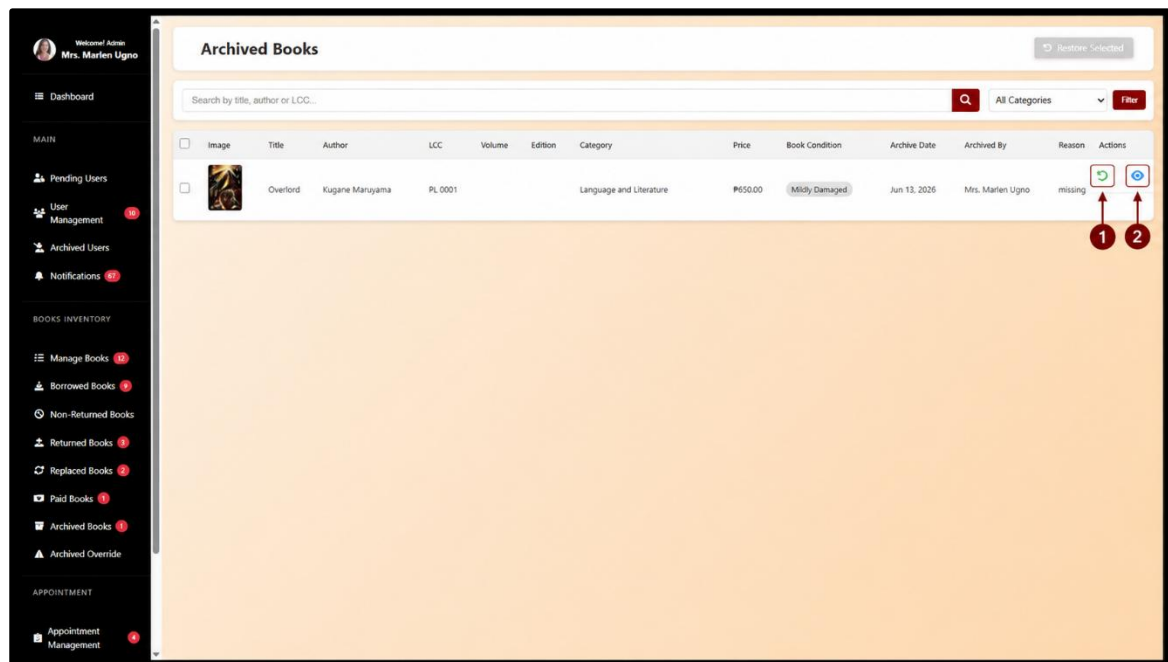
BORROWER NAME: PUP Bansud
BOOK LCC: PL 0001
DUE DATE: Feb 23, 2026, 06:07 PM
PAYMENT DATE: Feb 19, 2026, 06:09 PM

Payment Breakdown
Base Book Cost: ₱650.00



1. By clicking the View Details button in the Paid Books page, the system displays the Paid Book Details page, which provides comprehensive information about the completed payment transaction.

Archived Books

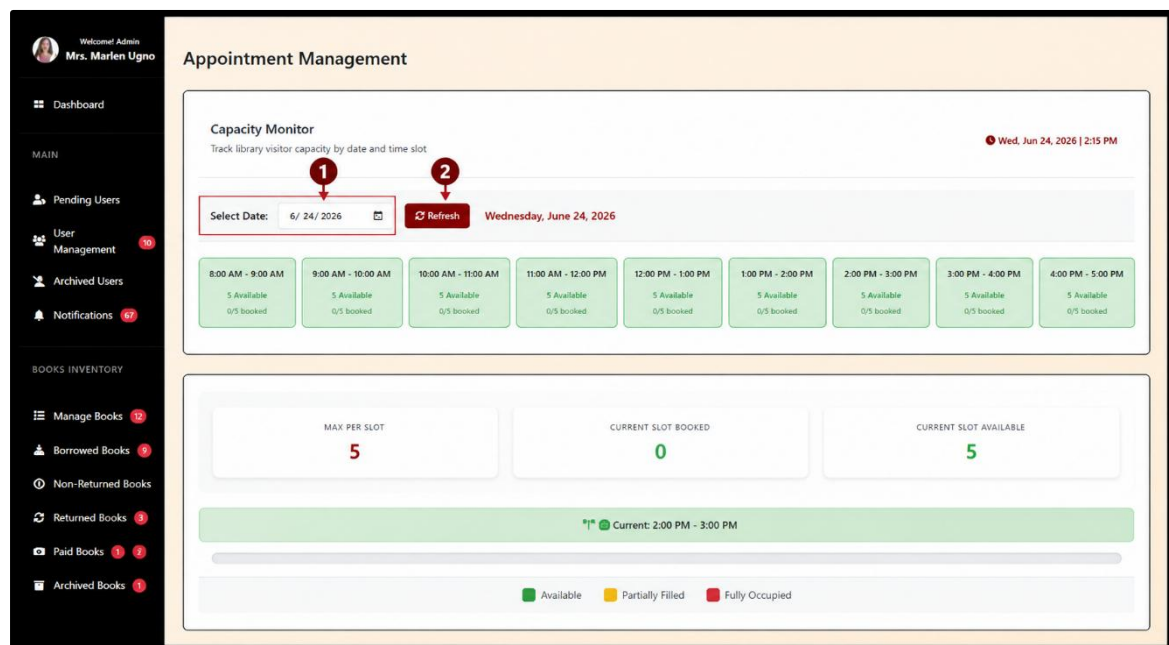


Upon clicking the Archived Books menu, the system redirects the administrator to the Archived Books page, which displays all books that have been marked as Archived by the administrator. Books may be archived for various reasons, such as having no remaining usable copies, being severely damaged, requiring replacement, or other library management concerns. Archiving a book temporarily removes it from the active Manage Books inventory, ensuring that unavailable books cannot be borrowed while they are undergoing further evaluation or processing. Each archived record includes a Reason field that specifies why the book was archived, providing proper documentation and improving inventory management. This feature promotes efficient book processing and helps maintain an organized and accurate library collection.



1. Return Icon – Restores the archived book to the Manage Books inventory, making it active and available for borrowing again.
2. View Details (Eye Icon) – Displays detailed information about the archived book, including its bibliographic information, archive reason, and other relevant records for the administrator's review.

Appointment Management



Upon clicking the Appointment Management module from the navigation menu, the administrator is redirected to the Appointment Management page, which allows the administrator to monitor and manage visitor appointment schedules and capacity for each available time slot. This page provides real-time information on appointment availability and supports the complete management of visitor appointments. The system automatically approves appointment requests whenever slots are available based on the configured capacity for each time slot. The administrator also has the authority to dynamically configure the appointment schedule by opening or closing available dates, modifying appointment time slots, and setting the maximum number of visitors allowed per time slot.



1. In the Capacity Monitor section, the administrator may select a specific date to view the appointment schedule and availability for that day. The administrator may also select future dates to monitor occupied and available appointment slots, allowing advance management of visitor appointments.
2. A Refresh button is also provided to reload the latest appointment data and ensure that the displayed booking information is up to date.

The appointment time slots are displayed as individual cards showing the scheduled appointment period, the number of available slots, and the number of booked appointments for each time slot. The administrator may select any time slot to view its corresponding capacity information, which is displayed in the Capacity Summary, including the maximum number of visitors allowed per time slot, the current number of booked appointments, and the remaining available slots. The page also indicates the currently selected appointment period, and selecting a different time slot automatically updates the displayed capacity information and appointment statistics. A Capacity Indicator provides a graphical representation of the occupancy level of the selected time slot, enabling the administrator to quickly determine whether the appointment schedule is available, partially filled, or fully occupied.



Override Control Panel

When the administrator scrolls down the Appointment Management page, the Override Control Panel becomes visible. This section allows the administrator to manually control the appointment calendar by temporarily closing appointment schedules for specific dates, durations, or custom time ranges.

1. Clicking the Toggle Calendar View shows or hide the calendar panel. This helps maximize workspace when the calendar is not needed.



Override Control Panel
Manually manage slot availability and restrictions with precision control

[Toggle Calendar View](#) [Capacity Settings](#)

Capacity Settings [Close](#)

Current Maximum Capacity: 50

Set Maximum Visitors Per Time Slot

Visitors/slot
Enter a value between 1 and 200

Quick Presets: 25 50 75 100 150

Impact Preview

Per Slot	Daily Total (9 slots)	Weekly Total
50	450	2,700

[Reset to Default \(50\)](#) [Apply Changes](#)

Visitor Appointments

- When the administrator clicks the Capacity Settings button in the Override Control Panel, the Capacity Settings module is displayed, allowing the administrator to configure the maximum number of visitors that can book each appointment time slot. The current appointment capacity is shown as a reference, while an input field and quick preset buttons (25, 50, 75, 100, and 150) enable the administrator to easily set a new capacity. As the value is changed, the Impact Preview section automatically updates to display the estimated number of visitors that can be accommodated per time slot, per day, and per week, helping the administrator evaluate the effect of the selected capacity. The Reset to Default button restores the original capacity setting, the Apply Changes button saves and implements the new appointment limit for future bookings, and the Close button exits the module and returns the administrator to the main Override Control Panel.
- After selecting a date from the calendar, the administrator may choose the duration for which appointments will be unavailable by selecting one of the available closure options: Full Day, Half Day, 1 Week, 1 Month, or Custom Time.
- If only specific hours need to be blocked, the administrator may select Custom Time, specify the start time and end time, and click Add Time Slot to save the selected period. Multiple custom time ranges may be added when necessary, allowing the administrator to accommodate temporary closures such as meetings, maintenance activities, or special events without affecting the entire day's appointment schedule.



Visitor Appointments

Visitor Appointments

Manage incoming appointment requests

1 Search by name, ID, or purpose... All Time Slots mm/dd/yyyy X Clear 2

VISITOR NAME	VISITOR ID	CONTACT	DATE	TIME SLOT	PURPOSE	STATUS	GRACE STATUS	ACTIONS
joshua javier	2C56-009	09518909530	Jun 29, 2026	8:00 AM - 9:00 AM, 2:00 PM - 3:00 PM	Study	PENDING	Starts in 11h 43m 34s	View Details, Edit, Print
ty ertert	2C56-005	09663097716	Jun 25, 2026	10:00 AM - 11:00 AM, 2:00 PM - 3:00 PM	Study	PENDING	Starts in 19h 43m 34s	View Details, Edit, Print
ty ertert	2C56-005	09663097716	Jun 23, 2026	3:00 PM - 4:00 PM	Research	CANCELLED	No-Show	View Details, Edit, Print
ty ertert	2C56-005	09663097716	Jun 23, 2026	11:00 AM - 12:00 PM, 2:00 PM - 3:00 PM	Study	FINISHED	Attended On time	View Details, Edit, Print
ty ertert	2C56-005	09663097716	Jun 23, 2026	8:00 AM - 9:00 AM	Research	CANCELLED	No-Show	View Details, Edit, Print

The Visitor Appointments section allows the administrator to monitor and manage all appointment requests submitted by visitors.

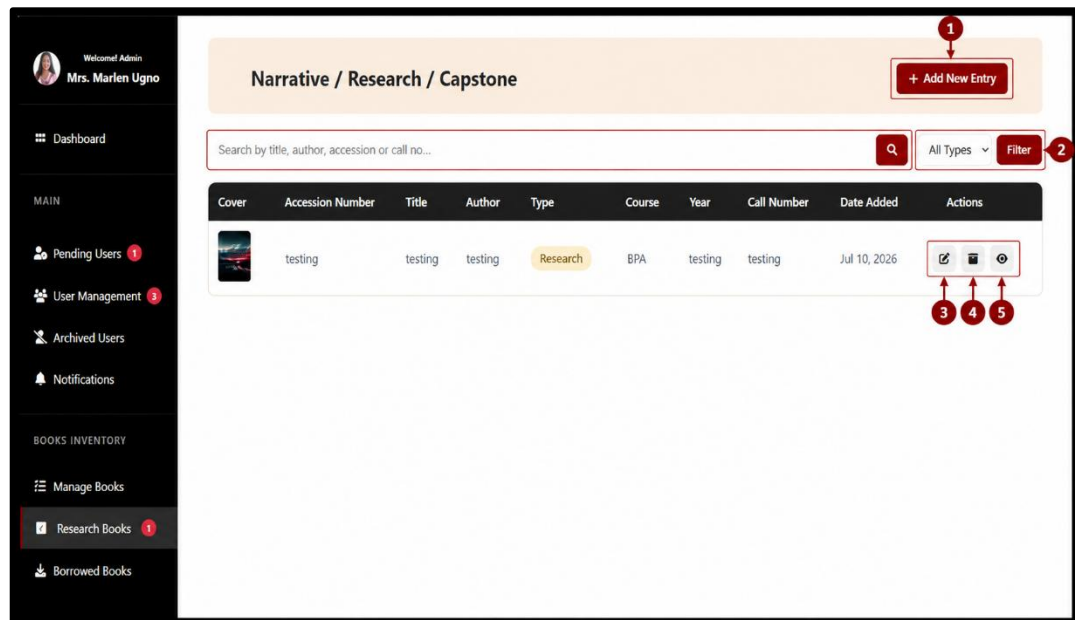
1. At the top of the page, the administrator can use the Search bar to locate appointments by visitor name, visitor ID, or purpose, filter records by Time Slot, select a specific Date.
2. Click Clear to remove all applied filters.

Each appointment record displays the visitor's name, visitor ID, contact number, appointment date, reserved time slot, purpose of the visit, appointment status, grace status, and available actions. The Status column indicates whether an appointment is Pending, Cancelled, or Finished, while the Grace Status monitors the visitor's attendance by displaying a countdown before the appointment begins, confirming when the visitor has Attended, or marking the appointment as a No-Show if the visitor fails to arrive within the allowed grace period. The Actions column provides management buttons that enable the administrator to approve, reject,



cancel, or otherwise manage appointment records depending on their current status, ensuring efficient monitoring and administration of all visitor appointments.

Narrative/Research/Capstone Page



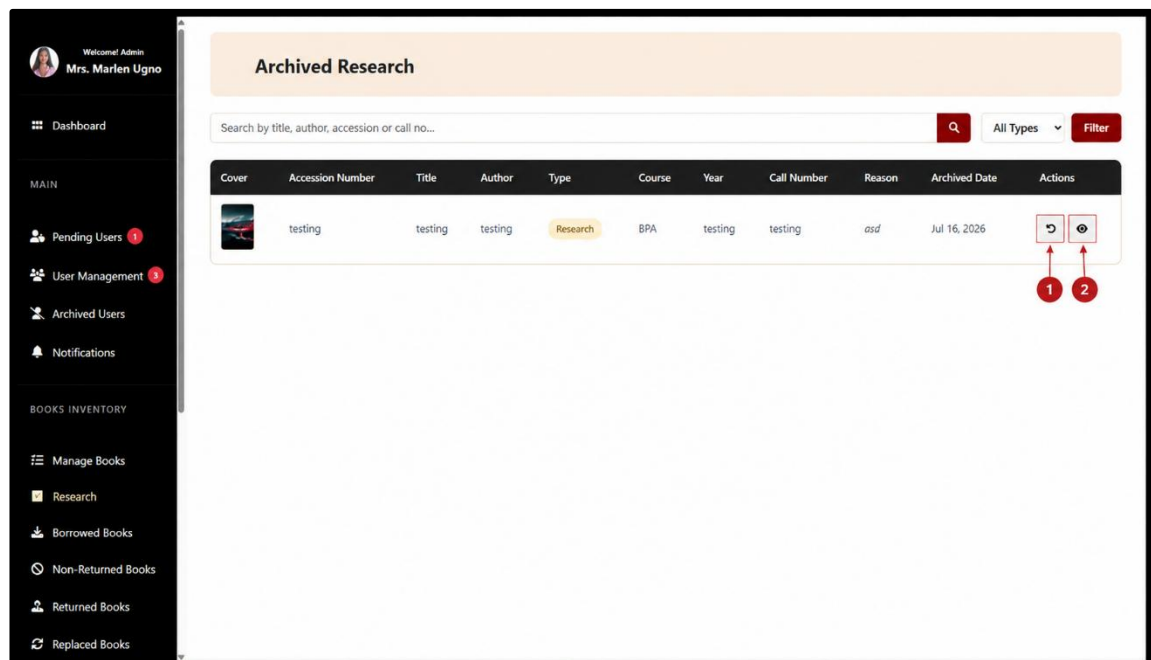
The Narrative/Research/Capstone page serves as the repository for encoding and managing all academic books and manuscripts under the Bachelor of Science in Information Technology (BSIT), Bachelor of Public Administration (BPA), Bachelor of Secondary Education major in English (BSEd English), and Bachelor of Secondary Education major in Mathematics (BSEd Mathematics) programs. This page displays essential information for each entry, including the book cover, accession number, title, author, type, course, year, call number, and date added. The page also provides the following functions:

1. Administrators can click the Add New Entry button to encode and register a new Narrative, Research, or Capstone book into the system.



2. The Search bar enables quick retrieval of records using the title, author, accession number, or call number, while the Type Filter allows books to be filtered according to their classification (Narrative, Research, or Capstone).
3. Under the Actions column, administrators can Edit existing book information.
4. Archive books by transferring them to the Research Archive page with a required reason for archiving.
5. View complete book details for verification and reference.

Archived Research

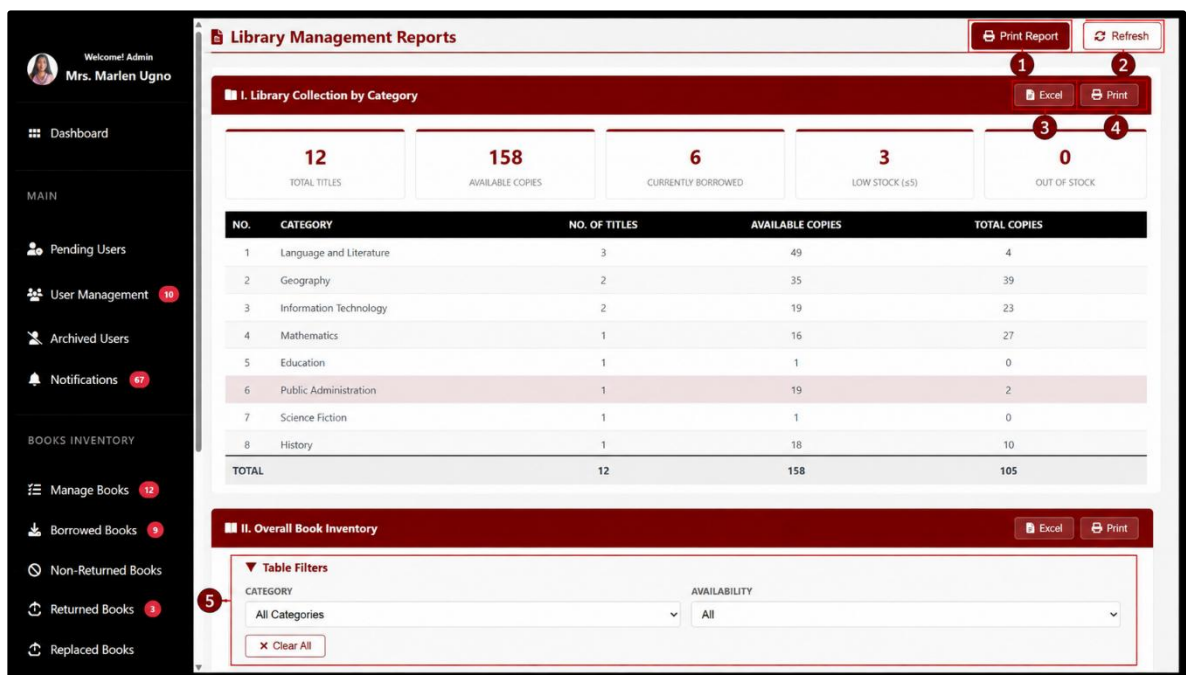


The Archived Research page serves as the repository for all Narrative, Research, and Capstone books that have been set aside due to discontinuation, loss, damage, or other archival reasons. This page allows administrators to review archived research materials while displaying essential information, including the book cover, accession number, title, author, type, course, year, call number, reason for archiving, archived date, and available actions. The page also provides the following functions:



1. The Return icon allows administrators to restore an archived Narrative, Research, or Capstone book to the Narrative/Research/Capstone page, making it active and available in the collection once again.
2. The View icon enables administrators to view the complete details of an archived research material for verification and reference.

Management Reports



The Reports page provides administrators with a comprehensive overview of the library's book inventory and collection statistics. It displays summarized information such as the total number of titles, available copies, currently borrowed books, low-stock books, and out-of-stock books, enabling administrators to monitor the status of the library collection. The page also categorizes books by subject and presents the number of titles, available copies, and total copies for each category. In addition, the Overall Book Inventory section allows administrators to filter records by Category and Availability for easier retrieval and analysis of inventory data. The page also provides the following functions:



1. The Print Report button allows administrators to generate and print a complete library inventory report.
2. The Refresh button updates the page to display the latest inventory records and statistics.
3. The Excel button exports the displayed report into an Excel spreadsheet for documentation and further analysis.
4. The Print button prints the selected report section directly from the system.
5. The Category and Availability filters enable administrators to narrow down inventory records based on book category and current availability status.



Appendix 13

PROGRAM CODE LISTING

```
Elib/
index.php
<?php
session_start();
// Database connection
require_once
'DBconnection/db_connection.php';

/**
 * Get Books with filters
 */

$search = isset($_GET['search']) ?
trim($_GET['search']) : '';

// Only fetch when searching (do not load all
books by default)
$books = [];
if (!empty($search)) {
    $sql = "SELECT * FROM add_books
WHERE is_archived = 0
AND (title LIKE :search OR book_id
LIKE :search OR control_no LIKE :search
OR author LIKE :search)";
    $params = [':search' => "%$search%"];
    $stmt = $pdo->prepare($sql);
    $stmt->execute($params);
    $books = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
}

?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    <title>E-Library Management
System</title>

    <link rel="icon" type="image/x-icon"
href="Pictures/PUP_LOGO.png">
    <link rel="stylesheet" href="Index.css">
</head>
<body>

<!-- Header -->
<header class="header">
    <div class="logo-section">
        <!-- use defined constant to avoid
path/case issues on deployment -->
        
        <span class="logo-text">E-LIB: A
WEB-BASED LIBRARY MANAGEMENT
INFORMATION SYSTEM (LMIS)</span>
    </div>
    <nav class="nav">
        <a href="Login_Path.php" class="login-
link">Login</a>
    </nav>
</header>

<!-- Hero -->
<section class="hero">
    <div class="hero-content">
        <h1>E-LIB: A WEB-BASED LIBRARY
MANAGEMENT INFORMATION SYSTEM
(LMIS)</h1>
        <p>Search and explore available
books</p>
        <div class="main-library-link">
            <a href="https://ils.pup.edu.ph/cgi-
bin/koha/opac-main.pl" target="_blank"
rel="noopener noreferrer">
                
                <div class="main-library-
caption">Visit PUP Main Campus
Library</div>
            </a>
        </div>
    </div>
</section>
</body>
</html>
```



```
</div>
</div>
</section>

<!-- Search -->
<section class="features">
  <div class="container">
    <h2>Search Book</h2>

    <form method="GET" class="search-
input-wrapper">
      <input type="text" name="search"
id="search-input" placeholder="Search by
Title, Author, Book ID, or Control No."
value="<?= htmlspecialchars($search) ?>">
      <button type="submit" id="search-
btn">Search</button>
      <button type="button" id="clear-
search-btn" class="btn-clear" <?=
empty($search) ? 'style="display:none;" : "
?>>Clear</button>
    </form>
  </div>
</section>

<!-- Book Results -->
<?php $showResults = !empty($search); ?>
<section class="book-results"
id="bookResults" <?php echo $showResults
? " : 'style="display:none;"; ?>>
  <div class="container">
    <?php if (empty($books)): ?>
      <p>No books found.</p>
    <?php else: ?>
      <?php foreach ($books as $book):
?>
        <?php
        // Fix image path - remove leading
../../ if present
        $imagePath =
$book['image_path'];
        if (strpos($imagePath, ' ../../') ===
0) {
          $imagePath =
substr($imagePath, 6); // Remove ' ../../'
        }
        ?>
        <div class="book-card">
```

```

    <h3><?=
htmlspecialchars($book['title']) ?></h3>
    <p><strong>Author:</strong>
<?= htmlspecialchars($book['author'])
?></p>
    <p><strong>Control
No.</strong> <?=
htmlspecialchars($book['control_no'])
?></p>
    <?php if
(!empty($book['description'])): ?>
      <p class="description"><?=
htmlspecialchars($book['description'])
?></p>
    <?php endif; ?>
    <?php
    $stock_qty =
isset($book['stock_quantity']) ?
intval($book['stock_quantity']) : 0;
    // Available only when more
than 1 copy remains; if 1 or less, not
available to borrow
    if ($stock_qty > 1) {
      $statusText = 'Available';
      $statusClass = 'available';
    } else {
      $statusText = 'Not Available
to Borrow';
      $statusClass = 'not-available-
to-borrow';
    }
    $stockText = ($stock_qty > 1) ?
"({$stock_qty} in stock)" : "";
    ?>
    <p class="status <?=
$statusClass ?>"><?=
htmlspecialchars($statusText) ?><?php if
(!empty($stockText)): ?> <span
class="stock-number"><?=
htmlspecialchars($stockText)
?></span><?php endif; ?></p>
  </div>
  <?php endforeach; ?>
<?php endif; ?>
</div>
</section>
```




```
<!-- Footer -->
<footer class="footer">
  <p>&copy; 2026 PUP Bansud
  Campus</p>
  <p>E-LIB: A WEB-BASED LIBRARY
  MANAGEMENT INFORMATION SYSTEM
  (LMIS)</p>
</footer>

<script>
document.addEventListener("DOMContentLoaded", () => {
  const bookResults =
document.getElementById("bookResults");
  const searchInput =
document.getElementById("search-input");
  const clearBtn =
document.getElementById("clear-search-
btn");
  const featuresSection =
document.querySelector('.features');

  // helper to show/hide the clear button
  function updateClearButtonVisibility() {
    if (!clearBtn || !searchInput) return;
    const hasText = searchInput.value &&
searchInput.value.trim() !== "";
    clearBtn.style.display = hasText ?
'inline-block' : 'none';
  }

  // Get URL params
  const urlParams = new
URLSearchParams(window.location.search)
;
  const searchValue =
urlParams.get("search");

  if (searchValue && searchValue.trim() !==
"") {
    bookResults.style.display = "block";
    if (featuresSection)
featuresSection.classList.add('search-
active');
    bookResults.scrollIntoView({ behavior:
"smooth" });
  } else {
    bookResults.style.display = "none";

    if (featuresSection)
featuresSection.classList.remove('search-
active');

    // Show/hide clear button based on initial
    value
    updateClearButtonVisibility();

    // Toggle clear button while typing
    if (searchInput) {
      searchInput.addEventListener('input',
updateClearButtonVisibility());
    }

    // Clear button handler
    if (clearBtn) {
      clearBtn.addEventListener('click', (e)
=> {
        if (searchInput) searchInput.value =
        "";
        if (bookResults)
bookResults.style.display = 'none';
        if (featuresSection)
featuresSection.classList.remove('search-
active');
        // Remove 'search' param from URL
        without reloading
        const newUrl = new
URL(window.location.href);
        newUrl.searchParams.delete('search
');
        window.history.replaceState({}, "",
newUrl.toString());
        updateClearButtonVisibility();
        if (searchInput) searchInput.focus();
      });
    }
  }
});
</script>
</body>
</html>

Login_Path.php
<?php
session_start();
?>
```



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>PUP Library Management System -
Login</title>
  <link rel="icon" type="image/x-icon"
href="Pictures/PUP_LOGO.png">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
  <style>
    :root {
      --primary-color: #8B0000; /* Dark
red */
      --secondary-color: #4A0404; /* Even
darker red */
      --accent-color: #FFB74D; /* Gold
(still available if needed) */
      --light-color: #F5F5F5;
      --dark-color: #333333;
      --shadow: 0 4px 12px
rgba(0,0,0,0.1);
      --transition: all 0.3s ease;
    }

    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: 'Segoe UI', Tahoma,
Geneva, Verdana, sans-serif;
    }

    body {
      background-color: #f9f9f9;
      background-image:
url('../Pictures/Background_Pattern.png');
    }

    /* Header Styles */
    .header {
      background-color: white;
      box-shadow: var(--shadow);
      padding: 1rem 2rem;
      position: fixed;
      width: 100%;
      top: 0;
      z-index: 1000;
      display: flex;
      justify-content: space-between;
      align-items: center;
    }

    .logo-section {
      display: flex;
      align-items: center;
      gap: 0.8rem;
    }

    .logo {
      width: 45px;
      height: 45px;
      border-radius: 50%;
      border: 2px solid var(--primary-
color);
      object-fit: cover;
      transition: var(--transition);
    }

    .logo:hover {
      transform: scale(1.05);
      box-shadow: 0 0 8px rgba(139, 0, 0,
0.3);
    }

    .logo-text {
      font-size: 1.3rem;
      font-weight: 700;
      color: var(--primary-color);
    }

    .nav {
      display: flex;
      gap: 1.5rem;
    }

    .nav a {
      color: var(--dark-color);
      text-decoration: none;
      font-weight: 500;
      padding: 0.5rem 1rem;
      border-radius: 4px;
      transition: var(--transition);
    }
```



```
.nav a:hover {
  background-color: rgba(139, 0, 0, 0.05);
  color: var(--primary-color);
}

/* Main Content */
.main-content {
  margin-top: 90px;
  padding: 2rem;
  min-height: calc(100vh - 90px);
  display: flex;
  justify-content: center;
  align-items: center;
}

.login-path-container {
  background-color: white;
  border-radius: 12px;
  box-shadow: var(--shadow);
  padding: 2.5rem;
  width: 100%;
  max-width: 500px;
  text-align: center;
  position: relative;
  overflow: hidden;
  animation: fadeIn 0.6s ease-in-out;
}

@keyframes fadeIn {
  from { opacity: 0; transform:
translateY(20px); }
  to { opacity: 1; transform:
translateY(0); }
}

.login-path-container::before {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  width: 100%;
  height: 5px;
  background: linear-gradient(90deg,
var(--primary-color), var(--secondary-color));
}

.login-title {
  color: var(--dark-color);
  margin-bottom: 1.5rem;
  font-size: 1.8rem;
  font-weight: 600;
}

.login-subtitle {
  color: #757575;
  margin-bottom: 2rem;
  font-size: 1rem;
}

.login-buttons {
  display: flex;
  flex-direction: column;
  gap: 1rem;
  margin-top: 2rem;
}

.login-btn {
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 0.8rem;
  border-radius: 8px;
  padding: 1rem;
  font-size: 1.1rem;
  font-weight: 500;
  cursor: pointer;
  transition: var(--transition);
  text-decoration: none;
}

.login-btn:hover {
  transform: translateY(-2px);
  box-shadow: 0 4px 8px
rgba(0,0,0,0.1);
}

/* Student / Faculty / Visitor Button -
PRIMARY */
.login-btn.visitor-btn {
  background-color: var(--primary-
color);
  color: white;
  border: none;
  font-weight: 600;
}

.login-btn.visitor-btn:hover {
```



```
background-color: var(--secondary-
color);
box-shadow: 0 4px 12px
rgba(0,0,0,0.2);
}

/* Admin button - SECONDARY */
.admin-btn {
background-color: #fff;
color: var(--dark-color);
border: 1px solid #ccc;
}

.admin-btn:hover {
background-color: #f5f5f5;
}

.admin-btn i {
color: var(--primary-color);
}

.divider {
display: flex;
align-items: center;
margin: 2rem 0;
}

.divider::before,
.divider::after {
content: "";
flex: 1;
border-bottom: 1px solid #e0e0e0;
}

.divider-text {
padding: 0 1rem;
color: #757575;
font-size: 0.9rem;
}

.footer {
margin-top: 2rem;
color: #757575;
font-size: 0.9rem;
}

/* Responsive styles */
@media (max-width: 600px) {
.login-path-container {
background-color: var(--secondary-
color);
padding: 2rem 1.5rem;
margin: 0 1rem;
}

.login-title {
font-size: 1.5rem;
}

.header {
padding: 1rem;
}

.logo-text {
font-size: 1.1rem;
}
}
</style>
</head>
<body>
<!-- Header -->
<header class="header">
<div class="logo-section">
<a href="index.php"></a>
<span class="logo-text">E-LIB: A
WEB-BASED LIBRARY MANAGEMENT
INFORMATION SYSTEM (LMIS)</span>
</div>
<nav class="nav">
<a href="index.php"><i class="fas fa-
home"></i> Home</a>
</nav>
</header>

<!-- Main Content -->
<main class="main-content">
<div class="login-path-container">
<h1 class="login-title">E-LIB: A
WEB-BASED LIBRARY MANAGEMENT
INFORMATION SYSTEM (LMIS)</h1>
<p class="login-subtitle">Please
select your login type to continue</p>

<div class="login-buttons">
<!-- Student / Faculty / Visitor
button (Primary) -->
<a href="Login Form/Login.php"
class="login-btn visitor-btn">
```



<pre><i class="fas fa-user-graduate"></i> Student / Faculty / Visitor Login <div class="divider"> OR </div> <!-- Admin button (Secondary) --> <i class="fas fa-user-shield"></i> Administrator Login </div> <div class="footer"> <p>&copy; <?php echo date("Y"); ?> PUP BANSUD CAMPUS</p> </div> </div> </main> <script> // Add a simple animation when hovering over buttons document.querySelectorAll('.login-btn').forEach(button => { button.addEventListener('mouseover', function() { this.style.transform = 'translateY(-3px)'; }); button.addEventListener('mouseout', function() { this.style.transform = 'translateY(0)'; }); }); </script> </body> </html></pre>	<pre>logout.php <?php session_start(); // Store user type for redirection \$userType = \$_SESSION['usertype'] ?? \$_SESSION['user_type'] ?? ""; // Full session destruction for security \$_SESSION = array(); // Destroy the session cookie if (ini_get("session.use_cookies")) { \$params = session_get_cookie_params(); setcookie(session_name(), "", time() - 42000, \$params["path"], \$params["domain"], \$params["secure"], \$params["httponly"]); } // Destroy the session session_destroy(); session_start(); // Set success message \$_SESSION['success'] = 'You have been successfully logged out.'; // Use the centralized login path file header("Location: ../Login Form/Login.php"); exit(); ?> Admin_Logout.php <?php session_start(); // Full session destruction for security \$_SESSION = array(); // Destroy the session cookie if (ini_get("session.use_cookies")) { \$params = session_get_cookie_params(); setcookie(session_name(), "", time() - 42000, \$params["path"], \$params["domain"], \$params["secure"], \$params["httponly"]); } }</pre>
---	---



```
session_destroy();

// Start new session for success message
session_start();
$_SESSION['success'] = 'You have been
successfully logged out.';

// Redirect to admin login
header("Location:
AdminDashboard/AdminLoginPage.php");
exit();
?>
Elib/AdminDasboard
Admin_Dashboard.php
<?php
// Only start session if one doesn't exist
already
if (session_status() ===
PHP_SESSION_NONE) {
    session_start();
}

require_once
'../Functiondb/cleanup_sessions.php';
require_once
'../Functiondb/AdminDb/check_admin_auth.
php';

// Check authentication - use loose
comparison for backward compatibility
if (!isset($_SESSION['admin_logged_in']) ||
!$_SESSION['admin_logged_in']) {
    header('Location: AdminLoginPage.php');
    exit();
}

// Get admin information
$admin_id = $_SESSION['admin_id'] ?? '';
$admin_username =
$_SESSION['admin_username'] ?? 'Admin';

// Get profile image if available
require_once
'../DBconnection/db_connection.php';
// Notification badges are loaded
if (!isset($badges) && isset($pdo)) {
    require_once __DIR__ . '/Notification
Badge/Notification Badge.php';
}

$profileImage =
'../Pictures/default_admin.png'; // Default
image path

try {
    $stmt = $pdo->prepare("SELECT
profile_image FROM admin WHERE
admin_id = ?");
    $stmt->execute([$admin_id]);
    $result = $stmt-
>fetch(PDO::FETCH_ASSOC);

    if ($result &&
!empty($result['profile_image'])) {
        $profileImage = $result['profile_image'];
    }
} catch (PDOException $e) {
    error_log("Error fetching admin profile: " .
$e->getMessage());
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    <title>Admin Dashboard - PUP Bansud
Campus</title>
    <link rel="stylesheet" href="Notification
Badge/badge.css">
    <link rel="stylesheet"
href="../Design/Admin.css">
    <link rel="stylesheet" href="../Design/A-
Dashboard_cards.css">
    <link rel="stylesheet"
href="../Design/AdminSidebar.css">
    <link rel="stylesheet"
href="Statistics_Dashboard/charts.css">
    <link rel="icon" type="image/x-icon"
href="../Pictures/PUP_LOGO.png">
    <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
    <script
src="https://cdn.jsdelivr.net/npm/chart.js"></
script>
</style>
```



```
.external-link::after {
    content: '\f35d';
    font-family: 'Font Awesome 5 Free';
    font-weight: 900;
    font-size: 0.75em;
    margin-left: 5px;
    opacity: 0.7;
}
</style>
</head>
<body>
    <!-- Hamburger Toggle & Overlay -->
    <?php include_once __DIR__ .
'/components/sidebar_toggle.php'; ?>

    <!-- Sidebar Navigation -->
    <nav class="sidebar" id="sidebar">
        <div class="sidebar-admin-profile"
id="sidebarAdminProfile"
style="cursor:pointer;"
onclick="window.location.href='AdminProfile.
php'">
            
            <div class="sidebar-admin-
username"><h6>Welcome!
Admin</h6><?php echo
htmlspecialchars($admin_username);
?></div>
            </div>
            <ul>
                <li><a href="<?php echo
$sidebarBasePath ?? ";
?>Admin_Dashboard.php" class="<?php
echo ($currentPage ?? ") === 'dashboard' ?
'active' : "; ?>"><i class="fa-solid fa-
grip"></i>
<span>Dashboard</span></a></li>
                <li class="section-header Main-
header">Main</li>
                <li><a href="<?php echo
$componentPath ?? ";
?>components/PendingApproval.php"
class="<?php echo ($currentPage ?? ") ===
'pending' ? 'active' : "; ?>"><i class="fas fa-
user-clock"></i> <span>Pending
Users</span><?php echo
renderBadge($badges['pending_users'] ??
0); ?></a></li>
                <li><a href="<?php echo
$componentPath ?? ";
?>components/UserManagement.php"
class="<?php echo ($currentPage ?? ") ===
'users' ? 'active' : "; ?>"><i class="fas fa-
users"></i> <span>User
Management</span><?php echo
renderBadge($badges['user_management']
?? 0); ?></a></li>
                <li><a href="<?php echo
$componentPath ?? ";
?>components/ArchivedUsers.php"
class="<?php echo ($currentPage ?? ") ===
'archivedusers' ? 'active' : "; ?>"><i
class="fas fa-user-slash"></i>
<span>Archived Users</span><?php echo
renderBadge($badges['archived_users'] ??
0); ?></a></li>
                <li><a href="<?php echo
$componentPath ?? ";
?>components/Admin_Notification.php"
class="<?php echo ($currentPage ?? ") ===
'notifications' ? 'active' : "; ?>"><i class="fas
fa-bell"></i>
<span>Notifications</span><?php echo
renderBadge($badges['notifications'] ?? 0);
?></a></li>
                <li class="section-header books-
inventory-header">Books Inventory</li>
                <li><a href="<?php echo
$componentPath ?? ";
?>components/AddBooks.php"
class="<?php echo ($currentPage ?? ") ===
'addbooks' ? 'active' : "; ?>"><i class="fa-
solid fa-list-check"></i><span>Manage
Books</span><?php echo
renderBadge($badges['manage_books'] ??
0); ?></a></li>
                <li><a href="<?php echo
$componentPath ?? './../Narrative-
Research-Capstone/'; ?>addBooks.php"
class="<?php echo ($currentPage ?? ") ===
'addbooks' ? 'active' : "; ?>"><i class="fa-
brands fa-
researchgate"></i><span>Research
Books</span><?php echo
```




```
renderBadge($badges['narrative_books'] ?? 0); ?></a></li>
    <li><a href="php echo $componentPath ?? "; ?&gt;components/BorrowedBooks.php" class="<?php echo ($currentPage ?? ") === 'borrowed' ? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-download"&gt;&lt;/i&gt; &lt;span&gt;Borrowed Books&lt;/span&gt;&lt;?php echo renderBadge($badges['borrowed_books'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? "; ?&gt;components/Non-ReturnedBooks.php" class="<?php echo ($currentPage ?? ") === 'nonreturned' ? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-ban"&gt;&lt;/i&gt; &lt;span&gt;Non-Returned Books&lt;/span&gt;&lt;?php echo renderBadge($badges['non_returned_books'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? "; ?&gt;components/ReturnedBooks.php" class="<?php echo ($currentPage ?? ") === 'returned' ? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-upload"&gt;&lt;/i&gt; &lt;span&gt;Returned Books&lt;/span&gt;&lt;?php echo renderBadge($badges['returned_books'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? "; ?&gt;components/ReplacedBooks.php" class="<?php echo ($currentPage ?? ") === 'replaced' ? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-sync"&gt;&lt;/i&gt; &lt;span&gt;Replaced Books&lt;/span&gt;&lt;?php echo renderBadge($badges['replaced_books'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? "; ?&gt;components/PaidBooks.php" class="<?php echo ($currentPage ?? ") === 'paidbook' ? 'active' : "; ?&gt;"&gt;&lt;i class="fa-solid fa-money-bill"&gt;&lt;/i&gt; &lt;span&gt;Paid Books&lt;/span&gt;&lt;?php echo renderBadge($badges['paid_book'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? "; ?&gt;components/ArchivedBooks.php" class="<?php echo ($currentPage ?? ") === 'archived' ? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-archive"&gt;&lt;/i&gt; &lt;span&gt;Archived Books&lt;/span&gt;&lt;?php echo renderBadge($badges['archived_books'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? '.../Narrative-Research-Capstone/'; ?&gt;archived_research_books/archived-research-books.php" class="<?php echo ($currentPage ?? ") === 'archivedresearch' ? 'active' : "; ?&gt;"&gt;&lt;i class="fa-solid fa-box-archive"&gt;&lt;/i&gt; &lt;span&gt;Archived Research&lt;/span&gt;&lt;?php echo renderBadge($badges['archived_research_books'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? "; ?&gt;components/ArchivedOverdueBooks.php" class="<?php echo ($currentPage ?? ") === 'archivedoverdue' ? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-exclamation-triangle"&gt;&lt;/i&gt; &lt;span&gt;Archived Overdue&lt;/span&gt;&lt;?php echo renderBadge($badges['archived_overdue'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li class="section-header appointment-header"&gt;Appointment&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? "; ?&gt;components/Appointment_Management.php" class="<?php echo ($currentPage ?? ") === 'appointments' ? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-calendar-check"&gt;&lt;/i&gt; &lt;span&gt;Appointment Management&lt;/span&gt;&lt;?php echo renderBadge($badges['appointments'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li class="section-header"&gt;Reports&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo $componentPath ?? "; ?&gt;components/ReportAnalytics.php" class="<?php echo ($currentPage ?? ") === 'reports' ? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-chart-line"&gt;&lt;/i&gt; &lt;span&gt;Reports&lt;/span&gt;&lt;/a&gt;&lt;/li&gt;</pre
```




```
</div>
<!-- Book Categories Distribution -
->
<div class="chart-container">
  <h3><i class="fas fa-chart-
pie"></i> Book Categories Distribution</h3>
  <div class="chart-wrapper">
    <canvas
id="categoryDistributionChart"></canvas>
  </div>
</div>

<!-- User Type Distribution -->
<div class="chart-container">
  <h3><i class="fas fa-user-
friends"></i> User Type Distribution</h3>
  <div class="chart-wrapper">
    <canvas
id="userDistributionChart"></canvas>
  </div>
</div>

<!-- Library Capacity Utilization -->
<div class="chart-container">
  <h3><i class="fas fa-
building"></i> Library Capacity
Utilization</h3>
  <div class="chart-wrapper">
    <canvas
id="capacityUtilizationChart"></canvas>
  </div>
</div>
</div>
</section>
</main>

<!-- Scripts -->
<script src=" ../Js/toast.js"></script>
<script src=" ../Js/Admin.js"></script>
<script
src=" ../Js/AdminProfile.js"></script>
<script
src=" ../Js/AdminNotification.js"></script>
<script
src="Statistics_Dashboard/charts.js"></scrip
t>
<script
src=" ../Js/SidebarToggle.js"></script>

<script>
  function handleLogout() {
    window.location.href =
"../logout.php";
  }
</script>
<script src="Notification
Badge/badge_refresh.js"></script>
</body>
</html>

AdminLoginPage.php
<?php
// Only start session if one doesn't exist
already
if (session_status() ===
PHP_SESSION_NONE) {
  session_start();
}

// Do NOT clear attempts on direct page
load
// This is what was causing your counter to
reset
// We only want to clear attempts after a
successful login

// Handle any error or success messages
$error = $_GET['error'] ?? "";
$success = $_GET['success'] ?? "";
?>

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>Admin Login - PUP Bansud
Campus</title>
  <link rel="stylesheet"
href=" ../Design/AdminLogin.css">
  <link rel="icon" type="image/x-icon"
href=" ../Pictures/PUP_LOGO.png">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
</style>
```




```
<!-- Add a hidden field to track
login attempts -->
<input type="hidden"
name="login_attempt" value="1">
</form>
</div>
</div>
</div>

<!-- Toast container for notifications -->
<div class="toast-container"></div>

<footer class="footer">
<p>© 2026 PUP Bansud Campus</p>
</footer>

<script src="..Js/AdminLogin.js"></script>
</body>
</html>

AdminOTP.php
<?php
session_start();

// Check if admin verification is pending
if
(!isset($_SESSION['admin_verification_pen
ding'])) ||
$_SESSION['admin_verification_pending']
!= true) {
    header("Location:
AdminLoginPage.php?error=Session
expired");
    exit();
}

// Initialize variables for the page
$errorMessage = "";
$successMessage = "";
$adminEmail = $_SESSION['admin_email']
?? "";
$adminPhone =
$_SESSION['admin_phone'] ?? "";
$smsSent = $_SESSION['otp_sms_sent'] ??
false;

// Mask email for display (e.g.,
p***d@gmail.com)
$maskedEmail = $adminEmail;
if (!empty($adminEmail)) {
    $parts = explode('@', $adminEmail);
    if (strlen($parts[0]) > 2) {
        $maskedEmail = substr($parts[0], 0, 1)
        . str_repeat('*', strlen($parts[0]) - 2) .
        substr($parts[0], -1) . '@' . $parts[1];
    }
}

// Mask phone for display (e.g., 09***09520)
$maskedPhone = "";
if (!empty($adminPhone)) {
    $cleanPhone = preg_replace('/[^0-9]/', '',
$adminPhone);
    if (strlen($cleanPhone) >= 7) {
        $maskedPhone = substr($cleanPhone,
0, 4) . str_repeat('*', strlen($cleanPhone) - 7)
        . substr($cleanPhone, -3);
    } else {
        $maskedPhone = $cleanPhone;
    }
}

// Handle error messages
if (isset($_GET['error'])) {
    $errorMessage = $_GET['error'];
}

// Handle success messages
if (isset($_GET['success'])) {
    $successMessage = $_GET['success'];
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    <title>Admin OTP Verification - PUP
Bansud Campus</title>
    <link rel="stylesheet"
href="..Design/AdminLogin.css">
    <link rel="icon" type="image/x-icon"
href="..Pictures/PUP_LOGO.png">
```



```
<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
<style>
  .otp-container {
    display: flex;
    justify-content: center;
    gap: 12px;
    margin: 25px 0;
  }

  .digit-input {
    width: 50px;
    height: 60px;
    font-size: 24px;
    text-align: center;
    border: 2px solid #d1d5db;
    border-radius: 8px;
    background-color: #f9f9f9;
    -moz-appearance: textfield; /* Firefox
*/
  }

  /* Remove arrows for number inputs */
  .digit-input::-webkit-outer-spin-button,
  .digit-input::-webkit-inner-spin-button {
    -webkit-appearance: none;
    margin: 0;
  }

  .digit-input:focus {
    border-color: #991b1b;
    outline: none;
    box-shadow: 0 0 0 3px rgba(153, 27,
27, 0.2);
    background-color: #fff;
  }

  .verification-info {
    text-align: center;
    margin-bottom: 20px;
  }

  .email-display {
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 8px;
    margin: 10px 0;
    padding: 10px;
    background-color: #f3f4f6;
    border-radius: 8px;
  }

  .submit-btn {
    margin-top: 20px;
  }

  .resend-link {
    display: block;
    text-align: center;
    margin-top: 15px;
    color: #991b1b;
    text-decoration: none;
  }

  .resend-link:hover {
    text-decoration: underline;
  }

  .alert {
    padding: 15px;
    margin: 20px 0;
    border-radius: 8px;
    display: flex;
    align-items: center;
    gap: 10px;
  }

  .alert-error {
    background: #f8d7da;
    color: #721c24;
    border: 1px solid #f5c6cb;
  }

  .alert-success {
    background: #d4edda;
    color: #155724;
    border: 1px solid #c3e6cb;
  }
</style>
</head>
<body>
  <!-- Fixed Header -->
  <header class="fixed-header">
    <div class="header-content">
      <div class="logo-container">
```




```
        value = value.slice(-1);
    }

    // Update the input value
    this.value = value;

    // If there's a value and this isn't
    the last input, move to the next input
    if (value && index <
    digitInputs.length - 1) {
        digitInputs[index + 1].focus();
    }

    // Update the combined OTP
    updateCombinedOtp();
    });

    // Handle backspace key
    input.addEventListener('keydown',
    function(e) {
        if (e.key === 'Backspace' ||
        e.key === 'Delete') {
            // If there's a value, clear it
            if (this.value !== "") {
                this.value = "";
                updateCombinedOtp();
            }
            // If there's no value and not
            the first input, move to the previous input
            else if (index > 0) {
                digitInputs[index -
                1].focus();
            }
        }

        // Arrow key navigation
        if (e.key === 'ArrowLeft' &&
        index > 0) {
            digitInputs[index - 1].focus();
        }
        if (e.key === 'ArrowRight' &&
        index < digitInputs.length - 1) {
            digitInputs[index + 1].focus();
        }
    });

    // Handle paste events for OTP
    input.addEventListener('paste',
    function(e) {
        e.preventDefault();
        const pasteData =
        (e.clipboardData ||
        window.clipboardData).getData('text');
        const numbersOnly =
        pasteData.replace(/[^0-9]/g, "").slice(0, 6);

        // Fill in the inputs with the
        pasted data
        for (let i = 0; i <
        Math.min(numbersOnly.length,
        digitInputs.length); i++) {
            digitInputs[i].value =
            numbersOnly[i];
        }

        // Focus on the appropriate
        input
        const focusIndex =
        Math.min(index + numbersOnly.length,
        digitInputs.length - 1);
        digitInputs[focusIndex].focus();

        updateCombinedOtp();
    });

    // Update the hidden combined OTP
    input
    function updateCombinedOtp() {
        let otp = "";
        digitInputs.forEach(input => {
            otp += input.value || ""; // Add the
            value or empty string if no value
        });
        combinedOtpInput.value = otp;

        // Enable the submit button only if
        all digits are filled
        document.getElementById('submit
        Btn').disabled = otp.length !== 6;
    }

    // Handle form submission
    otpForm.addEventListener('submit',
    function(e) {
        // Prevent submission if OTP is
        incomplete
```



```
        if (combinedOtpInput.value.length !== 6) {
            e.preventDefault();
            showToast('Please enter all 6 digits of the verification code', 'error');
        }
    });

    // Handle resend OTP button click with AJAX
    resendOtpBtn.addEventListener('click', function(e) {
        e.preventDefault();

        // Disable button to prevent multiple clicks
        this.disabled = true;
        const originalText = this.innerHTML;
        this.innerHTML = '<i class="fas fa-spinner fa-spin"></i> Sending...';

        // Make AJAX request to resend OTP
        fetch('../Functiondb/AdminDb/ResendAdminOTP.php')
            .then(response => response.json())
            .catch(error => {
                return { success: false, message: 'Network error' };
            })
            .then(data => {
                // Enable button again
                resendOtpBtn.disabled = false;
                resendOtpBtn.innerHTML = originalText;

                // Show success message
                showToast('Verification code has been sent!', 'success');

                // Clear and focus the first input
                digitInputs.forEach(input => input.value = "");
                digitInputs[0].focus();
                updateCombinedOtp();
            });
    });

    // Toast notification system
    function showToast(message, type = 'info') {
        // Create toast element
        const toast = document.createElement('div');
        toast.className = `toast ${type}`;

        // Create toast content
        const toastContent = document.createElement('div');
        toastContent.className = 'toast-content';

        // Add icon based on type
        const icon = document.createElement('i');
        switch(type) {
            case 'success':
                icon.className = 'fas fa-check-circle';
                break;
            case 'error':
                icon.className = 'fas fa-exclamation-circle';
                break;
            case 'warning':
                icon.className = 'fas fa-exclamation-triangle';
                break;
            default:
                icon.className = 'fas fa-info-circle';
        }

        toastContent.appendChild(icon);

        // Add message
        const messageSpan = document.createElement('span');
        messageSpan.className = 'toast-message';
        messageSpan.textContent = message;
        toastContent.appendChild(messageSpan);
    }
```



```
toast.appendChild(toastContent);
// Add close button
const closeBtn =
document.createElement('button');
closeBtn.className = 'toast-
close';
closeBtn.innerHTML = '<i
class="fas fa-times"></i>';
closeBtn.addEventListener('click',
() => {
    toast.classList.add('hide');
    setTimeout(() =>
toast.remove(), 300);
});

toast.appendChild(closeBtn);

// Add to container
const toastContainer =
document.querySelector('.toast-container');
toastContainer.appendChild(toast)
;

// Show toast after a small delay
setTimeout(() =>
toast.classList.add('show'), 10);

// Auto-hide after 4 seconds
setTimeout(() => {
    if (toast.parentElement) {
        toast.classList.add('hide');
        setTimeout(() =>
toast.remove(), 300);
    }
}, 4000);
});
</script>
</body>
</html>

AdminProfile.php
<?php
session_start();

// Security check - redirect if not logged in
if (!isset($_SESSION['admin_logged_in']) ||
$_SESSION['admin_logged_in'] !== true) {

    header('Location: AdminLoginPage.php');
    exit();
}

// Get admin information from session
$admin_id = $_SESSION['admin_id'] ?? '';
$admin_username =
$_SESSION['admin_username'] ?? '';
$admin_email = $_SESSION['admin_email']
?? '';
$admin_phone =
$_SESSION['admin_phone'] ?? '';

// Get profile image if available
require_once
'../DBconnection/db_connection.php';

// Notification badges are loaded
if (!isset($badges) && isset($pdo)) {
    require_once __DIR__ . '/Notification
Badge/Notification Badge.php';
}
$profileImage =
'../Pictures/PUP_LOGO.png'; // Default
image path

try {
    $stmt = $pdo->prepare("SELECT
profile_image, phone_number FROM admin
WHERE admin_id = ?");
    $stmt->execute([$admin_id]);
    $result = $stmt-
>fetch(PDO::FETCH_ASSOC);

    if ($result &&
!empty($result['profile_image'])) {
        $profileImage = $result['profile_image'];
    }
    if ($result &&
!empty($result['phone_number'])) {
        $admin_phone =
$result['phone_number'];
    }
} catch (PDOException $e) {
    error_log("Error fetching admin profile: " .
$e->getMessage());
}
?>
```



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>Admin Profile - PUP Bansud
Campus</title>
  <link rel="stylesheet"
href="../Design/Admin.css">
  <link rel="stylesheet"
href="../Design/AdminProfile.css">
  <link rel="stylesheet"
href="../Design/AdminSidebar.css">
  <link rel="stylesheet" href="Notification
Badge/badge.css">
  <link rel="icon" type="image/x-icon"
href="../Pictures/PUP_LOGO.png">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
</head>
<body>
  <!-- Hamburger Toggle & Overlay -->
  <?php include_once __DIR__ .
'/components/sidebar_toggle.php'; ?>

  <!-- Admin Header - Simplify to only logo
and title -->
  <header class="header">
    <div class="container header-content">
      <div class="logo-container">
        
        <h1>E-LIB: A WEB-BASED
LIBRARY MANAGEMENT INFORMATION
SYSTEM (LMIS)</h1>
      </div>
    </div>
  </header>

  <!-- Sidebar Navigation -->
  <nav class="sidebar" id="sidebar">
    <div class="sidebar-admin-profile">
      
      <div class="sidebar-admin-
username"> <h6>Welcome!
Admin</h6><?php echo
htmlspecialchars($admin_username);
?></div>
    </div>
    <ul>
      <li><a href="<?php echo
$sidebarBasePath ?? ";
?>Admin_Dashboard.php" class="<?php
echo ($currentPage ?? ") === 'dashboard' ?
'active' : "; ?>"><i class="fa-solid fa-
grip"></i>
<span>Dashboard</span></a></li>
      <li class="section-header Main-
header">Main</li>
      <li><a href="<?php echo
$componentPath ?? ";
?>components/PendingApproval.php"
class="<?php echo ($currentPage ?? ") ===
'pending' ? 'active' : "; ?>"><i class="fas fa-
user-clock"></i> <span>Pending
Users</span><?php echo
renderBadge($badges['pending_users'] ??
0); ?></a></li>
      <li><a href="<?php echo
$componentPath ?? ";
?>components/UserManagement.php"
class="<?php echo ($currentPage ?? ") ===
'users' ? 'active' : "; ?>"><i class="fas fa-
users"></i> <span>User
Management</span><?php echo
renderBadge($badges['user_management']
?? 0); ?></a></li>
      <li><a href="<?php echo
$componentPath ?? ";
?>components/ArchivedUsers.php"
class="<?php echo ($currentPage ?? ") ===
'archivedusers' ? 'active' : "; ?>"><i
class="fas fa-user-slash"></i>
<span>Archived Users</span><?php echo
renderBadge($badges['archived_users'] ??
0); ?></a></li>
      <li><a href="<?php echo
$componentPath ?? ";
?>components/Admin_Notification.php"
class="<?php echo ($currentPage ?? ") ===
```



```
'notifications' ? 'active' : ''; ?>"<i class="fas fa-bell"></i>
<span>Notifications</span><?php echo
renderBadge($badges['notifications'] ?? 0);
?></a></li>
    <li class="section-header books-
inventory-header">Books Inventory</li>
    <li><a href="<?php echo
$componentPath ?? ";
?>components/AddBooks.php"
class="<?php echo ($currentPage ?? ") ===
'addbooks' ? 'active' : ''; ?>"><i class="fa-
solid fa-list-check"></i><span>Manage
Books</span><?php echo
renderBadge($badges['manage_books'] ??
0); ?></a></li>
        <li><a href="<?php echo
$componentPath ?? './../Narrative-
Research-Capstone/'; ?>addBooks.php"
class="<?php echo ($currentPage ?? ") ===
'addbooks' ? 'active' : ''; ?>"><i class="fa-
brands fa-
researchgate"></i><span>Research
Books</span><?php echo
renderBadge($badges['narrative_books'] ??
0); ?></a></li>
            <li><a href="<?php echo
$componentPath ?? ";
?>components/BorrowedBooks.php"
class="<?php echo ($currentPage ?? ") ===
'borrowed' ? 'active' : ''; ?>"><i class="fas fa-
download"></i> <span>Borrowed
Books</span><?php echo
renderBadge($badges['borrowed_books'] ??
0); ?></a></li>
                <li><a href="<?php echo
$componentPath ?? "; ?>components/Non-
ReturnedBooks.php" class="<?php echo
($currentPage ?? ") === 'nonreturned' ?
'active' : ''; ?>"><i class="fas fa-ban"></i>
<span>Non-Returned Books</span><?php echo
renderBadge($badges['non_returned_books'
] ?? 0); ?></a></li>
                    <li><a href="<?php echo
$componentPath ?? ";
?>components/ReturnedBooks.php"
class="<?php echo ($currentPage ?? ") ===
'returned' ? 'active' : ''; ?>"><i class="fas fa-
upload"></i> <span>Returned
Books</span><?php echo
renderBadge($badges['returned_books'] ??
0); ?></a></li>
                        <li><a href="<?php echo
$componentPath ?? ";
?>components/ReplacedBooks.php"
class="<?php echo ($currentPage ?? ") ===
'replaced' ? 'active' : ''; ?>"><i class="fas fa-
sync"></i> <span>Replaced
Books</span><?php echo
renderBadge($badges['replaced_books'] ??
0); ?></a></li>
                            <li><a href="<?php echo
$componentPath ?? ";
?>components/PaidBooks.php"
class="<?php echo ($currentPage ?? ") ===
'paidbook' ? 'active' : ''; ?>"><i class="fa-
solid fa-money-bill"></i> <span>Paid
Books</span><?php echo
renderBadge($badges['paid_book'] ?? 0);
?></a></li>
                                <li><a href="<?php echo
$componentPath ?? ";
?>components/ArchivedBooks.php"
class="<?php echo ($currentPage ?? ") ===
'archived' ? 'active' : ''; ?>"><i class="fas fa-
archive"></i> <span>Archived
Books</span><?php echo
renderBadge($badges['archived_books'] ??
0); ?></a></li>
                                    <li><a href="<?php echo
$componentPath ?? './../Narrative-
Research-Capstone/';
?>archived_research_books/archived-
research-books.php" class="<?php echo
($currentPage ?? ") === 'archivedresearch'
? 'active' : ''; ?>"><i class="fa-solid fa-box-
archive"></i> <span>Archived
Research</span><?php echo
renderBadge($badges['archived_research_b
ooks'] ?? 0); ?></a></li>
                                        <li><a href="<?php echo
$componentPath ?? ";
?>components/ArchivedOverdueBooks.php"
class="<?php echo ($currentPage ?? ") ===
'archivedoverdue' ? 'active' : ''; ?>"><i
class="fas fa-exclamation-triangle"></i>
<span>Archived Overdue</span><?php
```




```
echo
renderBadge($badges['archived_overdue']
?? 0); ?></a></li>
    <li class="section-header
appointment-header">Appointment</li>
    <li><a href="<?php echo
$componentPath ?? ";
?>components/Appointment_Management.p
hp" class="<?php echo ($currentPage ?? ")
=== 'appointments' ? 'active' : "; ?>"><i
class="fas fa-calendar-check"></i>
<span>Appointment
Management</span><?php echo
renderBadge($badges['appointments'] ?? 0);
?></a></li>
    <li class="section-
header">Reports</li>
    <li><a href="<?php echo
$componentPath ?? ";
?>components/ReportAnalytics.php"
class="<?php echo ($currentPage ?? ") ===
'reports' ? 'active' : "; ?>"><i class="fas fa-
chart-line"></i>
<span>Reports</span></a></li>
    <li class="sidebar-separator"></li>
    <li><a href="<?php echo
$sidebarBasePath ?? './';
?>Admin_Logout.php"><i class="fas fa-sign-
out-alt"></i> <span>Logout</span></a></li>
</ul>
</nav>

<div class="profile-page-container">
    <div class="admin-profile-content">
        <div class="profile-header-section">
            <h2 class="profile-title">
                <i class="fas fa-user-
circle"></i> Admin Profile Management
            </h2>
        </div>

        <!-- Toast notification container -->
        <div id="toastContainer"
class="toast-container"></div>

        <div class="profile-content">
            <div class="profile-header">
                <div class="profile-image-
container">
                    
                    <button
id="editProfileImageBtn" class="edit-profile-
btn">
                        <i class="fas fa-
camera"></i>
                    </button>
                    <input type="file"
id="profileImageInput"
accept="image/jpeg,image/png,image/gif"
style="display: none;">
                </div>
                <div class="profile-name-
container">
                    <div class="profile-name-
info">
                        <h3
id="usernameDisplay"><?php echo
htmlspecialchars($admin_username);
?></h3>
                        <span class="profile-
role">System Administrator</span>
                    </div>
                    <div class="profile-name-
actions">
                        <button
id="editUsernameBtn" class="btn btn-icon">
                            <i class="fas fa-
edit"></i>
                        </button>
                    </div>
                </div>
                <div
id="editUsernameContainer" class="edit-
container" style="display: none;">
                    <div class="edit-field">
                        <input type="text"
id="usernameInput" placeholder="Enter new
username">
                        <button
id="cancelUsernameBtn" class="btn btn-
icon">
                            <i class="fas fa-
times"></i>
                        </button>
                    </div>
                </div>
            </div>
        </div>
    </div>
</div>
```




```
number (e.g., 09123456789)" pattern="09[0-9]{9}" maxlength="11" minlength="11">
    <button
id="cancelPhoneBtn" class="btn btn-icon">
    <i class="fas fa-
times"></i>
    </button>
</div>
<div class="field-
hint">Must be 11 digits starting with 09 (e.g.,
09123456789). Click "Save All Profile
Changes" to apply.</div>
</div>
</div>

<div class="admin-field">
    <label><i class="fas fa-
lock"></i> Password</label>
    <div class="field-content">
        <input type="password"
value="....." readonly class="read-only-
input">
        <button
id="editPasswordBtn" class="btn btn-icon">
        <i class="fas fa-
edit"></i>
        </button>
    </div>
</div>
id="editPasswordContainer" class="edit-
container" style="display: none;">
    <div class="edit-field">
        <div class="password-
input-container">
            <input
type="password" id="passwordInput"
placeholder="Enter new password">
            <span
id="togglePassword" class="toggle-
password" role="button" tabindex="0">
            <i class="fas fa-
eye"></i>
            </span>
        </div>
        <div class="password-
confirm-container">
            <input
type="password" id="confirmPasswordInput"
placeholder="Confirm new password">
            <span
id="toggleConfirmPassword" class="toggle-
password" role="button" tabindex="0">
            <i class="fas fa-
eye"></i>
            </span>
        </div>
    </div>
    <div class="field-
hint">Click "Save All Profile Changes" at the
bottom when finished to apply all
changes.</div>
</div>
</div>
</div>

<div class="profile-actions">
    <button id="saveProfileBtn"
class="btn btn-primary">
    <i class="fas fa-save"></i>
    Save All Profile Changes
    </button>
</div>
</div>
</div>

<!-- Script to handle sidebar toggle -->
<script>
    document.addEventListener('DOMCont
entLoaded', function() {
        // Sidebar toggle functionality
        const sidebarToggleBtn =
document.getElementById('sidebarToggleBt
n');
        const sidebar =
document.getElementById('sidebar');
        const profilePageContainer =
document.querySelector('.profile-page-
container');
        sidebarToggleBtn.addEventListener('
click', function() {
```



```
        sidebar.classList.toggle('collapsed'
    );

        // Adjust main content margin
        based on sidebar state
        if
        (sidebar.classList.contains('collapsed')) {
            profilePageContainer.style.mar
            ginLeft = '80px';
        } else {
            profilePageContainer.style.mar
            ginLeft = '260px';
        }
    });
    </script>

    <script
    src="../../Js/AdminProfile.js"></script>

    <?php

        if ($_SERVER['SERVER_NAME'] ===
        'localhost' || $_SERVER['SERVER_NAME']
        === '127.0.0.1'): ?>
            <div id="debugPanel" style="position:
            fixed; bottom: 10px; right: 10px; background:
            #f8f9fa; border: 1px solid #ddd; padding:
            10px; border-radius: 5px; font-size: 12px;
            max-width: 300px; display: none;">
                <h4>Debug Info</h4>
                <pre id="debugContent">No data
            yet</pre>
            </div>
            <script>
                // Add this to your existing
                DOMContentLoaded event or create a new
                one
                document.addEventListener("DOMCont
                entLoaded", function() {
                    // Add debug info
                    window.showDebug = function(info) {
                        const debugPanel =
                        document.getElementById('debugPanel');
                        const debugContent =
                        document.getElementById('debugContent');
                        debugPanel.style.display = 'block';
                        debugContent.textContent =
                        JSON.stringify(info, null, 2);

                    // Show changes when save button
                    is clicked
                    document.getElementById('saveProfi
                    leBtn').addEventListener('click', function() {
                        showDebug(changes);
                    });
                }
            </script>
            <script
            src="../../Js/SidebarToggle.js"></script>
            <script src="Notification
            Badge/badge_refresh.js"></script>
            <?php endif; ?>
        </body>
    </html>

    Elib/AdminDashboard/components
    AddBooks.php
    <?php

        if (session_status() ===
        PHP_SESSION_NONE) {
            session_start();
        }

        // Include database connection
        require_once
        '../../DBconnection/db_connection.php';
        //Badge Notification Function
        if (!isset($badges) && isset($pdo)) {
            require_once __DIR__ . '/../Notification
            Badge/Notification Badge.php';
        }

        $profileImage =
        '../../Pictures/PUP_LOGO.png'; // Default
        image path

        //////////////////////////////////////
        //////////////////////////////////////
        //////////////////////////////////////Fetching Profile Image from the
        Database////////////////////////////////////

        // Get admin_id from the session
        $admin_id = $_SESSION['admin_id'] ?? '';
```



```
$defaultImagePath =
'../Pictures/PUP_LOGO.png'; // Default
image path
$profileImage = $defaultImagePath; //
Initialize with default

// When fetching the profile image
try {
    $stmt = $pdo->prepare("SELECT
profile_image FROM admin WHERE
admin_id = ?");
    $stmt->execute([$admin_id]);
    $result = $stmt-
>fetch(PDO::FETCH_ASSOC);

    if ($result &&
!empty($result['profile_image'])) {
        $profileImage =
getImageUrl($result['profile_image']);
    } else {
        $profileImage =
DEFAULT_ADMIN_IMAGE;
    }
} catch (PDOException $e) {
    error_log("Error fetching admin profile: " .
$e->getMessage());
    $profileImage =
DEFAULT_ADMIN_IMAGE;
}
//////////

// Check if form was submitted for
adding/editing a book
if ($_SERVER['REQUEST_METHOD'] ===
'POST' && isset($_POST['action'])) {
    require_once
'../api_books/process_add_book.php';
}

// Fetch categories from database
try {
    $categoryStmt = $pdo-
>prepare("SELECT category_name FROM
categories WHERE is_active = 1 ORDER
BY category_name ASC");
    $categoryStmt->execute();
    $categories = $categoryStmt-
>fetchAll(PDO::FETCH_COLUMN);

    } catch (PDOException $e) {
        // Fallback to default categories if
        database query fails
        $categories = [
            'Information Technology',
            'English & Literature',
            'Mathematics',
            'Public Administration',
            'General References'
        ];
        error_log("Error fetching categories: " .
$e->getMessage());
    }

    // Search functionality
    $search = isset($_GET['search']) ?
trim($_GET['search']) : "";
    $category_filter =
isset($_GET['category_filter']) ?
$_GET['category_filter'] : "";

    // Fetch books for the table display
    try {
        $query = "SELECT * FROM add_books
WHERE is_archived = 0";
        $params = [];

        // Add search condition
        if (!empty($search)) {
            $query .= " AND (title LIKE ? OR author
LIKE ? OR control_no LIKE ?)";
            $search_param = "%{$search}%";
            $params = array_merge($params,
[$search_param, $search_param,
$search_param]);
        }

        // Add category filter
        if (!empty($category_filter)) {
            $query .= " AND category = ?";
            $params[] = $category_filter;
        }

        $query .= " ORDER BY date_added
DESC";

        $stmt = $pdo->prepare($query);
        $stmt->execute($params);
    }
}
```



```
$books = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
} catch (PDOException $e) {
    $error_message = "Database error: " . $e-
>getMessage();
}

?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    <title>Add Books - PUP Bansud
Campus</title>
    <link rel="stylesheet"
href=".../Design/Admin.css">
    <link rel="stylesheet"
href=".../Design/AdminSidebar.css">
    <link rel="stylesheet"
href=".../Design/AddBooks.css">
    <link rel="stylesheet" href=".../Notification
Badge/badge.css">
    <link rel="icon" type="image/x-icon"
href=".../Pictures/PUP_LOGO.png">
    <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
</head>
<body>
    <?php include_once __DIR__ .
'/sidebar_toggle.php'; ?>
    <!-- Sidebar Navigation -->
    <nav class="sidebar" id="sidebar">
        <div class="sidebar-admin-profile"
id="sidebarAdminProfile"
style="cursor:pointer;"
onclick="window.location.href='<?php echo
$sidebarBasePath ?? './';
?>AdminProfile.php'">
            '";">
        <div class="sidebar-admin-
username"><h6>Welcome!
Admin</h6><?php echo
htmlspecialchars($_SESSION['admin_usern
ame'] ?? 'admin_user'); ?></div>
        </div>
        <ul>
            <li><a href="<?php echo
$sidebarBasePath ?? './';
?>Admin_Dashboard.php" class="<?php
echo ($currentPage ?? '') === 'dashboard' ?
'active' : ''; ?>"><i class="fa-solid fa-
grip"></i>
<span>Dashboard</span></a></li>
            <li class="section-header Main-
header">Main</li>
            <li><a href="<?php echo
$componentPath ?? '';
?>PendingApproval.php" class="<?php
echo ($currentPage ?? '') === 'pending' ?
'active' : ''; ?>"><i class="fas fa-user-
clock"></i> <span>Pending
Users</span><?php echo
renderBadge($badges['pending_users'] ??
0); ?></a></li>
            <li><a href="<?php echo
$componentPath ?? '';
?>UserManagement.php" class="<?php
echo ($currentPage ?? '') === 'users' ?
'active' : ''; ?>"><i class="fas fa-users"></i>
<span>User Management</span><?php
echo
renderBadge($badges['user_management']
?? 0); ?></a></li>
            <li><a href="<?php echo
$componentPath ?? '';
?>ArchivedUsers.php" class="<?php echo
($currentPage ?? '') === 'archivedusers' ?
'active' : ''; ?>"><i class="fas fa-user-
slash"></i> <span>Archived
Users</span><?php echo
renderBadge($badges['archived_users'] ??
0); ?></a></li>
            <li><a href="<?php echo
$componentPath ?? '';
?>Admin_Notification.php" class="<?php
```



```
echo ($currentPage ?? "") === 'notifications'
? 'active' : ""; ?><i class="fas fa-bell"></i>
<span>Notifications</span><?php echo
renderBadge($badges['notifications'] ?? 0);
?></a></li>
<li class="section-header books-
inventory-header">Books Inventory</li>
<li><a href="<?php echo
$componentPath ?? ""; ?>AddBooks.php"
class="<?php echo ($currentPage ?? "") ===
'addbooks' ? 'active' : ""; ?>"><i class="fa-
solid fa-list-check"></i><span>Manage
Books</span><?php echo
renderBadge($badges['manage_books'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? '../Narrative-
Research-Capstone/'; ?>addBooks.php"
class="<?php echo ($currentPage ?? "") ===
'addbooks' ? 'active' : ""; ?>"><i class="fa-
brands fa-
researchgate"></i><span>Research
Books</span><?php echo
renderBadge($badges['narrative_books'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? "";
?>BorrowedBooks.php" class="<?php echo
($currentPage ?? "") === 'borrowed' ? 'active'
: ""; ?>"><i class="fas fa-download"></i>
<span>Borrowed Books</span><?php echo
renderBadge($badges['borrowed_books'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? ""; ?>Non-
ReturnedBooks.php" class="<?php echo
($currentPage ?? "") === 'nonreturned' ?
'active' : ""; ?>"><i class="fas fa-ban"></i>
<span>Non-Returned Books</span><?php echo
echo
renderBadge($badges['non_returned_books'
] ?? 0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? "";
?>ReturnedBooks.php" class="<?php echo
($currentPage ?? "") === 'returned' ? 'active' :
"; ?>"><i class="fas fa-upload"></i>
<span>Returned Books</span><?php echo
```

```
renderBadge($badges['returned_books'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? "";
?>ReplacedBooks.php" class="<?php echo
($currentPage ?? "") === 'replaced' ? 'active'
: ""; ?>"><i class="fas fa-sync"></i>
<span>Replaced Books</span><?php echo
renderBadge($badges['replaced_books'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? ""; ?>PaidBooks.php"
class="<?php echo ($currentPage ?? "") ===
'paidbook' ? 'active' : ""; ?>"><i class="fa-
solid fa-money-bill"></i><span>Paid
Books</span><?php echo
renderBadge($badges['paid_book'] ?? 0);
?></a></li>
<li><a href="<?php echo
$componentPath ?? "";
?>ArchivedBooks.php" class="<?php echo
($currentPage ?? "") === 'archived' ? 'active'
: ""; ?>"><i class="fas fa-archive"></i>
<span>Archived Books</span><?php echo
renderBadge($badges['archived_books'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? '../Narrative-
Research-Capstone/';
?>archived_research_books/archived-
research-books.php" class="<?php echo
($currentPage ?? "") === 'archivedresearch'
? 'active' : ""; ?>"><i class="fa-solid fa-box-
archive"></i> <span>Archived
Research</span><?php echo
renderBadge($badges['archived_research_b
ooks'] ?? 0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? "";
?>ArchivedOverdueBooks.php"
class="<?php echo ($currentPage ?? "") ===
'archivedoverdue' ? 'active' : ""; ?>"><i
class="fas fa-exclamation-triangle"></i>
<span>Archived Overdue</span><?php
echo
renderBadge($badges['archived_overdue']
?? 0); ?></a></li>
<li class="section-header appointment-
header">Appointment</li>
```



```
</li><a href="php echo
$componentPath ?? ";
?&gt;Appointment_Management.php"
class="<?php echo ($currentPage ?? ") ===
'appointments' ? 'active' : "; ?&gt;"&gt;&lt;i
class="fas fa-calendar-check"&gt;&lt;/i&gt;
&lt;span&gt;Appointment
Management&lt;/span&gt;&lt;?php echo
renderBadge($badges['appointments'] ?? 0);
?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li class="section-header"&gt;Reports&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? ";
?&gt;ReportAnalytics.php" class="<?php echo
($currentPage ?? ") === 'reports' ? 'active' :
"; ?&gt;"&gt;&lt;i class="fas fa-chart-line"&gt;&lt;/i&gt;
&lt;span&gt;Reports&lt;/span&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li class="sidebar-separator"&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$sidebarBasePath ?? './../';
?&gt;Admin_Logout.php"&gt;&lt;i class="fas fa-sign-
out-alt"&gt;&lt;/i&gt; &lt;span&gt;Logout&lt;/span&gt;&lt;/a&gt;&lt;/li&gt;
&lt;/ul&gt;
&lt;/nav&gt;

&lt;!-- Main Content Area --&gt;
&lt;div class="main-content"&gt;
&lt;header class="admin-header"&gt;
&lt;div class="header-content"&gt;
&lt;h1&gt;Book Management&lt;/h1&gt;
&lt;div class="header-actions"&gt;
&lt;button id="addCategoryBtn"
class="btn-primary"&gt;&lt;i class="fa-solid fa-
plus"&gt;&lt;/i&gt;Add New Category&lt;/button&gt;
&lt;button id="addBookBtn"
class="btn-primary"&gt;&lt;i class="fas fa-
plus"&gt;&lt;/i&gt; Add New Book&lt;/button&gt;
&lt;/div&gt;
&lt;/div&gt;
&lt;/header&gt;

&lt;!-- Success/Error Messages --&gt;
&lt;?php if
(isset($_SESSION['book_message'])): ?&gt;
&lt;div class="message &lt;?php echo
$_SESSION['book_message_type']; ?&gt;"&gt;
&lt;?php
echo
$_SESSION['book_message'];
unset($_SESSION['book_mess
age']);
unset($_SESSION['book_mess
age_type']);
?&gt;
&lt;/div&gt;
&lt;?php endif; ?&gt;

&lt;!-- Search and Filter --&gt;
&lt;div class="search-filter-container"&gt;
&lt;form method="GET" action=""
class="search-form"&gt;
&lt;div class="search-input-
container"&gt;
&lt;input type="text"
name="search" placeholder="Search by title,
author or Control No..." value="<?php echo
htmlspecialchars($search); ?&gt;"
class="search-input"&gt;
&lt;button type="submit"
class="search-button"&gt;&lt;i class="fas fa-
search"&gt;&lt;/i&gt;&lt;/button&gt;
&lt;/div&gt;
&lt;!--Filter--&gt;
&lt;div class="filter-container"&gt;
&lt;select name="category_filter"
class="filter-select"&gt;
&lt;option value=""&gt;All
Categories&lt;/option&gt;
&lt;?php foreach ($categories
as $category): ?&gt;
&lt;option value="<?php
echo htmlspecialchars($category); ?&gt;"
&lt;?php echo $category_filter === $category
? 'selected' : "; ?&gt;&gt;
&lt;?php echo
htmlspecialchars($category); ?&gt;
&lt;/option&gt;
&lt;?php endforeach; ?&gt;
&lt;/select&gt;
&lt;button type="submit"
class="filter-button"&gt;Filter&lt;/button&gt;
&lt;?php if (!empty($search) ||
!empty($category_filter)): ?&gt;
&lt;a href="AddBooks.php"
class="reset-button"&gt;Reset&lt;/a&gt;
&lt;?php endif; ?&gt;
&lt;/div&gt;
&lt;/form&gt;</pre
```




```
<label for="volume">Volume
(optional)</label>
<input type="text"
id="volume" name="volume"
placeholder="e.g. Vol. 1, Volume II">
</div>

<!-- Add Edition field -->
<div class="form-group">
<label for="edition">Edition
(optional)</label>
<input type="text" id="edition"
name="edition" placeholder="e.g. 2nd
Edition, Revised Edition">
</div>

<div class="form-group">
<label
for="category">Category*</label>
<select id="category"
name="category" required>
<option value="">Select a
category</option>
<?php foreach
($categories as $category): ?>
<option value="<?php
echo $category; ?>"><?php echo
htmlspecialchars($category); ?></option>
<?php endforeach; ?>
</select>
</div>

<div class="form-group">
<label
for="description">Description</label>
<textarea id="description"
name="description" rows="4"></textarea>
</div>

<!-- Add TOC Editor Field -->
<div class="form-group">
<label
for="table_of_contents">Table of Contents
(HTML)</label>
<textarea
id="table_of_contents"
name="table_of_contents" rows="6"
placeholder="Enter table of contents in
HTML format..."></textarea>
```

```
<small class="form-text text-
muted">You can use HTML formatting.
Leave empty if not applicable.</small>

<!-- Read-only rendered TOC
(shown when viewing ) -->
<div id="toc_display"
class="toc-display" style="display:none;
border:1px solid #ddd; padding:10px;
border-radius:4px; background:#fff; max-
height:200px; overflow:auto; margin-
top:8px;"></div>
</div>

<!-- Add URL References
Editor Field -->
<div class="form-group">
<label
for="references">References (URLs /
HTML)</label>
<textarea id="references"
name="references" rows="6"
placeholder="Enter references: HTML
allowed, or put one URL per
line..."></textarea>
<small class="form-text text-
muted">You can enter HTML (<a> tags) or
one URL per line. Leave empty if not
applicable.</small>

<!-- Read-only rendered
References (shown when viewing book
details) -->
<div id="references_display"
class="toc-display" style="display:none;
border:1px solid #ddd; padding:10px;
border-radius:4px; background:#fff; max-
height:200px; overflow:auto; margin-
top:8px;"></div>
</div>

<div class="form-group">
<label
for="stock_quantity">Stock
Quantity*</label>
<input type="number"
id="stock_quantity" name="stock_quantity"
min="0">
```



```
<small class="form-text text-muted stock-qty-note" style="display:none; color:#dc3545;">
```

Note: Stock quantity cannot be changed when editing a book.

```
</small>
</div>
```

```
<div class="form-group">
  <label
for="book_price">Book Price (₱)*</label>
  <input type="number"
id="book_price" name="book_price" min="0"
step="0.01" placeholder="Enter book price">
  <small class="form-text text-muted">
```

This price will be used as the replacement cost if the book is damaged or lost.

```
</small>
</div>
```

```
<div class="form-group">
  <label for="status">Status
(Auto Not Available when it reaches
1)</label>
  <select id="status"
name="status" class="form-control"
required>
    <option
value="Available">Available</option>
    <option value="Not
Available">Not Available to Borrow</option>
  </select>
  <small class="form-text text-muted">
```

Note: Books with 1 remaining stock will always be "Not Available".

```
</small>
</div>
```

```
<div class="form-group">
  <label
for="book_condition">Book
Condition</label>
  <select id="book_condition"
name="book_condition" class="form-control"
required>
```

```
<option value="">Select
Condition</option>
```

```
<option value="Good
Condition">Good Condition</option>
```

```
<option value="Mildly
Damaged">Mildly Damaged (Readable
Condition)</option>
```

```
<option value="Minor
Damage">Minor Damage – Good for
Reading</option>
```

```
<option value="Slight
Wear">Slight Wear and Tear</option>
```

```
<option
value="Used">Used – With Minor
Damage</option>
```

```
<option value="Fair
Condition">Fair Condition (Has Minor
Flaws)</option>
```

```
</select>
</div>
```

```
<!-- replaced duplicate input
with condition image -->
```

```
<div class="form-group">
  <label
for="book_condition_image">Book
Condition Image (optional)</label>
  <input type="file"
id="book_condition_image"
name="book_condition_image"
accept="image/*">
  <div
id="condition_image_preview"></div>
</div>
```

```
<div class="form-group">
  <label
for="book_image">Book Cover</label>
  <input type="file"
id="book_image" name="book_image"
accept="image/*">
  <div
id="image_preview"></div>
</div>
```

```
<div class="form-actions">
  <button type="submit"
class="btn-save">Save</button>
```



```
<button type="button"
class="btn-cancel"
id="cancelBtn">Cancel</button>
</div>
</form>
</div>
</div>
```

```
<!-- Books Table -->
<div class="books-container">
<table class="books-table">
<thead>
<tr>
<th>Image</th>
<th>Title</th>
<th>Author</th>
<!-- New Additions for the
Database and the pages -->
<th>Control No.</th>
<th>Accession NO.</th>
<th>Copyright</th>
<th>Publisher</th>
<th>Publishing Add</th>
<th>ISBN</th>
<th>Width</th>
<th>Height</th>
<th>Length</th>
<th>Type of Trade
Purchased</th>
```

```
<!-- New Addition -->
<th>Volume</th>
<th>Edition</th>
<th>Category</th>
<th>Stock</th>
<th>Price</th>
<th>Status</th>
<th>Book Condition</th>
<th>Date Added</th>
<th>Actions</th>
```

```
</tr>
</thead>
<tbody>
<?php if (isset($books) &&
count($books) > 0): ?>
<?php foreach ($books as $book):
?>
```

```
<tr>
<td class="col-image">
```

```
<?php if
(!empty($book['image_path'])): ?>

<?php else: ?>
<div class="no-
image">No Image</div>
<?php endif; ?>
</td>
<td class="col-title"
title="<?php echo
htmlspecialchars($book['title']); ?>">
<?php echo
htmlspecialchars($book['title']); ?>
</td>
<td class="col-author"
title="<?php echo
htmlspecialchars($book['author']); ?>">
<?php echo
htmlspecialchars($book['author']); ?>
</td>
<td class="col-lcc"><?php
echo htmlspecialchars($book['control_no']);
?></td>
<td class="col-
accession"><?php echo
htmlspecialchars($book['accession_no'] ??
"); ?></td>
<td class="col-
copyright"><?php echo
htmlspecialchars($book['copyright'] ?? ");
?></td>
<td class="col-
publisher"><?php echo
htmlspecialchars($book['publisher'] ?? ");
?></td>
<td class="col-publishing-
add"><?php echo
htmlspecialchars($book['publishing_add'] ??
"); ?></td>
<td class="col-isbn"><?php
echo htmlspecialchars($book['isbn'] ?? ");
?></td>
<td class="col-width"><?php
echo htmlspecialchars($book['width'] ?? ");
?></td>
```



```
<td class="col-height"><?php
echo htmlspecialchars($book['height'] ?? "");
?></td>
<td class="col-lenght"><?php
echo htmlspecialchars($book['lenght'] ?? "");
?></td>
<td class="col-trade-
purchased"><?php echo
htmlspecialchars($book['type_of_trade_purc
hased'] ?? ""); ?></td>
<td class="col-
volume"><?php echo
htmlspecialchars($book['volume'] ?? "");
?></td>
<td class="col-
edition"><?php echo
htmlspecialchars($book['edition'] ?? "");
?></td>
<td class="col-category"
title="<?php echo
htmlspecialchars($book['category'] ??
'Uncategorized'); ?>">
<?php echo
htmlspecialchars($book['category'] ??
'Uncategorized'); ?>
</td>
<td class="col-stock"><?php
echo
htmlspecialchars($book['stock_quantity']);
?></td>
<td class="col-price"><?php
echo number_format($book['book_price'] ??
0, 2); ?></td>
<td class="col-status">
<?php
$statusClass =
($book['stock_quantity'] <= 1) ? 'not-
available' : 'available';
$statusText =
($book['stock_quantity'] <= 1) ? 'Not
Available to Borrow' : 'Available';
?>
<span class="status-
badge <?php echo $statusClass; ?>">
<?php echo
$statusText; ?>
</span>
</td>
```

```
<td class="col-
bookcondition"><?php echo
htmlspecialchars($book['book_condition'] ??
'N/A'); ?></td>
<td class="col-date"><?php
echo date('M d, Y',
strtotime($book['date_added'])); ?></td>
<td class="col-actions">
<div class="actions">
<button class="btn-edit"
data-id="<?php echo $book['book_id']; ?>"
title="Edit"><i class="fas fa-
edit"></i></button>
<button class="btn-
archive" data-id="<?php echo
$book['book_id']; ?>" title="Archive"><i
class="fas fa-archive"></i></button>
<button class="btn-
view" data-id="<?php echo $book['book_id'];
?>" title="View Details"><i class="fas fa-
eye"></i></button>
</div>
</td>
</tr>
<?php endforeach; ?>
<?php else: ?>
<tr>
<td colspan="22" class="no-
data">
<?php if (!empty($search) ||
!empty($category_filter)): ?>
No books found matching
your search criteria. Try different keywords
or filters.
<?php else: ?>
No books available. Add a
new book to get started.
<?php endif; ?>
</td>
</tr>
<?php endif; ?>
</tbody>
</table>
</div>
</div>
<script src="../../../Js/toast.js"></script>
<script
src="../../../Js/AddBooks.js"></script>
```



```
<script
src="../../../Js/SidebarToggle.js"></script>
<script src="../../../Notification
Badge/badge_refresh.js"></script>
</body>
</html>
Admin_Notification.php
<?php

if(session_status() ===
PHP_SESSION_NONE) {
    session_start();
}

require_once
'../../DBconnection/db_connection.php';
//Notification Badge Function
if (!isset($badges) && isset($pdo)) {
    require_once __DIR__ . '/../Notification
Badge/Notification Badge.php';
}

//Check if Admin is logged in
if(isset($_SESSION['admin_logged_in']) ||
$_SESSION['admin_logged_in'] !== true) {
    header('Location:
../../AdminLoginPage.php');
    exit();
}

// Get admin profile image
$admin_id = $_SESSION['admin_id'] ?? '';
$defaultImagePath =
'../../Pictures/PUP_LOGO.png';
$profileImage = $defaultImagePath;

try{
    if(!empty($admin_id)) {
        $stmt = $pdo->prepare("SELECT
profile_image FROM admin WHERE
admin_id = ?");
        $stmt->execute([$admin_id]);
        $result = $stmt-
>fetch(PDO::FETCH_ASSOC);

        if ($result &&
!empty($result['profile_image'])) {
            $dbPath = $result['profile_image'];

            if (strpos($dbPath, 'http://') === 0 ||
strpos($dbPath, 'https://') === 0) {
                $profileImage = $dbPath;
            }
            else if (strpos($dbPath, '../uploads/')
=== 0) {
                $profileImage = '../' . $dbPath;
            }
            else if (strpos($dbPath, 'uploads/')
=== 0) {
                $profileImage = '../..' . $dbPath;
            }
            else {
                $profileImage =
'../../uploads/admin_profile/' . $dbPath;
            }
        }
    } catch (PDOException $e) {
        error_log("Error fetching admin profile: " .
$e->getMessage());
    }

    // ===== FETCH
    NOTIFICATIONS / PENDING REQUESTS
    =====
    $notifications = [];
    $unreadCount = 0;
    $queryErrors = [];

    // Helper function to check if table exists
    function tableExists($pdo, $table) {
        try {
            $result = $pdo->query("SELECT 1
FROM {$table} LIMIT 1");
            return $result !== false;
        } catch (Exception $e) {
            return false;
        }
    }

    // Create admin_notification_reads table if it
    doesn't exist
    try {
        $pdo->exec("
        CREATE TABLE IF NOT EXISTS
        admin_notification_reads (
            id INT AUTO_INCREMENT
            PRIMARY KEY,
```



```
admin_id VARCHAR(50) NOT
NULL,
notification_id VARCHAR(100) NOT
NULL,
read_at TIMESTAMP DEFAULT
CURRENT_TIMESTAMP,
UNIQUE KEY unique_read
(admin_id, notification_id)
);
} catch (PDOException $e) {
    $queryErrors[] = "Could not create
admin_notification_reads table: " . $e-
>getMessage();
}

// Create admin_notification_dismissed table
— tracks cleared/dismissed notifications so
they don't reappear
try {
    $pdo->exec("
CREATE TABLE IF NOT EXISTS
admin_notification_dismissed (
    id INT AUTO_INCREMENT
PRIMARY KEY,
    admin_id VARCHAR(50) NOT
NULL,
    notification_id VARCHAR(100) NOT
NULL,
    dismissed_at TIMESTAMP
DEFAULT CURRENT_TIMESTAMP,
    UNIQUE KEY unique_dismissed
(admin_id, notification_id)
);
} catch (PDOException $e) {
    $queryErrors[] = "Could not create
admin_notification_dismissed table: " . $e-
>getMessage();
}

// Create admin_notifications table if it
doesn't exist (persistent storage)
try {
    $pdo->exec("
CREATE TABLE IF NOT EXISTS
admin_notifications (
    id INT AUTO_INCREMENT
PRIMARY KEY,
    notification_id VARCHAR(100) NOT
NULL,
    notification_type VARCHAR(50)
NOT NULL,
    message TEXT NOT NULL,
    source VARCHAR(50) NOT NULL,
    full_name VARCHAR(255)
DEFAULT NULL,
    user_type VARCHAR(50) DEFAULT
NULL,
    identifier VARCHAR(255) DEFAULT
NULL,
    course VARCHAR(100) DEFAULT
NULL,
    book_id VARCHAR(50) DEFAULT
NULL,
    book_title VARCHAR(255)
DEFAULT NULL,
    user_id VARCHAR(100) DEFAULT
NULL,
    appointment_date DATE DEFAULT
NULL,
    start_time VARCHAR(10) DEFAULT
NULL,
    end_time VARCHAR(10) DEFAULT
NULL,
    contact_number VARCHAR(20)
DEFAULT NULL,
    booking_type VARCHAR(100)
DEFAULT NULL,
    purpose VARCHAR(255) DEFAULT
NULL,
    status VARCHAR(50) DEFAULT
NULL,
    return_condition VARCHAR(50)
DEFAULT NULL,
    replacement_status VARCHAR(50)
DEFAULT NULL,
    is_read TINYINT(1) DEFAULT 0,
    admin_id VARCHAR(50) NOT NULL
DEFAULT 'admin',
    created_at DATETIME DEFAULT
CURRENT_TIMESTAMP,
    UNIQUE KEY unique_notification
(notification_id, admin_id),
    INDEX idx_admin_read (admin_id,
is_read),
    INDEX idx_created (created_at
DESC)
```



```
)
");
} catch (PDOException $e) {
    $queryErrors[] = "Could not create
admin_notifications table: " . $e-
>getMessage();
}

$current_admin_id =
$_SESSION['admin_id'] ?? 'admin';

// Get read notification IDs for this admin
$readNotificationIds = [];
try {
    $stmt = $pdo->prepare("SELECT
notification_id FROM
admin_notification_reads WHERE admin_id
= ?");
    $stmt->execute([$current_admin_id]);
    $readNotificationIds = $stmt-
>fetchAll(PDO::FETCH_COLUMN);
} catch (PDOException $e) {
    $queryErrors[] = "Error fetching read
notifications: " . $e->getMessage();
}

// Get dismissed notification IDs for this
admin
$dismissedNotificationIds = [];
try {
    $stmt = $pdo->prepare("SELECT
notification_id FROM
admin_notification_dismissed WHERE
admin_id = ?");
    $stmt->execute([$current_admin_id]);
    $dismissedNotificationIds = $stmt-
>fetchAll(PDO::FETCH_COLUMN);
} catch (PDOException $e) {
    $queryErrors[] = "Error fetching dismissed
notifications: " . $e->getMessage();
}

// Check if tables exist before querying
$tables = ['pending_users',
'pending_borrows', 'appointments', 'visitor',
'returned_books', 'book_borrowings',
'non_returned_books', 'paid_books',
'archived_overdue_books'];
$existingTables = [];
```

```
foreach ($tables as $table) {
    if (tableExists($pdo, $table)) {
        $existingTables[] = $table;
    } else {
        $queryErrors[] = "Table {$table} does
not exist";
    }
}

try {
    $liveNotifications = [];

    // ===== STEP 1: Gather ALL user
transactions from source tables =====

    // 1. Query pending_users - ALL statuses
(pending, approved, rejected)
    if (in_array('pending_users',
$existingTables)) {
        try {
            $stmt = $pdo->query("
SELECT
'user_registration' AS
notification_type,
CASE
    WHEN status = 'pending'
THEN CONCAT('New user registration: ',
full_name)
    WHEN status = 'approved'
THEN CONCAT('User approved: ',
full_name)
    WHEN status = 'rejected'
THEN CONCAT('User rejected: ', full_name)
    ELSE CONCAT('User
registration: ', full_name)
END AS message,
email AS identifier,
full_name,
user_type,
course,
NULL AS book_id,
NULL AS book_title,
NULL AS user_id,
NULL AS appointment_date,
NULL AS start_time,
NULL AS end_time,
NULL AS contact_number,
NULL AS booking_type,
NULL AS purpose,
```




```
        registration_date AS
created_at,
        'user' AS source,
        0 AS is_read,
        CONCAT('user_', id) AS id,
        status,
        NULL AS return_condition,
        NULL AS replacement_status
FROM pending_users
ORDER BY registration_date
DESC
        LIMIT 200
    ");
    $userNotifications = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
    $liveNotifications =
array_merge($liveNotifications,
$userNotifications);
    } catch (PDOException $e) {
        $queryErrors[] = "Error querying
pending_users: " . $e->getMessage();
    }
}

// 2. Query pending_borrows - ALL
statuses (pending, approved, rejected,
cancelled, expired)
if (in_array('pending_borrows',
$existingTables)) {
    try {
        $stmt = $pdo->query("
        SELECT
            'book_borrow' AS
notification_type,
            CASE
                WHEN pb.status = 'pending'
THEN CONCAT('New borrow request: ',
COALESCE(ab.title, 'Unknown Book'), ' by ',
pb.user_id)
                WHEN pb.status = 'approved'
THEN CONCAT('Borrow approved: ',
COALESCE(ab.title, 'Unknown Book'), ' for ',
pb.user_id)
                WHEN pb.status = 'rejected'
THEN CONCAT('Borrow rejected: ',
COALESCE(ab.title, 'Unknown Book'), ' for ',
pb.user_id)
                WHEN pb.status =
'cancelled' THEN CONCAT('Borrow
cancelled: ', COALESCE(ab.title, 'Unknown
Book'), ' by ', pb.user_id)
                WHEN pb.status = 'expired'
THEN CONCAT('Borrow request expired: ',
COALESCE(ab.title, 'Unknown Book'), ' for ',
pb.user_id)
            ELSE CONCAT('Borrow
request: ', COALESCE(ab.title, 'Unknown
Book'), ' by ', pb.user_id)
        END AS message,
        NULL AS identifier,
        NULL AS full_name,
        pb.user_type,
        NULL AS course,
        pb.book_id,
        ab.title AS book_title,
        pb.user_id,
        NULL AS appointment_date,
        NULL AS start_time,
        NULL AS end_time,
        NULL AS contact_number,
        NULL AS booking_type,
        NULL AS purpose,
        pb.request_date AS created_at,
        'borrow' AS source,
        0 AS is_read,
        CONCAT('borrow_',
pb.request_id) AS id,
        pb.status,
        NULL AS return_condition,
        NULL AS replacement_status
FROM pending_borrows pb
LEFT JOIN add_books ab ON
pb.book_id = ab.book_id
ORDER BY pb.request_date
DESC
        LIMIT 200
    ");
    $borrowNotifications = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
    $liveNotifications =
array_merge($liveNotifications,
$borrowNotifications);
    } catch (PDOException $e) {
        $queryErrors[] = "Error querying
pending_borrows: " . $e->getMessage();
    }
}
```




```
// 3. Query appointments - ALL statuses
(Pending, Cancelled, Finished)
if (in_array('appointments',
$existingTables) && in_array('visitor',
$existingTables)) {
    try {
        $stmt = $pdo->query("
            SELECT
            CASE
                WHEN a.status = 'Pending'
            THEN 'appointment_requested'
                WHEN a.status = 'Cancelled'
            THEN 'appointment_cancelled'
                WHEN a.status = 'Finished'
            THEN 'appointment_completed'
            ELSE
            'appointment_requested'
            END AS notification_type,
            CASE
                WHEN a.status = 'Pending'
            THEN CONCAT('Pending appointment from
            ', COALESCE(CONCAT(v.first_name, ' ',
            v.last_name), a.visitor_id))
                WHEN a.status = 'Cancelled'
            THEN CONCAT('Appointment cancelled: ',
            COALESCE(CONCAT(v.first_name, ' ',
            v.last_name), a.visitor_id))
                WHEN a.status = 'Finished'
            THEN CONCAT('Appointment completed: ',
            COALESCE(CONCAT(v.first_name, ' ',
            v.last_name), a.visitor_id))
            ELSE
            CONCAT('Appointment from ',
            COALESCE(CONCAT(v.first_name, ' ',
            v.last_name), a.visitor_id))
            END AS message,
            a.visitor_id AS identifier,
            CONCAT(v.first_name, ' ',
            v.last_name) AS full_name,
            v.visitor_type AS user_type,
            NULL AS course,
            NULL AS book_id,
            NULL AS book_title,
            a.visitor_id AS user_id,
            a.appointment_date,
            TIME_FORMAT(a.grace_period
            _start, '%H:%i') AS start_time,
            TIME_FORMAT(a.grace_period
            _end, '%H:%i') AS end_time,
```

```
        a.contact_number,
        a.purpose AS booking_type,
        a.purpose,
        a.created_at,
        'appointment' AS source,
        0 AS is_read,
        CONCAT('appointment_',
        a.appointment_id) AS id,
        a.status,
        NULL AS return_condition,
        NULL AS replacement_status
    FROM appointments a
    LEFT JOIN visitor v ON
    a.visitor_id = v.email
    ORDER BY a.created_at DESC
    LIMIT 200
    ");
    $appointmentNotifications = $stmt-
    >fetchAll(PDO::FETCH_ASSOC);
    $liveNotifications =
    array_merge($liveNotifications,
    $appointmentNotifications);
    } catch (PDOException $e) {
        $queryErrors[] = "Error querying
        appointments: " . $e->getMessage();
    }
}

// 4. Query book_borrowings - ALL
statuses (pending, borrowed, due,
pending_return, returned, cancelled,
lost/damage, ready_for_pickup)
if (in_array('book_borrowings',
$existingTables)) {
    try {
        $stmt = $pdo->query("
            SELECT
            CASE
                WHEN bb.status =
                'pending_return' THEN 'book_return'
                WHEN bb.status = 'borrowed'
            THEN 'book_borrowed'
                WHEN bb.status = 'due'
            THEN 'book_due'
                WHEN bb.status = 'returned'
            THEN 'book_returned'
                WHEN bb.status =
                'lost/damage' THEN 'book_lost_damage'
```



```
        WHEN bb.status =
'ready_for_pickup' THEN
'book_ready_pickup'
        ELSE 'book_borrow'
        END AS notification_type,
        CASE
        WHEN bb.status =
'pending_return' THEN CONCAT('Return
request: ', COALESCE(ab.title, 'Unknown
Book'), ' from ', bb.user_id)
        WHEN bb.status = 'borrowed'
THEN CONCAT('Book borrowed: ',
COALESCE(ab.title, 'Unknown Book'), ' by ',
bb.user_id)
        WHEN bb.status = 'due'
THEN CONCAT('Book due: ',
COALESCE(ab.title, 'Unknown Book'), ' from
', bb.user_id)
        WHEN bb.status = 'returned'
THEN CONCAT('Book returned: ',
COALESCE(ab.title, 'Unknown Book'), ' by ',
bb.user_id)
        WHEN bb.status =
'lost/damage' THEN CONCAT('Book
DAMAGE/LOST: ', COALESCE(ab.title,
'Unknown Book'), ' by ', bb.user_id)
        WHEN bb.status =
'ready_for_pickup' THEN CONCAT('Ready
for pickup: ', COALESCE(ab.title, 'Unknown
Book'), ' for ', bb.user_id)
        WHEN bb.status =
'cancelled' THEN CONCAT('Borrowing
cancelled: ', COALESCE(ab.title, 'Unknown
Book'), ' for ', bb.user_id)
        ELSE CONCAT('Book
transaction: ', COALESCE(ab.title,
'Unknown Book'), ' - ', bb.user_id)
        END AS message,
        NULL AS identifier,
        NULL AS full_name,
        bb.user_type,
        NULL AS course,
        bb.book_id,
        ab.title AS book_title,
        bb.user_id,
        NULL AS appointment_date,
        NULL AS start_time,
        NULL AS end_time,
        NULL AS contact_number,
        NULL AS booking_type,
        NULL AS purpose,
        COALESCE(bb.returned_date,
bb.borrow_date) AS created_at,
        CASE
        WHEN bb.status =
'pending_return' THEN 'return'
        WHEN bb.status IN
('borrowed', 'due', 'ready_for_pickup') THEN
'borrow'
        WHEN bb.status =
'lost/damage' THEN 'replacement'
        ELSE 'borrow'
        END AS source,
        0 AS is_read,
        CONCAT('borrowing_',
bb.borrow_id) AS id,
        bb.status,
        NULL AS return_condition,
        NULL AS replacement_status
FROM book_borrowings bb
LEFT JOIN add_books ab ON
bb.book_id = ab.book_id
ORDER BY bb.borrow_date
DESC
LIMIT 200
");
$borrowingNotifications = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
$liveNotifications =
array_merge($liveNotifications,
$borrowingNotifications);
} catch (PDOException $e) {
$queryErrors[] = "Error querying
book_borrowings: " . $e->getMessage();
}
}

// 5. Query returned_books - ALL returned
books (all conditions and replacement
statuses)
if (in_array('returned_books',
$existingTables)) {
try {
$stmt = $pdo->query("
SELECT
CASE
WHEN rb.return_condition IN
('damaged', 'lost', 'damaged_lost') AND
```



```
rb.replacement_status IN ('pending',
'pending_validation') THEN
    replacement_pending
        WHEN rb.return_condition IN
('damaged', 'lost', 'damaged_lost') AND
rb.replacement_status = 'completed' THEN
    replacement_completed
        WHEN rb.return_condition IN
('damaged', 'lost', 'damaged_lost') AND
rb.replacement_status = 'overdue' THEN
    replacement_overdue
        WHEN rb.return_condition =
'good' THEN 'book_returned_good'
        ELSE 'book_returned'
    END AS notification_type,
CASE
    WHEN rb.return_condition IN
('damaged', 'lost', 'damaged_lost') AND
rb.replacement_status IN ('pending',
'pending_validation')
        THEN
CONCAT('Replacement pending: ',
COALESCE(ab.title, 'Unknown Book'), ' from
', rb.user_id)
        WHEN rb.return_condition IN
('damaged', 'lost', 'damaged_lost') AND
rb.replacement_status = 'completed'
        THEN
CONCAT('Replacement completed: ',
COALESCE(ab.title, 'Unknown Book'), ' from
', rb.user_id)
        WHEN rb.return_condition IN
('damaged', 'lost', 'damaged_lost') AND
rb.replacement_status = 'overdue'
        THEN
CONCAT('Replacement overdue: ',
COALESCE(ab.title, 'Unknown Book'), ' from
', rb.user_id)
        WHEN rb.return_condition IN
('damaged', 'lost', 'damaged_lost') AND
rb.replacement_status = 'archived'
        THEN
CONCAT('Replacement archived: ',
COALESCE(ab.title, 'Unknown Book'), ' from
', rb.user_id)
        WHEN rb.return_condition =
'good'
```

```
        THEN CONCAT('Book
returned (Good): ', COALESCE(ab.title,
'Unknown Book'), ' by ', rb.user_id)
        ELSE CONCAT('Book
returned: ', COALESCE(ab.title, 'Unknown
Book'), ' by ', rb.user_id)
    END AS message,
    NULL AS identifier,
    NULL AS full_name,
    rb.user_type,
    NULL AS course,
    rb.book_id,
    ab.title AS book_title,
    rb.user_id,
    NULL AS appointment_date,
    NULL AS start_time,
    NULL AS end_time,
    NULL AS contact_number,
    NULL AS booking_type,
    NULL AS purpose,
    rb.returned_date AS
created_at,
CASE
    WHEN rb.return_condition IN
('damaged', 'lost', 'damaged_lost') THEN
'replacement'
        ELSE 'return'
    END AS source,
    0 AS is_read,
    CONCAT('returned_',
rb.return_id) AS id,
    COALESCE(rb.replacement_st
atus, 'returned') AS status,
    rb.return_condition,
    rb.replacement_status
FROM returned_books rb
LEFT JOIN add_books ab ON
rb.book_id = ab.book_id
ORDER BY rb.returned_date
DESC
LIMIT 200
");
$returnedNotifications = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
$liveNotifications =
array_merge($liveNotifications,
$returnedNotifications);
} catch (PDOException $e) {
```



```
$queryErrors[] = "Error querying
returned_books: " . $e->getMessage();
}
}

// 6. Query appointment_notifications for
admin - ALL notifications (read and unread)
if (tableExists($pdo,
'appointment_notifications')) {
    try {
        $stmt = $pdo->query("
            SELECT
                an.notification_type,
                an.message,
                NULL AS identifier,
                NULL AS full_name,
                NULL AS user_type,
                NULL AS course,
                NULL AS book_id,
                NULL AS book_title,
                NULL AS user_id,
                a.appointment_date,
                TIME_FORMAT(a.grace_period
_start, '%H:%i') AS start_time,
                TIME_FORMAT(a.grace_period
_end, '%H:%i') AS end_time,
                NULL AS contact_number,
                NULL AS booking_type,
                NULL AS purpose,
                an.created_at,
                'appointment' AS source,
                an.is_read,
                CONCAT('notif_', an.id) AS id,
                COALESCE(a.status, 'N/A') AS
status,
                NULL AS return_condition,
                NULL AS replacement_status
            FROM appointment_notifications
        an
            LEFT JOIN appointments a ON
an.appointment_id = a.appointment_id
            WHERE an.recipient_type =
'admin'

            ORDER BY an.created_at DESC
            LIMIT 200
        ");
        $adminPersistentNotifications =
$stmt->fetchAll(PDO::FETCH_ASSOC);
```

```
$liveNotifications =
array_merge($liveNotifications,
$adminPersistentNotifications);
    } catch (PDOException $e) {
        $queryErrors[] = "Error querying
appointment_notifications: " . $e-
>getMessage();
    }
}

// 7. Query non_returned_books -
Overdue books not yet returned
if (in_array('non_returned_books',
$existingTables)) {
    try {
        $stmt = $pdo->query("
            SELECT
                'non_returned' AS
notification_type,
                CONCAT('Non-returned book: ',
COALESCE(ab.title, 'Unknown Book'), ' by ',
nr.user_id,
                ' (Fine: ₱',
FORMAT(COALESCE(nr.total_fine, 0), 2),
                ')') AS message,
                NULL AS identifier,
                NULL AS full_name,
                nr.user_type,
                NULL AS course,
                nr.book_id,
                ab.title AS book_title,
                nr.user_id,
                NULL AS appointment_date,
                NULL AS start_time,
                NULL AS end_time,
                NULL AS contact_number,
                NULL AS booking_type,
                NULL AS purpose,
                nr.moved_to_non_returned_dat
e AS created_at,
                'overdue' AS source,
                0 AS is_read,
                CONCAT('nonreturned_',
nr.borrow_id) AS id,
                COALESCE(nr.settlement_stat
us, 'unsettled') AS status,
                NULL AS return_condition,
                NULL AS replacement_status
            FROM non_returned_books nr
```



```
LEFT JOIN add_books ab ON
nr.book_id = ab.book_id
ORDER BY
nr.moved_to_non_returned_date DESC
LIMIT 200
");
$nonReturnedNotifs = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
$liveNotifications =
array_merge($liveNotifications,
$nonReturnedNotifs);
} catch (PDOException $e) {
$queryErrors[] = "Error querying
non_returned_books: " . $e->getMessage();
}
}

// 8. Query paid_books - Books that have
been paid for
if (in_array('paid_books',
$existingTables)) {
try {
$stmt = $pdo->query("
SELECT
'book_paid' AS
notification_type,
CONCAT('Book paid: ',
COALESCE(ab.title, 'Unknown Book'), ' by ',
pb.user_id,
' (P',
FORMAT(COALESCE(pb.payment_amount,
0), 2), ')') AS message,
NULL AS identifier,
NULL AS full_name,
pb.user_type,
NULL AS course,
pb.book_id,
ab.title AS book_title,
pb.user_id,
NULL AS appointment_date,
NULL AS start_time,
NULL AS end_time,
NULL AS contact_number,
NULL AS booking_type,
NULL AS purpose,
pb.payment_date AS
created_at,
'payment' AS source,
0 AS is_read,
CONCAT('paid_', pb.paid_id)
AS id,
'paid' AS status,
pb.original_condition AS
return_condition,
NULL AS replacement_status
FROM paid_books pb
LEFT JOIN add_books ab ON
pb.book_id = ab.book_id
ORDER BY pb.payment_date
DESC
LIMIT 200
");
$paidNotifs = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
$liveNotifications =
array_merge($liveNotifications, $paidNotifs);
} catch (PDOException $e) {
$queryErrors[] = "Error querying
paid_books: " . $e->getMessage();
}
}

// 9. Query archived_overdue_books -
Archived overdue settlement records
if (in_array('archived_overdue_books',
$existingTables)) {
try {
$stmt = $pdo->query("
SELECT
'archived_overdue' AS
notification_type,
CONCAT('Archived overdue: ',
COALESCE(ab.title, 'Unknown Book'), ' by ',
ao.user_id,
' - ',
UPPER(ao.settlement_status)) AS
message,
NULL AS identifier,
NULL AS full_name,
ao.user_type,
NULL AS course,
ao.book_id,
ab.title AS book_title,
ao.user_id,
NULL AS appointment_date,
NULL AS start_time,
NULL AS end_time,
NULL AS contact_number,
```



```
NULL AS booking_type,
NULL AS purpose,
ao.archived_date AS
created_at,
'overdue' AS source,
0 AS is_read,
CONCAT('archivedoverdue_',
ao.id) AS id,
ao.settlement_status AS status,
ao.return_condition,
NULL AS replacement_status
FROM archived_overdue_books
ao
LEFT JOIN add_books ab ON
ao.book_id = ab.book_id
ORDER BY ao.archived_date
DESC
LIMIT 200
");
    $archivedOverdueNotifs = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
    $liveNotifications =
array_merge($liveNotifications,
$archivedOverdueNotifs);
    } catch (PDOException $e) {
        $queryErrors[] = "Error querying
archived_overdue_books: " . $e-
>getMessage();
    }
}

// ===== STEP 2: Persist live notifications
to admin_notifications table =====
// Skip notifications that have been
dismissed by this admin
$insertStmt = $pdo->prepare("
    INSERT INTO admin_notifications
(notification_id, notification_type,
message, source, full_name, user_type,
identifier, course,
book_id, book_title, user_id,
appointment_date, start_time, end_time,
contact_number,
booking_type, purpose, status,
return_condition, replacement_status,
admin_id, created_at)
VALUES
(?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?,
?, ?, ?, ?, ?, ?, ?, ?)
ON DUPLICATE KEY UPDATE
    status = VALUES(status),
    message = VALUES(message),
    return_condition =
VALUES(return_condition),
    replacement_status =
VALUES(replacement_status)
");

foreach ($liveNotifications as $notif) {
    // Skip if this notification was
    dismissed/cleared by the admin
    if (in_array($notif['id'],
    $dismissedNotificationIds)) {
        continue;
    }
    try {
        $insertStmt->execute([
            $notif['id'],
            $notif['notification_type'],
            $notif['message'],
            $notif['source'],
            $notif['full_name'] ?? null,
            $notif['user_type'] ?? null,
            $notif['identifier'] ?? null,
            $notif['course'] ?? null,
            $notif['book_id'] ?? null,
            $notif['book_title'] ?? null,
            $notif['user_id'] ?? null,
            $notif['appointment_date'] ?? null,
            $notif['start_time'] ?? null,
            $notif['end_time'] ?? null,
            $notif['contact_number'] ?? null,
            $notif['booking_type'] ?? null,
            $notif['purpose'] ?? null,
            $notif['status'] ?? null,
            $notif['return_condition'] ?? null,
            $notif['replacement_status'] ??
            null,
            $current_admin_id,
            $notif['created_at'] ?? date('Y-m-d
            H:i:s')
        ]);
    } catch (PDOException $e) {
        // Silently skip duplicates or log error
        error_log("Error persisting
notification {$notif['id']}: " . $e-
>getMessage());
    }
}
```




```
}

// ===== STEP 3: Update status of stored
notifications to reflect current source status
=====
// Only update the status field to keep it in
sync with source tables — do NOT mark as
"processed"
foreach ($liveNotifications as $notif) {
    // Skip dismissed notifications
    if (in_array($notif['id'],
$dismissedNotificationIds)) {
        continue;
    }
    try {
        $updateStmt = $pdo->prepare("
            UPDATE admin_notifications
            SET status = ?, message = ?,
return_condition = COALESCE(?,
return_condition),
            replacement_status =
COALESCE(?, replacement_status)
            WHERE notification_id = ? AND
admin_id = ?
        ");
        $updateStmt->execute([
            $notif['status'] ?? null,
            $notif['message'] ?? null,
            $notif['return_condition'] ?? null,
            $notif['replacement_status'] ??
null,
            $notif['id'],
            $current_admin_id
        ]);
    } catch (PDOException $e) {
        // Silently skip errors
    }
}

// ===== STEP 4: Read from
admin_notifications table (persistent history)
=====
$stmt = $pdo->prepare("
    SELECT
        notification_id AS id,
        notification_type,
        message,
        source,
        full_name,
```

```
user_type,
identifier,
course,
book_id,
book_title,
user_id,
appointment_date,
start_time,
end_time,
contact_number,
booking_type,
purpose,
status,
return_condition,
replacement_status,
is_read,
created_at
FROM admin_notifications
WHERE admin_id = ?
ORDER BY
    CASE WHEN is_read = 0 THEN 0
ELSE 1 END,
    created_at DESC
LIMIT 500
");
$stmt->execute([$current_admin_id]);
$allNotifications = $stmt-
>fetchAll(PDO::FETCH_ASSOC);

// Sync read state: if it's in
admin_notification_reads, ensure is_read =
1
foreach ($allNotifications as
&$notification) {
    if (in_array($notification['id'],
$readNotificationIds)) {
        $notification['is_read'] = 1;
    }
}
unset($notification);

$notifications = $allNotifications;

// Count only unread notifications
$unreadCount = 0;
foreach ($notifications as $notification) {
    if (empty($notification['is_read']) ||
$notification['is_read'] == 0) {
        $unreadCount++;
    }
}
```



```
    }
}

} catch (PDOException $e) {
    $queryErrors[] = "Error fetching combined
notifications: " . $e->getMessage();
    $notifications = [];
}

$sisDevMode = false;

// Helper function to get icon class based on
notification type
function getIconClass($source) {
    switch($source) {
        case 'user': return 'icon-user';
        case 'borrow': return 'icon-borrow';
        case 'appointment': return 'icon-
appointment';
        case 'return': return 'icon-return';
        case 'replacement': return 'icon-
replacement';
        case 'overdue': return 'icon-overdue';
        case 'payment': return 'icon-payment';
        default: return 'icon-user';
    }
}

// Helper function to get icon based on
notification type
function getIcon($source) {
    switch($source) {
        case 'user': return 'fa-user-plus';
        case 'borrow': return 'fa-book';
        case 'appointment': return 'fa-calendar-
check';
        case 'return': return 'fa-undo';
        case 'replacement': return 'fa-
exchange-alt';
        case 'overdue': return 'fa-exclamation-
circle';
        case 'payment': return 'fa-money-bill-
wave';
        default: return 'fa-bell';
    }
}

// Helper function to get view link based on
notification type
function getViewLink($notification) {
    switch($notification['source']) {
        case 'user': return
'PendingApproval.php';
        case 'borrow': return
'BorrowedBooks.php';
        case 'appointment': return
'Appointment_Management.php';
        case 'return': return
'ReturnedBooks.php';
        case 'replacement': return
'ReplacedBooks.php';
        case 'overdue': return 'Non-
ReturnedBooks.php';
        case 'payment': return 'PaidBooks.php';
        default: return '#';
    }
}

// Helper function to format time ago
function timeAgo($datetime) {
    $time = strtotime($datetime);
    $now = time();
    $diff = $now - $time;

    if ($diff < 60) {
        return 'Just now';
    } elseif ($diff < 3600) {
        $mins = floor($diff / 60);
        return $mins . ' minute' . ($mins > 1 ? 's'
: '') . ' ago';
    } elseif ($diff < 86400) {
        $hours = floor($diff / 3600);
        return $hours . ' hour' . ($hours > 1 ? 's'
: '') . ' ago';
    } elseif ($diff < 604800) {
        $days = floor($diff / 86400);
        return $days . ' day' . ($days > 1 ? 's' :
'') . ' ago';
    } else {
        return date('M d, Y', $time);
    }
}

?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
```




```
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<title>Admin || Notification</title>
<link rel="stylesheet"
href="../../Design/Admin.css">
<link rel="stylesheet"
href="../../Design/AdminSidebar.css">
<link rel="stylesheet"
href="../../Design/Admin_Notification.css">
<link rel="stylesheet" href="../../Notification
Badge/badge.css">
<link rel="icon" type="image/x-icon"
href="../../Pictures/PUP_LOGO.png">
<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
</head>
<body>
<?php include_once __DIR__ .
'/sidebar_toggle.php'; ?>
<!-- Sidebar Navigation -->
<nav class="sidebar" id="sidebar">
<div class="sidebar-admin-profile"
id="sidebarAdminProfile"
style="cursor:pointer;"
onclick="window.location.href='../AdminProfil
e.php';">
';">
<div class="sidebar-admin-
username"><h6>Welcome!
Admin</h6><?php echo
htmlspecialchars($_SESSION['admin_usern
ame']) ?? 'admin_user'; ?></div>
</div>
<ul>
<li><a href="<?php echo
$sidebarBasePath ?? '../';
?>Admin_Dashboard.php" class="<?php
echo ($currentPage ?? '') === 'dashboard' ?
'active' : ''; ?>"><i class="fa-solid fa-
grip"></i>
<span>Dashboard</span></a></li>
```

```
<li class="section-header Main-
header">Main</li>
<li><a href="<?php echo
$componentPath ?? '';
?>PendingApproval.php" class="<?php
echo ($currentPage ?? '') === 'pending' ?
'active' : ''; ?>"><i class="fas fa-user-
clock"></i> <span>Pending
Users</span><?php echo
renderBadge($badges['pending_users'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? '';
?>UserManagement.php" class="<?php
echo ($currentPage ?? '') === 'users' ?
'active' : ''; ?>"><i class="fas fa-users"></i>
<span>User Management</span><?php
echo
renderBadge($badges['user_management']
?? 0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? '';
?>ArchivedUsers.php" class="<?php echo
($currentPage ?? '') === 'archivedusers' ?
'active' : ''; ?>"><i class="fas fa-user-
slash"></i> <span>Archived
Users</span><?php echo
renderBadge($badges['archived_users'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? '';
?>Admin_Notification.php" class="<?php
echo ($currentPage ?? '') === 'notifications'
? 'active' : ''; ?>"><i class="fas fa-bell"></i>
<span>Notifications</span><?php echo
renderBadge($badges['notifications'] ?? 0);
?></a></li>
<li class="section-header books-
inventory-header">Books Inventory</li>
<li><a href="<?php echo
$componentPath ?? ''; ?>AddBooks.php"
class="<?php echo ($currentPage ?? '') ===
'addbooks' ? 'active' : ''; ?>"><i class="fa-
solid fa-list-check"></i><span>Manage
Books</span><?php echo
renderBadge($badges['manage_books'] ??
0); ?></a></li>
<li><a href="<?php echo
$componentPath ?? '../Narrative-
```



Research-Capstone/"; ?>addBooks.php" class="<?php echo (\$currentPage ?? ") === 'addbooks' ? 'active' : "; ?>"><i class="fa-brands fa-researchgate"></i>Research Books<?php echo renderBadge(\$badges['narrative_books'] ?? 0); ?>

<a href="<?php echo \$componentPath ?? "; ?>BorrowedBooks.php" class="<?php echo (\$currentPage ?? ") === 'borrowed' ? 'active' : "; ?>"><i class="fas fa-download"></i>Borrowed Books<?php echo renderBadge(\$badges['borrowed_books'] ?? 0); ?>

<a href="<?php echo \$componentPath ?? "; ?>Non-ReturnedBooks.php" class="<?php echo (\$currentPage ?? ") === 'nonreturned' ? 'active' : "; ?>"><i class="fas fa-ban"></i>Non-Returned Books<?php echo renderBadge(\$badges['non_returned_books'] ?? 0); ?>

<a href="<?php echo \$componentPath ?? "; ?>ReturnedBooks.php" class="<?php echo (\$currentPage ?? ") === 'returned' ? 'active' : "; ?>"><i class="fas fa-upload"></i>Returned Books<?php echo renderBadge(\$badges['returned_books'] ?? 0); ?>

<a href="<?php echo \$componentPath ?? "; ?>ReplacedBooks.php" class="<?php echo (\$currentPage ?? ") === 'replaced' ? 'active' : "; ?>"><i class="fas fa-sync"></i>Replaced Books<?php echo renderBadge(\$badges['replaced_books'] ?? 0); ?>

<a href="<?php echo \$componentPath ?? "; ?>PaidBooks.php" class="<?php echo (\$currentPage ?? ") === 'paidbook' ? 'active' : "; ?>"><i class="fas fa-solid fa-money-bill"></i>Paid Books<?php echo renderBadge(\$badges['paid_book'] ?? 0); ?>

<a href="<?php echo \$componentPath ?? "; ?>ArchivedBooks.php" class="<?php echo (\$currentPage ?? ") === 'archived' ? 'active' : "; ?>"><i class="fas fa-archive"></i>Archived Books<?php echo renderBadge(\$badges['archived_books'] ?? 0); ?>

<a href="<?php echo \$componentPath ?? './../Narrative-Research-Capstone/"; ?>archived_research_books/archived-research-books.php" class="<?php echo (\$currentPage ?? ") === 'archivedresearch' ? 'active' : "; ?>"><i class="fa-solid fa-box-archive"></i>Archived Research<?php echo renderBadge(\$badges['archived_research_books'] ?? 0); ?>

<a href="<?php echo \$componentPath ?? "; ?>ArchivedOverdueBooks.php" class="<?php echo (\$currentPage ?? ") === 'archivedoverdue' ? 'active' : "; ?>"><i class="fas fa-exclamation-triangle"></i>Archived Overdue<?php echo renderBadge(\$badges['archived_overdue'] ?? 0); ?>

<li class="section-header appointment-header">Appointment

<a href="<?php echo \$componentPath ?? "; ?>Appointment_Management.php" class="<?php echo (\$currentPage ?? ") === 'appointments' ? 'active' : "; ?>"><i class="fas fa-calendar-check"></i>Appointment Management<?php echo renderBadge(\$badges['appointments'] ?? 0); ?>

<li class="section-header">Reports

<a href="<?php echo \$componentPath ?? "; ?>ReportAnalytics.php" class="<?php echo (\$currentPage ?? ") === 'reports' ? 'active' : "; ?>"><i class="fas fa-chart-line"></i>Reports



```
<li class="sidebar-separator"></li>
<li><a href="php echo
$sidebarBasePath ?? '../..';
?&gt;Admin_Logout.php"&gt;&lt;i class="fas fa-sign-
out-alt"&gt;&lt;/i&gt; &lt;span&gt;Logout&lt;/span&gt;&lt;/a&gt;&lt;/li&gt;
&lt;/ul&gt;
&lt;/nav&gt;

&lt;!--Main Content || Admin_Notification --&gt;
&lt;div class="content-area" style="display:
flex; flex-direction: column; padding-bottom:
0;"&gt;
  &lt;div class="notification-container"
style="flex-grow: 1; margin-bottom: 0;"&gt;
    &lt;div class="notification-header"&gt;
      &lt;h1 class="page-title"&gt;
        &lt;i class="fas fa-bell"&gt;&lt;/i&gt;
Notifications
      &lt;?php if($unreadCount &gt; 0): ?&gt;
        &lt;span class="notification-
badge" id="notificationBadge"&gt;&lt;?php echo
$unreadCount; ?&gt;&lt;/span&gt;
        &lt;?php endif; ?&gt;
      &lt;/h1&gt;

      &lt;?php if(!empty($notifications)):
?&gt;
        &lt;div class="notification-actions"&gt;
          &lt;button id="markAllReadBtn"
onclick="markAllAsRead()" &lt;?php echo
$unreadCount == 0 ? 'disabled
style="opacity:0.6;cursor:not-allowed;" : "";
?&gt;&gt;
            &lt;i class="fas fa-check-
double"&gt;&lt;/i&gt; Mark All as Read
          &lt;/button&gt;
          &lt;button id="clearHistoryBtn"
onclick="clearNotificationHistory()"
class="btn-clear-history" title="Clear all
read/processed notifications"&gt;
            &lt;i class="fas fa-trash-
alt"&gt;&lt;/i&gt; Clear History
          &lt;/button&gt;
        &lt;/div&gt;
        &lt;?php endif; ?&gt;
      &lt;/div&gt;

    &lt;!-- Filter Tabs --&gt;</pre
```

```
<div class="notification-filter-tabs">
  <button class="filter-tab active"
data-filter="all">All <span class="tab-
count"><?php echo count($notifications);
?></span></button>
  <button class="filter-tab" data-
filter="unread">Unread <span class="tab-
count"><?php echo $unreadCount;
?></span></button>
  <button class="filter-tab" data-
filter="read">Read <span class="tab-
count"><?php echo count($notifications) -
$unreadCount; ?></span></button>
</div>

Appointment_Management.php
<?php
if (session_status() ===
PHP_SESSION_NONE) {
  session_start();
}

// Include database connection
require_once
'../..DBconnection/db_connection.php';
//Notification Badge Function
if (!isset($badges) && isset($pdo)) {
  require_once __DIR__ . '/../Notification
Badge/Notification Badge.php';
}
$profileImage =
'../..Pictures/PUP_LOGO.png'; // Default
image path

// Define constant for default image if not
already defined
if (!defined('DEFAULT_ADMIN_IMAGE')) {
  define('DEFAULT_ADMIN_IMAGE',
'../..Pictures/PUP_LOGO.png');
}

/**
 * Convert time slots from 24-hour format to
12-hour format with AM/PM
 * @param string $timeSlots - Time slots in
format "HH:MM-HH:MM" or comma-
separated
 * @return string - Formatted time slots with
AM/PM
```



```
*/
function formatTimeSlots($timeSlots) {
    if (empty($timeSlots) || $timeSlots ===
'N/A') {
        return 'N/A';
    }

    // Split by comma if multiple time slots
    $slots = explode(',', $timeSlots);
    $formattedSlots = [];

    foreach ($slots as $slot) {
        $slot = trim($slot);

        // Check if slot contains a range
        (HH:MM-HH:MM)
        if (strpos($slot, '-') !== false) {
            $times = explode('-', $slot);
            if (count($times) === 2) {
                $startTime =
formatSingleTime(trim($times[0]));
                $endTime =
formatSingleTime(trim($times[1]));
                $formattedSlots[] = $startTime . ' -
' . $endTime;
            } else {
                $formattedSlots[] = $slot;
            }
        } else {
            // Single time
            $formattedSlots[] =
formatSingleTime($slot);
        }
    }

    return implode(',', $formattedSlots);
}

/**
 * Convert a single time from 24-hour to 12-
hour format
 * @param string $time - Time in HH:MM
format
 * @return string - Time in h:mm AM/PM
format
 */
function formatSingleTime($time) {
    $time = trim($time);

    // Try to parse the time
    $timestamp = strtotime($time);
    if ($timestamp !== false) {
        return date('g:i A', $timestamp);
    }

    // Fallback: manual conversion if strtotime
fails
    if (preg_match('/^\d{1,2}:\d{2}$/', $time,
$matches)) {
        $hour = (int)$matches[1];
        $minute = $matches[2];

        $period = ($hour >= 12) ? 'PM' : 'AM';

        if ($hour == 0) {
            $hour = 12;
        } elseif ($hour > 12) {
            $hour -= 12;
        }

        return $hour . ':' . $minute . ' ' . $period;
    }

    return $time; // Return original if can't
parse
}

////////////////////////////////////
/////
////////////////////////////////////Fetching Profile Image from the
Database////////////////////////////////////

// Get admin_id from the session
$admin_id = $_SESSION['admin_id'] ?? '';
$defaultImagePath =
'../Pictures/PUP_LOGO.png'; // Default
image path
$profileImage = $defaultImagePath; //
Initialize with default

try {
    if (!empty($admin_id)) {
        $stmt = $pdo->prepare("SELECT
profile_image FROM admin WHERE
admin_id = ?");
        $stmt->execute([$admin_id]);
        $result = $stmt-
>fetch(PDO::FETCH_ASSOC);
    }
}
```



```
if ($result &&
!empty($result['profile_image'])) {
    // Get the raw path from database
    $dbPath = $result['profile_image'];

    // Adjust path based on different
    possible formats
    if (strpos($dbPath, 'http://') === 0 ||
    strpos($dbPath, 'https://') === 0) {
        // It's a full URL, use as is
        $profileImage = $dbPath;
    }
    else if (strpos($dbPath, '../uploads/')
    === 0) {
        // Path is relative to
        AdminDashboard - add another "../" for
        components subdirectory
        $profileImage = '../' . $dbPath; //
        Makes it "../uploads/..."
    }
    else if (strpos($dbPath, 'uploads/')
    === 0) {
        // Path starts with uploads/ - add
        ../
        $profileImage = '../..' . $dbPath;
    }
    else {
        // Assume it's just a filename
        $profileImage =
        '../uploads/admin_profile/' . $dbPath;
    }
}
} catch (PDOException $e) {
    error_log("Error fetching admin profile: " .
    $e->getMessage());
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
    content="width=device-width, initial-
    scale=1.0">
    <title>Appointment Management - PUP
    Bansud Campus</title>
```

```
<link rel="stylesheet" href="../Notification
Badge/badge.css">
<link rel="stylesheet"
href="../Design/Admin.css">
<link rel="stylesheet"
href="../Design/AdminSidebar.css">
<link rel="stylesheet"
href="../Design/AdminAppointment/Admi
n_Appointment_Main-Content.css">
<link rel="stylesheet"
href="../Design/AdminAppointment/overri
de_control.css">
<link rel="stylesheet"
href="../Design/AdminAppointment/capa
city_settings.css">
<link rel="stylesheet"
href="../Design/AdminAppointment/moda
l-view.css">
<link rel="icon" type="image/x-icon"
href="../Pictures/PUP_LOGO.png">
<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
</head>
<body>
    <?php include_once __DIR__ .
    '/sidebar_toggle.php'; ?>
    <!-- Sidebar Navigation -->
    <nav class="sidebar" id="sidebar">
        <div class="sidebar-admin-profile"
        id="sidebarAdminProfile"
        style="cursor:pointer;"
        onclick="window.location.href='../AdminProfil
        e.php">
            ';">
            <div class="sidebar-admin-
            username"><h6>Welcome!
            Admin</h6><?php echo
            htmlspecialchars($_SESSION['admin_usern
            ame'] ?? 'admin_user'); ?></div>
            </div>
            <ul>
                <li><a href="<?php echo
                $sidebarBasePath ?? '../";
```



```
<?php echo ($currentPage ?? "") === 'dashboard' ?  
'active' : ""; ?>><i class="fa-solid fa-grip"></i>  
<span>Dashboard</span></a></li>  
    <li class="section-header Main-header">Main</li>  
    <li><a href="<?php echo $componentPath ?? ">  
<?php echo ($currentPage ?? "") === 'pending' ?  
'active' : ""; ?>><i class="fas fa-user-clock"></i> <span>Pending  
Users</span><?php echo  
renderBadge($badges['pending_users'] ??  
0); ?></a></li>  
    <li><a href="<?php echo $componentPath ?? ">  
<?php echo ($currentPage ?? "") === 'users' ?  
'active' : ""; ?>><i class="fas fa-users"></i>  
<span>User Management</span><?php echo  
renderBadge($badges['user_management']  
?? 0); ?></a></li>  
    <li><a href="<?php echo $componentPath ?? ">  
<?php echo ($currentPage ?? "") === 'archivedusers' ?  
'active' : ""; ?>><i class="fas fa-user-slash"></i> <span>Archived  
Users</span><?php echo  
renderBadge($badges['archived_users'] ??  
0); ?></a></li>  
    <li><a href="<?php echo $componentPath ?? ">  
<?php echo ($currentPage ?? "") === 'notifications'  
? 'active' : ""; ?>><i class="fas fa-bell"></i>  
<span>Notifications</span><?php echo  
renderBadge($badges['notifications'] ?? 0);  
?></a></li>  
    <li class="section-header books-  
inventory-header">Books Inventory</li>  
    <li><a href="<?php echo $componentPath ?? "><?php echo ($currentPage ?? "") ===  
'addbooks' ? 'active' : ""; ?>><i class="fa-solid fa-list-check"></i><span>Manage
```

```
Books</span><?php echo  
renderBadge($badges['manage_books'] ??  
0); ?></a></li>  
    <li><a href="<?php echo $componentPath  
?? '../Narrative-Research-Capstone/';  
<?php echo ($currentPage ?? "") === 'addbooks' ?  
'active' : ""; ?>><i class="fa-brands fa-researchgate"></i><span>Research  
Books</span><?php echo  
renderBadge($badges['narrative_books'] ??  
0); ?></a></li>  
    <li><a href="<?php echo $componentPath ?? ">  
<?php echo ($currentPage ?? "") === 'borrowed' ? 'active'  
: ""; ?>><i class="fas fa-download"></i>  
<span>Borrowed Books</span><?php echo  
renderBadge($badges['borrowed_books'] ??  
0); ?></a></li>  
    <li><a href="<?php echo $componentPath ?? "><?php echo ($currentPage ?? "") === 'nonreturned' ?  
'active' : ""; ?>><i class="fas fa-ban"></i>  
<span>Non-Returned Books</span><?php  
echo  
renderBadge($badges['non_returned_books']  
?? 0); ?></a></li>  
    <li><a href="<?php echo $componentPath ?? ">  
<?php echo ($currentPage ?? "") === 'returned' ? 'active' :  
"; ?>><i class="fas fa-upload"></i>  
<span>Returned Books</span><?php echo  
renderBadge($badges['returned_books'] ??  
0); ?></a></li>  
    <li><a href="<?php echo $componentPath ?? ">  
<?php echo ($currentPage ?? "") === 'replaced' ? 'active'  
: ""; ?>><i class="fas fa-sync"></i>  
<span>Replaced Books</span><?php echo  
renderBadge($badges['replaced_books'] ??  
0); ?></a></li>  
    <li><a href="<?php echo $componentPath ?? "><?php echo ($currentPage ?? "") ===  
'paidbook' ? 'active' : ""; ?>><i class="fa-
```




```
solid fa-money-bill"></i><span>Paid
Books</span><?php echo
renderBadge($badges['paid_book'] ?? 0);
?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ArchivedBooks.php" class="<?php echo
($currentPage ?? ") === 'archived' ? 'active'
: "; ?>"><i class="fas fa-archive"></i>
<span>Archived Books</span><?php echo
renderBadge($badges['archived_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? '../Narrative-
Research-Capstone/';
?>archived_research_books/archived-
research-books.php" class="<?php echo
($currentPage ?? ") === 'archivedresearch'
? 'active' : "; ?>"><i class="fa-solid fa-box-
archive"></i> <span>Archived
Research</span><?php echo
renderBadge($badges['archived_research_b
ooks'] ?? 0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ArchivedOverdueBooks.php"
class="<?php echo ($currentPage ?? ") ===
'archivedoverdue' ? 'active' : "; ?>"><i
class="fas fa-exclamation-triangle"></i>
<span>Archived Overdue</span><?php
echo
renderBadge($badges['archived_overdue']
?? 0); ?></a></li>
    <li class="section-header appointment-
header">Appointment</li>
    <li><a href="<?php echo
$componentPath ?? ";
?>Appointment_Management.php"
class="<?php echo ($currentPage ?? ") ===
'appointments' ? 'active' : "; ?>"><i
class="fas fa-calendar-check"></i>
<span>Appointment
Management</span><?php echo
renderBadge($badges['appointments'] ?? 0);
?></a></li>
    <li class="section-header">Reports</li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ReportAnalytics.php" class="<?php echo
```

```
($currentPage ?? ") === 'reports' ? 'active' :
"; ?>"><i class="fas fa-chart-line"></i>
<span>Reports</span></a></li>
    <li class="sidebar-separator"></li>
    <li><a href="<?php echo
$sidebarBasePath ?? '../';
?>Admin_Logout.php"><i class="fas fa-sign-
out-alt"></i> <span>Logout</span></a></li>
</ul>
</nav>
```

```
<!-- Main Content -->
<main class="main-content" data-admin-
id="<?php echo
htmlspecialchars($_SESSION['admin_id'] ??
 ""); ?>">
```

```
    <h1 class="page-title">Appointment
Management</h1>
```

```
<!-- Add this right after the opening
<main> tag -->
    <input type="hidden" id="adminIdField"
value="<?php echo
htmlspecialchars($_SESSION['admin_id'] ??
 ""); ?>">
```

```
<!-- Capacity Monitor -->
    <div class="card" style="margin-
bottom: 30px;">
        <div class="card-header">
            <div>
                <h2 class="card-title">Capacity
Monitor</h2>
                <p class="card-subtitle">Track
library visitor capacity by date and time
slot</p>
```

```
            </div>
            <div class="time-header-display">
                <span
id="currentTimeDisplay">Current Time:
<?php echo date('g:i A'); ?></span>
            </div>
        </div>
```

```
<!-- Date Filter for Capacity -->
    <div class="capacity-date-filter"
style="padding: 20px; border-bottom: 1px
solid #e0e0e0;">
```




```
<span class="halfday-
icon"> 🌅 </span>
<div>
  <span
class="halfday-label">Morning (AM)</span>
  <span
class="halfday-time">8:00 AM - 12:00
PM</span>
</div>
</label>
<label class="halfday-
option" onclick="selectHalfday('pm')">
  <input type="radio"
name="halfday" value="pm">
  <span class="halfday-
icon"> 🌆 </span>
  <div>
    <span
class="halfday-label">Afternoon
(PM)</span>
    <span
class="halfday-time">1:00 PM - 5:00
PM</span>
  </div>
</label>
</div>

<!-- Custom Time Section -->
<div class="custom-time-
section active" id="customTimeSection">
  <div class="custom-time-
header">
    <h4><i class="fas fa-
clock"></i> Set Custom Time Range</h4>
  </div>

  <div class="time-slot-
inputs">
    <div class="time-input-
group">
      <label>Start
Time</label>
      <div class="time-
input-row">
        <select
class="time-select" id="startTimeHour">
          <option
value="08:00">08:00</option>
          <option
value="09:00" selected>09:00</option>
          <option
value="10:00">10:00</option>
          <option
value="11:00">11:00</option>
          <option
value="12:00">12:00</option>
          <option
value="13:00">01:00</option>
          <option
value="14:00">02:00</option>
          <option
value="15:00">03:00</option>
          <option
value="16:00">04:00</option>
        </select>
        <select
class="time-select" id="startTimePeriod">
          <option
value="AM">AM</option>
          <option
value="PM">PM</option>
        </select>
      </div>
    </div>

    <div class="time-input-
group">
      <label>End
Time</label>
      <div class="time-
input-row">
        <select
class="time-select" id="endTimeHour">
          <option
value="09:00">09:00</option>
          <option
value="10:00">10:00</option>
          <option
value="11:00">11:00</option>
          <option
value="12:00">12:00</option>
          <option
value="13:00">01:00</option>
          <option
value="14:00">02:00</option>
          <option
value="15:00">03:00</option>
          <option
value="16:00">04:00</option>
        </select>
        <select
class="time-select" id="endTimePeriod">
          <option
value="AM">AM</option>
          <option
value="PM">PM</option>
        </select>
      </div>
    </div>
  </div>
</div>
```




```
placeholder="Search by name, ID, or
purpose...">
</div>
<div class="filter-group">
  <select id="timeSlotFilter">
    <option value="">All Time
Slots</option>
    <option value="8:00
AM">8:00 AM - 9:00 AM</option>
    <option value="9:00
AM">9:00 AM - 10:00 AM</option>
    <option value="10:00
AM">10:00 AM - 11:00 AM</option>
    <option value="11:00
AM">11:00 AM - 12:00 PM</option>
    <option value="12:00
PM">12:00 PM - 1:00 PM</option>
    <option value="1:00
PM">1:00 PM - 2:00 PM</option>
    <option value="2:00
PM">2:00 PM - 3:00 PM</option>
    <option value="3:00
PM">3:00 PM - 4:00 PM</option>
    <option value="4:00
PM">4:00 PM - 5:00 PM</option>
  </select>
  <input type="date"
id="appointmentDateFilter">
  <button class="btn-clear-
filters" onclick="clearAppointmentFilters()">
    <i class="fas fa-
times"></i> Clear
  </button>
</div>
</div>

<!-- Appointments Table -->
<table class="appointments-
table">
  <thead>
    <tr>
      <th>VISITOR NAME</th>
      <th>VISITOR ID</th>
      <th>CONTACT</th>
      <th>DATE</th>
      <th>TIME SLOT</th>
      <th>PURPOSE</th>
      <th>STATUS</th>
      <th>GRACE
STATUS</th>
      <th>ACTIONS</th>
    </tr>
  </thead>
  <tbody
id="appointmentsTableBody">
    <?php
// Fetch appointments with
time slots AND profile picture
try {
  $query = "SELECT
    a.appointment_id,
    a.visitor_id,
    a.contact_number,
    a.appointment_dat
    e,
    a.purpose,
    a.status,
    a.additional_notes,
    a.grace_period_sta
    rt,
    a.cancel_available
    _at,
    a.grace_period_en
    d,
    v.first_name,
    v.last_name,
    v.booking_id,
    v.profile_picture,
    GROUP_CONCAT
(ats.time_slot ORDER BY ats.time_slot
SEPARATOR ', ') as time_slots
FROM appointments
a
LEFT JOIN visitor v
ON a.visitor_id = v.email
LEFT JOIN
appointment_time_slots ats ON
a.appointment_id = ats.appointment_id
GROUP BY
a.appointment_id
ORDER BY
a.appointment_date DESC, a.created_at
DESC";

  $stmt = $pdo-
>prepare($query);
  $stmt->execute();
```




```
$appointments = $stmt-
>fetchAll(PDO::FETCH_ASSOC);

if (empty($appointments)) {
    echo "<tr><td
colspan='9' class='no-appointments'
style='text-align: center; padding: 40px;'>";
    echo "<i class='fas fa-
calendar-times' style='font-size: 48px; color:
#ddd; margin-bottom: 15px; display:
block;'></i>";
    echo "No appointments
found</td></tr>";
} else {
    foreach ($appointments
as $appointment) {
        $visitorName =
htmlspecialchars(($appointment['first_name']
?? ") . ' ' . ($appointment['last_name'] ?? "));
        $visitorId =
htmlspecialchars($appointment['booking_id']
?? 'N/A');
        $contactNumber =
htmlspecialchars($appointment['contact_nu
mber'] ?? 'N/A');
        $appointmentDate =
date('M d, Y',
strtotime($appointment['appointment_date'])
);
        $timeSlots =
formatTimeSlots($appointment['time_slots']
?? 'N/A');
        $purpose =
htmlspecialchars($appointment['purpose'] ??
'N/A');
        $status =
$appointment['status'];
        $appointmentId =
$appointment['appointment_id'];
        $profilePicture =
$appointment['profile_picture'] ?? "";

        // Build profile image
HTML
        $profileImgHtml = "";
        if
(!empty($profilePicture)) {
            $profileImgHtml =
src='.././uploads/profile_pictures/{ $profilePic
ture}' alt='Profile' class='visitor-profile-img'
onerror=\"this.onerror=null;
this.src='.././Pictures/default-
avatar.png';\">";
        } else {
            $profileImgHtml =
"<div class='visitor-profile-placeholder'><i
class='fas fa-user'></i></div>";
        }

        // Determine status
        badge class
        $statusClass =
        $statusText =
        'PENDING';
        if ($status ===
        $statusClass =
        $statusText =
        'Finished') {
            $statusClass =
            $statusText =
            'FINISHED';
        } elseif ($status ===
        $statusClass =
        $statusText =
        'Cancelled') {
            $statusClass =
            $statusText =
            'status-cancelled';
        }
        'CANCELLED';
    }

    echo "<tr data-
appointment-id='{ $appointmentId }'>";
    // Visitor Name Cell
    with Profile Picture and View Details Button
    echo "<td
class='visitor-name-cell'>";
    echo "<div
class='visitor-info-container'>";
    echo "<div
class='visitor-name-wrapper'>";
    echo $profileImgHtml;
    echo "<span
class='visitor-name-text'
title='{ $visitorName }'>{ $visitorName }</span>
";
    echo "</div>";
    echo "<button
class='btn-view-details-small'

```




```
onclick='openVisitorModal({$appointmentId})'><i class='fas fa-eye'></i> View
Details</button>";
        echo "</div>";
        echo "</td>";
        echo
" <td>{$visitorId}</td>";
        echo
" <td>{$contactNumber}</td>";
        echo
" <td>{$appointmentDate}</td>";
        echo " <td class='time-
slot-cell'
title='{$timeSlots}'>{$timeSlots}</td>";
        echo
" <td>{$purpose}</td>";
        echo " <td><span
class='status-badge
{$statusClass}'>{$statusText}</span></td>";
        echo " <td
class='grace-status-cell' id='grace-status-
{$appointmentId}'><span class='grace-
loading'><i class='fas fa-spinner fa-
spin'></i></span></td>";
        // Actions Cell - Now
only for Mark Attended button
        echo " <td
class='actions-cell' id='admin-actions-
{$appointmentId}'>";
        echo " <div
class='action-buttons-wrapper'>";
        echo " <span
class='waiting-action'><i class='fas fa-
hourglass-half'></i></span>";
        echo "</div>";
        echo "</td>";
        echo "</tr>";
    }
    }
    } catch (PDOException $e) {
        echo " <tr><td colspan='9'
style='text-align: center; color: red;'>Error
loading appointments: " .
htmlspecialchars($e->getMessage()) .
" </td></tr>";
    }
    ?>
</tbody>
</table>
```

```
</div>
</div>
<!--End of The Appointment Card--
---->
</main>

<!-- Include External JavaScript -->
<script src="../../Js/toast.js"></script>
<script
src="../../Appointment/Appointment_Admin
_Functions/admin_appointment.js"></script>

<!-- Include Override Control JavaScript -
ADD THIS LINE -->
<script
src="../../Appointment/Appointment_Admin_
Functions/override_control.js"></script>

<!-- Add before </body> after other script
includes -->
<script
src="../../Appointment/Appointment_Admin_
Functions/capacity_settings.js"></script>

<!--Add modal-view.js-->
<script
src="../../Appointment/Appointment_Admin_
Functions/modal-view.js"></script>

<!-- Add impact preview update script -->
<script>
document.addEventListener('DOMContentLoaded', function() {
    const capacityInput =
document.getElementById('maxCapacityInp
ut');

    if (capacityInput) {
        capacityInput.addEventListener('input',
updateImpactPreview);
        updateImpactPreview();
    }

    function updateImpactPreview() {
        const value =
parseInt(capacityInput.value) || 50;
        const perSlot =
document.getElementById('impactPerSlot');
```



```
const dailyTotal =
document.getElementById('impactDailyTotal
');
const weeklyTotal =
document.getElementById('impactWeeklyTo
tal');

if (perSlot) perSlot.textContent = value;
if (dailyTotal) dailyTotal.textContent =
(value * 9).toLocaleString();
if (weeklyTotal)
weeklyTotal.textContent = (value * 9 *
6).toLocaleString(); // 6 days (no Sunday)
}
});
</script>

<script>
// Capacity Monitor Functions
document.addEventListener('DOMContentLoaded', function() {
  console.log('Page loaded, initializing
capacity monitor...');

  // Update current time immediately with
date
updateCurrentTime();

  // Get the date input element
const dateInput =
document.getElementById('capacityDateFilt
er');
if (dateInput) {
  console.log('Date input found, value:',
dateInput.value);

  // Load initial data
loadCapacityByDate(dateInput.value);

  // Update display
updateSelectedDateDisplay();

  // Listen for date changes
dateInput.addEventListener('change',
function() {
  console.log('Date changed to:',
this.value);
  updateSelectedDateDisplay();
  refreshCapacityByDate();
});
});
});
} else {
  console.error('capacityDateFilter
element not found!');
}

// Update time every minute
setInterval(updateCurrentTime, 60000);

// Auto-refresh every 2 minutes
setInterval(() => {
  const currentDate =
document.getElementById('capacityDateFilt
er')?.value;
  if (currentDate) {
    loadCapacityByDate(currentDate);
  }
}, 120000);
});

function refreshCapacityByDate() {
  const selectedDate =
document.getElementById('capacityDateFilt
er').value;
  console.log('Refreshing capacity for:',
selectedDate);
  loadCapacityByDate(selectedDate);
}

function getCurrentTimeSlot() {
  const now = new Date();
  const currentHour = now.getHours();

  // Define time slots (24-hour format)
  const timeSlots = [
    { start: 8, end: 9, label: '8:00 AM - 9:00
AM', key: '08:00-09:00' },
    { start: 9, end: 10, label: '9:00 AM -
10:00 AM', key: '09:00-10:00' },
    { start: 10, end: 11, label: '10:00 AM -
11:00 AM', key: '10:00-11:00' },
    { start: 11, end: 12, label: '11:00 AM -
12:00 PM', key: '11:00-12:00' },
    { start: 12, end: 13, label: '12:00 PM -
1:00 PM', key: '12:00-13:00' },
    { start: 13, end: 14, label: '1:00 PM -
2:00 PM', key: '13:00-14:00' },
    { start: 14, end: 15, label: '2:00 PM -
3:00 PM', key: '14:00-15:00' },
  ]
}
```



```
{ start: 15, end: 16, label: '3:00 PM -
4:00 PM', key: '15:00-16:00' },
{ start: 16, end: 17, label: '4:00 PM -
5:00 PM', key: '16:00-17:00' }
];

for (const slot of timeSlots) {
  if (currentHour >= slot.start &&
currentHour < slot.end) {
    return slot;
  }
}

// If outside operating hours
if (currentHour < 8) {
  return { ...timeSlots[0], isUpcoming:
true, isClosed: false };
} else {
  return { ...timeSlots[timeSlots.length -
1], isClosed: true, isUpcoming: false };
}
}

function getDateStatus(selectedDate) {
  const today = new Date();
  today.setHours(0, 0, 0, 0);

  const selected = new Date(selectedDate
+ 'T00:00:00');
  selected.setHours(0, 0, 0, 0);

  if (selected.getTime() ===
today.getTime()) {
    return 'today';
  } else if (selected.getTime() >
today.getTime()) {
    return 'future';
  } else {
    return 'past';
  }
}

function loadCapacityByDate(date) {
  console.log('Loading capacity for date:',
date);

  const apiUrl =
`../Appointment/Appointment_Admin_Func
```

```
tions/get_capacity_by_date.php?date=${date}`;
  console.log('Fetching from:', apiUrl);

  // Show loading state
  const timeline =
document.getElementById('timeline');
  if (timeline) {
    timeline.innerHTML = `
    <div style="text-align: center;
padding: 40px; color: #999;">
      <i class="fas fa-spinner fa-spin"
style="font-size: 24px;"></i>
      <p>Loading capacity data...</p>
    </div>
  `;
  }

  fetch(apiUrl)
    .then(response => {
      console.log('Response received:',
response.status, response.statusText);
      if (!response.ok) {
        throw new Error(`HTTP
${response.status}:
${response.statusText}`);
      }
      return response.text();
    })
    .then(text => {
      console.log('Raw response:',
text.substring(0, 200));
      try {
        const data = JSON.parse(text);
        console.log('Parsed data:', data);

        if (data.success) {
          updateCapacityDisplay(data);
        } else {
          showErrorMessage(data.messa
ge || 'Error loading capacity data');
        }
      } catch (e) {
        console.error('JSON parse error:',
e);

        console.error('Full response:',
text);

        showErrorMessage('Invalid server
response. Check console for details.');
```



```
    }  
  })  
  .catch(error => {  
    console.error('Fetch error:', error);  
    showErrorMessage('Failed to load  
capacity: ' + error.message);  
  });  
}  
  
function updateCapacityDisplay(data) {  
  console.log('Updating display with data:',  
data);  
  
  const timeline =  
document.getElementById('timeline');  
  if (!timeline) {  
    console.error('Timeline element not  
found');  
    return;  
  }  
  
  timeline.innerHTML = "";  
  
  // Get current time slot and date status  
  const currentSlot = getCurrentTimeSlot();  
  const selectedDate =  
document.getElementById('capacityDateFilt  
er').value;  
  const dateStatus =  
getDateStatus(selectedDate);  
  
  // Get max capacity from data (dynamic)  
  const maxCapacity = data.max_capacity  
|| 50;  
  
  let currentSlotBooked = 0;  
  let currentSlotAvailable = maxCapacity;  
  let currentSlotLabel = 'No Active Slot';  
  
  // Update the Max Per Slot display  
  const maxCapacityDisplay =  
document.querySelector('.stat-box.max  
.stat-value');  
  if (maxCapacityDisplay) {  
    maxCapacityDisplay.textContent =  
maxCapacity;  
  }  
  
  // Create timeline slots
```

```
Object.keys(data.slots).forEach(slotKey  
=> {  
  const slot = data.slots[slotKey];  
  const slotDiv =  
document.createElement('div');  
  slotDiv.className = 'time-slot';  
  
  // Use dynamic max_capacity from  
each slot  
  const slotMaxCapacity =  
slot.max_capacity || maxCapacity;  
  const available = slot.available;  
  const booked = slot.booked;  
  const percentage = slot.percentage;  
  
  // Determine slot status  
  let statusClass = 'available';  
  if (slot.is_full || available <= 0) {  
    statusClass = 'occupied';  
  } else if (percentage >= 70) {  
    statusClass = 'partial';  
  }  
  
  slotDiv.classList.add(statusClass);  
  
  // Check if this is the current slot (for  
today)  
  const isCurrentSlot = dateStatus ===  
'today' && currentSlot && slotKey ===  
currentSlot.key;  
  if (isCurrentSlot) {  
    slotDiv.classList.add('current');  
    currentSlotBooked = booked;  
    currentSlotAvailable = available;  
    currentSlotLabel = slot.label;  
  }  
  
  slotDiv.innerHTML = `  
    <div class="slot-  
time">${slot.label}</div>  
    <div class="slot-status">${available}  
Available</div>  
    <div class="slot-  
count">${booked}/${slotMaxCapacity}  
booked</div>  
  `;  
  
  timeline.appendChild(slotDiv);  
});
```



```
// Update summary stats
const currentCountEl =
document.getElementById('currentCount');
const availableSlotsEl =
document.getElementById('availableSlots');
const capacityProgressEl =
document.getElementById('capacityProgress');
const currentSlotLabelEl =
document.getElementById('currentSlotLabel');

// Format selected date for display
const selectedDateObj = new
Date(selectedDate + 'T00:00:00');
const formattedDate =
selectedDateObj.toLocaleDateString('en-
US', {
  weekday: 'long',
  month: 'long',
  day: 'numeric',
  year: 'numeric'
});

if (dateStatus === 'future') {
  // Future date - show totals
  if (currentCountEl)
currentCountEl.textContent =
data.summary.total_booked;
  if (availableSlotsEl)
availableSlotsEl.textContent =
data.summary.total_available;
  if (currentSlotLabelEl)
currentSlotLabelEl.innerHTML = `<i
class="fas fa-calendar-day"></i>
${formattedDate}`;

  // Update progress bar
  if (capacityProgressEl) {
    const percentage =
data.summary.percentage || 0;
    capacityProgressEl.style.width =
percentage + '%';
    capacityProgressEl.className =
'progress-fill';
    if (percentage >= 90) {
      capacityProgressEl.classList.add('
danger');
    } else if (percentage >= 70) {
      capacityProgressEl.classList.add('
warning');
    } else {
      capacityProgressEl.classList.add('
available');
    }
  }
} else if (dateStatus === 'past') {
  // Past date
  if (currentCountEl)
currentCountEl.textContent =
data.summary.total_booked;
  if (availableSlotsEl)
availableSlotsEl.textContent = '-';
  if (currentSlotLabelEl)
currentSlotLabelEl.innerHTML = `<i
class="fas fa-history"></i> ${formattedDate}
(Past)`;

  if (capacityProgressEl) {
    capacityProgressEl.style.width =
'100%';
    capacityProgressEl.className =
'progress-fill closed';
  }
} else if (dateStatus === 'today') {
  // Today - show current slot info
  if (currentCountEl)
currentCountEl.textContent =
currentSlotBooked;
  if (availableSlotsEl)
availableSlotsEl.textContent =
currentSlotAvailable;

  if (currentSlot && !currentSlot.isClosed)
{
    if (currentSlotLabelEl)
currentSlotLabelEl.innerHTML = `<i
class="fas fa-clock"></i> Current:
${currentSlotLabel}`;
  } else {
    if (currentSlotLabelEl)
currentSlotLabelEl.innerHTML = `<i
class="fas fa-moon"></i> Closed for Today`;
  }

  // Update progress bar based on
current slot
  if (capacityProgressEl) {
```



```
const percentage = maxCapacity > 0
? (currentSlotBooked / maxCapacity) * 100 :
0;
capacityProgressEl.style.width =
percentage + '%';
capacityProgressEl.className =
'progress-fill';
if (percentage >= 90) {
    capacityProgressEl.classList.add('
danger');
} else if (percentage >= 70) {
    capacityProgressEl.classList.add('
warning');
} else {
    capacityProgressEl.classList.add('
available');
}
}
}
</script>

<script>
function showErrorMessage(message) {
    const timeline =
document.getElementById('timeline');
    if (timeline) {
        timeline.innerHTML = `
        <div style="text-align: center;
padding: 40px; color: #dc3545;">
        <i class="fas fa-exclamation-
circle" style="font-size: 24px; margin-bottom:
15px;"></i>
        <p style="margin: 10px 0; font-
weight: 600;">${message}</p>
        <button
onclick="refreshCapacityByDate()"
style="margin-top: 15px;
padding: 10px 20px; background: #800000;
color: white; border: none; border-radius:
4px; cursor: pointer; font-size: 14px;">
        <i class="fas fa-sync"></i> Try
Again
        </button>
        </div>
        `;
    }
}

function updateSelectedDateDisplay() {
    const dateInput =
document.getElementById('capacityDateFilt
er');
    const displayEl =
document.getElementById('selectedDateDis
play');

    if (dateInput && displayEl) {
        const selectedDate = new
Date(dateInput.value + 'T00:00:00');
        const options = { weekday: 'long', year:
'numeric', month: 'long', day: 'numeric' };
        const formattedDate =
selectedDate.toLocaleDateString('en-US',
options);
        displayEl.textContent = formattedDate;
    }
}

function updateCurrentTime() {
    const timeEl =
document.getElementById('currentTimeDisp
lay');
    if (timeEl) {
        const now = new Date();
        const timeString =
now.toLocaleTimeString('en-US', {
            hour: 'numeric',
            minute: '2-digit',
            hour12: true
        });
        const dateString =
now.toLocaleDateString('en-US', {
            weekday: 'short',
            month: 'short',
            day: 'numeric',
            year: 'numeric'
        });
        timeEl.innerHTML = `<i class="fas fa-
clock"></i> ${dateString} | ${timeString}`;
    }

    // Also refresh capacity to update current
    slot (only if viewing today)
    const currentDate =
document.getElementById('capacityDateFilt
er')?.value;
```



```
const today = new Date().toISOString().split('T')[0];
if (currentDate === today) {
    loadCapacityByDate(currentDate);
}
}

// Toggle Calendar View
function toggleCalendarView() {
    const calendarSection = document.getElementById('calendarSection');
    const isVisible = calendarSection.style.display !== 'none';

    if (isVisible) {
        // Hide calendar
        calendarSection.style.display = 'none';
        document.querySelector('.btn-close-calendar')?.classList.remove('show');
    } else {
        // Show calendar
        calendarSection.style.display = 'block';
        document.querySelector('.btn-close-calendar')?.classList.add('show');

        // Load calendar if not already loaded
        if (!calendarSection.classList.contains('initialized')) {
            loadCalendarDays();
            calendarSection.classList.add('initialized');
        }
    }
}

// Toggle Capacity Settings Panel
function toggleCapacitySettings() {
    const panel = document.getElementById('capacitySettingsPanel');
    const isVisible = panel.style.display !== 'none';

    if (isVisible) {
        // Hide panel
        panel.style.display = 'none';
    } else {
        // Show panel
        panel.style.display = 'block';

        // Load current capacity settings
        loadCurrentCapacitySettings();

        // Simulate loading
        label.textContent = 'Loading...';
        input.value = '';

        // Fetch current settings from server (simulate with timeout)
        setTimeout(() => {
            // For demo, set static values
            const currentMaxCapacity = 50;

            label.textContent = currentMaxCapacity;
            input.value = currentMaxCapacity;
        }, 1000);

        // Preset button click
        document.querySelectorAll('.capacity-preset-btn').forEach(button => {
            button.addEventListener('click', function() {
                const value = this.getAttribute('data-value');

                // Remove active class from all buttons
                document.querySelectorAll('.capacity-preset-btn').forEach(btn => btn.classList.remove('active'));

                // Add active class to the clicked button
            });
        });
    }
}
```




```
this.classList.add('active');

// Set the input value
const input =
document.getElementById('maxCapacityInput');
if (input) {
    input.value = value;
}

// Update impact preview
updateImpactPreview(value);
});

// Update impact preview based on capacity
function updateImpactPreview(capacity) {
    const perSlot =
document.getElementById('impactPerSlot');
    const dailyTotal =
document.getElementById('impactDailyTotal');
    const weeklyTotal =
document.getElementById('impactWeeklyTotal');

    if (perSlot && dailyTotal && weeklyTotal) {
        const slotValue = parseInt(capacity, 10);
        const dailyValue = slotValue * 9;
        const weeklyValue = dailyValue * 7;

        perSlot.textContent = slotValue;
        dailyTotal.textContent = dailyValue;
        weeklyTotal.textContent = weeklyValue;
    }
}

// Reset capacity to default
document.getElementById('resetCapacityBtn').addEventListener('click', function() {
    const input =
document.getElementById('maxCapacityInput');
    if (input) {
        input.value = 50;

        // Update active preset button
        document.querySelectorAll('.capacity-preset-btn').forEach(btn =>
        btn.classList.remove('active'));
        document.querySelector('.capacity-preset-btn[data-value="50"]')?.classList.add('active');

        // Update impact preview
        updateImpactPreview(50);
    }
});

// Apply capacity changes
document.getElementById('applyCapacityBtn').addEventListener('click', function() {
    const input =
document.getElementById('maxCapacityInput');
    const value = input.value;

    if (value && !isNaN(value)) {
        // Send update request to server
        // For demo, just log the value
        console.log('New max capacity:', value);

        // Show success message (demo)
        showToast('Capacity updated successfully!', 'success');

        // Close the panel
        toggleCapacitySettings();
    } else {
        showToast('Please enter a valid number for capacity.', 'warning');
    }
});
</script>

<!-- Include admin grace control -->
<script
src="../../../Appointment/js/admin_grace_control.js"></script>
<script
src="../../../Js/SidebarToggle.js"></script>
<script src="../../../NotificationBadge/badge_refresh.js"></script>
</body>
</html>
```




```
ArchivedBooks.php
<?php
if (session_status() ===
PHP_SESSION_NONE) {
    session_start();
}

// Include database connection
require_once
'../../DBconnection/db_connection.php';
//Notification Badge Function
if (!isset($badges) && isset($pdo)) {
    require_once __DIR__ . '/../Notification
Badge/Notification Badge.php';
}

$profileImage =
'../../Pictures/PUP_LOGO.png'; // Default
image path

////////////////////
/////
////////////////////Fetching Profile Image from the
Database////////////////////

// Get admin_id from the session
$admin_id = $_SESSION['admin_id'] ?? '';
$defaultImagePath =
'../../Pictures/PUP_LOGO.png'; // Default
image path
$profileImage = $defaultImagePath; //
Initialize with default

// When fetching the profile image
try {
    $stmt = $pdo->prepare("SELECT
profile_image FROM admin WHERE
admin_id = ?");
    $stmt->execute([$admin_id]);
    $result = $stmt-
>fetch(PDO::FETCH_ASSOC);

    if ($result &&
!empty($result['profile_image'])) {
        $profileImage =
getImageUrl($result['profile_image']);
    } else {
        $profileImage =
DEFAULT_ADMIN_IMAGE;
    }
} catch (PDOException $e) {
    error_log("Error fetching admin profile: " .
$e->getMessage());
    $profileImage =
DEFAULT_ADMIN_IMAGE;
}

// Search functionality
$search = isset($_GET['search']) ?
trim($_GET['search']) : '';
$category_filter =
isset($_GET['category_filter']) ?
$_GET['category_filter'] : '';

// Fetch categories dynamically from the
database
try {
    $categoryStmt = $pdo->query("SELECT
category_name FROM categories WHERE
is_active = 1 ORDER BY category_name
ASC");
    $categories = $categoryStmt-
>fetchAll(PDO::FETCH_COLUMN);
} catch (PDOException $e) {
    error_log("Error fetching categories: " .
$e->getMessage());
    $categories = [];
}

// Fetch archived books
try {
    $query = "SELECT * FROM
archived_books";
    $params = [];

    // Add search condition
    if (!empty($search)) {
        $query .= " WHERE (title LIKE ? OR
author LIKE ? OR control_no LIKE ?)";
        $search_param = "%{$search}%";
        $params = array_merge($params,
[$search_param, $search_param,
$search_param]);
    }

    // Add category filter
    if (!empty($category_filter)) {

```



```
$query .= empty($search) ? " WHERE
category = ?" : " AND category = ?";
$params[] = $category_filter;
}

$query .= " ORDER BY archived_date
DESC";

$stmt = $pdo->prepare($query);
$stmt->execute($params);
$archived_books = $stmt-
>fetchAll(PDO::FETCH_ASSOC);
} catch (PDOException $e) {
    $error_message = "Database error: " . $e-
>getMessage();
}
?>
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    <title>Archived Books - PUP Bansud
Campus</title>
    <link rel="stylesheet"
href="../../Design/Admin.css">
    <link rel="stylesheet"
href="../../Design/AdminProfile.css">
    <link rel="stylesheet"
href="../../Design/AdminNotification.css">
    <link rel="stylesheet"
href="../../Design/AdminSidebar.css">
    <link rel="stylesheet"
href="../../Design/ArchiveBooks.css">
    <link rel="stylesheet" href="../../Notification
Badge/badge.css">
    <link rel="icon" type="image/x-icon"
href="../../Pictures/PUP_LOGO.png">
    <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
</head>
<body>
    <?php include_once __DIR__ .
'/sidebar_toggle.php'; ?>
    <!-- Sidebar Navigation -->
```

```
<nav class="sidebar" id="sidebar">
    <div class="sidebar-admin-profile"
id="sidebarAdminProfile"
style="cursor:pointer;"
onclick="window.location.href='../AdminProfil
e.php'">
        ';">
        <div class="sidebar-admin-
username"><h6>Welcome!
Admin</h6><?php echo
htmlspecialchars($_SESSION['admin_usern
ame'] ?? 'admin_user'); ?></div>
        </div>
        <ul>
            <li><a href="<?php echo
$sidebarBasePath ?? '../';
?>Admin_Dashboard.php" class="<?php
echo ($currentPage ?? '') === 'dashboard' ?
'active' : ''; ?>"><i class="fa-solid fa-
grip"></i>
<span>Dashboard</span></a></li>
            <li class="section-header Main-
header">Main</li>
            <li><a href="<?php echo
$componentPath ?? '';
?>PendingApproval.php" class="<?php
echo ($currentPage ?? '') === 'pending' ?
'active' : ''; ?>"><i class="fas fa-user-
clock"></i> <span>Pending
Users</span><?php echo
renderBadge($badges['pending_users'] ??
0); ?></a></li>
            <li><a href="<?php echo
$componentPath ?? '';
?>UserManagement.php" class="<?php
echo ($currentPage ?? '') === 'users' ?
'active' : ''; ?>"><i class="fas fa-users"></i>
<span>User Management</span><?php
echo
renderBadge($badges['user_management']
?? 0); ?></a></li>
            <li><a href="<?php echo
$componentPath ?? '';
?>ArchivedUsers.php" class="<?php echo
```



```
($currentPage ?? ") === 'archivedusers' ?
'active' : "; ?>"<i class="fas fa-user-
slash"></i> <span>Archived
Users</span><?php echo
renderBadge($badges['archived_users'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>Admin_Notification.php" class="<?php
echo ($currentPage ?? ") === 'notifications'
? 'active' : "; ?>"><i class="fas fa-bell"></i>
<span>Notifications</span><?php echo
renderBadge($badges['notifications'] ?? 0);
?></a></li>
    <li class="section-header books-
inventory-header">Books Inventory</li>
    <li><a href="<?php echo
$componentPath ?? "; ?>AddBooks.php"
class="<?php echo ($currentPage ?? ") ===
'addbooks' ? 'active' : "; ?>"><i class="fa-
solid fa-list-check"></i><span>Manage
Books</span><?php echo
renderBadge($badges['manage_books'] ??
0); ?></a></li>
    <li><a href="<?php echo $componentPath
?? '../Narrative-Research-Capstone/';
?>addBooks.php" class="<?php echo
($currentPage ?? ") === 'addbooks' ?
'active' : "; ?>"><i class="fa-brands fa-
researchgate"></i><span>Research
Books</span><?php echo
renderBadge($badges['narrative_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>BorrowedBooks.php" class="<?php echo
($currentPage ?? ") === 'borrowed' ? 'active'
: "; ?>"><i class="fas fa-download"></i>
<span>Borrowed Books</span><?php echo
renderBadge($badges['borrowed_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? "; ?>Non-
ReturnedBooks.php" class="<?php echo
($currentPage ?? ") === 'nonreturned' ?
'active' : "; ?>"><i class="fas fa-ban"></i>
<span>Non-Returned Books</span><?php
echo
renderBadge($badges['non_returned_books'
] ?? 0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ReturnedBooks.php" class="<?php echo
($currentPage ?? ") === 'returned' ? 'active' :
"; ?>"><i class="fas fa-upload"></i>
<span>Returned Books</span><?php echo
renderBadge($badges['returned_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ReplacedBooks.php" class="<?php echo
($currentPage ?? ") === 'replaced' ? 'active'
: "; ?>"><i class="fas fa-sync"></i>
<span>Replaced Books</span><?php echo
renderBadge($badges['replaced_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? "; ?>PaidBooks.php"
class="<?php echo ($currentPage ?? ") ===
'paidbook' ? 'active' : "; ?>"><i class="fa-
solid fa-money-bill"></i><span>Paid
Books</span><?php echo
renderBadge($badges['paid_book'] ?? 0);
?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ArchivedBooks.php" class="<?php echo
($currentPage ?? ") === 'archived' ? 'active'
: "; ?>"><i class="fas fa-archive"></i>
<span>Archived Books</span><?php echo
renderBadge($badges['archived_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? '../Narrative-
Research-Capstone/';
?>archived_research_books/archived-
research-books.php" class="<?php echo
($currentPage ?? ") === 'archivedresearch'
? 'active' : "; ?>"><i class="fa-solid fa-box-
archive"></i> <span>Archived
Research</span><?php echo
renderBadge($badges['archived_research_b
ooks'] ?? 0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ArchivedOverdueBooks.php"
class="<?php echo ($currentPage ?? ") ===
```



```
'archivedoverdue' ? 'active' : ''; ?>"<i
class="fas fa-exclamation-triangle"></i>
<span>Archived Overdue</span><?php
echo
renderBadge($badges['archived_overdue']
?? 0); ?></a></li>
<li class="section-header appointment-
header">Appointment</li>
<li><a href="<?php echo
$componentPath ?? ";
?>Appointment_Management.php"
class="<?php echo ($currentPage ?? ") ===
'appointments' ? 'active' : ''; ?>"><i
class="fas fa-calendar-check"></i>
<span>Appointment
Management</span><?php echo
renderBadge($badges['appointments'] ?? 0);
?></a></li>
<li class="section-header">Reports</li>
<li><a href="<?php echo
$componentPath ?? ";
?>ReportAnalytics.php" class="<?php echo
($currentPage ?? ") === 'reports' ? 'active' :
"; ?>"><i class="fas fa-chart-line"></i>
<span>Reports</span></a></li>
<li class="sidebar-separator"></li>
<li><a href="<?php echo
$sidebarBasePath ?? './../';
?>Admin_Logout.php"><i class="fas fa-sign-
out-alt"></i> <span>Logout</span></a></li>
</ul>
</nav>

<!-- Main Content Area -->
<div class="main-content">
<header class="admin-header">
<div class="header-content">
<h1>Archived Books</h1>
<div class="header-actions">
<button id="restoreSelectedBtn"
class="btn-primary" disabled><i class="fas
fa-undo"></i> Restore Selected</button>
</div>
</div>
</header>

<!-- Success/Error Messages -->
<?php if
(isset($_SESSION['book_message'])): ?>
```

```
<div class="message <?php echo
htmlspecialchars($_SESSION['book_messa
ge_type'], ENT_QUOTES, 'UTF-8'); ?>">
<?php
echo
htmlspecialchars($_SESSION['book_messa
ge'], ENT_QUOTES, 'UTF-8');
unset($_SESSION['book_mess
age']);
unset($_SESSION['book_mess
age_type']);
?>
</div>
<?php endif; ?>

<!-- Search and Filter -->
<div class="search-filter-container">
<form method="GET" action=""
class="search-form">
<div class="search-input-
container">
<input type="text"
name="search" placeholder="Search by title,
author or Control No..." value="<?php echo
htmlspecialchars($search); ?>"
class="search-input">
<button type="submit"
class="search-button"><i class="fas fa-
search"></i></button>
</div>
<div class="filter-container">
<select name="category_filter"
class="filter-select">
<option value="">All
Categories</option>
<?php foreach ($categories
as $category): ?>
<option value="<?php
echo htmlspecialchars($category); ?>"
<?php echo $category_filter === $category
? 'selected' : ''; ?>>
<?php echo
htmlspecialchars($category); ?>
</option>
<?php endforeach; ?>
</select>
<button type="submit"
class="filter-button">Filter</button>
```



```

<?php if (!empty($search) ||
!empty($category_filter)): ?>
    <a href="ArchivedBooks.php"
class="reset-button">Reset</a>
    <?php endif; ?>
</div>
</form>
</div>

<!-- Archived Books Table -->
<div class="books-container">
    <table class="books-table">
        <thead>
            <tr>
                <th><input type="checkbox"
id="selectAll"></th>
                <th>Image</th>
                <th>Title</th>
                <th>Author</th>
                <th>Control No.</th>
                <th>Volume</th>
                <th>Edition</th>
                <th>Category</th>
                <th>Price</th>
                <th>Book Condition</th>
                <th>Archive Date</th>
                <th>Archived By</th>
                <th>Reason</th>
                <th>Actions</th>
            </tr>
        </thead>
        <tbody>
            <?php if
(isset($archived_books) &&
count($archived_books) > 0): ?>
                <?php foreach
($archived_books as $book): ?>
                    <tr>
                        <td><input
type="checkbox" class="book-select" data-
id="<?php echo $book['archive_id'];
?>"></td>
                        <td>
                            <?php if
(!empty($book['image_path'])): ?>
                                
                                <?php else: ?>
                                    <div class="no-
image">No Image</div>
                                <?php endif; ?>
                            </td>
                            <td><?php echo
htmlspecialchars($book['title']); ?></td>
                            <td><?php echo
htmlspecialchars($book['author']); ?></td>
                            <td><?php echo
htmlspecialchars($book['control_no'] ??
'N/A'); ?></td>
                            <td><?php echo
htmlspecialchars($book['volume'] ?? 'N/A');
?></td>
                            <td><?php echo
htmlspecialchars($book['edition'] ?? 'N/A');
?></td>
                            <td><?php echo
htmlspecialchars($book['category'] ??
'Uncategorized'); ?></td>
                            <td><?php echo
!empty($book['book_price']) ? 'P' .
number_format($book['book_price'], 2) :
'N/A'; ?></td>
                            <td>
                                <?php
                                    $condition =
$book['book_condition'] ?? 'N/A';
                                    $condClass = "";
                                    switch(strtolower($
condition)) {
                                        case 'good':
                                            $condClass = 'condition-good'; break;
                                        case 'damaged':
                                            $condClass = 'condition-damaged'; break;
                                        case 'poor':
                                            $condClass = 'condition-poor'; break;
                                        default:
                                            $condClass = 'condition-default'; break;
                                    }
                                <?>
                                <span class="book-
condition-badge <?php echo $condClass;
?>"><?php echo
htmlspecialchars(ucfirst($condition));
?></span>
                            </td>
                        </tr>
                    </?php endforeach; ?>
                </?php if; ?>
            </?php if; ?>
        </tbody>
    </table>
</div>

```



<pre><td><?php echo date('M d, Y', strtotime(\$book['archived_date'])); ?></td> <td><?php echo htmlspecialchars(\$book['archived_by']); ?></td> <td> <?php \$reason = !empty(\$book['archive_reason']) ? \$book['archive_reason'] : 'No reason provided'; echo strlen(\$reason) > 50 ? htmlspecialchars(substr(\$reason, 0, 50)) . '...' : htmlspecialchars(\$reason); ?> </td> <td class="actions"> <button class="btn- restore" data-id="<?php echo \$book['archive_id']; ?>" title="Restore Book"><i class="fas fa-undo"></i></button> <button class="btn- view" data-id="<?php echo \$book['archive_id']; ?>" title="View Details"><i class="fas fa-eye"></i></button> </td> </tr> <?php endforeach; ?> <?php else: ?> <tr> <td colspan="14" class="no-data"> <?php if (!empty(\$search) !empty(\$category_filter)): ?> No archived books found matching your search criteria. Try different keywords or filters. <?php else: ?> No archived books available. <?php endif; ?> </td> </tr> <?php endif; ?> </tbody> </table></pre>	<pre></div> <!-- Book Details Modal --> <div id="bookDetailsModal" class="modal"> <div class="modal-content"> &times; <h2>Book Details</h2> <div id="bookDetails"></div> </div> </div> </div> <script src="../../Js/toast.js"></script> <script src="../../Js/ArchiveBooks.js"></script> <script src="../../Js/SidebarToggle.js"></script> <script src="../../Notification Badge/badge_refresh.js"></script> </body> </html> ArchivedOverdueBooks.php <?php require_once '../../DBconnection/Config.php'; require_once '../../DBconnection/db_connection.php'; require_once '../../DBconnection/ensure_tables.php'; // Include settlement mailer require_once __DIR__ . '../../api_books/archived-overdue- settlement-mailer.php'; require_once __DIR__ . '../../api_books/book_sms_helper.php'; // Badge Notification Function if (!isset(\$badges) && isset(\$pdo)) { require_once __DIR__ . '../../Notification Badge/Notification Badge.php'; } if (session_status() === PHP_SESSION_NONE) { session_start(); }</pre>
---	---



```
// Only define the constant if it's not already defined
if (!defined('DEFAULT_ADMIN_IMAGE')) {
    define('DEFAULT_ADMIN_IMAGE',
        './../Pictures/PUP_LOGO.png');
}

// Get admin_id from the session
$admin_id = $_SESSION['admin_id'] ?? '';
$profileImage = DEFAULT_ADMIN_IMAGE;

// Fetch admin profile image
try {
    $stmt = $pdo->prepare("SELECT
profile_image FROM admin WHERE
admin_id = ?");
    $stmt->execute([$admin_id]);
    $result = $stmt-
>fetch(PDO::FETCH_ASSOC);

    if ($result &&
!empty($result['profile_image'])) {
        $profileImage =
getImageUrl($result['profile_image'],
DEFAULT_ADMIN_IMAGE);
    }
} catch (PDOException $e) {
    error_log("Error fetching admin profile: " .
$e->getMessage());
}

// Handle AJAX requests
if (isset($_POST['action'])) {
    header('Content-Type: application/json');

    switch ($_POST['action']) {
        case 'get_archived_overdue':
            try {
                $search = isset($_POST['search'])
? trim($_POST['search']) : '';
                $userType =
isset($_POST['user_type']) ?
$_POST['user_type'] : 'all';
                $status = isset($_POST['status'])
? $_POST['status'] : 'all';

                $query = "
                    SELECT aob.*,
                    CASE
                        WHEN aob.user_type =
'student' THEN CONCAT(s.first_name, ' ',
s.last_name)
                        WHEN aob.user_type =
'faculty' THEN CONCAT(f.first_name, ' ',
f.last_name)
                        ELSE aob.user_id
                    END as user_name,
                    CASE
                        WHEN aob.user_type =
'student' THEN s.email
                        WHEN aob.user_type =
'faculty' THEN f.email
                        ELSE NULL
                    END as user_email,
                    b.image_path
                FROM
                archived_overdue_books aob
                LEFT JOIN student s ON
aob.user_id = s.student_id AND
aob.user_type = 'student'
                LEFT JOIN faculty f ON
aob.user_id = f.faculty_id AND
aob.user_type = 'faculty'
                LEFT JOIN add_books b ON
aob.book_id = b.book_id
                WHERE 1=1
                ";

                $params = [];

                if (!empty($search)) {
                    $searchTerm = '%' . $search .
'%';

                    $query .= " AND (aob.book_title
LIKE ? OR aob.user_id LIKE ? OR
aob.book_author LIKE ?)";
                    $params =
array_merge($params, [$searchTerm,
$searchTerm, $searchTerm]);
                }

                if ($userType !== 'all') {
                    $query .= " AND aob.user_type
= ?";
                    $params[] = $userType;
                }

                if ($status !== 'all') {
```



```
$query .= " AND
aob.settlement_status = ?";
$params[] = $status;
}

$query .= " ORDER BY
aob.archived_date DESC";

$stmt = $pdo->prepare($query);
$stmt->execute($params);
$records = $stmt->fetchAll(PDO::FETCH_ASSOC);

// Calculate days since archived
foreach ($records as &$record) {
    $archivedDate = new
DateTime($record['archived_date']);
    $now = new DateTime();
    $interval = $archivedDate-
>diff($now);
    $record['days_since_archived']
= $interval->days;

    // Format replacement type
    if ($record['replacement_type']
=== 'replace_whole_book') {
        $record['replacement_type_d
isplay'] = 'Replace Book';
    } elseif
($record['replacement_type'] ===
'pay_for_book') {
        $record['replacement_type_d
isplay'] = 'Pay for Book';
    } else {
        $record['replacement_type_d
isplay'] = 'Not specified';
    }

    // Prepare display fields
    $original_total =
isset($record['total_accumulated_penalty']) ?
floatval($record['total_accumulated_penalty']
) : 0.00;
    $base_cost =
isset($record['base_replacement_cost']) ?
floatval($record['base_replacement_cost']) :
0.00;

    if ($record['replacement_type']
=== 'replace_whole_book') {
        // For replace_whole_book:
Maximum overdue penalty is capped at
base cost
        // Total = base_cost +
min(overdue, base_cost)
        // So max liability =
base_cost * 2, but user only pays overdue
(not base since book is replaced)
        $overdueAmount = max(0,
$original_total - $base_cost);
        $cappedOverdue =
min($overdueAmount, $base_cost);
        $record['display_total_accum
ulated_penalty'] = $cappedOverdue; // Only
the overdue amount for replace_whole_book
        $record['maximum_liability_di
splay'] = $base_cost; // Maximum they could
owe in overdue penalties
    } else {
        // For pay_for_book: Total =
base_cost + overdue (capped at base_cost)
        $record['display_total_accum
ulated_penalty'] = min($original_total,
$base_cost * 2);
        $record['maximum_liability_di
splay'] = $original_total;
    }

    // Always show DAMAGE/LOST
for condition display
    $record['condition_display'] =
'DAMAGE/LOST';
}

echo json_encode([
    'success' => true,
    'data' => $records,
    'count' => count($records)
]);
} catch (PDOException $e) {
    echo json_encode(['success' =>
false, 'message' => $e->getMessage()]);
}
exit;

case 'settle_archived':
try {
```




```
if (!isset($_SESSION['admin_id']))
{
    echo json_encode(['success' =>
false, 'message' => 'Unauthorized']);
    exit;
}

$archiveId =
isset($_POST['archive_id']) ?
intval($_POST['archive_id']) : 0;
$notes = isset($_POST['notes']) ?
trim($_POST['notes']) : "";

if (!$archiveId) {
    echo json_encode(['success' =>
false, 'message' => 'Invalid archive ID']);
    exit;
}

$db->beginTransaction();

// Get archive record
$stmt = $db->prepare("SELECT
* FROM archived_overdue_books WHERE
archive_id = ?");
$stmt->execute([$archiveId]);
$archive = $stmt-
>fetch(PDO::FETCH_ASSOC);

if (!$archive) {
    $db->rollBack();
    echo json_encode(['success' =>
false, 'message' => 'Archive record not
found']);
    exit;
}

if ($archive['settlement_status']
=== 'settled') {
    $db->rollBack();
    echo json_encode(['success' =>
false, 'message' => 'This record is already
settled']);
    exit;
}

// Update archive record
$stmt = $db->prepare("

UPDATE
archived_overdue_books
SET settlement_status =
'settled',
    settlement_date = NOW(),
    settlement_processed_by =
?,
    settlement_notes = ?
WHERE archive_id = ?
");
$stmt-
>execute([$SESSION['admin_id'], $notes,
$archiveId]);

// Update original returned_books
record
// For replace_whole_book:
fine_amount should be overdue-only (base
cost waived since book was physically
replaced)
$settleFineAmount =
floatval($archive['total_accumulated_penalty
']);
if ($archive['replacement_type']
=== 'replace_whole_book') {
    $settleFineAmount = max(0,
$settleFineAmount -
floatval($archive['base_replacement_cost']));
}
$stmt = $db->prepare("
UPDATE returned_books
SET replacement_status =
'completed',
    replacement_completed_dat
e = NOW(),
    fine_amount = ?,
    notes =
CONCAT(IFNULL(notes, ''), ' | Settled from
archived on ', NOW(), ' ', ?)
WHERE return_id = ?
");
$stmt-
>execute([$settleFineAmount, $notes,
$archive['return_id']]);

// If it was a replacement type,
increase book stock
if ($archive['replacement_type']
=== 'replace_whole_book') {
```



```
$stmt = $pdo->prepare("
    UPDATE add_books
    SET stock_quantity =
stock_quantity + 1,
        status = 'Available'
    WHERE book_id = ?
");
$stmt->execute([$archive['book_id']]);

// Create notification for user
$notificationMessage = "Your
archived overdue obligation for
\"{$archive['book_title']}\" has been settled. "
.
        "Amount paid: ₱" .
number_format($archive['total_accumulated
_penalty'], 2) . ". " .
        "Your borrowing
privileges have been restored.";

$stmt = $pdo->prepare("
    INSERT INTO
book_notifications (user_id, user_type,
message, notification_type, book_id,
is_read, created_at)
    VALUES (?, ?, ?,
'settlement_confirmed', ?, 0, NOW())
");
$stmt->execute([
    $archive['user_id'],
    $archive['user_type'],
    $notificationMessage,
    $archive['book_id']
]);

$pdo->commit();

// Send settlement email receipt
$receiptNumber = null;
try {
    $emailResult =
sendArchivedOverdueSettlementEmail($pdo
, $archive, $_SESSION['admin_id'], $notes);
    if ($emailResult['success']) {
        $receiptNumber =
$emailResult['receipt_number'] ?? null;
        error_log("Settlement email
sent successfully - Receipt: " .
$receiptNumber);
    } else {
        error_log("Failed to send
settlement email: " .
$emailResult['message']);
    }
} catch (Exception $emailError) {
    error_log("Email sending error
(non-critical): " . $emailError-
->getMessage());
}

// Send SMS to user about
archived overdue settlement
try {
    $archSmsMsg = 'Your archived
overdue obligation for "' .
($archive['book_title'] ?? 'a book') . '" has
been settled. Your borrowing privileges are
restored. Check your email for the receipt.';
    sendBookSMSToUser($pdo,
    $archive['user_id'], $archive['user_type'],
    $archSmsMsg, ['book_id' =>
    $archive['book_id']]);
} catch (Exception $smsErr) {
    error_log("Archived overdue
settlement SMS failed (non-critical): " .
    $smsErr->getMessage());
}

echo json_encode([
    'success' => true,
    'message' => 'Obligation settled
successfully. User borrowing privileges
restored.',
    'receipt_number' =>
$receiptNumber,
    'email_sent' =>
($receiptNumber !== null)
]);
} catch (PDOException $e) {
    if ($pdo->inTransaction()) {
        $pdo->rollBack();
    }
    echo json_encode(['success' =>
false, 'message' => $e->getMessage()]);
}
```



```
exit;

case 'get_archive_details':
    try {
        $archiveId =
isset($_POST['archive_id']) ?
intval($_POST['archive_id']) : 0;

        if (!$archiveId) {
            echo json_encode(['success' =>
false, 'message' => 'Invalid archive ID']);
            exit;
        }

        $stmt = $pdo->prepare("
            SELECT aob.*,
            CASE
                WHEN aob.user_type =
'student' THEN CONCAT(s.first_name, ' ',
s.last_name)
                WHEN aob.user_type =
'faculty' THEN CONCAT(f.first_name, ' ',
f.last_name)
                ELSE aob.user_id
            END as user_name,
            CASE
                WHEN aob.user_type =
'student' THEN s.email
                WHEN aob.user_type =
'faculty' THEN f.email
            END as user_email,
            b.image_path,
            b.category,
            b.control_no AS lcc
        FROM
        archived_overdue_books aob
        LEFT JOIN student s ON
aob.user_id = s.student_id AND
aob.user_type = 'student'
        LEFT JOIN faculty f ON
aob.user_id = f.faculty_id AND
aob.user_type = 'faculty'
        LEFT JOIN add_books b ON
aob.book_id = b.book_id
        WHERE aob.archive_id = ?
    ");
        $stmt->execute([$archiveId]);
        $archive = $stmt-
>fetch(PDO::FETCH_ASSOC);

        // Prepare display fields for single
        archive result
        if ($archive) {
            $original_total =
isset($archive['total_accumulated_penalty'])
?
floatval($archive['total_accumulated_penalty
']) : 0.00;
            $base_cost =
isset($archive['base_replacement_cost']) ?
floatval($archive['base_replacement_cost']) :
0.00;

            if ($archive['replacement_type']
=== 'replace_whole_book') {
                // For replace_whole_book:
                Maximum overdue penalty is capped at
                base cost
                // User only pays overdue
                penalty (not base since book is replaced)
                $overdueAmount = max(0,
                $original_total - $base_cost);
                $cappedOverdue =
min($overdueAmount, $base_cost);
                $archive['display_total_accu
                mulated_penalty'] = $cappedOverdue;
                $archive['maximum_liability_
                display'] = $base_cost;
            } else {
                // For pay_for_book: Total =
                base_cost + overdue (capped at base_cost)
                $archive['display_total_accu
                mulated_penalty'] = min($original_total,
                $base_cost * 2);
                $archive['maximum_liability_
                display'] = $original_total;
            }
            $archive['condition_display'] =
'DAMAGE/LOST';
        }

        if (!$archive) {
            echo json_encode(['success' =>
false, 'message' => 'Archive record not
found']);
            exit;
        }
    }
}
```




```
renderBadge($badges['manage_books']) ??
0); ?></a></li>
    <li><a href="php echo
$componentPath ?? '../Narrative-
Research-Capstone/'; ?&gt;addBooks.php"
class="<?php echo ($currentPage ?? '') ===
'addbooks' ? 'active' : ''; ?&gt;"&gt;&lt;i class="fa-
brands fa-
researchgate"&gt;&lt;/i&gt;&lt;span&gt;Research
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['narrative_books']) ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? '';
?&gt;BorrowedBooks.php"&gt;&lt;i class="fas fa-
download"&gt;&lt;/i&gt; &lt;span&gt;Borrowed
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['borrowed_books']) ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? ''; ?&gt;Non-
ReturnedBooks.php"&gt;&lt;i class="fas fa-
ban"&gt;&lt;/i&gt; &lt;span&gt;Non-Returned
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['non_returned_books'
] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? '';
?&gt;ReturnedBooks.php"&gt;&lt;i class="fas fa-
upload"&gt;&lt;/i&gt; &lt;span&gt;Returned
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['returned_books']) ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? '';
?&gt;ReplacedBooks.php"&gt;&lt;i class="fas fa-
sync"&gt;&lt;/i&gt; &lt;span&gt;Replaced
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['replaced_books']) ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? ''; ?&gt;PaidBooks.php"&gt;&lt;i
class="fa-solid fa-money-
bill"&gt;&lt;/i&gt;&lt;span&gt;Paid Books&lt;/span&gt;&lt;?php
echo renderBadge($badges['paid_book']) ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? '';
?&gt;ArchivedBooks.php"&gt;&lt;i class="fas fa-
archive"&gt;&lt;/i&gt; &lt;span&gt;Archived
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['archived_books']) ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? '../Narrative-
Research-Capstone/';
?&gt;archived_research_books/archived-
research-books.php" class="<?php echo
($currentPage ?? '') === 'archivedresearch'
? 'active' : ''; ?&gt;"&gt;&lt;i class="fa-solid fa-box-
archive"&gt;&lt;/i&gt; &lt;span&gt;Archived
Research&lt;/span&gt;&lt;?php echo
renderBadge($badges['archived_research_b
ooks']) ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? '';
?&gt;ArchivedOverdueBooks.php"
class="active"&gt;&lt;i class="fas fa-exclamation-
triangle"&gt;&lt;/i&gt; &lt;span&gt;Archived
Overdue&lt;/span&gt;&lt;?php echo
renderBadge($badges['archived_overdue']
?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li class="section-header
appointment-header"&gt;Appointment&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? '';
?&gt;Appointment_Management.php"&gt;&lt;i
class="fas fa-calendar-check"&gt;&lt;/i&gt;
&lt;span&gt;Appointment
Management&lt;/span&gt;&lt;?php echo
renderBadge($badges['appointments']) ?? 0);
?&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li class="section-
header"&gt;Reports&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$componentPath ?? '';
?&gt;ReportAnalytics.php"&gt;&lt;i class="fas fa-
chart-line"&gt;&lt;/i&gt;
&lt;span&gt;Reports&lt;/span&gt;&lt;/a&gt;&lt;/li&gt;
    &lt;li class="sidebar-separator"&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="<?php echo
$sidebarBasePath ?? '../';
?&gt;Admin_Logout.php"&gt;&lt;i class="fas fa-sign-
out-alt"&gt;&lt;/i&gt; &lt;span&gt;Logout&lt;/span&gt;&lt;/a&gt;&lt;/li&gt;
&lt;/ul&gt;
&lt;/nav&gt;

&lt;!-- Main Content --&gt;</pre
```




```
<button class="btn-details"
onclick="closeSettleModal()">Cancel</button>

<button class="btn-settle"
id="confirm-settle-btn"
onclick="confirmSettle()">
    <i class="fas fa-check"></i>
Confirm Settlement
</button>
</div>
</div>
</div>

<!-- Details Modal -->
<div class="modal-overlay" id="details-modal">
    <div class="modal-content">
        <div class="modal-header">
            <h3><i class="fas fa-info-circle"></i> Archive Details</h3>
            <button class="modal-close"
onclick="closeDetailsModal()">&times;</button>
        </div>
        <div class="modal-body" id="details-content">
            <!-- Details will be loaded here -->
        </div>
        <div class="modal-footer">
            <button class="btn-details"
onclick="closeDetailsModal()">Close</button>
        </div>
    </div>
</div>

<script src="../../Js/toast.js"></script>
<script
src="../../Js/AdminArchivedOverdueBooks.js"></script>
<script src="../../NotificationBadge/badge_refresh.js"></script>
</body>
</html>
ArchivedUsers.php
<?php
if (session_status() === PHP_SESSION_NONE) {
    session_start();
    }
    }

    }

    // Check if admin is logged in
    if(!isset($_SESSION['admin_logged_in']) ||
    $_SESSION['admin_logged_in'] !== true) {
        header("Location:
        ../AdminLoginPage.php");
        exit();
    }

    // Include database connection
    require_once
    '../DBconnection/db_connection.php';

    // Badge Notification Function
    if (!isset($badges) && isset($pdo)) {
        require_once __DIR__ . '/../Notification
        Badge/Notification Badge.php';
    }

    $profileImage =
    '../Pictures/PUP_LOGO.png';
    $admin_id = $_SESSION['admin_id'] ?? '';
    $defaultImagePath =
    '../Pictures/PUP_LOGO.png';

    try {
        if (!empty($admin_id)) {
            $stmt = $pdo->prepare("SELECT
            profile_image FROM admin WHERE
            admin_id = ?");
            $stmt->execute([$admin_id]);
            $result = $stmt-
            >fetch(PDO::FETCH_ASSOC);

            if ($result &&
            !empty($result['profile_image'])) {
                $dbPath = $result['profile_image'];
                if (strpos($dbPath, '../uploads/') ===
                0) {
                    $profileImage = '../' . $dbPath;
                } else if (strpos($dbPath, 'uploads/')
                === 0) {
                    $profileImage = '../..' . $dbPath;
                } else {
                    $profileImage =
                    '../..../uploads/admin_profile/' . $dbPath;
                }
            }
        }
    }
```



```
}
} catch (PDOException $e) {
    error_log("Error fetching admin profile: " .
    $e->getMessage());
}

// Ensure archived_users table exists
try {
    $pdo->exec("
        CREATE TABLE IF NOT EXISTS
        `archived_users` (
            `archive_id` int(11) NOT NULL
            AUTO_INCREMENT,
            `original_id` int(11) NOT NULL,
            `user_type`
            enum('student','faculty','visitor') NOT NULL,
            `user_identifier` varchar(50)
            DEFAULT NULL,
            `library_id` varchar(50) DEFAULT
            NULL,
            `email` varchar(255) DEFAULT
            NULL,
            `first_name` varchar(100) DEFAULT
            NULL,
            `last_name` varchar(100) DEFAULT
            NULL,
            `middle_name` varchar(100)
            DEFAULT NULL,
            `suffix` varchar(20) DEFAULT NULL,
            `birthdate` date DEFAULT NULL,
            `age` int(11) DEFAULT NULL,
            `sex` enum('male','female','other')
            DEFAULT NULL,
            `profile_picture` varchar(255)
            DEFAULT NULL,
            `course` varchar(100) DEFAULT
            NULL,
            `department` varchar(100)
            DEFAULT NULL,
            `visitor_type` varchar(50) DEFAULT
            NULL,
            `institution` varchar(255) DEFAULT
            NULL,
            `original_status` varchar(20)
            DEFAULT NULL,
            `registration_date` datetime
            DEFAULT NULL,
            `archived_date` datetime DEFAULT
            CURRENT_TIMESTAMP,
            `archived_by` varchar(50) DEFAULT
            NULL,
            `archive_reason` text DEFAULT
            NULL,
            `is_active` tinyint(1) DEFAULT 0,
            PRIMARY KEY (`archive_id`)
        ) ENGINE=InnoDB DEFAULT
        CHARSET=utf8mb4
        COLLATE=utf8mb4_general_ci
    ");
} catch (Exception $e) {
    error_log("Error creating archived_users
    table: " . $e->getMessage());
}

// Get archived users
$archivedStudents = [];
$archivedFaculty = [];
$archivedVisitors = [];

try {
    $stmt = $pdo->query("SELECT * FROM
    archived_users WHERE is_active = 0
    ORDER BY archived_date DESC");
    $allArchived = $stmt-
    >fetchAll(PDO::FETCH_ASSOC);

    foreach ($allArchived as $user) {
        switch ($user['user_type']) {
            case 'student':
                $archivedStudents[] = $user;
                break;
            case 'faculty':
                $archivedFaculty[] = $user;
                break;
            case 'visitor':
                $archivedVisitors[] = $user;
                break;
        }
    }
} catch (PDOException $e) {
    error_log("Error fetching archived users: "
    . $e->getMessage());
}

$totalArchived = count($archivedStudents) +
count($archivedFaculty) +
count($archivedVisitors);
?>
```




```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>Archived Users - PUP Bansud
Campus</title>
  <link rel="stylesheet" href="../../Notification
Badge/badge.css">
  <link rel="stylesheet"
href="../../Design/Admin.css">
  <link rel="stylesheet"
href="../../Design/AdminSidebar.css">
  <link rel="stylesheet"
href="../../Design/ArchivedUsers.css">
  <link rel="icon" type="image/x-icon"
href="../../Pictures/PUP_LOGO.png">
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">
</head>
<body>
  <?php include_once __DIR__ .
'/sidebar_toggle.php'; ?>
  <!-- Sidebar Navigation -->
  <nav class="sidebar" id="sidebar">
    <div class="sidebar-admin-profile"
id="sidebarAdminProfile"
style="cursor:pointer;"
onclick="window.location.href='../AdminProfil
e.php'">
      ';">
      <div class="sidebar-admin-
username"><h6>Welcome!
Admin</h6><?php echo
htmlspecialchars($_SESSION['admin_usern
ame']) ?? 'admin_user'; ?></div>
    </div>
  </ul>
```

```
    <li><a href="<?php echo
$sidebarBasePath ?? '../';
?>Admin_Dashboard.php"><i class="fa-
solid fa-grip"></i>
<span>Dashboard</span></a></li>
    <li class="section-header Main-
header">Main</li>
    <li><a href="<?php echo
$componentPath ?? ";
?>PendingApproval.php"><i class="fas fa-
user-clock"></i> <span>Pending
Users</span><?php echo
renderBadge($badges['pending_users']) ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>UserManagement.php"><i class="fas fa-
users"></i> <span>User
Management</span><?php echo
renderBadge($badges['user_management']
?? 0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ArchivedUsers.php" class="active"><i
class="fas fa-user-slash"></i>
<span>Archived Users</span><?php echo
renderBadge($badges['archived_users']) ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>Admin_Notification.php"><i class="fas fa-
bell"></i> <span>Notifications</span><?php
echo renderBadge($badges['notifications']
?? 0); ?></a></li>
    <li class="section-header books-
inventory-header">Books Inventory</li>
    <li><a href="<?php echo
$componentPath ?? "; ?>AddBooks.php"><i
class="fa-solid fa-list-
check"></i><span>Manage
Books</span><?php echo
renderBadge($badges['manage_books']) ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? '../Narrative-
Research-Capstone/'; ?>addBooks.php"
class="<?php echo ($currentPage ?? '') ==
'addbooks' ? 'active' : ''; ?>"><i class="fa-
brands fa-
```



```
researchgate"></i><span>Research
Books</span><?php echo
renderBadge($badges['narrative_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>BorrowedBooks.php"><i class="fas fa-
download"></i> <span>Borrowed
Books</span><?php echo
renderBadge($badges['borrowed_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? "; ?>Non-
ReturnedBooks.php"><i class="fas fa-
ban"></i> <span>Non-Returned
Books</span><?php echo
renderBadge($badges['non_returned_books'
] ?? 0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ReturnedBooks.php"><i class="fas fa-
upload"></i> <span>Returned
Books</span><?php echo
renderBadge($badges['returned_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ReplacedBooks.php"><i class="fas fa-
sync"></i> <span>Replaced
Books</span><?php echo
renderBadge($badges['replaced_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? "; ?>PaidBooks.php"><i
class="fa-solid fa-money-
bill"></i><span>Paid Books</span><?php
echo renderBadge($badges['paid_book'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ArchivedBooks.php"><i class="fas fa-
archive"></i> <span>Archived
Books</span><?php echo
renderBadge($badges['archived_books'] ??
0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? '../Narrative-
Research-Capstone/";
?>archived_research_books/archived-
```

```
research-books.php" class="<?php echo
($currentPage ?? ") === 'archivedresearch'
? 'active' : "; ?>"><i class="fa-solid fa-box-
archive"></i> <span>Archived
Research</span><?php echo
renderBadge($badges['archived_research_b
ooks'] ?? 0); ?></a></li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ArchivedOverdueBooks.php"><i
class="fas fa-exclamation-triangle"></i>
<span>Archived Overdue</span><?php
echo
renderBadge($badges['archived_overdue']
?? 0); ?></a></li>
    <li class="section-header
appointment-header">Appointment</li>
    <li><a href="<?php echo
$componentPath ?? ";
?>Appointment_Management.php"><i
class="fas fa-calendar-check"></i>
<span>Appointment
Management</span><?php echo
renderBadge($badges['appointments'] ?? 0);
?></a></li>
    <li class="section-
header">Reports</li>
    <li><a href="<?php echo
$componentPath ?? ";
?>ReportAnalytics.php"><i class="fas fa-
chart-line"></i>
<span>Reports</span></a></li>
    <li class="sidebar-separator"></li>
    <li><a href="<?php echo
$sidebarBasePath ?? '../';
?>Admin_Logout.php"><i class="fas fa-sign-
out-alt"></i> <span>Logout</span></a></li>
</ul>
</nav>

<!-- Main Content -->
<div class="main-content">
    <div class="main-card">
        <a href="UserManagement.php"
class="back-btn">
            <i class="fas fa-arrow-left"></i>
Back to User Management
    </a>
```



```
<section id="archived-users">
  <h2><i class="fas fa-user-
slash"></i> Archived Users</h2>
  <p style="color: #666; margin-
bottom: 20px;">These user accounts are
frozen and cannot login. You can restore
them to reactivate access.</p>

  <div class="filter-tabs">
    <button class="filter-tab active"
data-type="all">
      All <span
class="count"><?php echo $totalArchived;
?></span>
    </button>
    <button class="filter-tab" data-
type="student">
      <i class="fas fa-user-
graduate"></i> Students
      <span class="count"><?php
echo count($archivedStudents); ?></span>
    </button>
    <button class="filter-tab" data-
type="faculty">
      <i class="fas fa-chalkboard-
teacher"></i> Faculty
      <span class="count"><?php
echo count($archivedFaculty); ?></span>
    </button>
    <button class="filter-tab" data-
type="visitor">
      <i class="fas fa-user-tie"></i>
Visitors
      <span class="count"><?php
echo count($archivedVisitors); ?></span>
    </button>
  </div>

  <div class="search-box">
    <input type="text"
id="searchInput" placeholder="Search by
name, ID...">
    <button type="button"
id="searchButton">
      <i class="fas fa-search"></i>
Search
    </button>
  </div>
```

```
<div class="approved-users-
container">
  <?php if($totalArchived > 0): ?>
  <div class="user-list">
    <?php
      $allUsers = array_merge(
        array_map(function($u) {
          $u['_type'] = 'student'; return $u; },
          $archivedStudents),
        array_map(function($u) {
          $u['_type'] = 'faculty'; return $u; },
          $archivedFaculty),
        array_map(function($u) {
          $u['_type'] = 'visitor'; return $u; },
          $archivedVisitors)
      );

      foreach($allUsers as $user):
        $photoPath =
        '../uploads/profile_pictures/' .
        ($user['profile_picture'] ?? 'default.jpg');
        $photoUrl =
        file_exists($photoPath) ? $photoPath :
        '../uploads/profile_pictures/default.jpg';

        $fullName =
        ($user['last_name'] ?? ' ') . ', ' .
        ($user['first_name'] ?? ' ');
        if
        (!empty($user['middle_name'])) {
          $fullName .= ' ' .
          substr($user['middle_name'], 0, 1) . ' ';
        }

        $additionalInfo = "";
        if ($user['_type'] ===
        'student' && !empty($user['course'])) {
          $additionalInfo = '(' .
          htmlspecialchars($user['course']) . ')';
        } elseif ($user['_type'] ===
        'faculty' && !empty($user['department'])) {
          $additionalInfo = '(' .
          htmlspecialchars($user['department']) . ')';
        } elseif ($user['_type'] ===
        'visitor' && !empty($user['visitor_type'])) {
          $additionalInfo = '(' .
          htmlspecialchars($user['visitor_type']) . ')';
        }
      endforeach;
    </?php>
  </div>
```



```
// Prepare user data for
JavaScript
    $userData =
json_encode([
    'archive_id' =>
$user['archive_id'],
    'user_type' =>
$user['_type'],
    'user_identifier' =>
$user['user_identifier'] ?? "",
    'library_id' =>
$user['library_id'] ?? "",
    'email' => $user['email']
    ?? "",
    'first_name' =>
$user['first_name'] ?? "",
    'last_name' =>
$user['last_name'] ?? "",
    'middle_name' =>
$user['middle_name'] ?? "",
    'suffix' => $user['suffix']
    ?? "",
    'full_name' =>
$fullName,
    'birthdate' =>
$user['birthdate'] ?? "",
    'age' => $user['age'] ??
    "",
    'sex' => $user['sex'] ??
    "",
    'course' =>
$user['course'] ?? "",
    'department' =>
$user['department'] ?? "",
    'visitor_type' =>
$user['visitor_type'] ?? "",
    'institution' =>
$user['institution'] ?? "",
    'registration_date' =>
$user['registration_date'] ?? "",
    'archived_date' =>
$user['archived_date'] ?? "",
    'archived_by' =>
$user['archived_by'] ?? "",
    'archive_reason' =>
$user['archive_reason'] ?? "",
    'photo_url' => $photoUrl
    ], JSON_HEX_APOS |
JSON_HEX_QUOT);

?>
<div class="user-item" data-
type="<?php echo $user['_type']; ?>">
    <div class="user-photo">
        
    </div>
    <div class="user-info">
        <div class="user-
id"><?php echo
htmlspecialchars($user['user_identifier'] ??
$user['library_id'] ?? 'No ID'); ?></div>
        <div class="user-
name">- <?php echo
htmlspecialchars($fullName); ?> <?php
echo $additionalInfo; ?></div>
        <div class="archive-
info">
            Archived: <?php echo
date('M j, Y g:i A',
strtotime($user['archived_date'])); ?>
            by <?php echo
htmlspecialchars($user['archived_by'] ??
'Admin'); ?>
        </div>
        <?php if
(!empty($user['archive_reason'])); ?>
        <div class="archive-
reason">
            Reason: <?php echo
htmlspecialchars($user['archive_reason']);
?>
        </div>
    <?php endif; ?>
</div>
<div class="user-status">
    <span class="archived-
badge"><i class="fas fa-lock"></i>
Archived</span>
</div>
<div class="user-actions">
    <button class="view-
details-btn" onclick="viewUserDetails(<?php
echo $userData; ?>)">
        <i class="fas fa-
eye"></i> View
    </button>
    <button class="restore-
btn" onclick="restoreUser(<?php echo
```



```
$user['archive_id']; ?>, '<?php echo
htmlspecialchars(addslashes($fullName));
?>')">
        <i class="fas fa-
undo"></i> Restore
    </button>
</div>
</div>
<?php endforeach; ?>
</div>
<?php else: ?>
<div class="empty-state">
    <i class="fas fa-user-
check"></i>
    <p>No archived users found.
All user accounts are active.</p>
</div>
<?php endif; ?>
</div>
</section>
</div>
</div>

<!-- Toast Message -->
<div id="toast" class="toast"></div>

<!-- View Details Modal -->
<div id="detailsModal" class="details-
modal-overlay">
    <div class="details-modal">
        <div class="details-modal-header">
            <h3><i class="fas fa-user-
slash"></i> Archived User Details</h3>
            <button class="details-modal-
close" onclick="closeDetailsModal()">
                <i class="fas fa-times"></i>
            </button>
        </div>
        <div class="details-modal-body">
            <!-- User Profile Header -->
            <div class="user-profile-header">
                <div class="user-profile-photo">
                    
                </div>
                <div class="user-profile-info">
                    <h4
id="modalUserName">User Name</h4>
```

```
        <span id="modalUserType"
class="user-type-badge
student">Student</span>
        </div>
    </div>

    <!-- User Details Grid -->
    <div class="details-grid">
        <div class="detail-item">
            <div class="detail-
label">User ID</div>
            <div class="detail-value"
id="modalUserId"></div>
        </div>
        <div class="detail-item">
            <div class="detail-
label">Email Address</div>
            <div class="detail-value"
id="modalEmail"></div>
        </div>
        <div class="detail-item">
            <div class="detail-
label">Birthdate</div>
            <div class="detail-value"
id="modalBirthdate"></div>
        </div>
        <div class="detail-item">
            <div class="detail-
label">Age</div>
            <div class="detail-value"
id="modalAge"></div>
        </div>
        <div class="detail-item">
            <div class="detail-
label">Sex</div>
            <div class="detail-value"
id="modalSex"></div>
        </div>
        <div class="detail-item">
            <div class="detail-
label">Course/Department</div>
            <div class="detail-value"
id="modalCourseOrDept"></div>
        </div>
        <div class="detail-item hidden"
id="institutionRow">
            <div class="detail-
label">Institution</div>
```



```

        <div class="detail-value"
id="modalInstitution">-</div>
        </div>
        <div class="detail-item">
            <div class="detail-
label">Registration Date</div>
            <div class="detail-value"
id="modalRegDate">-</div>
            </div>
        </div>

        <!-- Archive Information Section --
>
        <div class="archive-section">
            <h5><i class="fas fa-
exclamation-triangle"></i> Archive
Information</h5>
            <div class="details-grid">
                <div class="detail-item">
                    <div class="detail-
label">Archived Date</div>
                    <div class="detail-value"
id="modalArchiveDate">-</div>
                    </div>
                <div class="detail-item">
                    <div class="detail-
label">Archived By</div>
                    <div class="detail-value"
id="modalArchivedBy">-</div>
                    </div>
                <div class="detail-item full-
width">
                    <div class="detail-
label">Reason for Archiving</div>
                    <div class="detail-value"
id="modalArchiveReason">-</div>
                    </div>
                </div>
            </div>
            <div class="details-modal-footer">
                <button class="modal-btn modal-
btn-close" onclick="closeDetailsModal()">
                    <i class="fas fa-times"></i>
Close
                </button>
                <button id="modalRestoreBtn"
class="modal-btn modal-btn-restore"
onclick="restoreFromModal()">

```

```

        </button>
    </div>
</div>
</div>

<script src="../../../Js/toast.js"></script>
<script
src="../../../Js/AdminArchivedUser.js"></script>
>
    <script
src="../../../Js/SidebarToggle.js"></script>
    <script src="../../../Notification
Badge/badge_refresh.js"></script>
</body>
</html>

BorrowedBooks.php
<?php
require_once '../../../DBconnection/Config.php';
require_once
'../../../DBconnection/db_connection.php';
//Badge Notification Function
if (!isset($badges) && isset($pdo)) {
    require_once __DIR__ . '../../../Notification
Badge/Notification Badge.php';
}

if (session_status() ===
PHP_SESSION_NONE) {
    session_start();
}

// Only define the constant if it's not already
defined
if (!defined('DEFAULT_ADMIN_IMAGE')) {
    define('DEFAULT_ADMIN_IMAGE',
'../../../Pictures/PUP_LOGO.png');
}

// Get admin_id from the session
$admin_id = $_SESSION['admin_id'] ?? "";
$profileImage = DEFAULT_ADMIN_IMAGE;
// Initialize with default

// When fetching the profile image
try {
    $stmt = $pdo->prepare("SELECT
profile_image FROM admin WHERE
admin_id = ?");

```




```
$stmt->execute([$admin_id]);
$result = $stmt-
>fetch(PDO::FETCH_ASSOC);

if ($result &&
!empty($result['profile_image'])) {
    $profileImage =
getImageUrl($result['profile_image'],
DEFAULT_ADMIN_IMAGE);
} else {
    $profileImage =
DEFAULT_ADMIN_IMAGE;
}
} catch (PDOException $e) {
    error_log("Error fetching admin profile: " .
    $e->getMessage());
    $profileImage =
DEFAULT_ADMIN_IMAGE;
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    <title>Borrowed Books - PUP Bansud
Campus</title>
    <link rel="stylesheet"
href=".../Design/Admin.css">
    <link rel="stylesheet"
href=".../Design/AdminBorrowed.css">
    <link rel="stylesheet"
href=".../Design/AdminSidebar.css">
    <link rel="stylesheet"
href=".../Design/AdminBooks/AdminB-
Tabs.css">
    <link rel="stylesheet"
href=".../Design/AdminBooks/AdminB-
BooksModal.css">
    <link rel="stylesheet" href=".../Notification
Badge/badge.css">
    <link rel="icon" type="image/x-icon"
href=".../Pictures/PUP_LOGO.png">
    <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.0/css/all.min.css">

</head>
<body>
    <!-- Sidebar profile -->
    <nav class="sidebar" id="sidebar">
        <div class="sidebar-admin-profile"
id="sidebarAdminProfile"
style="cursor:pointer;"
onclick="window.location.href='../AdminProfil
e.php">
            ';">
            <div class="sidebar-admin-
username"><h6>Welcome!
Admin</h6><?php echo
htmlspecialchars($_SESSION['admin_usern
ame'] ?? 'admin_user'); ?></div>
        </div>
        <ul>
            <li><a href="<?php echo
$sidebarBasePath ?? '../';
?>Admin_Dashboard.php" class="<?php
echo ($currentPage ?? '') === 'dashboard' ?
'active' : ''; ?>"><i class="fa-solid fa-
grip"></i>
<span>Dashboard</span></a></li>
            <li class="section-header Main-
header">Main</li>
            <li><a href="<?php echo
$componentPath ?? '';
?>PendingApproval.php" class="<?php
echo ($currentPage ?? '') === 'pending' ?
'active' : ''; ?>"><i class="fas fa-user-
clock"></i> <span>Pending
Users</span><?php echo
renderBadge($badges['pending_users'] ??
0); ?></a></li>
            <li><a href="<?php echo
$componentPath ?? '';
?>UserManagement.php" class="<?php
echo ($currentPage ?? '') === 'users' ?
'active' : ''; ?>"><i class="fas fa-users"></i>
<span>User Management</span><?php
echo
renderBadge($badges['user_management']
?? 0); ?></a></li>
```



```
</li><a href="php echo
$componentPath ?? ";
?&gt;ArchivedUsers.php" class="<?php echo
($currentPage ?? ") === 'archivedusers' ?
'active' : "; ?&gt;"&gt;&lt;i class="fas fa-user-
slash"&gt;&lt;/i&gt; &lt;span&gt;Archived
Users&lt;/span&gt;&lt;?php echo
renderBadge($badges['archived_users'] ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? ";
?&gt;Admin_Notification.php" class="<?php
echo ($currentPage ?? ") === 'notifications'
? 'active' : "; ?&gt;"&gt;&lt;i class="fas fa-bell"&gt;&lt;/i&gt;
&lt;span&gt;Notifications&lt;/span&gt;&lt;?php echo
renderBadge($badges['notifications'] ?? 0);
?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li class="section-header books-
inventory-header"&gt;Books Inventory&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? "; ?&gt;AddBooks.php"
class="<?php echo ($currentPage ?? ") ===
'addbooks' ? 'active' : "; ?&gt;"&gt;&lt;i class="fa-
solid fa-list-check"&gt;&lt;/i&gt;&lt;span&gt;Manage
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['manage_books'] ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? './../Narrative-
Research-Capstone/"; ?&gt;addBooks.php"
class="<?php echo ($currentPage ?? ") ===
'addbooks' ? 'active' : "; ?&gt;"&gt;&lt;i class="fa-
brands fa-
researchgate"&gt;&lt;/i&gt;&lt;span&gt;Research
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['narrative_books'] ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? ";
?&gt;BorrowedBooks.php" class="<?php echo
($currentPage ?? ") === 'borrowed' ? 'active'
: "; ?&gt;"&gt;&lt;i class="fas fa-download"&gt;&lt;/i&gt;
&lt;span&gt;Borrowed Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['borrowed_books'] ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? "; ?&gt;Non-
ReturnedBooks.php" class="<?php echo
($currentPage ?? ") === 'nonreturned' ?
'active' : "; ?&gt;"&gt;&lt;i class="fas fa-ban"&gt;&lt;/i&gt;
&lt;span&gt;Non-Returned Books&lt;/span&gt;&lt;?php
echo
renderBadge($badges['non_returned_books'
] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? ";
?&gt;ReturnedBooks.php" class="<?php echo
($currentPage ?? ") === 'returned' ? 'active' :
"; ?&gt;"&gt;&lt;i class="fas fa-upload"&gt;&lt;/i&gt;
&lt;span&gt;Returned Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['returned_books'] ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? ";
?&gt;ReplacedBooks.php" class="<?php echo
($currentPage ?? ") === 'replaced' ? 'active'
: "; ?&gt;"&gt;&lt;i class="fas fa-sync"&gt;&lt;/i&gt;
&lt;span&gt;Replaced Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['replaced_books'] ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? "; ?&gt;PaidBooks.php"
class="<?php echo ($currentPage ?? ") ===
'paidbook' ? 'active' : "; ?&gt;"&gt;&lt;i class="fa-
solid fa-money-bill"&gt;&lt;/i&gt;&lt;span&gt;Paid
Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['paid_book'] ?? 0);
?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? ";
?&gt;ArchivedBooks.php" class="<?php echo
($currentPage ?? ") === 'archived' ? 'active'
: "; ?&gt;"&gt;&lt;i class="fas fa-archive"&gt;&lt;/i&gt;
&lt;span&gt;Archived Books&lt;/span&gt;&lt;?php echo
renderBadge($badges['archived_books'] ??
0); ?&gt;&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="<?php echo
$componentPath ?? './../Narrative-
Research-Capstone/";
?&gt;archived_research_books/archived-
research-books.php" class="<?php echo
($currentPage ?? ") === 'archivedresearch'
? 'active' : "; ?&gt;"&gt;&lt;i class="fa-solid fa-box-
archive"&gt;&lt;/i&gt; &lt;span&gt;Archived
Research&lt;/span&gt;&lt;?php echo
renderBadge($badges['archived_research_b
ooks'] ?? 0); ?&gt;&lt;/a&gt;&lt;/li&gt;</pre
```




```
</li><a href="<?php echo
$componentPath ?? ";
?>ArchivedOverdueBooks.php"
class="<?php echo ($currentPage ?? ") ===
'archivedoverdue' ? 'active' : "; ?>"><i
class="fas fa-exclamation-triangle"></i>
<span>Archived Overdue</span><?php echo
renderBadge($badges['archived_overdue']
?? 0); ?></a></li>
<li class="section-header
appointment-header">Appointment</li>
<li><a href="<?php echo
$componentPath ?? ";
?>Appointment_Management.php"
class="<?php echo ($currentPage ?? ") ===
'appointments' ? 'active' : "; ?>"><i
class="fas fa-calendar-check"></i>
<span>Appointment
Management</span><?php echo
renderBadge($badges['appointments'] ?? 0);
?></a></li>
<li class="section-
header">Reports</li>
<li><a href="<?php echo
$componentPath ?? ";
?>ReportAnalytics.php" class="<?php echo
($currentPage ?? ") === 'reports' ? 'active' :
"; ?>"><i class="fas fa-chart-line"></i>
<span>Reports</span></a></li>
<li class="sidebar-separator"></li>
<li><a href="<?php echo
$sidebarBasePath ?? './../';
?>Admin_Logout.php"><i class="fas fa-sign-
out-alt"></i> <span>Logout</span></a></li>
</ul>
</nav>

<!-- Main Content -->
<div class="main-content">
<!-- Page Header -->
<div class="page-header">
<h1><i class="fas fa-book"></i>
Book Borrowing Management</h1>
<p>Manage book borrowing
requests and borrowed books</p>
</div>

<!-- Tabs Navigation - All 3 Tabs -->
```

```
<div class="tabs-container">
<div class="tabs-header">
<button class="tab-button active"
data-tab="pending-tab">
<i class="fas fa-clock"></i>
Pending Requests
<span class="badge"
id="pending-count">0</span>
</button>

<button class="tab-button" data-
tab="approved-tab">
<i class="fas fa-check-
circle"></i> Borrowed
<span class="badge"
id="approved-count">0</span>
</button>

<button class="tab-button" data-
tab="replace-tab">
<i class="fas fa-sync-alt"></i>
Replacements
<span class="badge"
id="replace-count">0</span>
</button>
</div>

<!-- Pending Requests Tab -->
<div id="pending-tab" class="tab-
content active">
<div class="filter-row">
<div class="search-container">
<input type="text"
id="pending-search" placeholder="Search
by title, name, ID or any keyword...">
<button id="pending-search-
btn"><i class="fas fa-search"></i></button>
</div>
<div class="filter-container">
<select id="pending-user-
type">
<option value="all">All
User Types</option>
<option
value="student">Students</option>
<option
value="faculty">Faculty</option>
</select>
```



```

</div>

<div class="info-banner">
  <i class="fas fa-info-circle"></i>
  <div>
    <strong>Pending
Request:</strong><br>
    Books listed here require
approval first before issuing. Upon lapsing of
the 24 Hours (Twenty Four Hours)
timeframe the Requested Book will be
Available again for Requesting.<br>
  </div>
</div>

<!-- This container will hold the
book cards -->
<div id="pending-container"
class="books-container">
  <!-- Cards will be generated
here by JavaScript -->
</div>
</div>

<!-- Approved/Borrowed Tab -->
<div class="tab-content"
id="approved-tab">
  <div class="filter-row">
    <div class="search-container">
      <input type="text"
id="approved-search" placeholder="Search
by title, name, ID...">
      <button type="button"><i
class="fas fa-search"></i></button>
    </div>
    <div class="filter-container">
      <select id="approved-user-
type">
        <option value="all">All
User Types</option>
        <option
value="student">Student</option>
        <option
value="faculty">Faculty</option>
      </select>
    </div>
    <div class="filter-container">
      <select id="approved-
status">
        <option value="all">All
Status</option>
        <option
value="borrowed">Borrowed</option>
        <option
value="returned">Returned</option>
        <option
value="lost/damage">DAMAGE/LOST</opti
on>
      </select>
    </div>
  </div>

  <div class="info-banner">
    <i class="fas fa-info-circle"></i>
    <div>
      <strong>Borrowed:</strong><br>
      Books listed here Require
Validation Based on the Physical Condition
Set by the Returning Borrower (Good,
Damage/Lost).
      Penalties Will Halt during
Non-Working Days such as Sunday and
Continue after it ended.<br>
      <strong>Fine
System:</strong> Fines start after the due
date at ₱2.09/hour (The 3-day grace period
is the borrowing duration).
      Fines are capped at the book
price. Once the cap is reached, the book
moves to Non-Returned Books for
settlement.<br>
    </div>
  </div>

  <!-- Replace table with cards
container -->
  <div id="approved-container"
class="books-container">
    <!-- Cards will be generated
here by JavaScript -->
  </div>
</div>

<!-- Replacements Tab -->

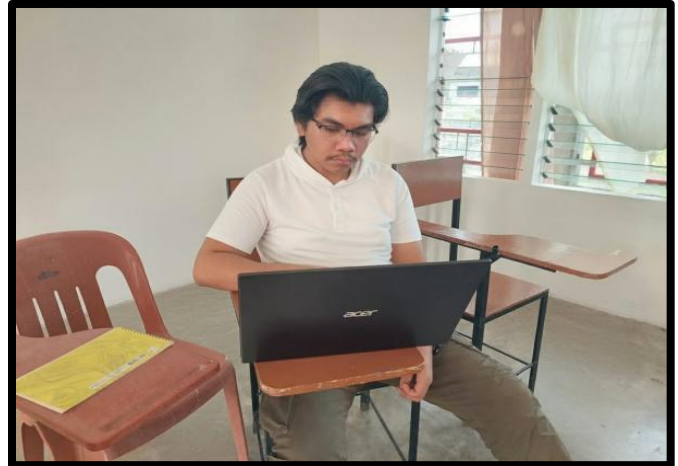
```




<img id="approval-book-condition-image"
src="" alt="Book Condition"

**Appendix 14****PHOTO DOCUMENTATION**

Edil Con L. Gorospe served as the Lead Programmer. He was primarily responsible for the design, development, implementation, and integration of the overall system. He developed the core functionalities, integrated all system modules, ensured



that each component operated according to the specified project requirements, and resolved technical issues encountered throughout the development process. He also supervised the programming activities to ensure the reliability, functionality, and successful implementation of the system.



Joshua D. Javier served as the Assistant Programmer. He supported the lead programmer in the development and implementation of the system. He was responsible for developing assigned system modules, implementing enhancements and modifications, integrating program components, fixing identified system errors, and assisting in testing and debugging activities. He also contributed to ensuring the system's reliability,

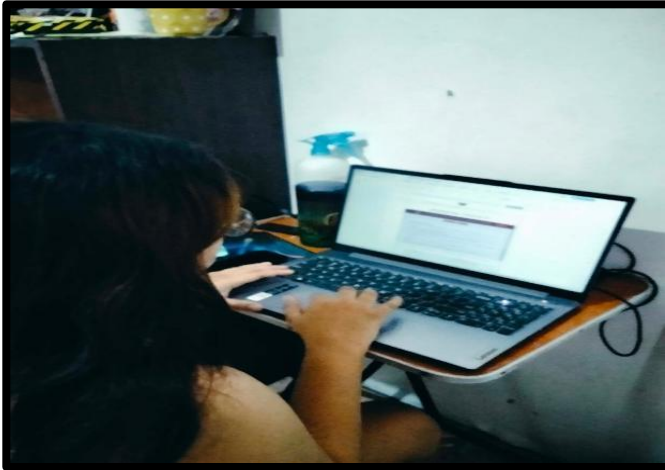


functionality, and overall performance by verifying that implemented features operated according to the project requirements.



Kristine C. Rodeo served as the Technical Writer and was responsible for preparing, organizing, compiling, and maintaining the capstone project's documentation. She wrote, revised, and edited the manuscript, documented each phase of the system development

process, and ensured that all project documentation was accurate, complete, consistent, and properly formatted in accordance with the prescribed institutional guidelines. She was primarily responsible for the overall compilation and finalization of the manuscript, ensuring that all chapters, appendices, and supporting materials were complete and properly arranged for submission. In addition, she prepared and managed the official project documents required by the institution, including letters, questionnaires, UREC application documents, copyright application requirements, and other administrative documents necessary for project approval and completion. She also reviewed all documentation for clarity and correctness and ensured that all required documents were submitted within the established project timeline.



Pearl Lorane M. Rafol served as the System Analyst, was responsible for gathering and analyzing the project requirements and evaluating the existing library processes to determine the system's functional and technical needs. She prepared the necessary

system analysis and design documents, including the Gantt Chart, System Flowchart for User, Entity Relationship Diagram (ERD), and User's Guide for User. She also documented the system requirements, validated the proposed design with the development team, and ensured that the system architecture and specifications aligned with the project's objectives and user requirements.

Mariadel B. Falculan served as the Project Manager, was responsible for leading and supervising the development team throughout the project. In collaboration with the System Analyst, she participated in the planning, preparation, and development of the System Flowchart for the



Administrator side, ensuring that the proposed system processes aligned with the project requirements and supported the effective implementation of the administrative functions.



TITLE DEFENSE



PRE-ORAL DEFENSE





FINAL DEFENSE





Appendix 15

CURRICULUM VITAE

MARIADEL B. FALCULAN

Proper Bansud, Bansud, Oriental Mindoro

+639919160571

falculanmariadelb@gmail.com

OBJECTIVES: To gain hands-on research experience by conducting ethical and meaningful studies that benefit society and advance knowledge in my field, while always protecting the rights and well-being of research participants.

PERSONAL DATA BACKGROUND

Date of Birth: October 25, 2003
Place of Birth: Tambo, Ligao City, Albay
Sex: Female
Civil Status: Single
Citizenship: Filipino
Religion: Roman Catholic
Height: 4'9
Weight: 62 kg.
Present Address: Tambong, Proper Bansud, Oriental Mindoro

EDUCATIONAL BACKGROUND

Elementary: **Proper Bansud Elementary School**
Proper Bansud, Bansud, Oriental Mindoro

Junior & Senior
High School: **Pag-Asa National High School**
Pag-Asa, Bansud, Oriental Mindoro

Tertiary: **Polytechnic University of the Philippines**
Poblacion, Bansud, Oriental Mindoro
Undergraduate

I do hereby certify that the above information is true and correct to the best of my knowledge and belief.

MARIADEL B. FALCULAN
Student

**EDIL CON L. GOROSPE**

Nabuslot, St. Bernadette Phase 1,
Pinamalayan, Oriental Mindoro
+639291706231
gorospeedilcon@gmail.com



OBJECTIVES: Seeking to leverage hands-on research experience to produce socially beneficial studies, ensuring all methodologies remain ethically sound and participant-centered.

PERSONAL DATA BACKGROUND

Date of Birth:	December 20, 2003
Place of Birth:	Perpetual Succor, Sampaloc, Manila
Sex:	Male
Civil Status:	Single
Citizenship:	Filipino
Religion:	Roman Catholic
Height:	5'1
Weight:	76 kg
Present Address:	Nabuslot, Pinamalayan, Oriental Mindoro

EDUCATIONAL BACKGROUND

Elementary:	Nabuslot Central School Nabuslot, Pinamalayan, Oriental Mindoro
Junior High School:	Immaculate Heart of Mary Academy (IHMA) Madrid Avenue, Pinamalayan, Oriental Mindoro
Senior High School:	Divine Word College of Calapan Pinamalayan Campus Sta. Rita, Pinamalayan, Oriental Mindoro
Tertiary:	Polytechnic University of the Philippines Poblacion, Bansud, Oriental Mindoro Undergraduate

I do hereby certify that the above information is true and correct to the best of my knowledge and belief.

EDIL CON L. GOROSPE
Student

**JOSHUA D. JAVIER**

Proper Tiguisan, Bansud, Oriental Mindoro

+639858909520

langjok9@gmail.com

OBJECTIVES: To develop practical skills in networking and other IT fields by applying technical knowledge to real-world systems and maintaining secure, efficient, and reliable technology environments.

PERSONAL DATA BACKGROUND

Date of Birth: March 21, 2003
Place of Birth: Proper Tiguisan, Bansud, Oriental Mindoro
Sex: Male
Civil Status: Single
Citizenship: Filipino
Religion: Iglesia Ni Cristo
Height: 5'3
Weight: 70 kg
Present Address: Proper Tiguisan, Bansud, Oriental Mindoro

EDUCATIONAL BACKGROUND

Elementary: **Policarpio Hernandez Memorial Elementary School**
Proper Tiguisan, Bansud, Oriental Mindoro

Junior & Senior High School: **Alcadesma National High School**
Alcadesma, Bansud, Oriental Mindoro

Tertiary: **Polytechnic University of the Philippines**
Poblacion, Bansud, Oriental Mindoro
Undergraduate

I do hereby certify that the above information is true and correct to the best of my knowledge and belief.

JOSHUA D. JAVIER
Student

**PEARL LORANE M. RAFOL**

Banus, Gloria, Oriental Mindoro

+639368525009

rafolpearl@gmail.com

OBJECTIVES: To gain hands-on research experience by conducting ethical and meaningful studies that benefit society and advance knowledge in my field, while always protecting the rights and well-being of research participants.

PERSONAL DATA BACKGROUND

Date of Birth: December 14, 2003
Place of Birth: Banus, Gloria, Oriental Mindoro
Sex: Female
Civil Status: Single
Citizenship: Filipino
Religion: Roman Catholic
Height: 5'0
Weight: 58 kg
Present Address: Banus, Gloria, Oriental Mindoro

EDUCATIONAL BACKGROUND

Elementary: **Banus Elementary School**
Banus, Gloria, Oriental Mindoro

Junior & Senior
High School: **Pag-Asa National High School**
Pag-Asa, Bansud, Oriental Mindoro

Tertiary: **Polytechnic University of the Philippines**
Poblacion, Bansud, Oriental Mindoro
Undergraduate

I do hereby certify that the above information is true and correct to the best of my knowledge and belief.

PEARL LORANE M. RAFOL
Student

**KRISTINE C. RODEO**

Malamig, Salcedo, Bansud, Oriental Mindoro

+639939395098

kristinerodeo915@gmail.com

OBJECTIVES: To contribute to meaningful and ethical research within a forward-thinking organization that values innovation, continuous learning, and professional growth, while supporting initiatives that advance knowledge and benefit society.

PERSONAL DATA BACKGROUND

Date of Birth: February 14, 2003
Place of Birth: Malamig, Salcedo, Bansud, Oriental Mindoro
Sex: Female
Civil Status: Single
Citizenship: Filipino
Religion: Iglesia Ni Cristo
Height: 5'2
Weight: 47 kg
Present Address: Malamig, Salcedo, Bansud, Oriental Mindoro

EDUCATIONAL BACKGROUND

Elementary: **Tiguisan Elementary School**
Alcadesma, Bansud, Oriental Mindoro

Junior & Senior High School: **Alcadesma National High School**
Alcadesma, Bansud, Oriental Mindoro

Tertiary: **Polytechnic University of the Philippines**
Poblacion, Bansud, Oriental Mindoro
Undergraduate

I do hereby certify that the above information is true and correct to the best of my knowledge and belief.

KRISTINE C. RODEO
Student